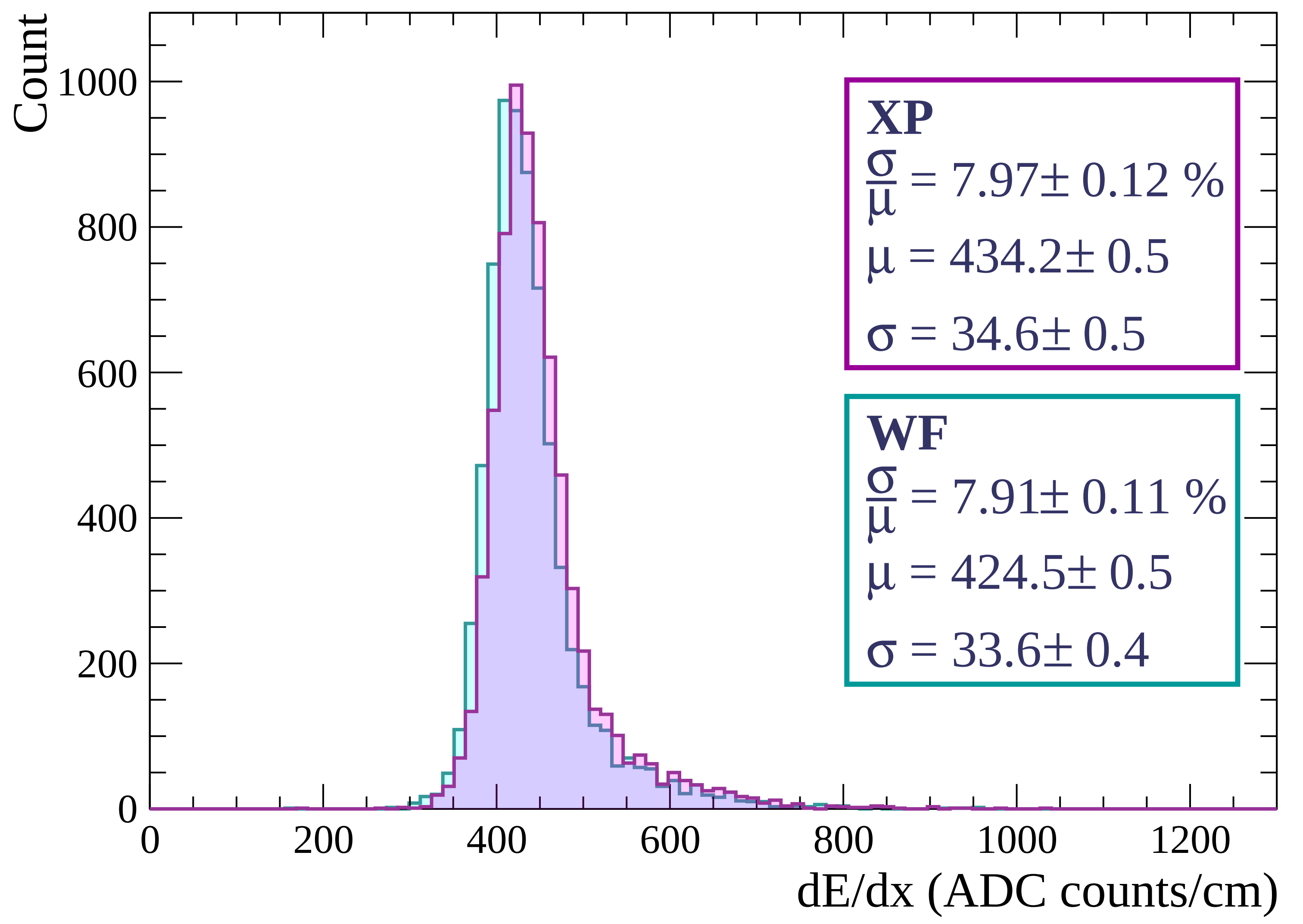
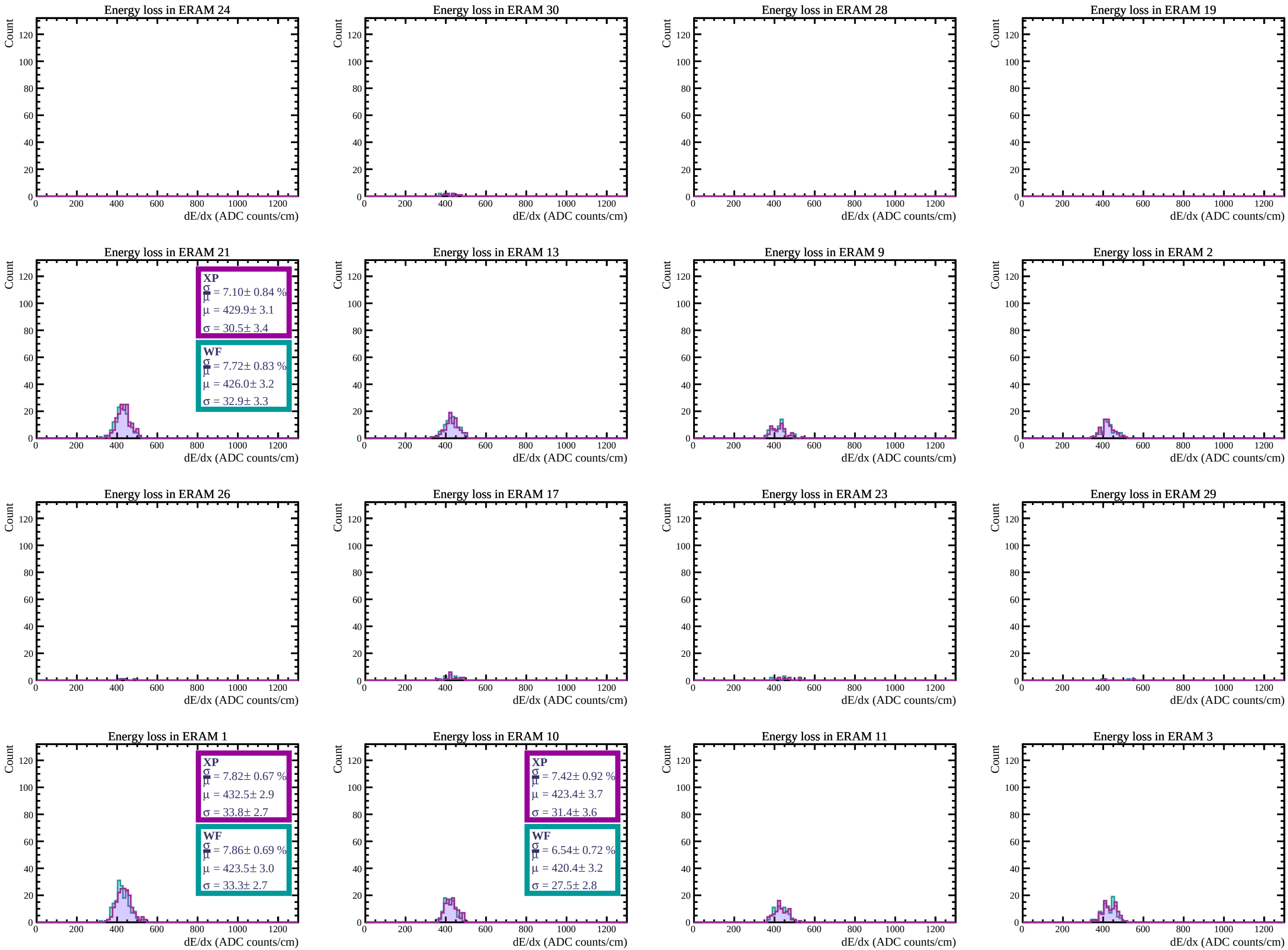
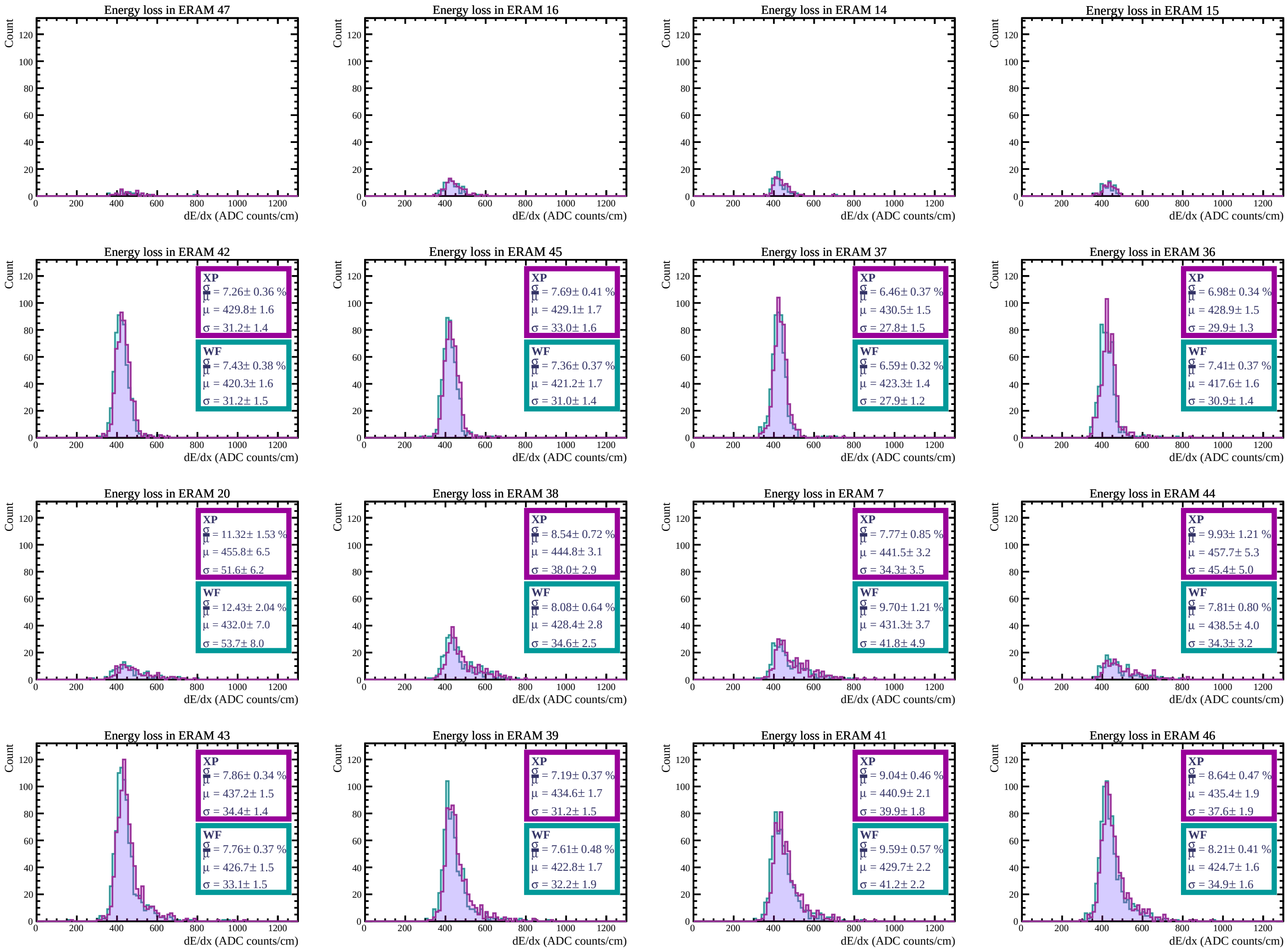
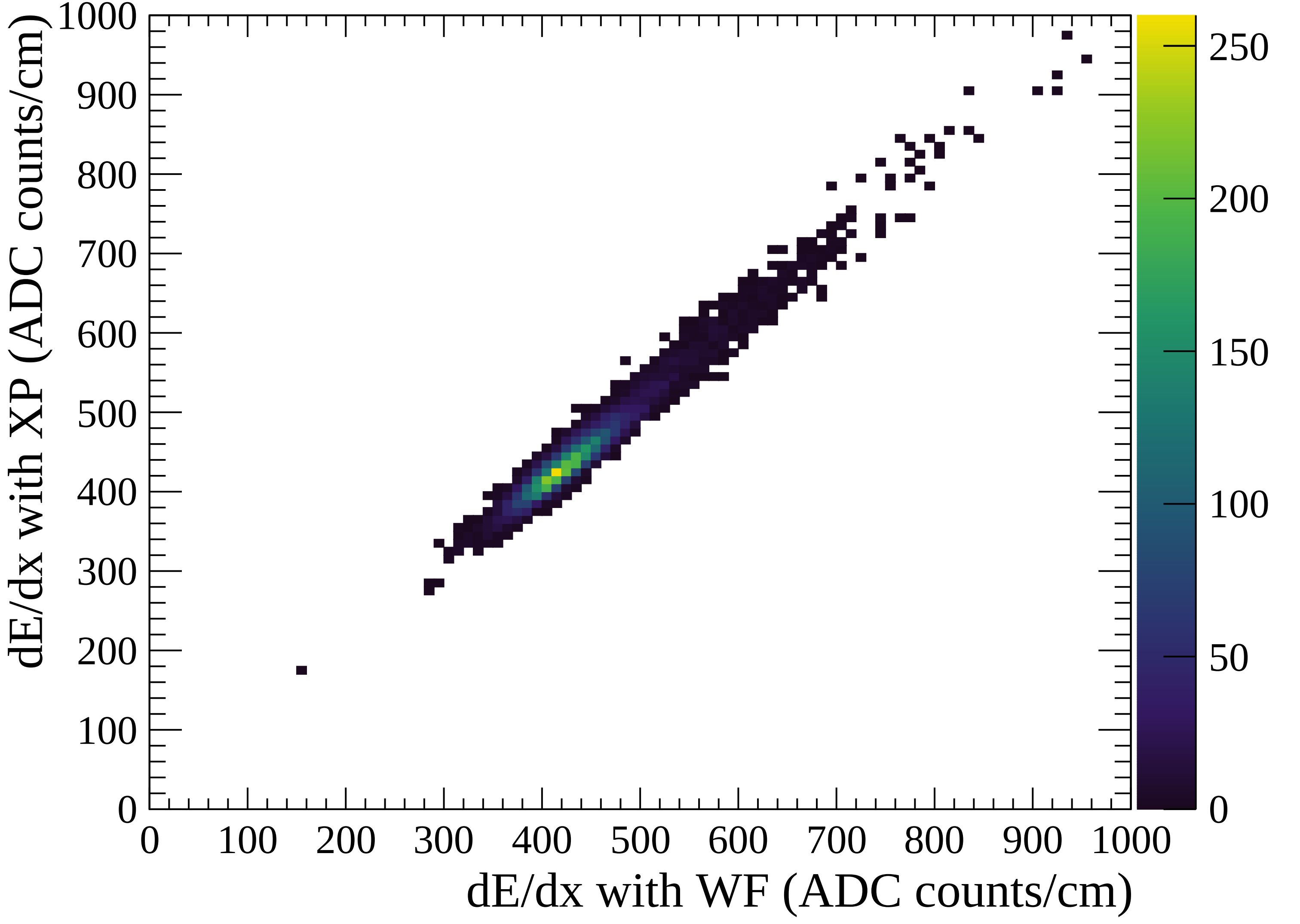

Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.97± 0.12 %
μ = 434.2± 0.5
σ = 34.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 7.91± 0.11 %
μ = 424.5± 0.5
σ = 33.6± 0.4



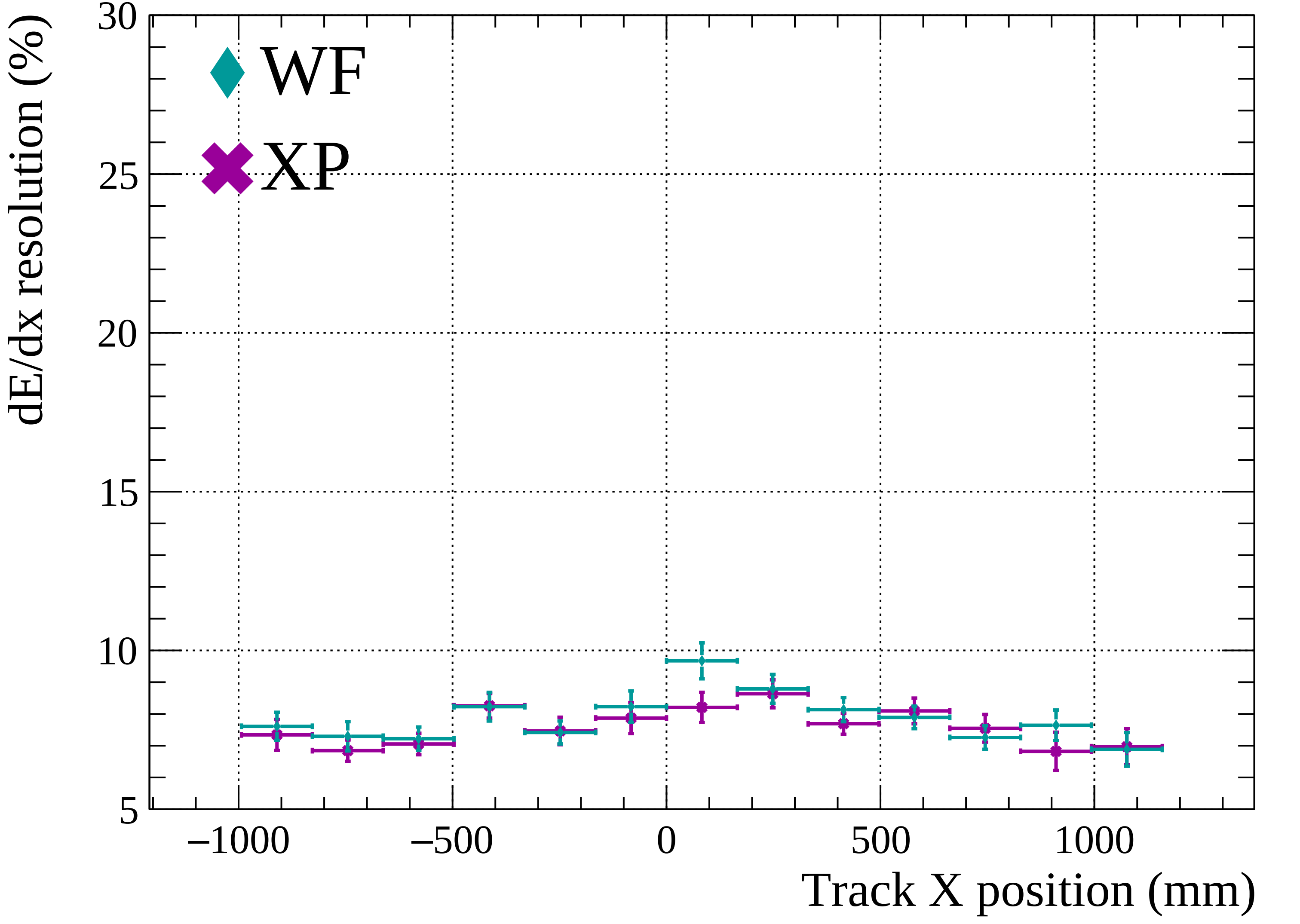
Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
XP
σ/μ = 7.10± 0.84 %
μ = 429.9± 3.1
σ = 30.5± 3.4
WF
σ/μ = 7.72± 0.83 %
μ = 426.0± 3.2
σ = 32.9± 3.3
Energy loss in ERAM 13
Energy loss in ERAM 9
Energy loss in ERAM 2
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
Energy loss in ERAM 1
XP
σ/μ = 7.82± 0.67 %
μ = 432.5± 2.9
σ = 33.8± 2.7
WF
σ/μ = 7.86± 0.69 %
μ = 423.5± 3.0
σ = 33.3± 2.7
Energy loss in ERAM 10
XP
σ/μ = 7.42± 0.92 %
μ = 423.4± 3.7
σ = 31.4± 3.6
WF
σ/μ = 6.54± 0.72 %
μ = 420.4± 3.2
σ = 27.5± 2.8
Energy loss in ERAM 11
Energy loss in ERAM 3
Count
dE/dx (ADC counts/cm)



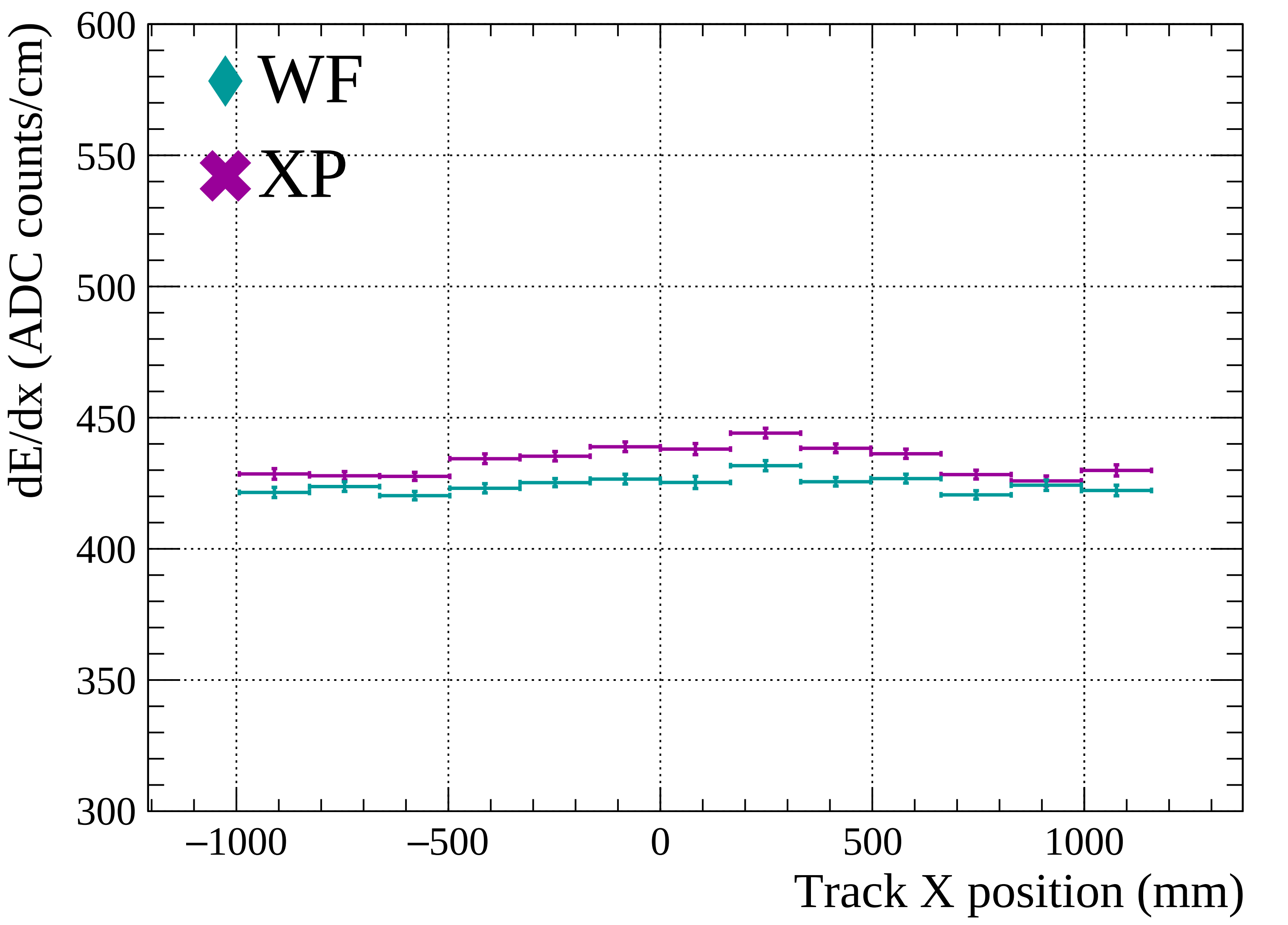
Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
XP
σ/μ = 7.26± 0.36 %
μ = 429.8± 1.6
σ = 31.2± 1.4
WF
σ/μ = 7.43± 0.38 %
μ = 420.3± 1.6
σ = 31.2± 1.5
Energy loss in ERAM 45
XP
σ/μ = 7.69± 0.41 %
μ = 429.1± 1.7
σ = 33.0± 1.6
WF
σ/μ = 7.36± 0.37 %
μ = 421.2± 1.7
σ = 31.0± 1.4
Energy loss in ERAM 37
XP
σ/μ = 6.46± 0.37 %
μ = 430.5± 1.5
σ = 27.8± 1.5
WF
σ/μ = 6.59± 0.32 %
μ = 423.3± 1.4
σ = 27.9± 1.2
Energy loss in ERAM 36
XP
σ/μ = 6.98± 0.34 %
μ = 428.9± 1.5
σ = 29.9± 1.3
WF
σ/μ = 7.41± 0.37 %
μ = 417.6± 1.6
σ = 30.9± 1.4
Energy loss in ERAM 20
XP
σ/μ = 11.32± 1.53 %
μ = 455.8± 6.5
σ = 51.6± 6.2
WF
σ/μ = 12.43± 2.04 %
μ = 432.0± 7.0
σ = 53.7± 8.0
Energy loss in ERAM 38
XP
σ/μ = 8.54± 0.72 %
μ = 444.8± 3.1
σ = 38.0± 2.9
WF
σ/μ = 8.08± 0.64 %
μ = 428.4± 2.8
σ = 34.6± 2.5
Energy loss in ERAM 7
XP
σ/μ = 7.77± 0.85 %
μ = 441.5± 3.2
σ = 34.3± 3.5
WF
σ/μ = 9.70± 1.21 %
μ = 431.3± 3.7
σ = 41.8± 4.9
Energy loss in ERAM 44
XP
σ/μ = 9.93± 1.21 %
μ = 457.7± 5.3
σ = 45.4± 5.0
WF
σ/μ = 7.81± 0.80 %
μ = 438.5± 4.0
σ = 34.3± 3.2
Energy loss in ERAM 43
XP
σ/μ = 7.86± 0.34 %
μ = 437.2± 1.5
σ = 34.4± 1.4
WF
σ/μ = 7.76± 0.37 %
μ = 426.7± 1.5
σ = 33.1± 1.5
Energy loss in ERAM 39
XP
σ/μ = 7.19± 0.37 %
μ = 434.6± 1.7
σ = 31.2± 1.5
WF
σ/μ = 7.61± 0.48 %
μ = 422.8± 1.7
σ = 32.2± 1.9
Energy loss in ERAM 41
XP
σ/μ = 9.04± 0.46 %
μ = 440.9± 2.1
σ = 39.9± 1.8
WF
σ/μ = 9.59± 0.57 %
μ = 429.7± 2.2
σ = 41.2± 2.2
Energy loss in ERAM 46
XP
σ/μ = 8.64± 0.47 %
μ = 435.4± 1.9
σ = 37.6± 1.9
WF
σ/μ = 8.21± 0.41 %
μ = 424.7± 1.6
σ = 34.9± 1.6
Count
dE/dx (ADC counts/cm)



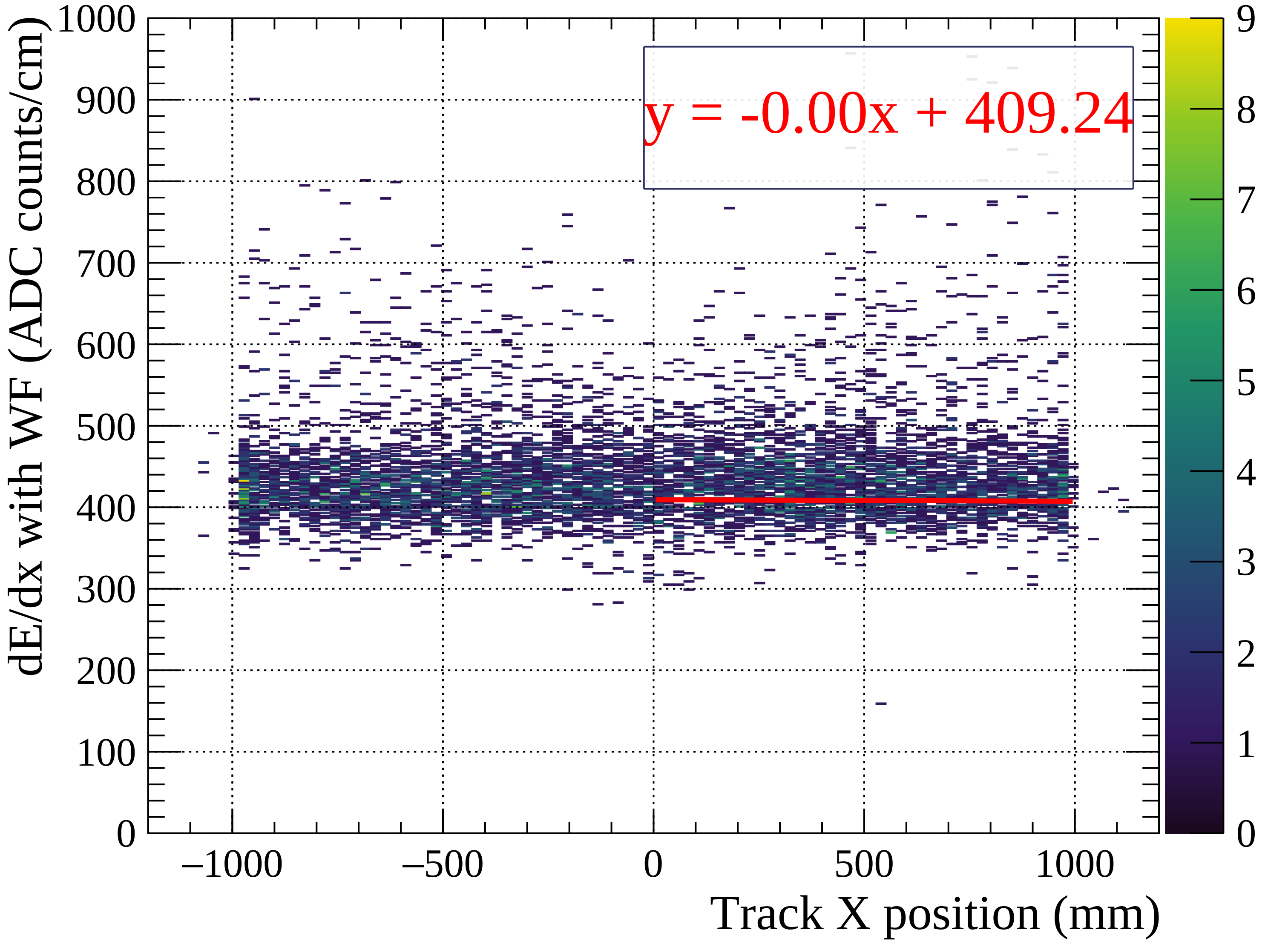
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
0
50
100
150
200
250



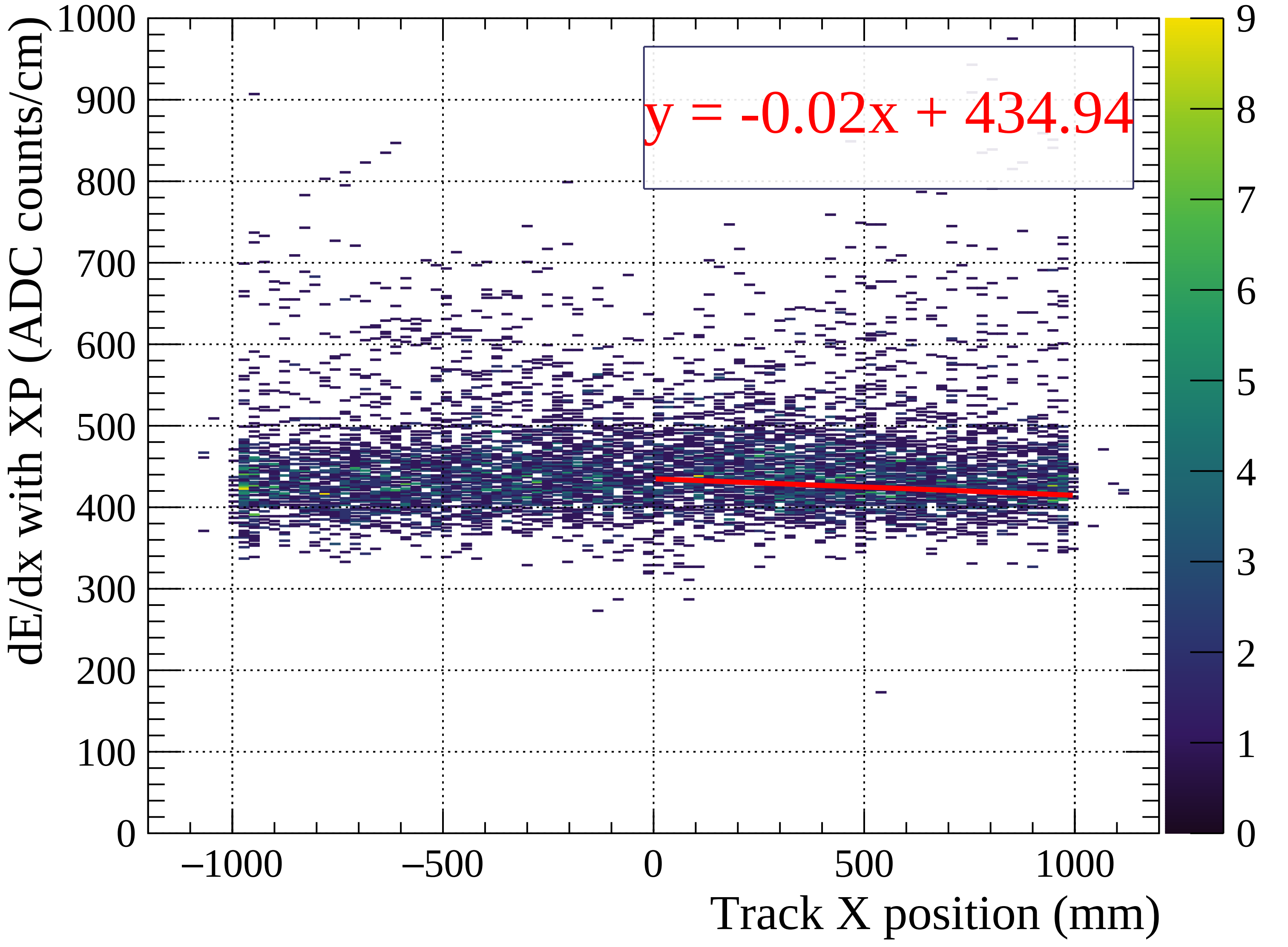
WF
XP
dE/dx resolution (%)
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)



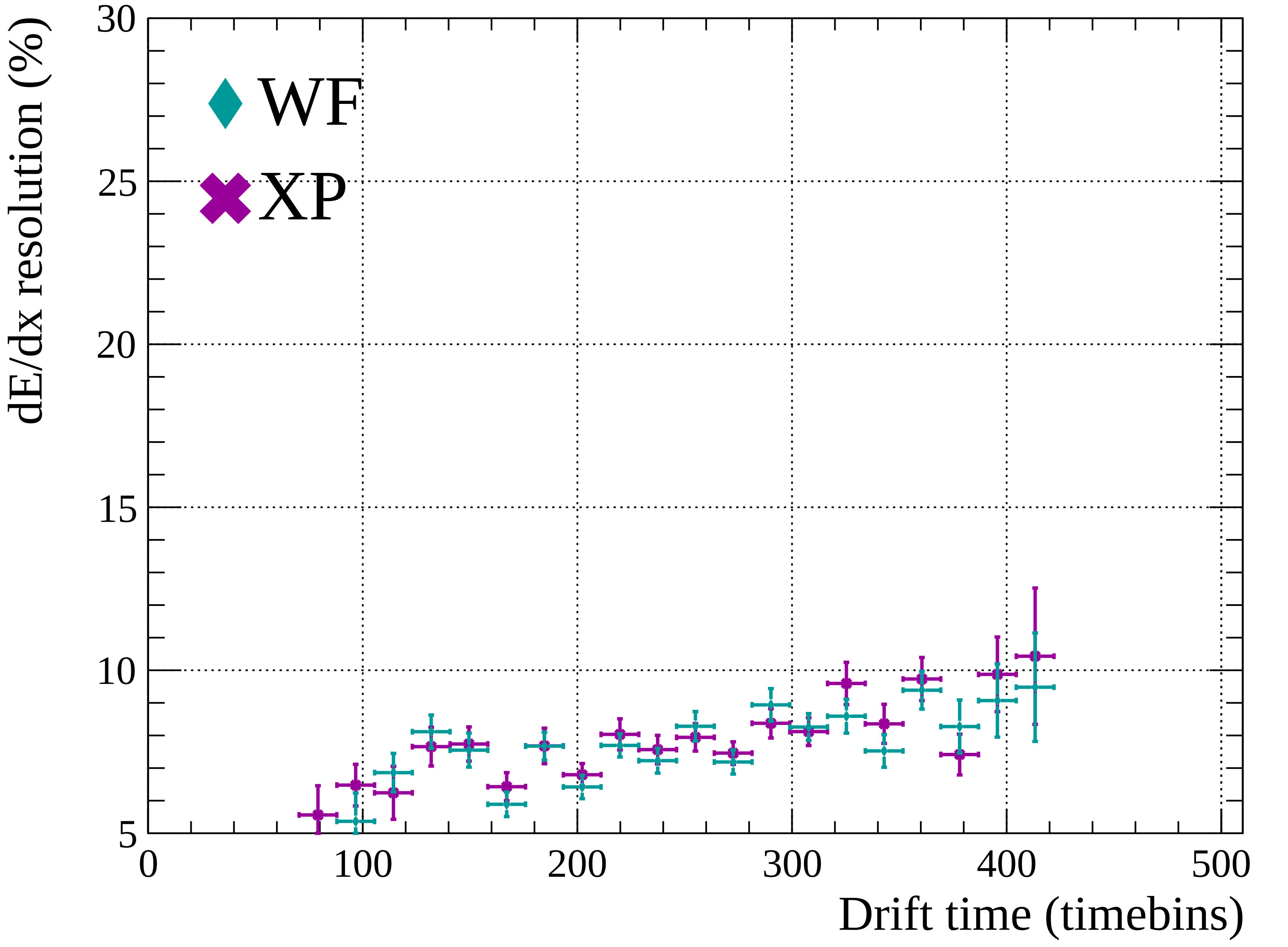
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)



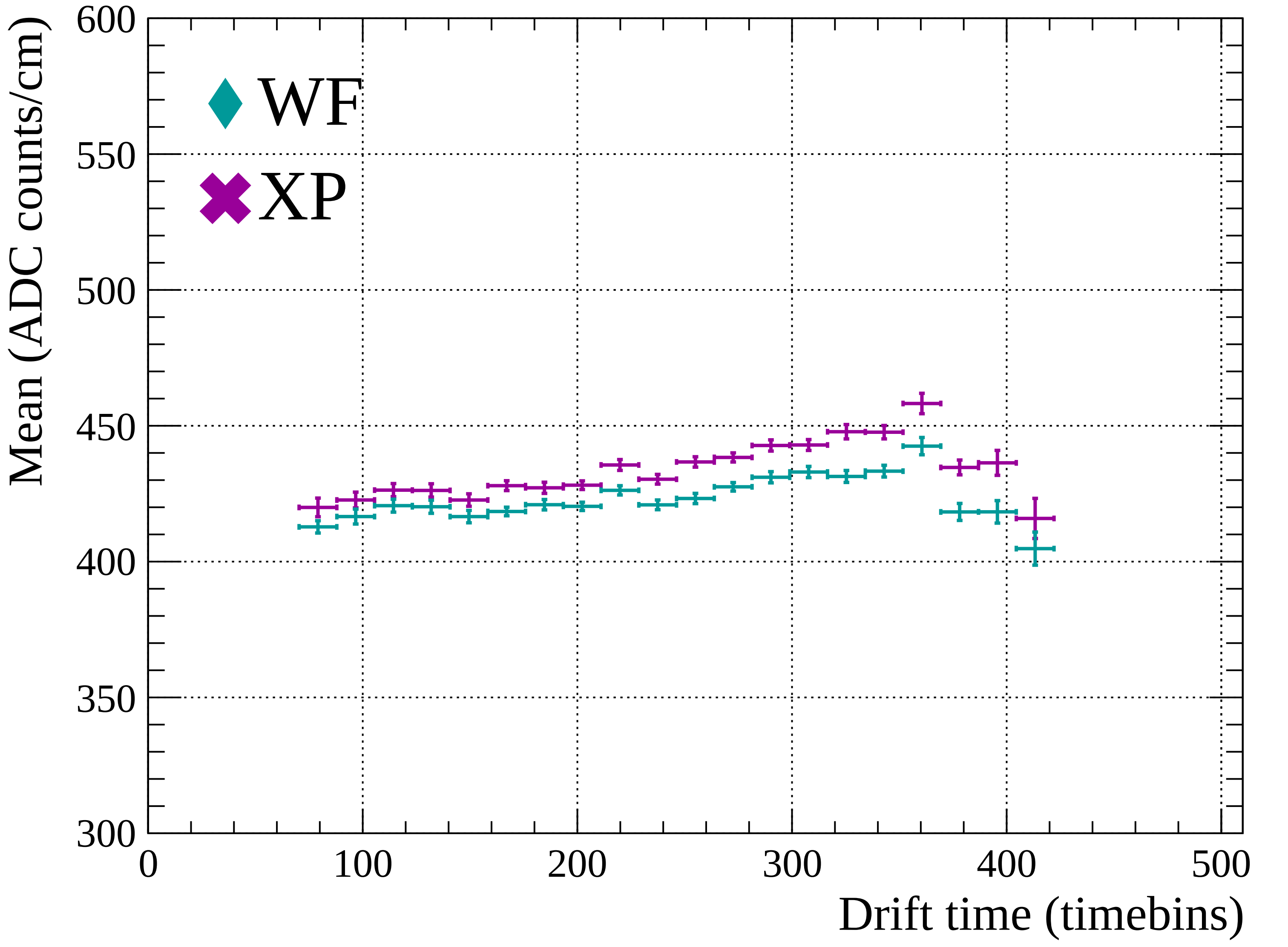
y = -0.00x + 409.24
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6
7
8
9



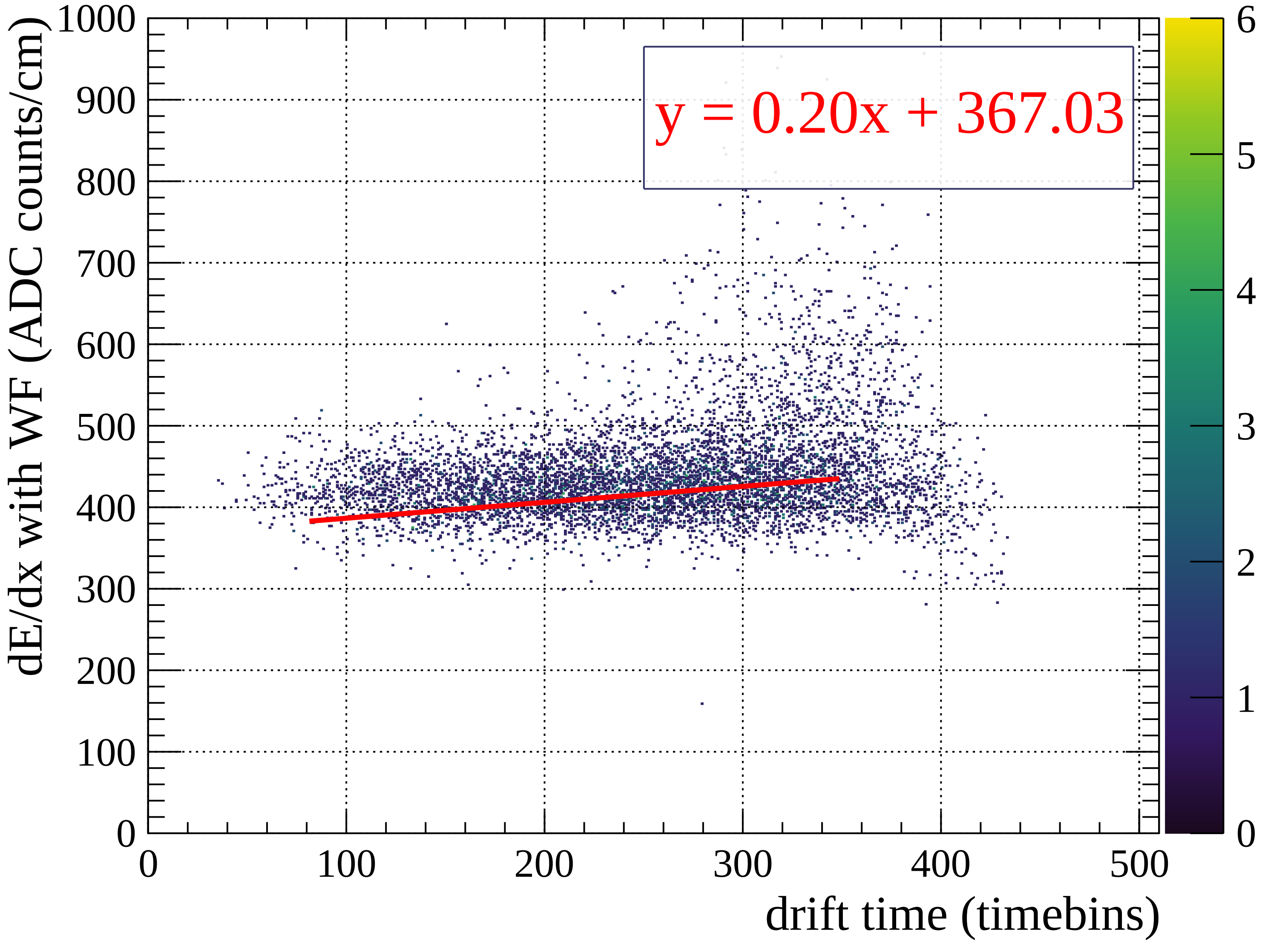
y = -0.02x + 434.94
dE/dx with XP (ADC counts/cm)
Track X position (mm)



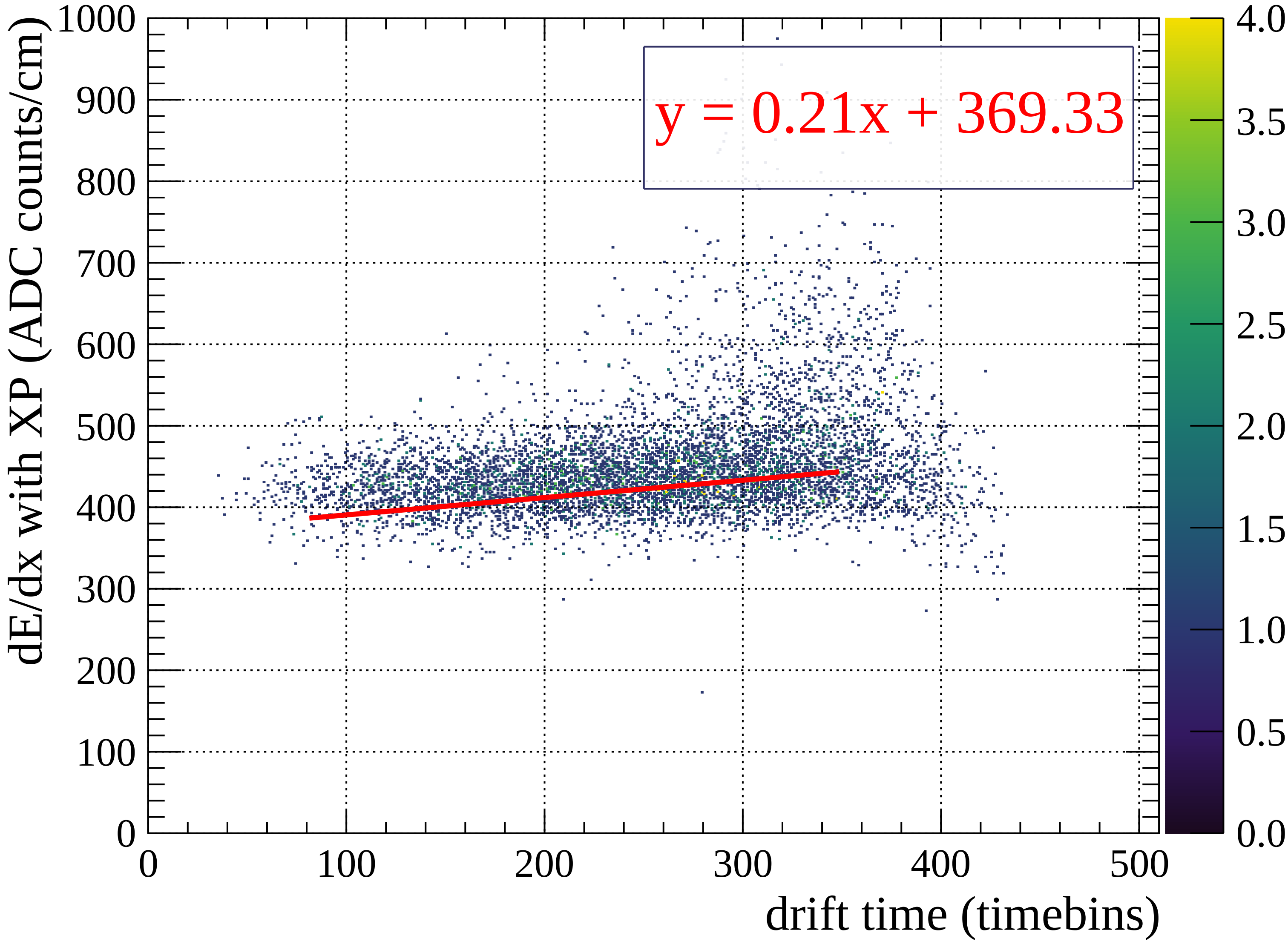
WF
XP
dE/dx resolution (%)
Drift time (timebins)
30
25
20
15
10
5
0
100
200
300
400
500



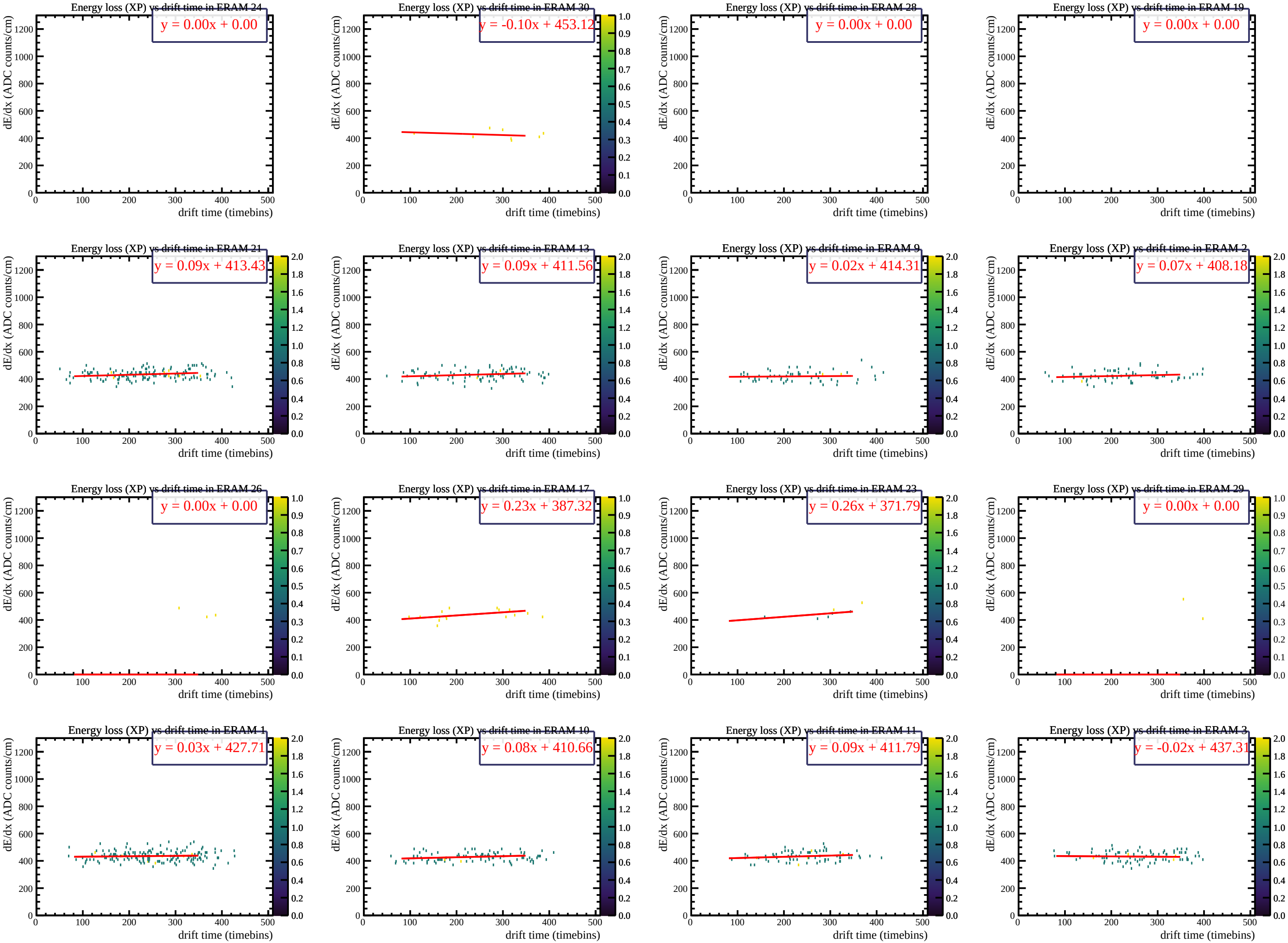
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)
WF
XP



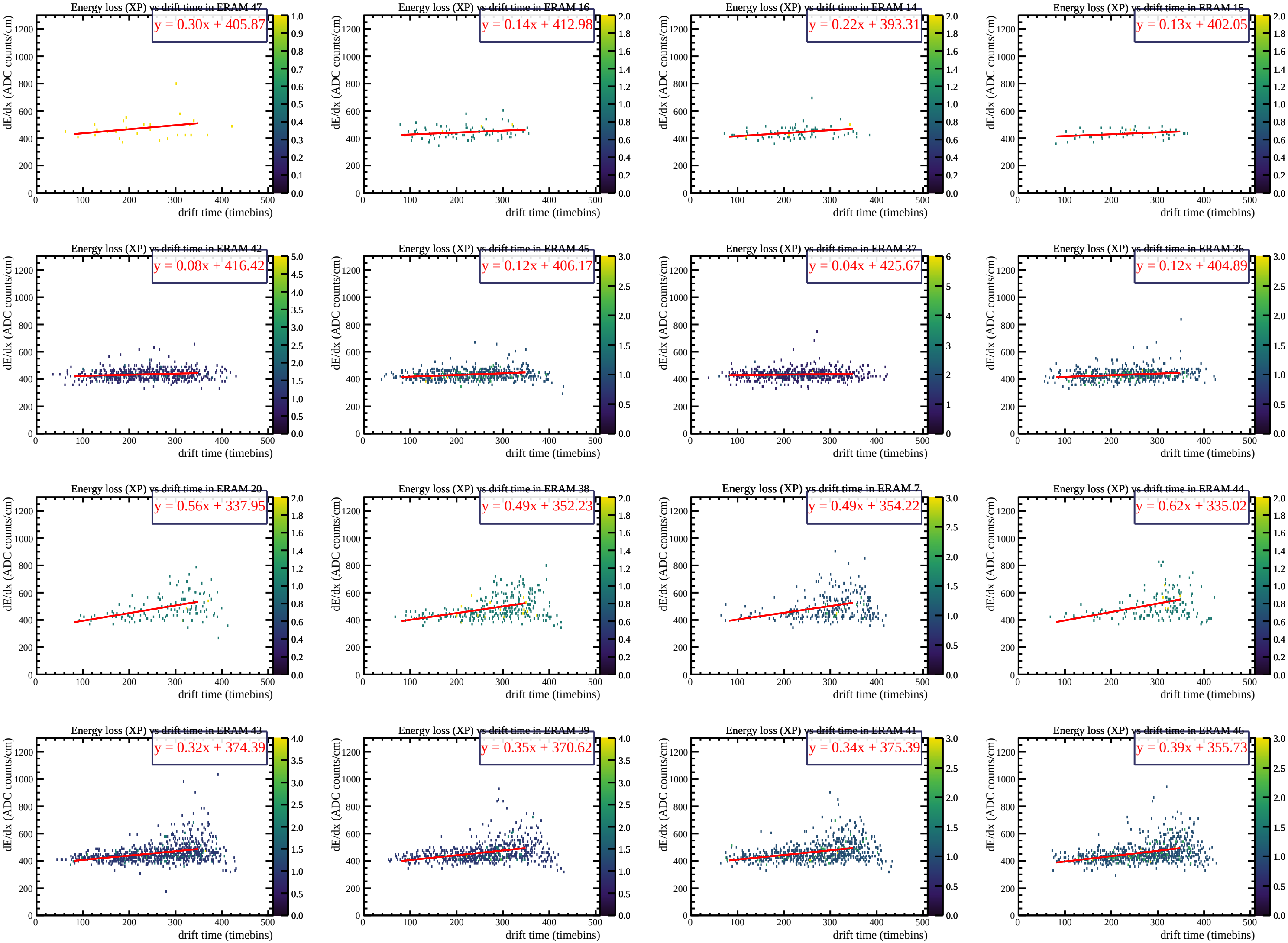
y = 0.20x + 367.03
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6



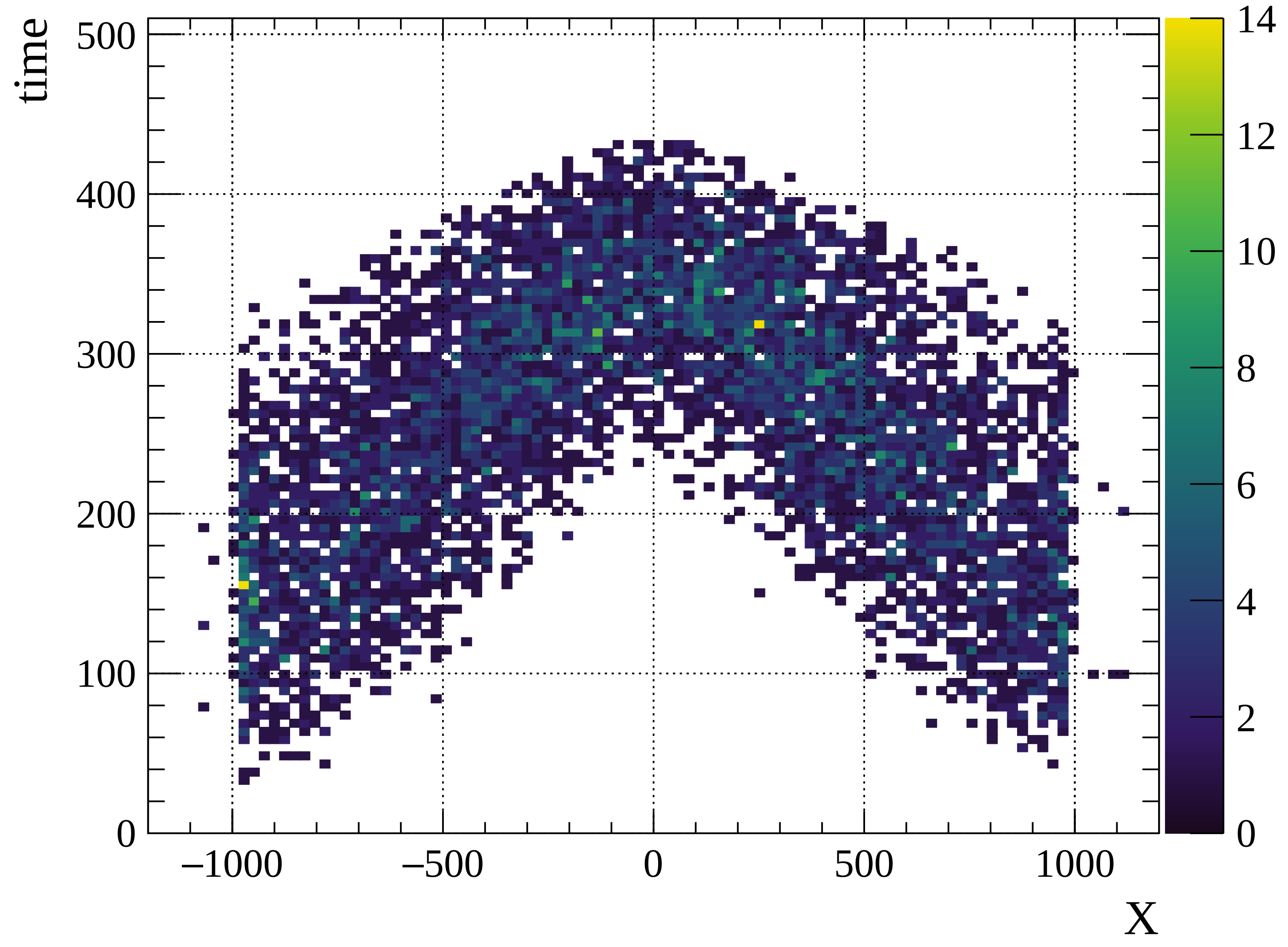
y = 0.21x + 369.33
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0



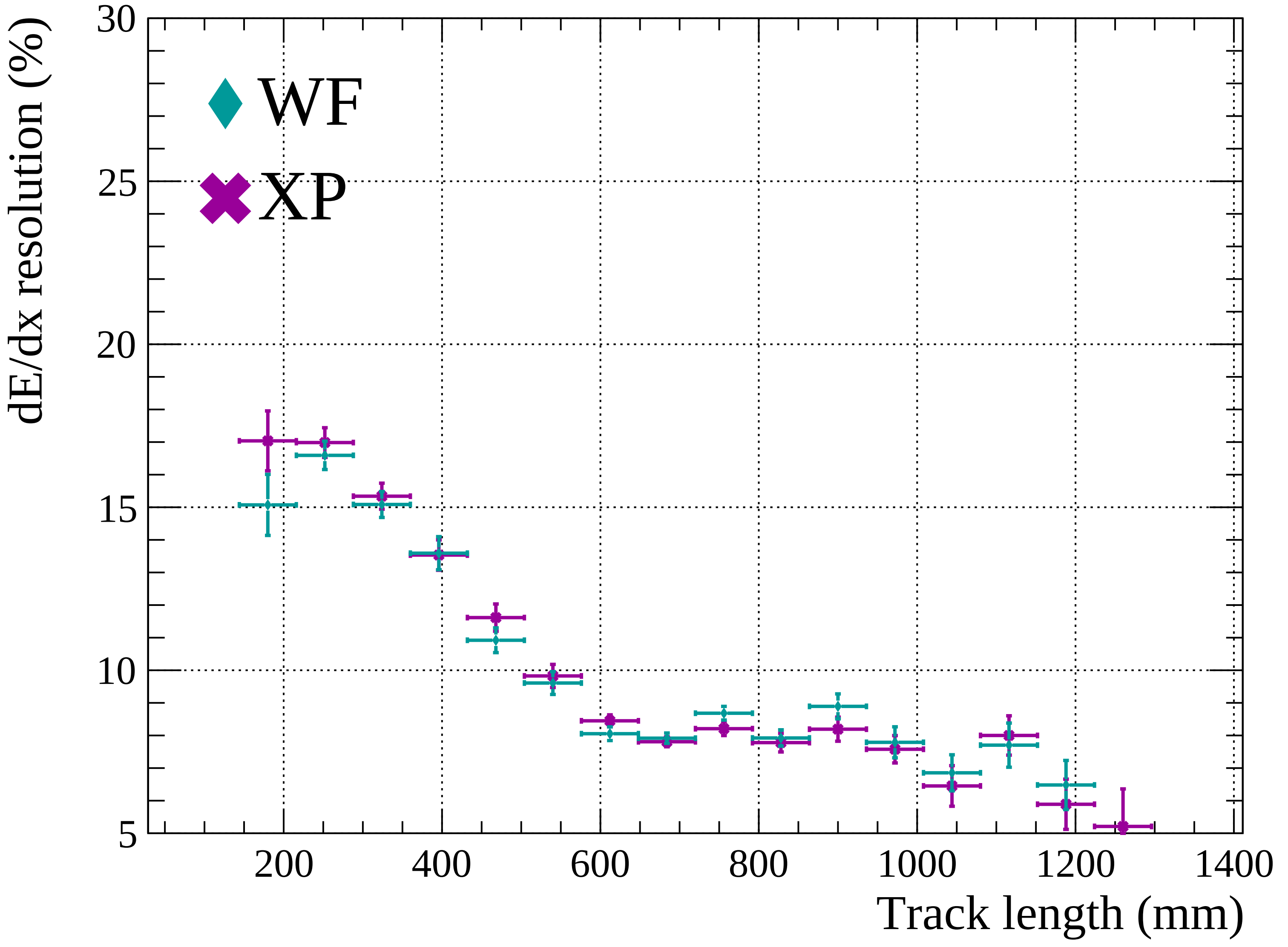
Energy loss (XP) vs drift time in ERAM 24
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 30
y = -0.10x + 453.12
Energy loss (XP) vs drift time in ERAM 28
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 19
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 21
y = 0.09x + 413.43
Energy loss (XP) vs drift time in ERAM 13
y = 0.09x + 411.56
Energy loss (XP) vs drift time in ERAM 9
y = 0.02x + 414.31
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 408.18
Energy loss (XP) vs drift time in ERAM 26
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 17
y = 0.23x + 387.32
Energy loss (XP) vs drift time in ERAM 23
y = 0.26x + 371.79
Energy loss (XP) vs drift time in ERAM 29
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 1
y = 0.03x + 427.71
Energy loss (XP) vs drift time in ERAM 10
y = 0.08x + 410.66
Energy loss (XP) vs drift time in ERAM 11
y = 0.09x + 411.79
Energy loss (XP) vs drift time in ERAM 3
y = -0.02x + 437.31
dE/dx (ADC counts/cm)
drift time (timebins)

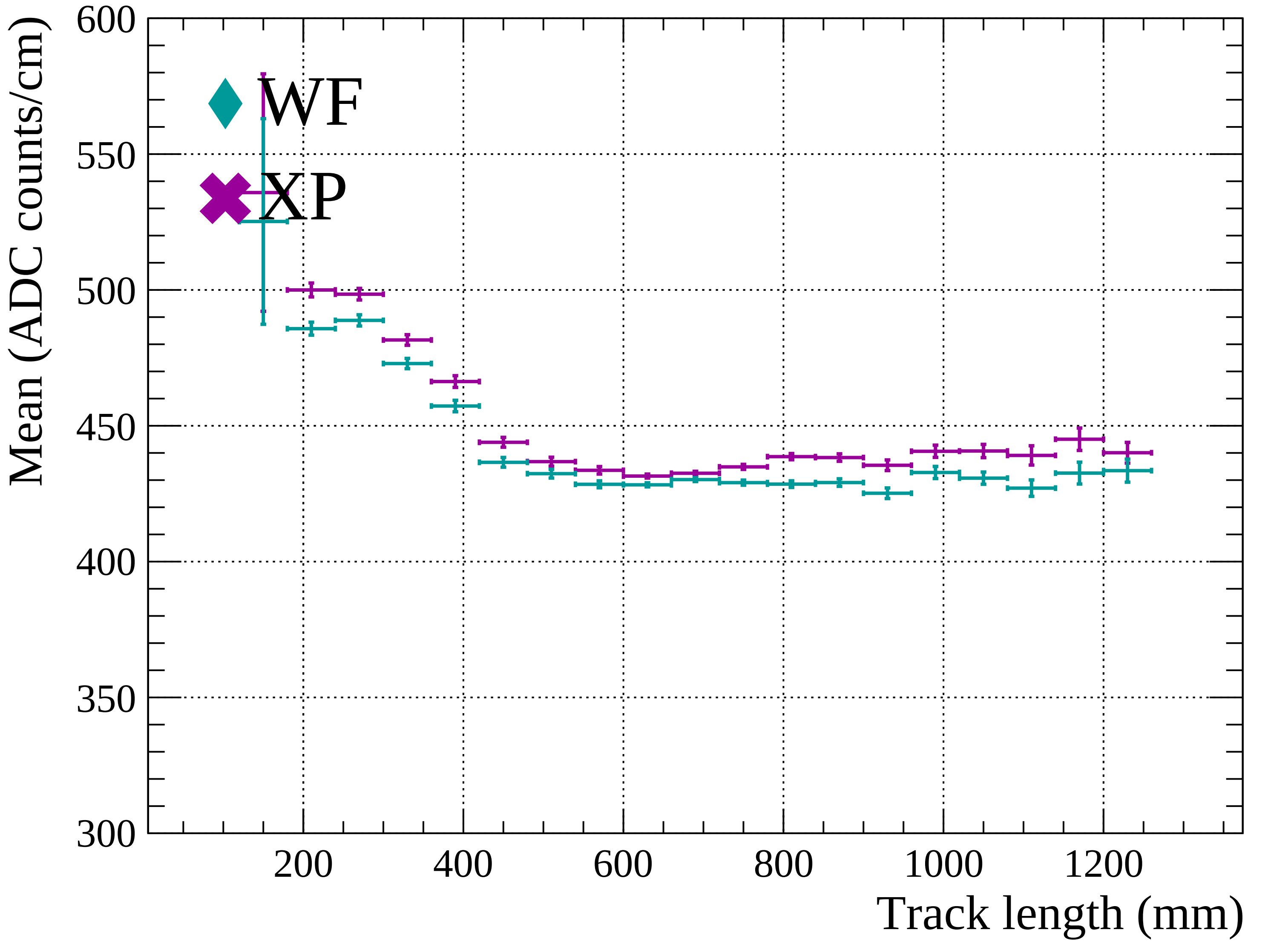


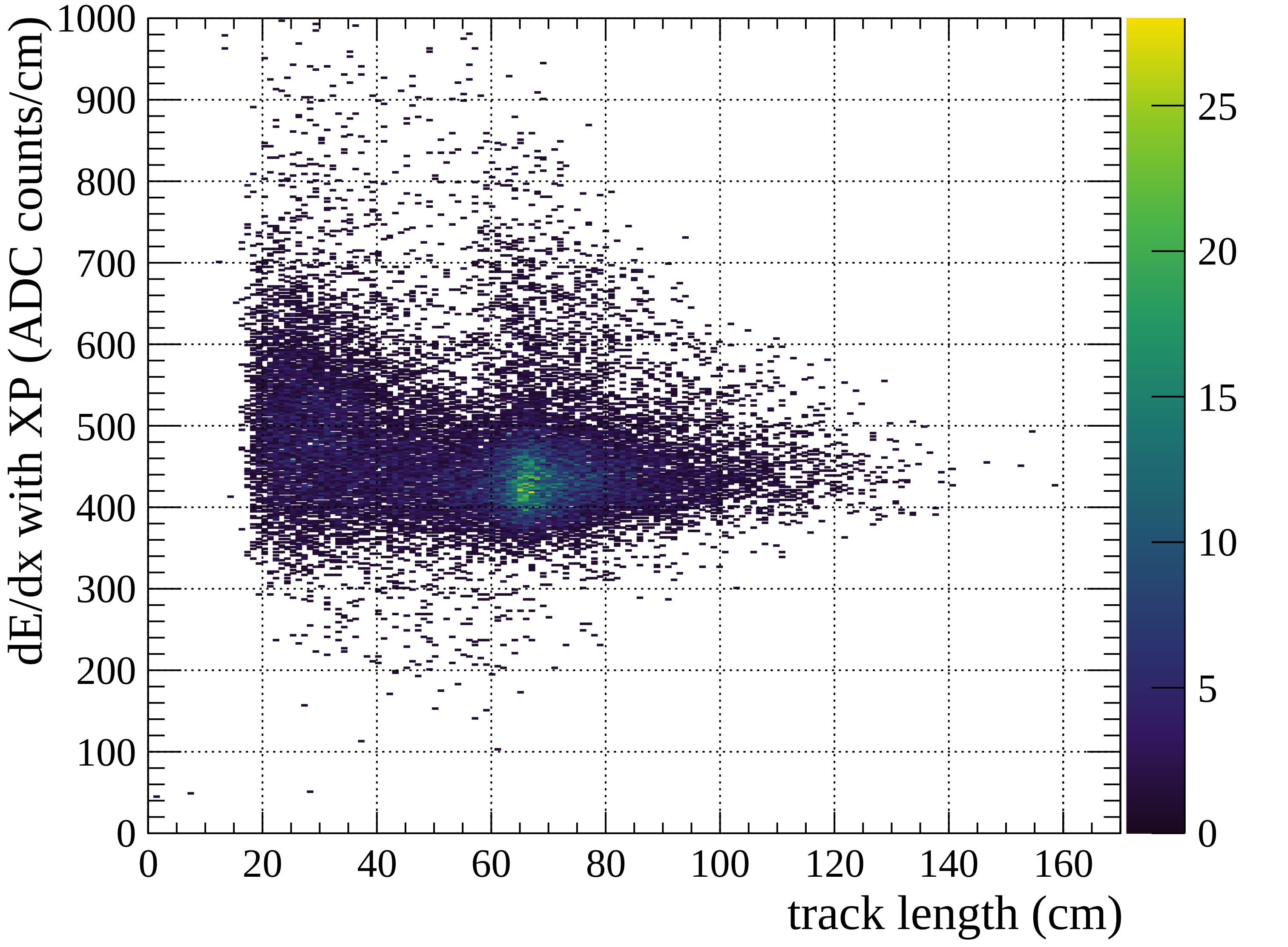
Energy loss (XP) vs drift time in ERAM 47
y = 0.30x + 405.87
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 16
y = 0.14x + 412.98
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 14
y = 0.22x + 393.31
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 15
y = 0.13x + 402.05
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 42
y = 0.08x + 416.42
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 45
y = 0.12x + 406.17
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 37
y = 0.04x + 425.67
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 36
y = 0.12x + 404.89
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 20
y = 0.56x + 337.95
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 38
y = 0.49x + 352.23
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 7
y = 0.49x + 354.22
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 44
y = 0.62x + 335.02
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 43
y = 0.32x + 374.39
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 39
y = 0.35x + 370.62
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 41
y = 0.34x + 375.39
dE/dx (ADC counts/cm)
drift time (timebins)
Energy loss (XP) vs drift time in ERAM 46
y = 0.39x + 355.73
dE/dx (ADC counts/cm)
drift time (timebins)

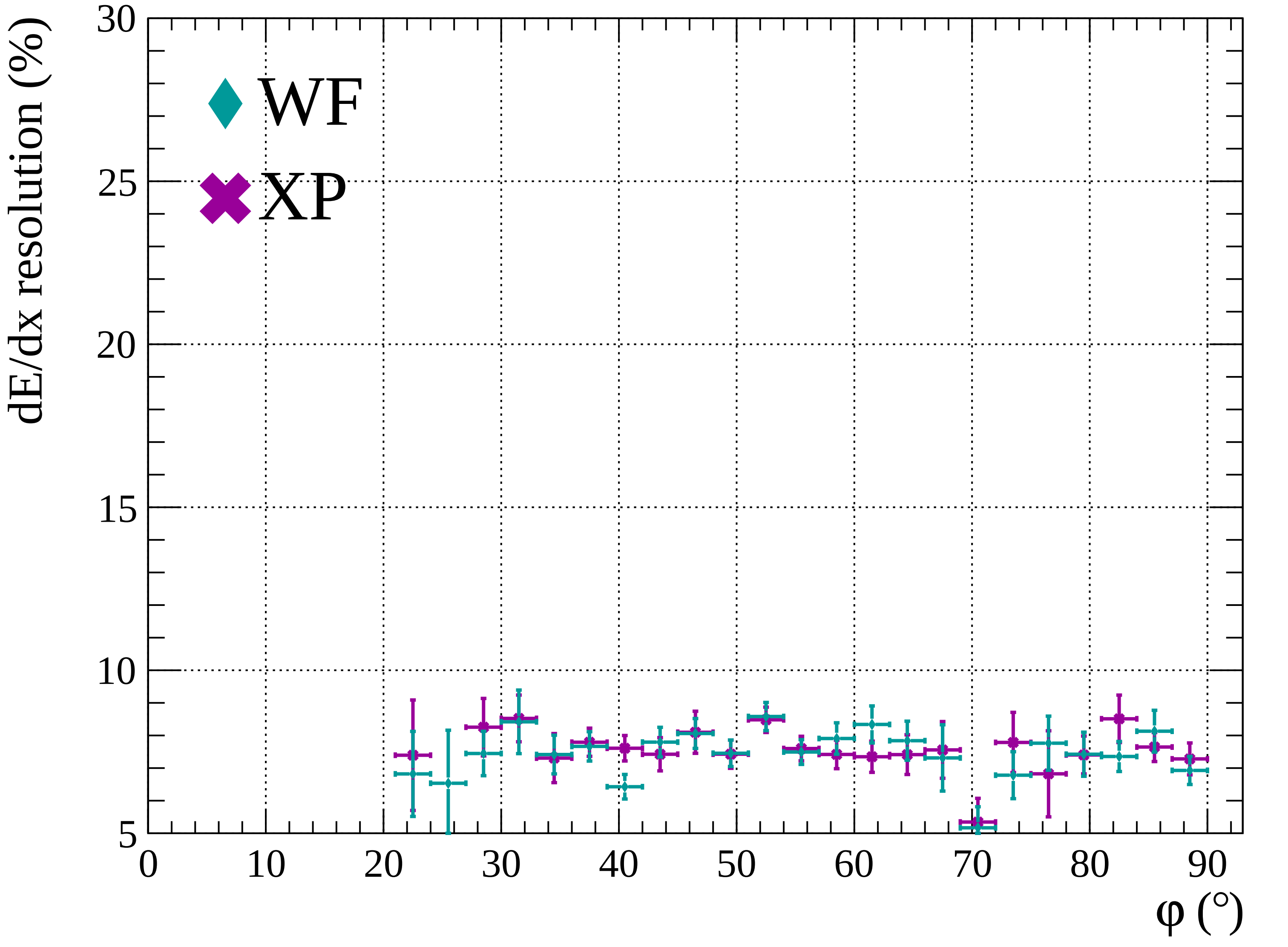


time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
14
12
10
8
6
4
2
0

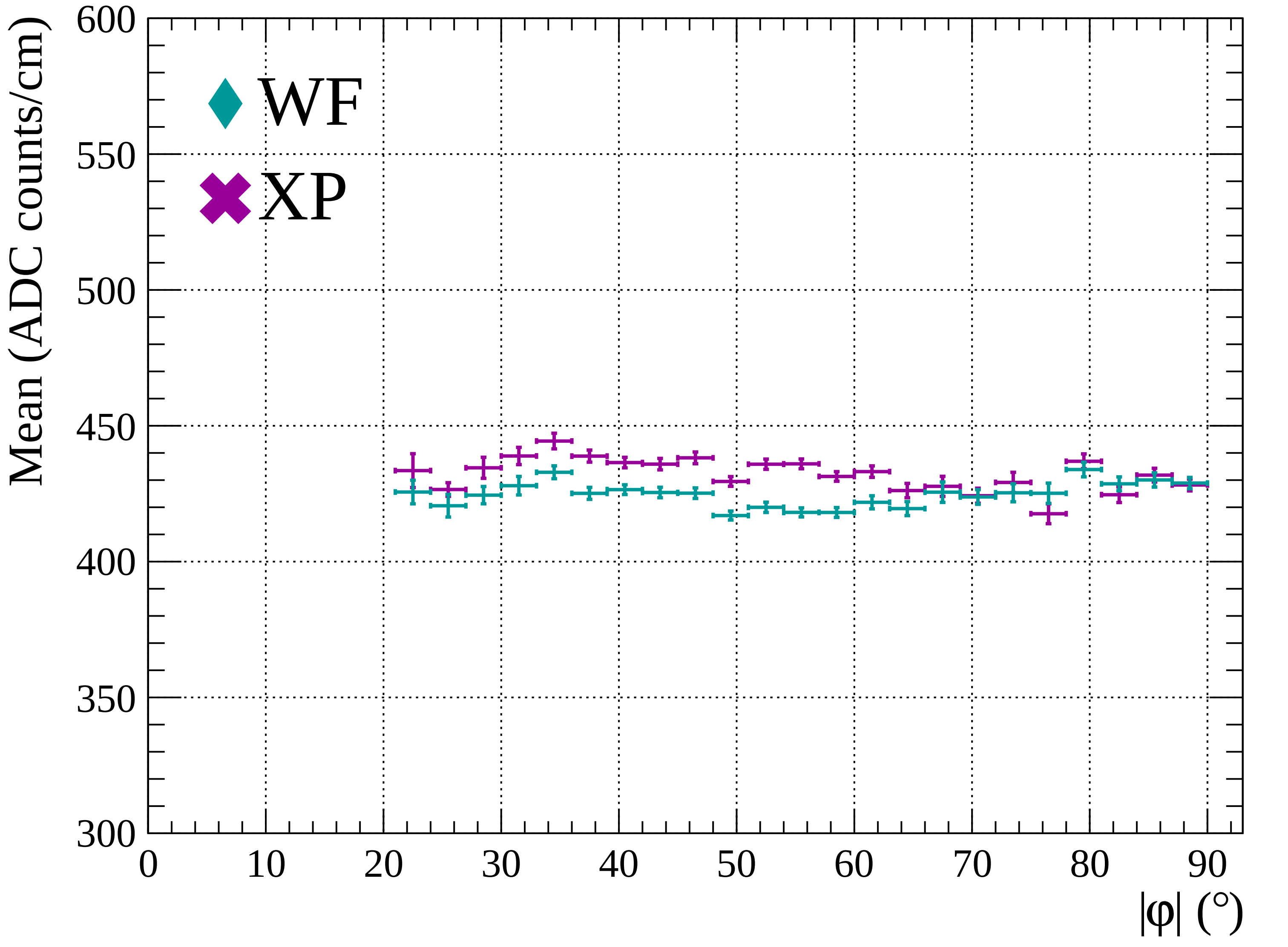

WF
XP
dE/dx resolution (%)
Track length (mm)
5
10
15
20
25
30
200
400
600
800
1000
1200
1400


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
200
400
600
800
1000
1200
Track length (mm)

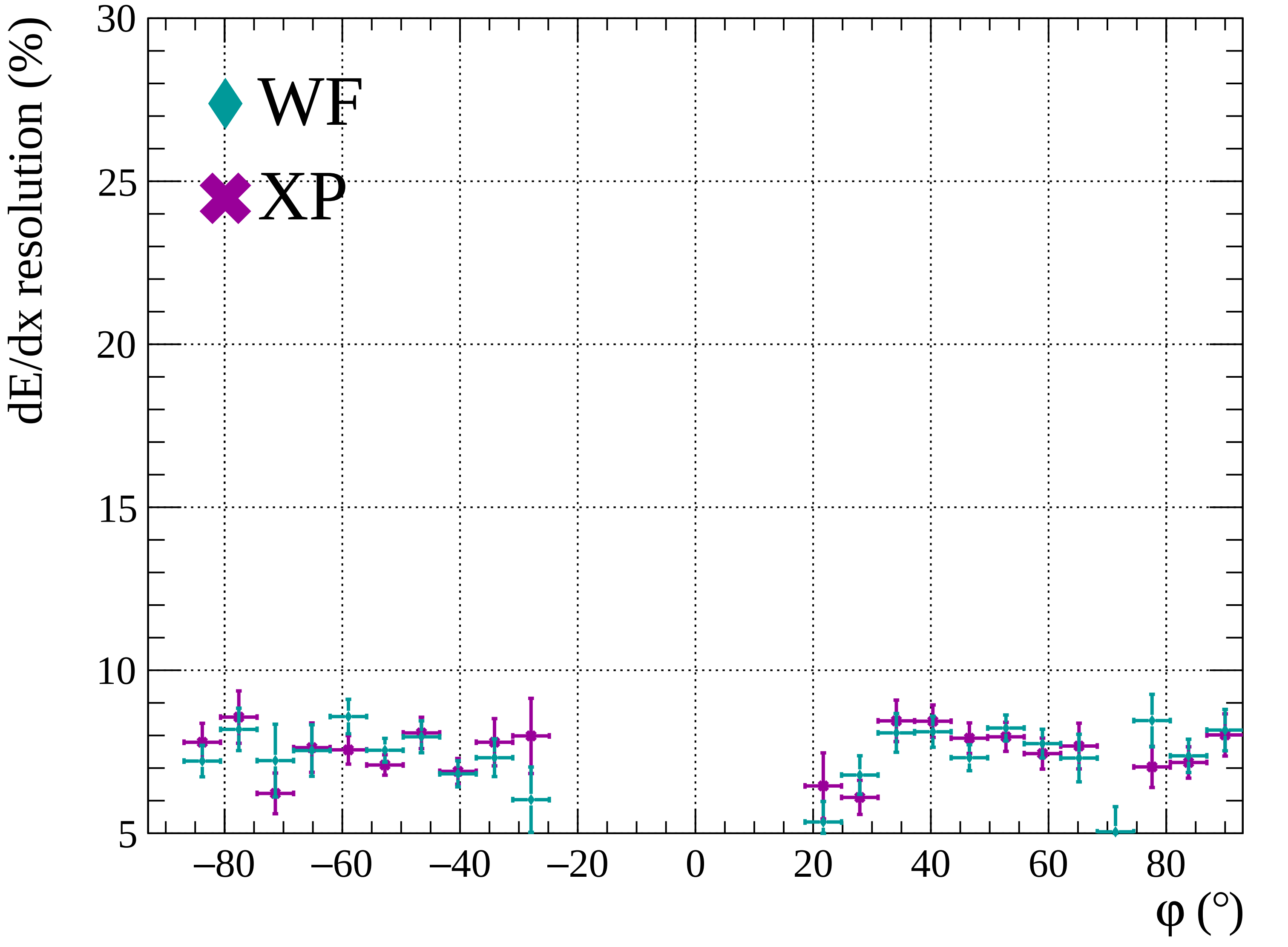

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
5
10
15
20
25



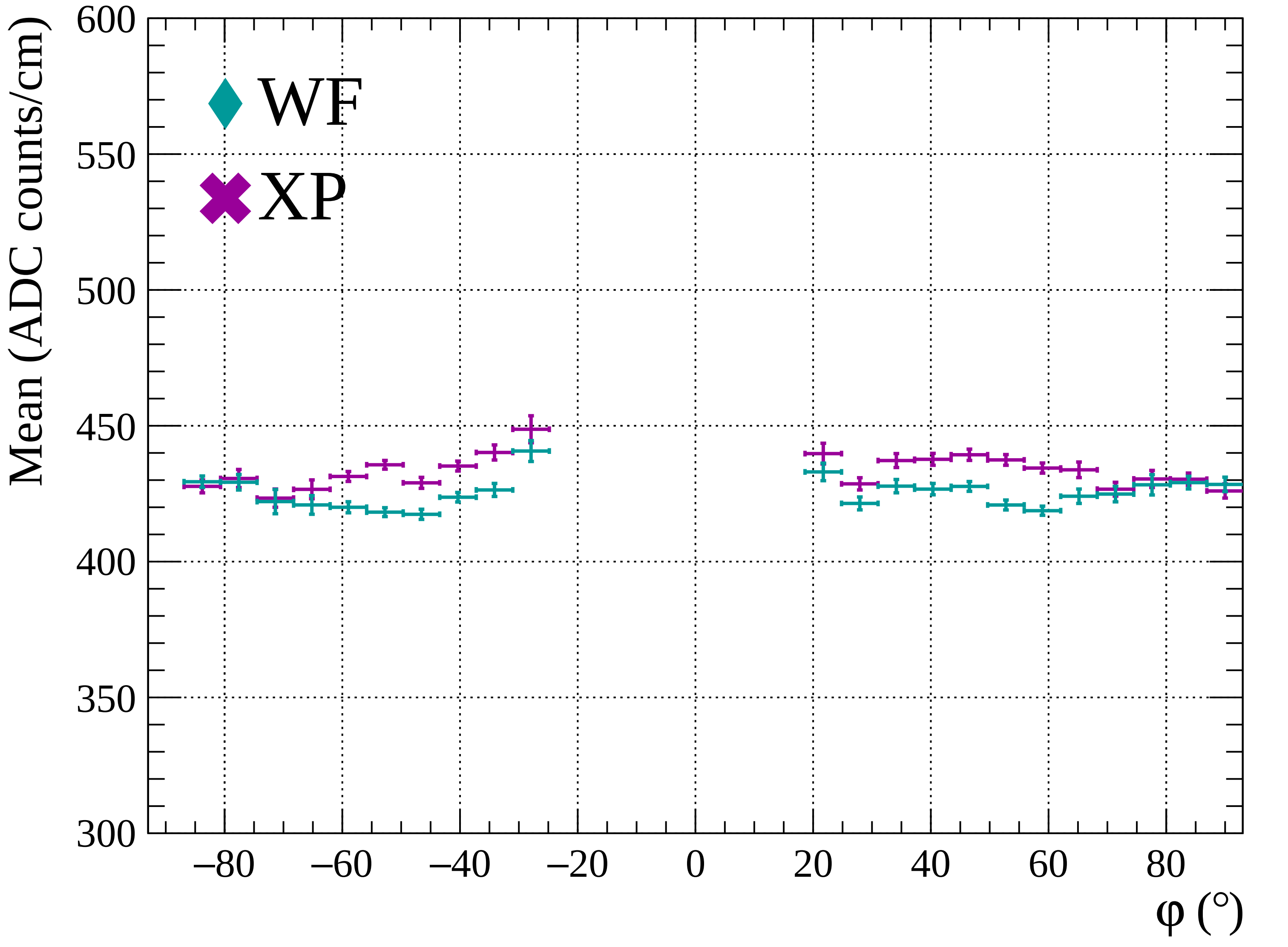
dE/dx resolution (%)
WF
XP
φ (°)
0
10
20
30
40
50
60
70
80
90
5
10
15
20
25
30



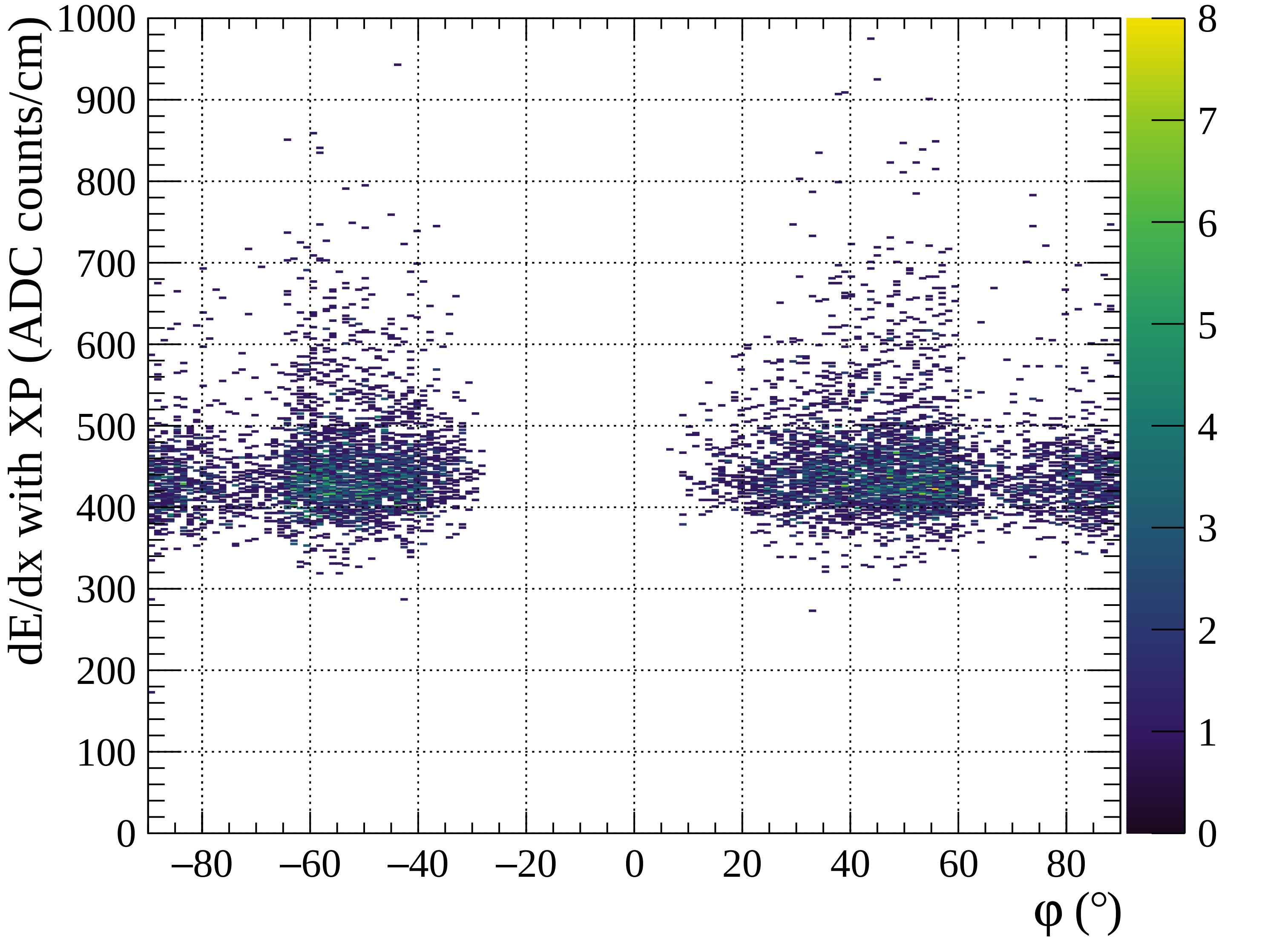
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
|φ| (°)



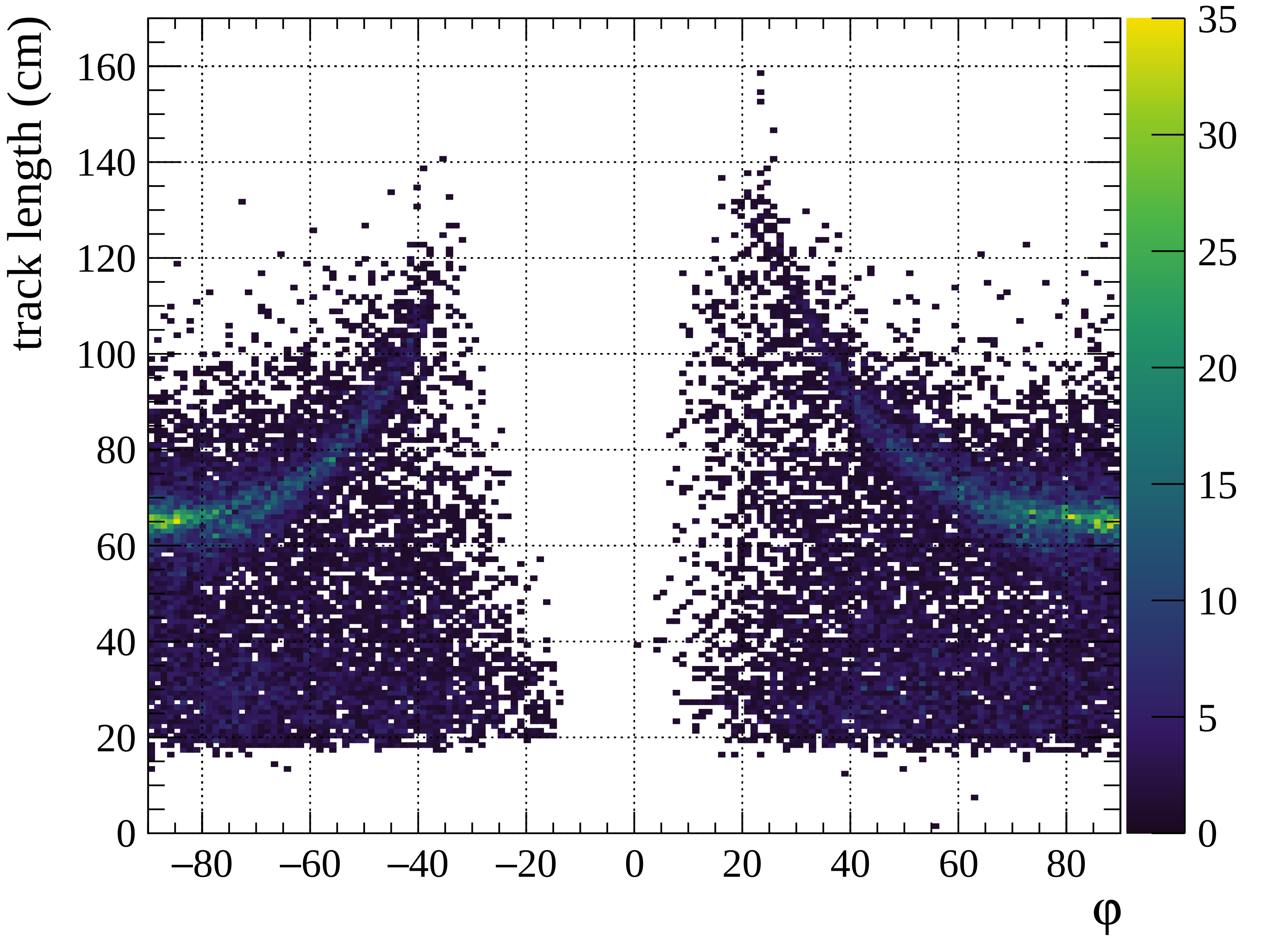
dE/dx resolution (%)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
φ (°)
WF
XP



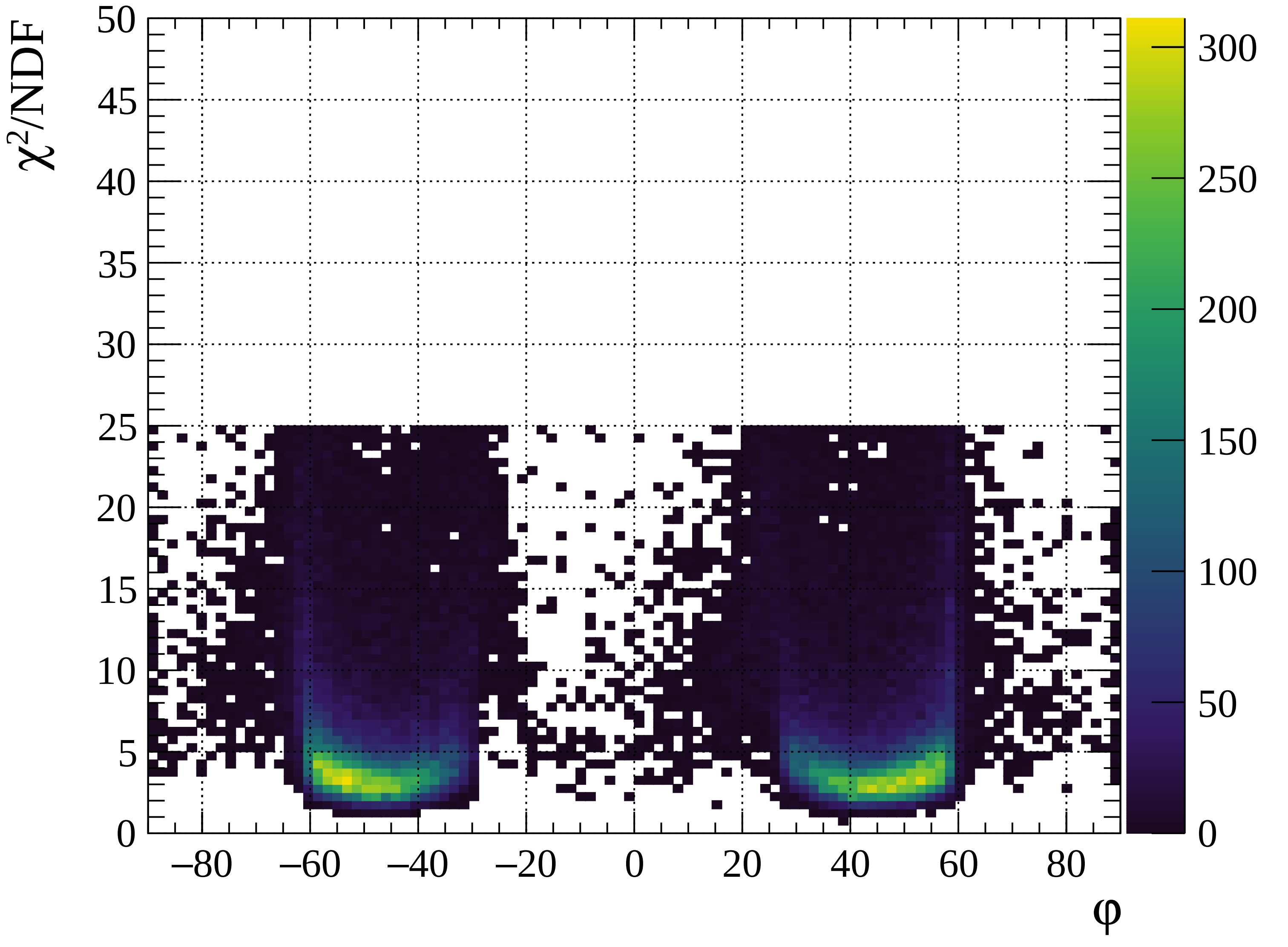
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
φ (°)

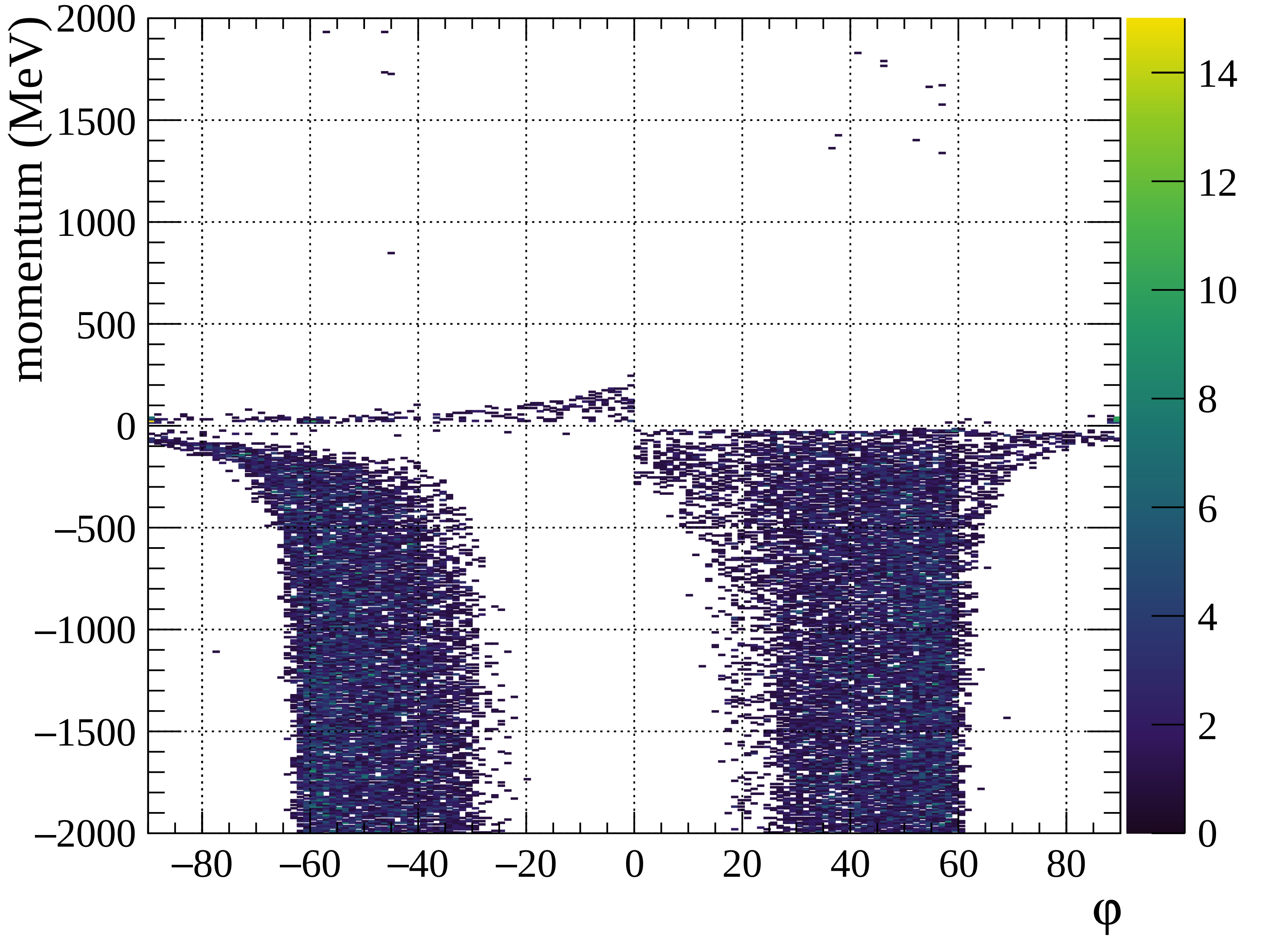


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
8
7
6
5
4
3
2
1
0

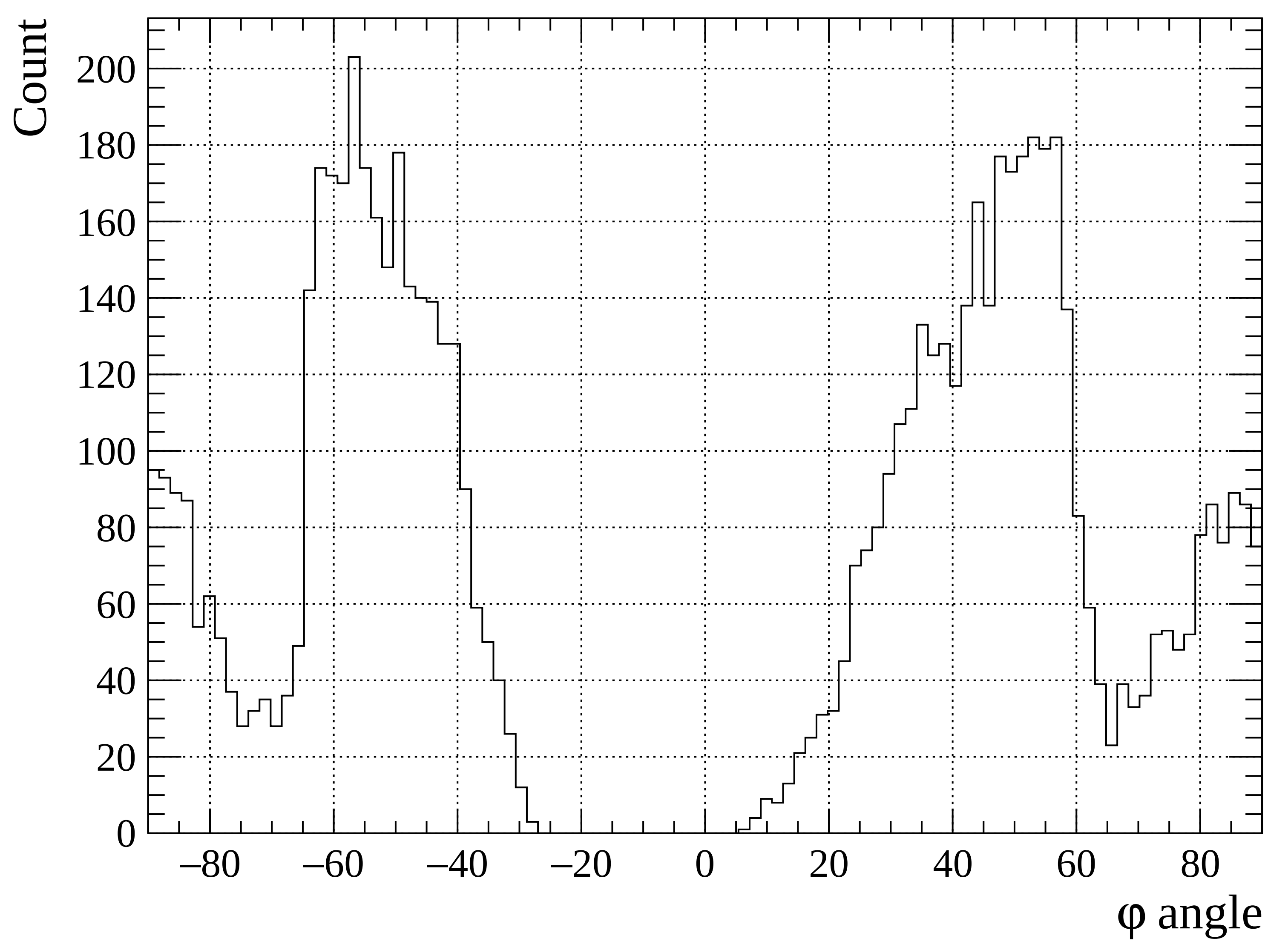


track length (cm)
φ
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25
30
35

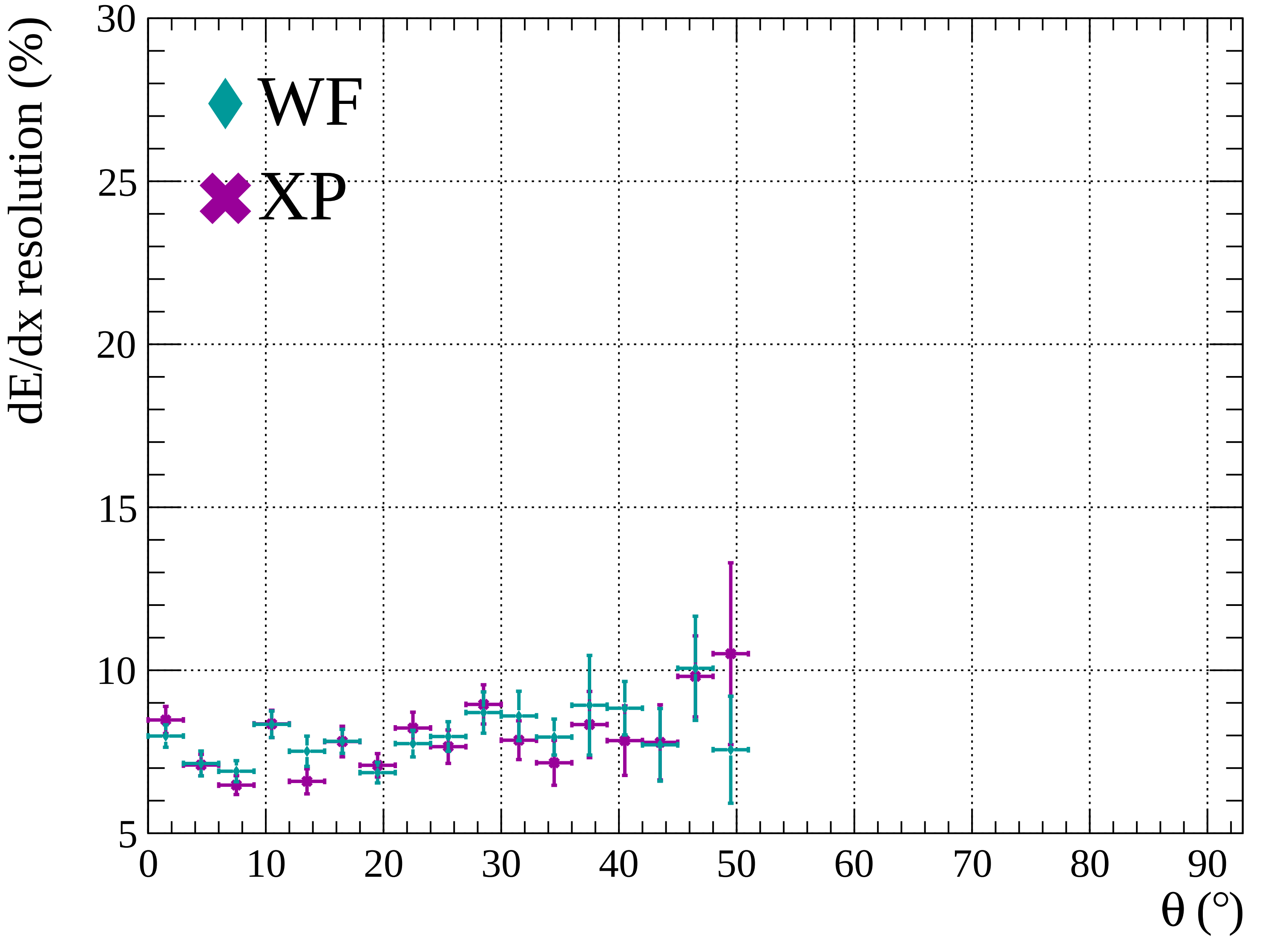

χ²/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
300
250
200
150
100
50
0



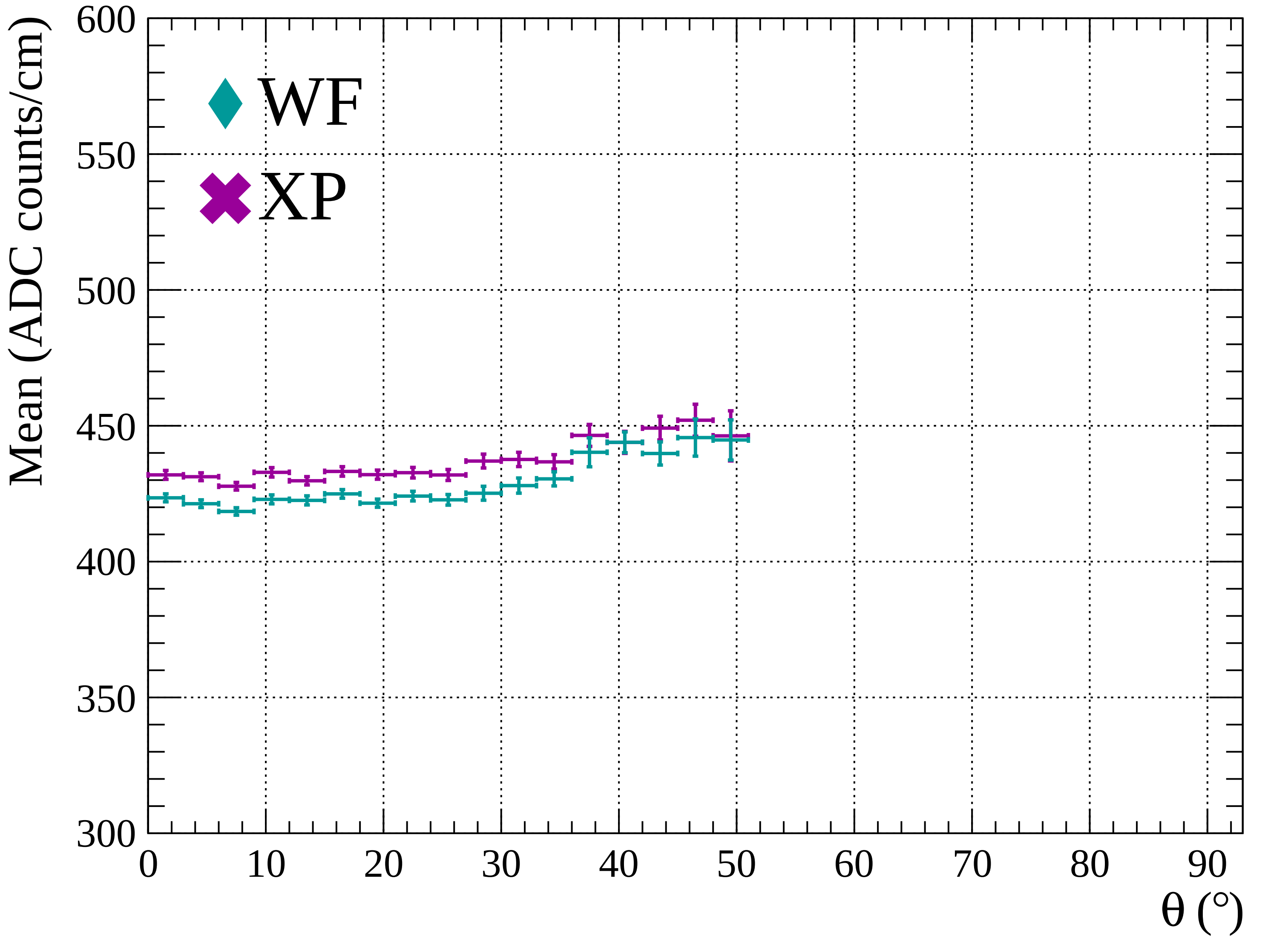
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
14
12
10
8
6
4
2
0



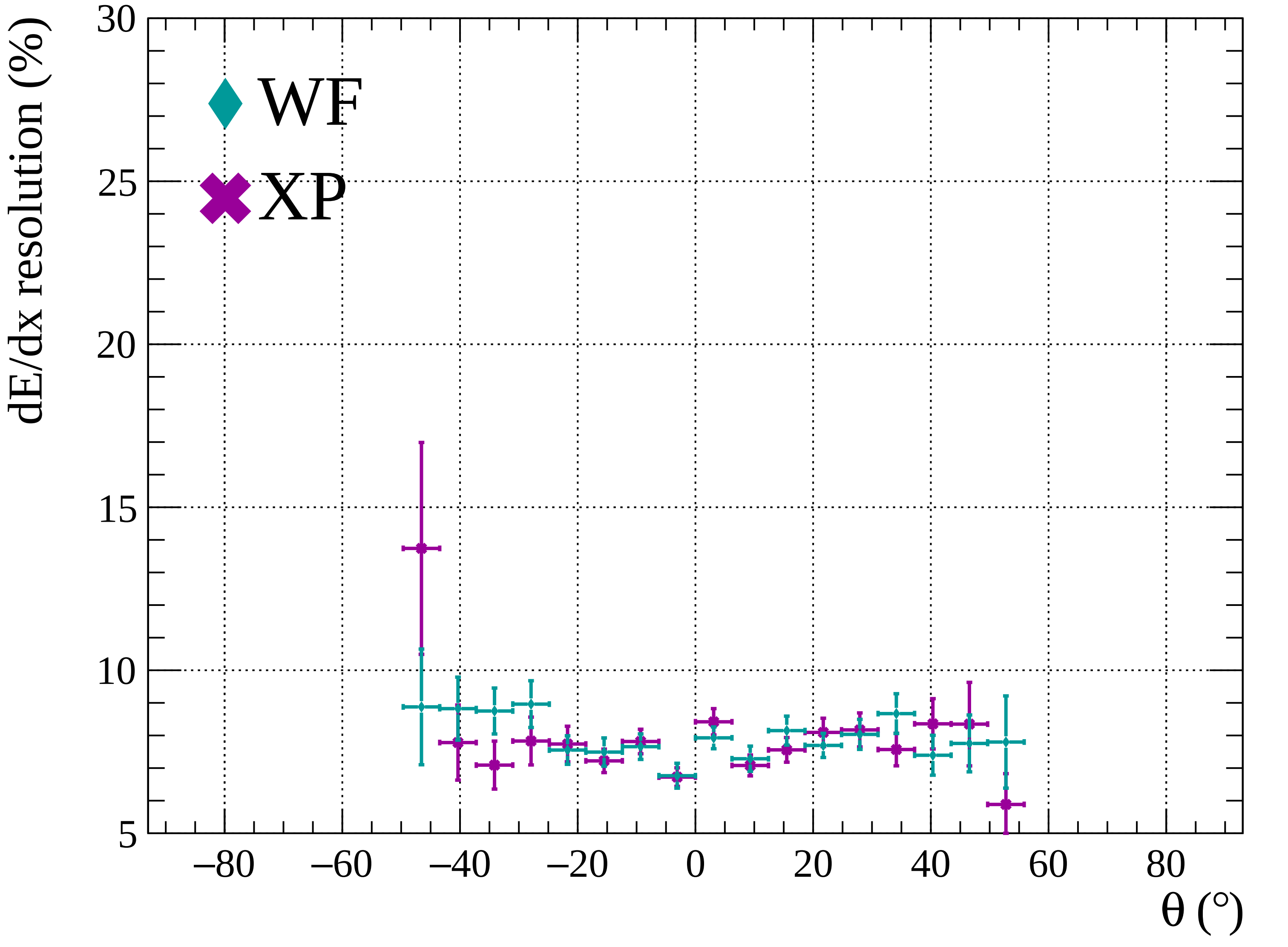
Count
200
180
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



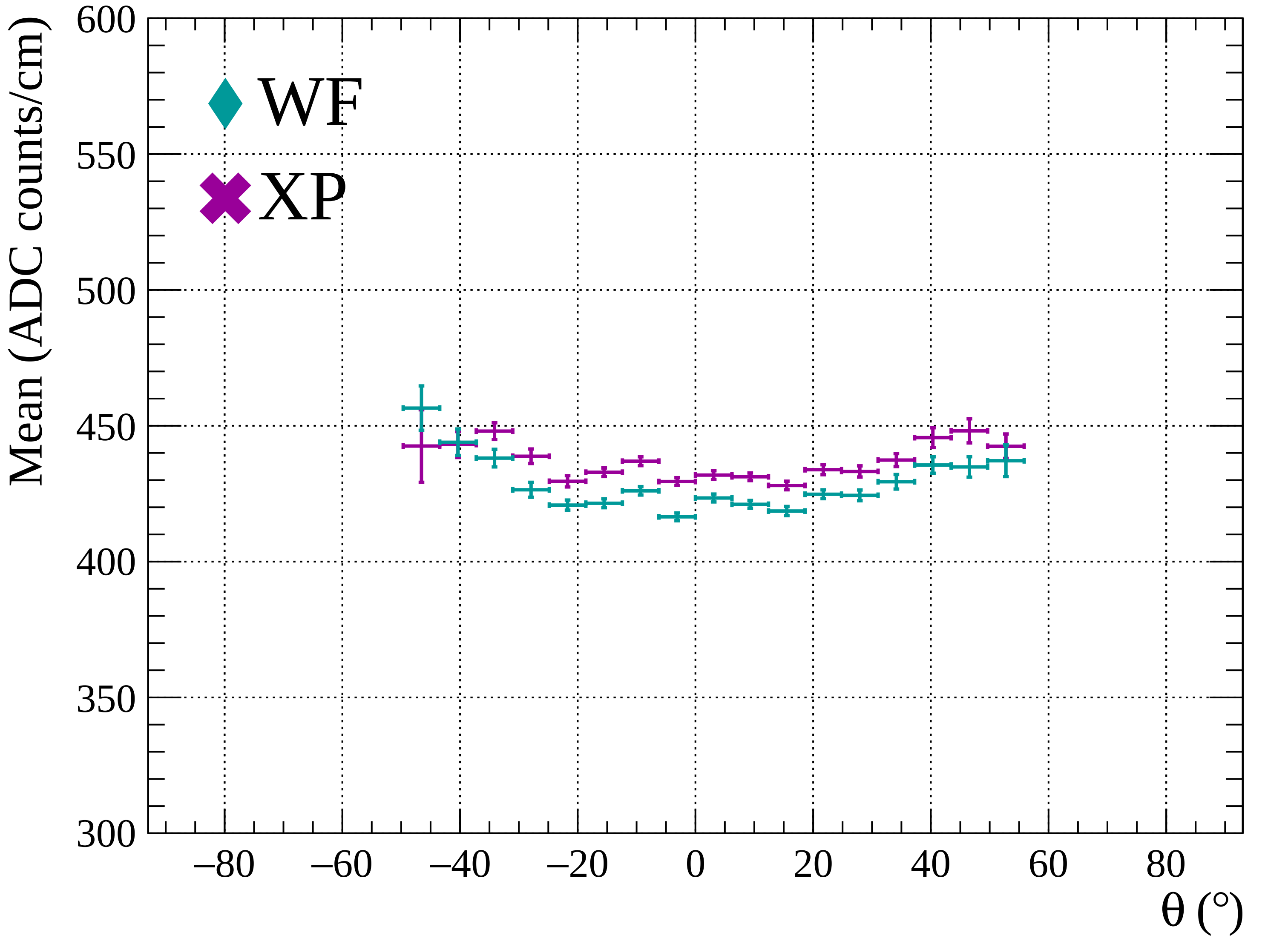
WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90

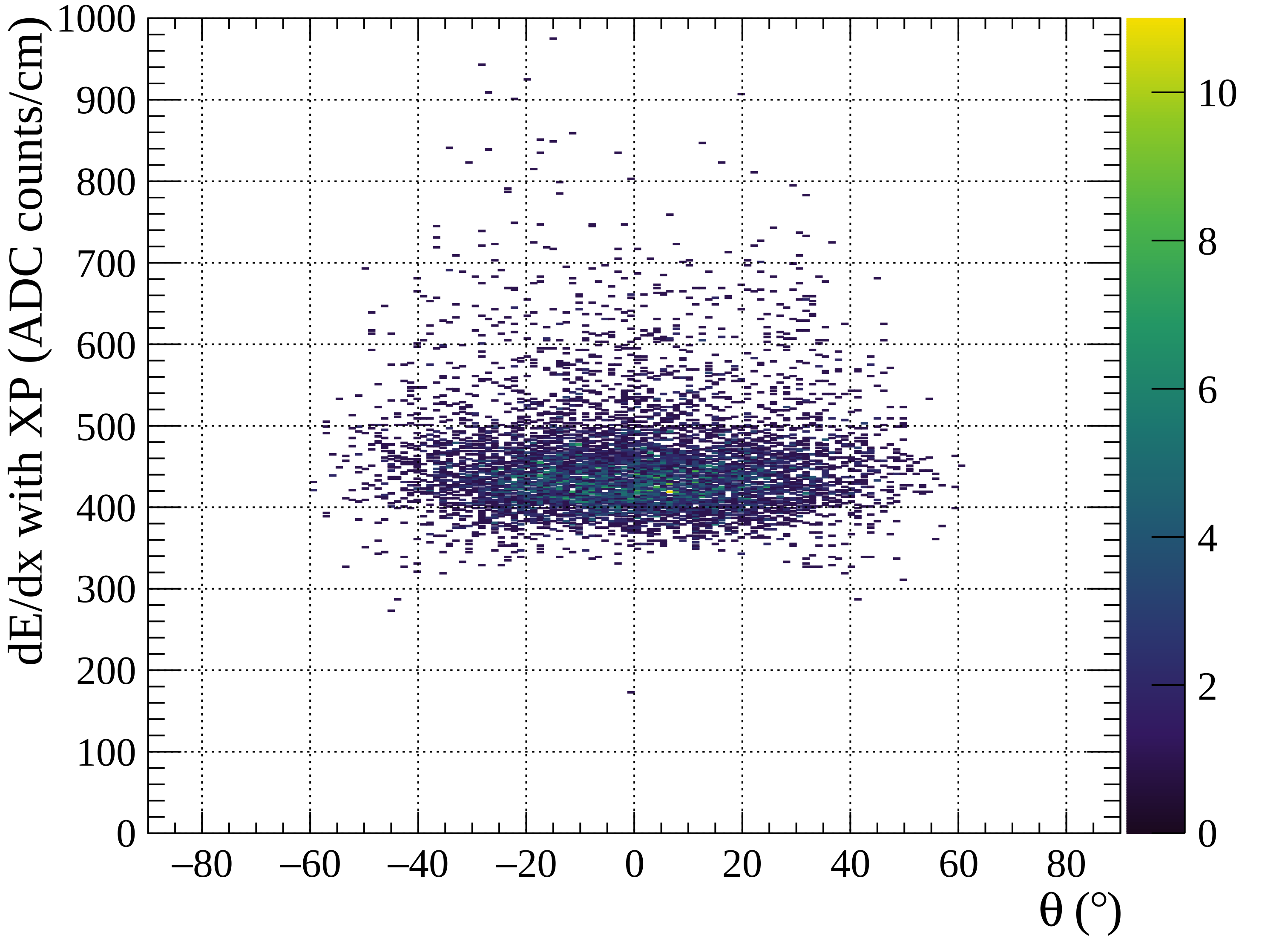


WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90

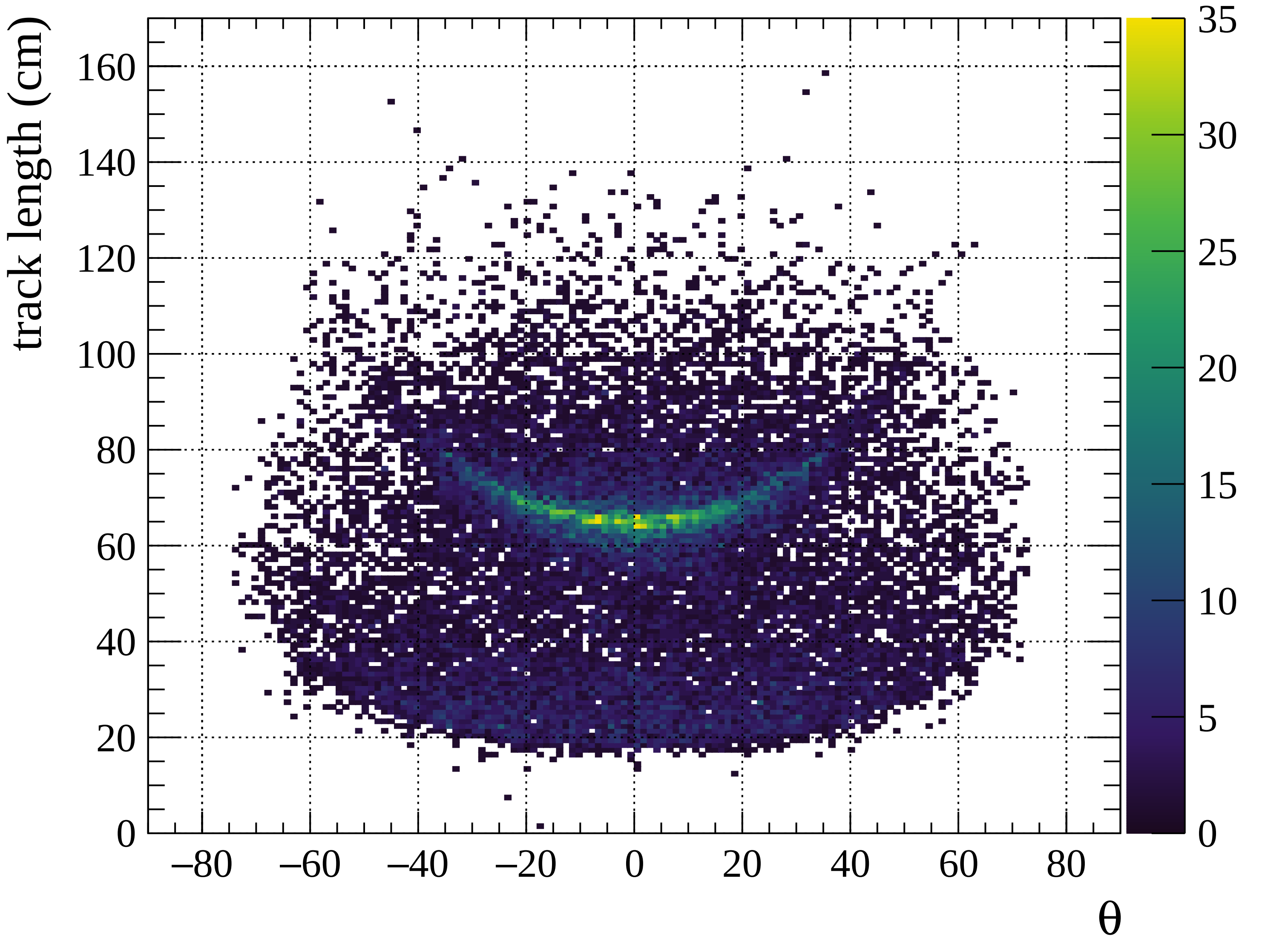


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

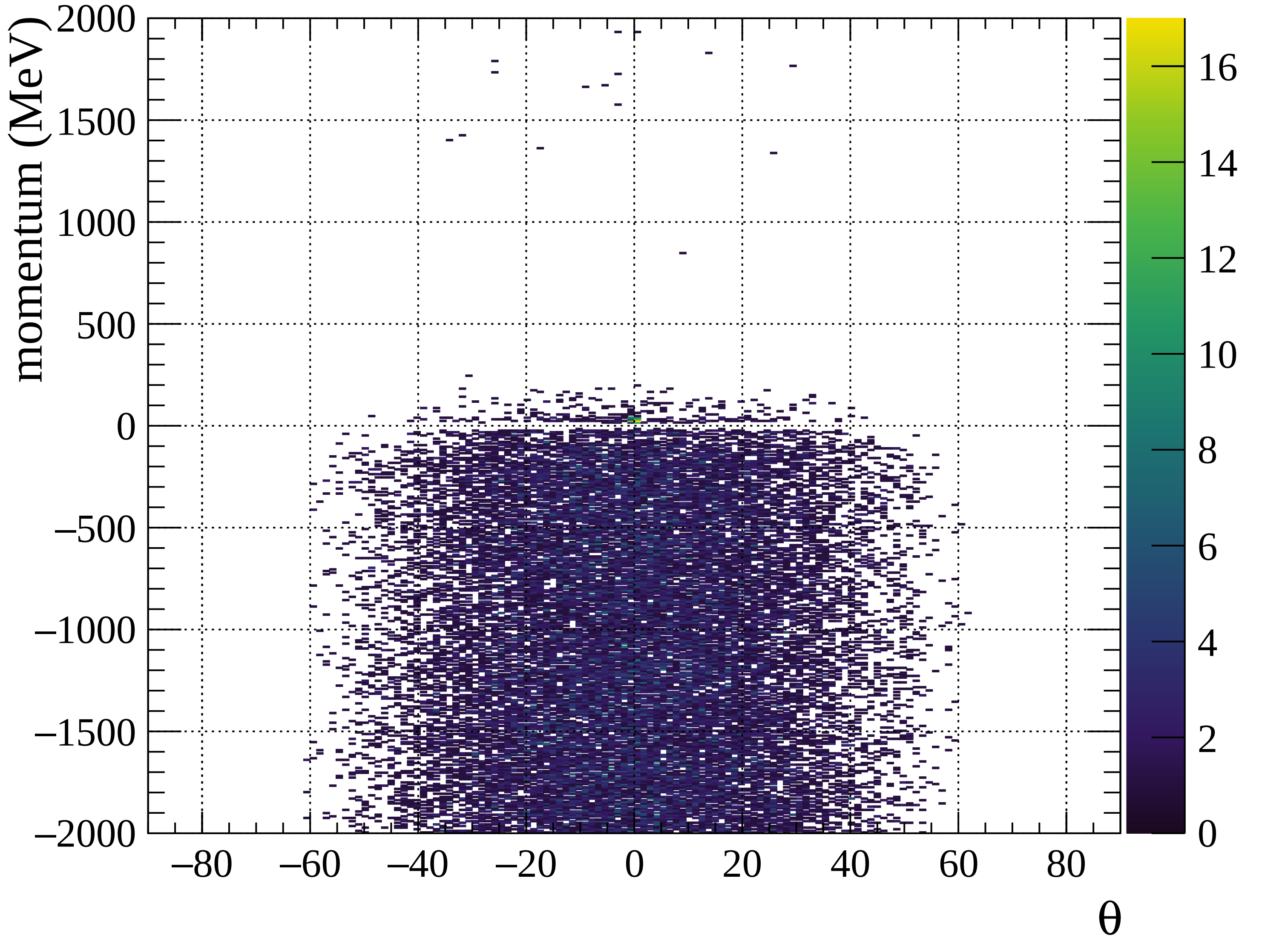

WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80



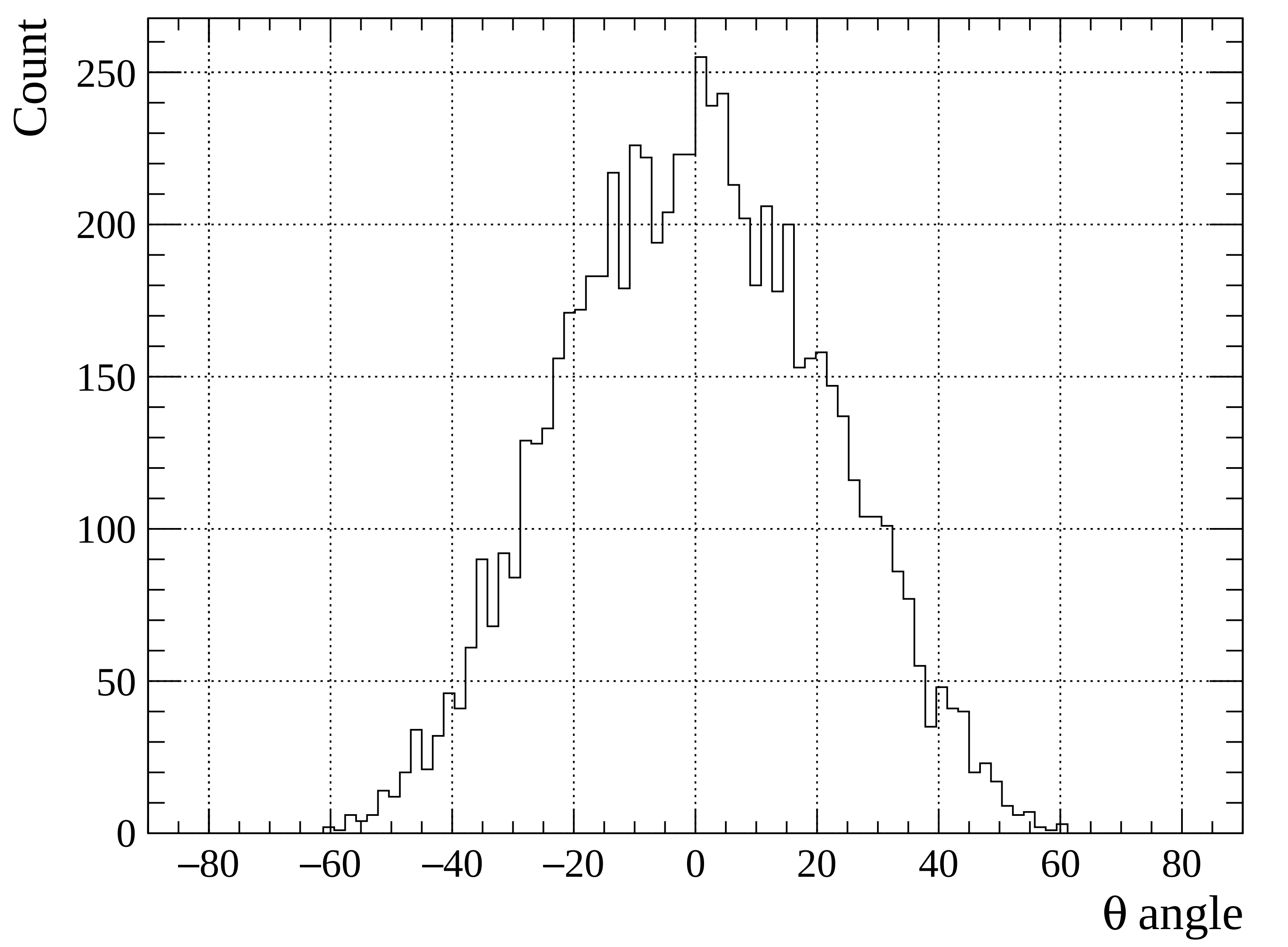
dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
2
4
6
8
10



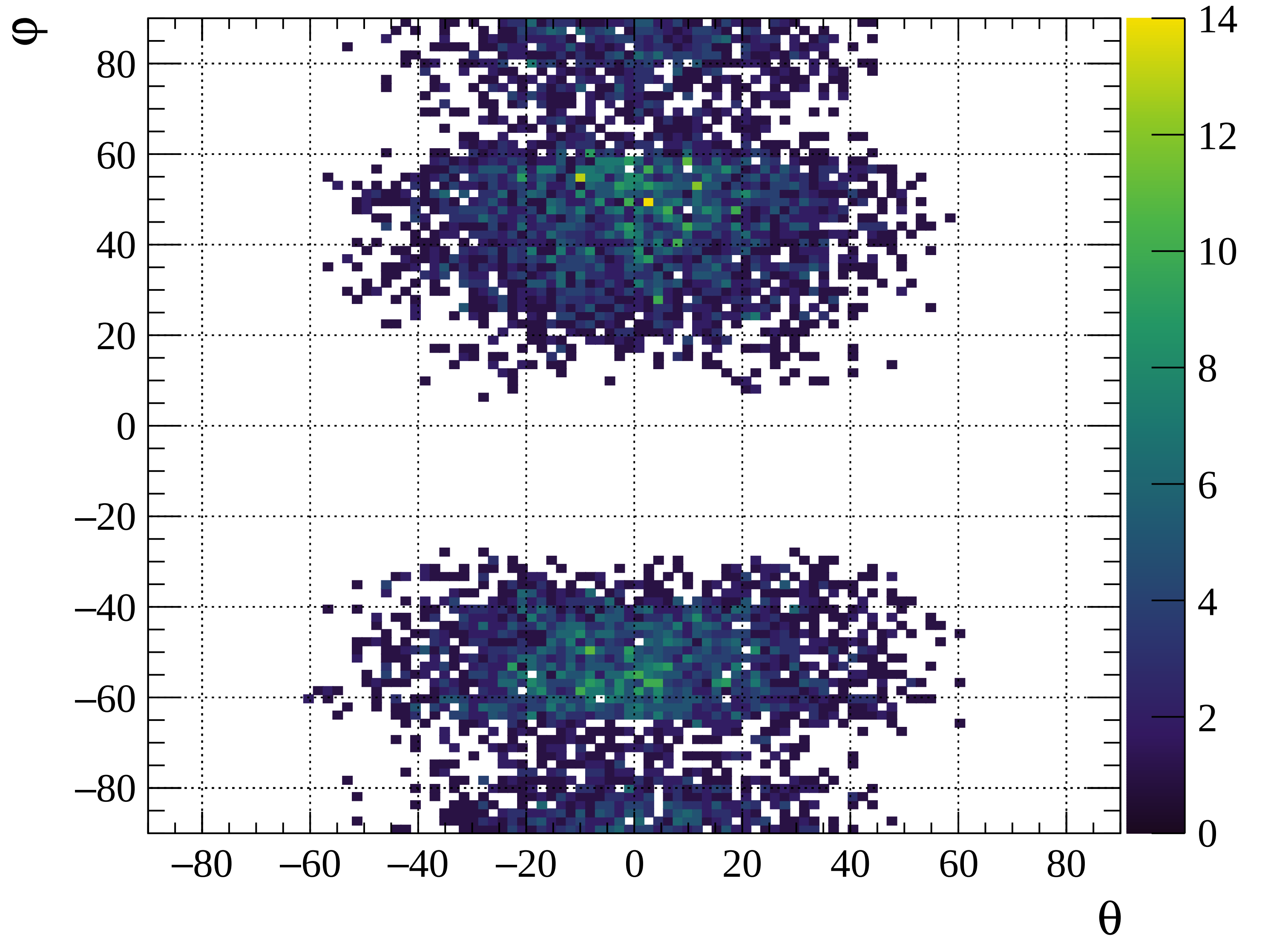
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
35
30
25
20
15
10
5
0



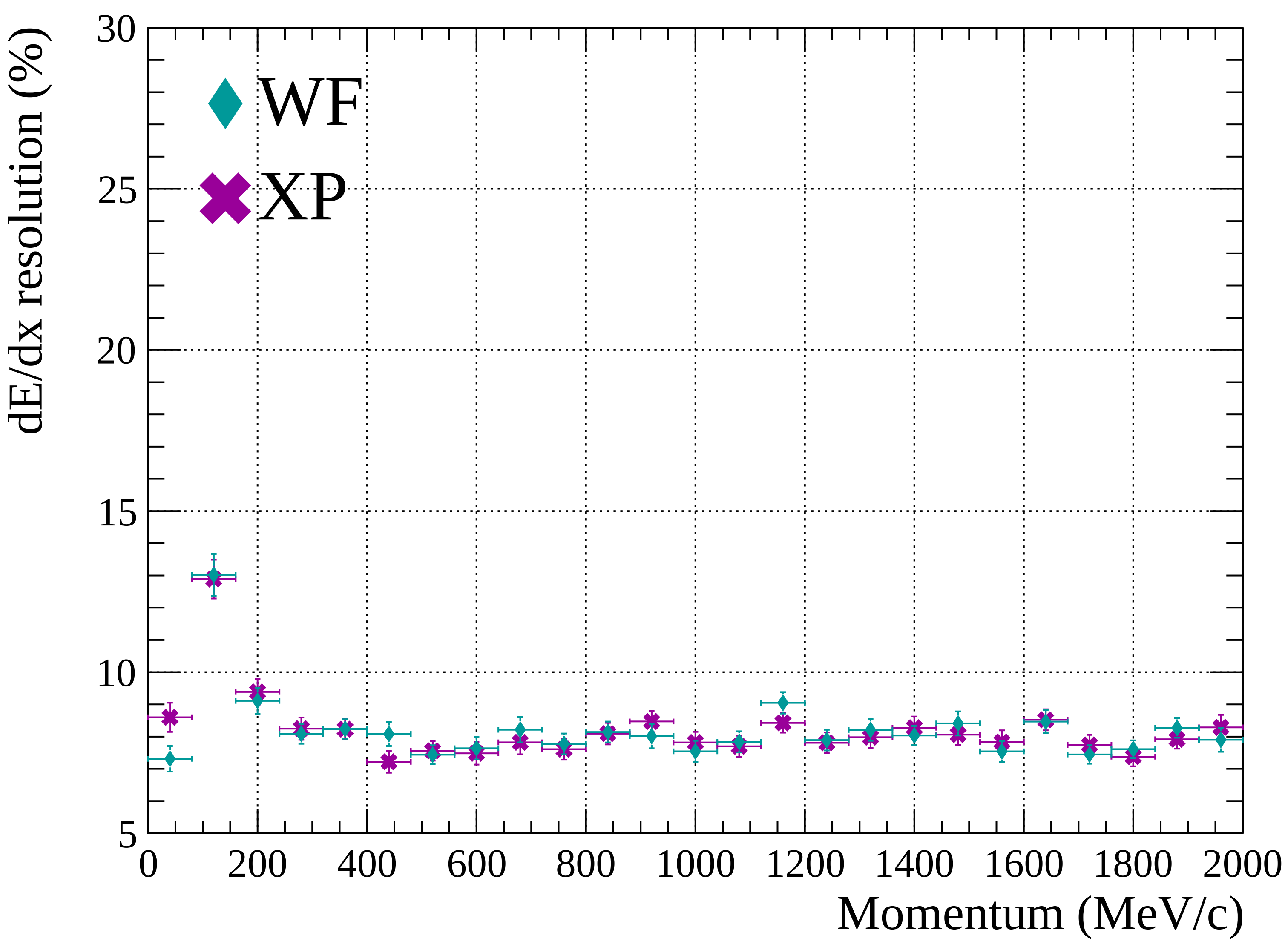
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
16
14
12
10
8
6
4
2
0

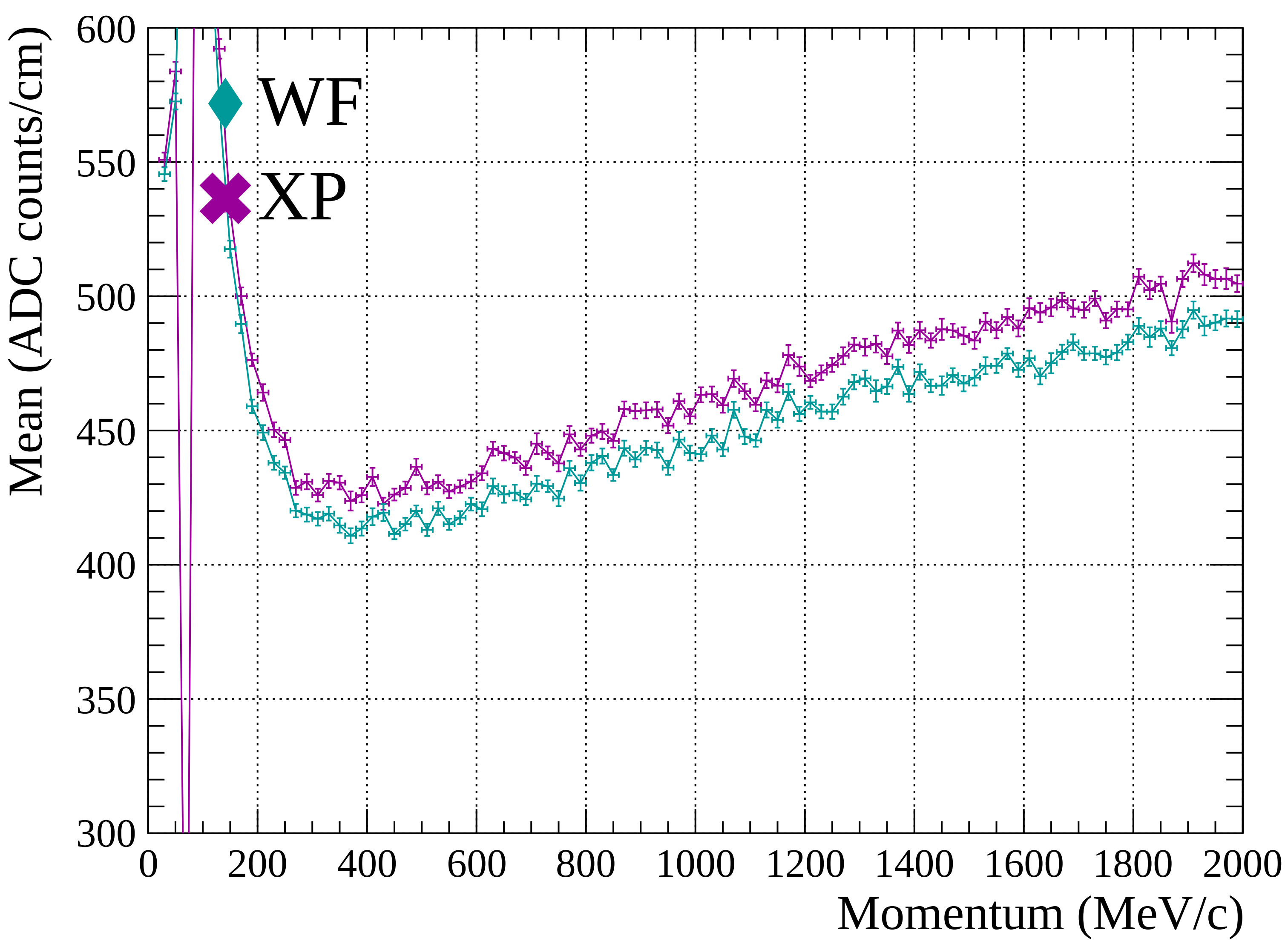


Count
250
200
150
100
50
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

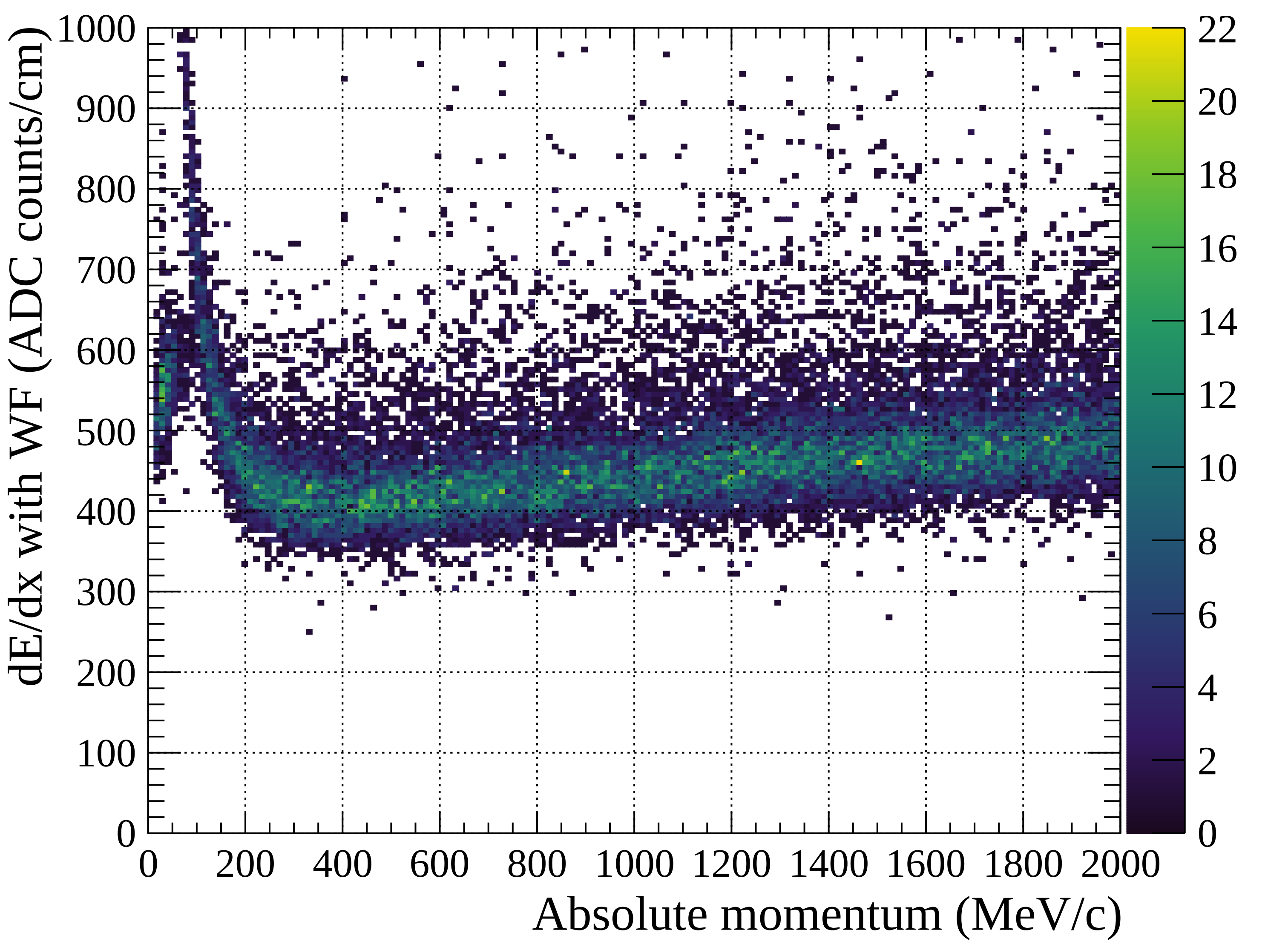


φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
14
12
10
8
6
4
2
0

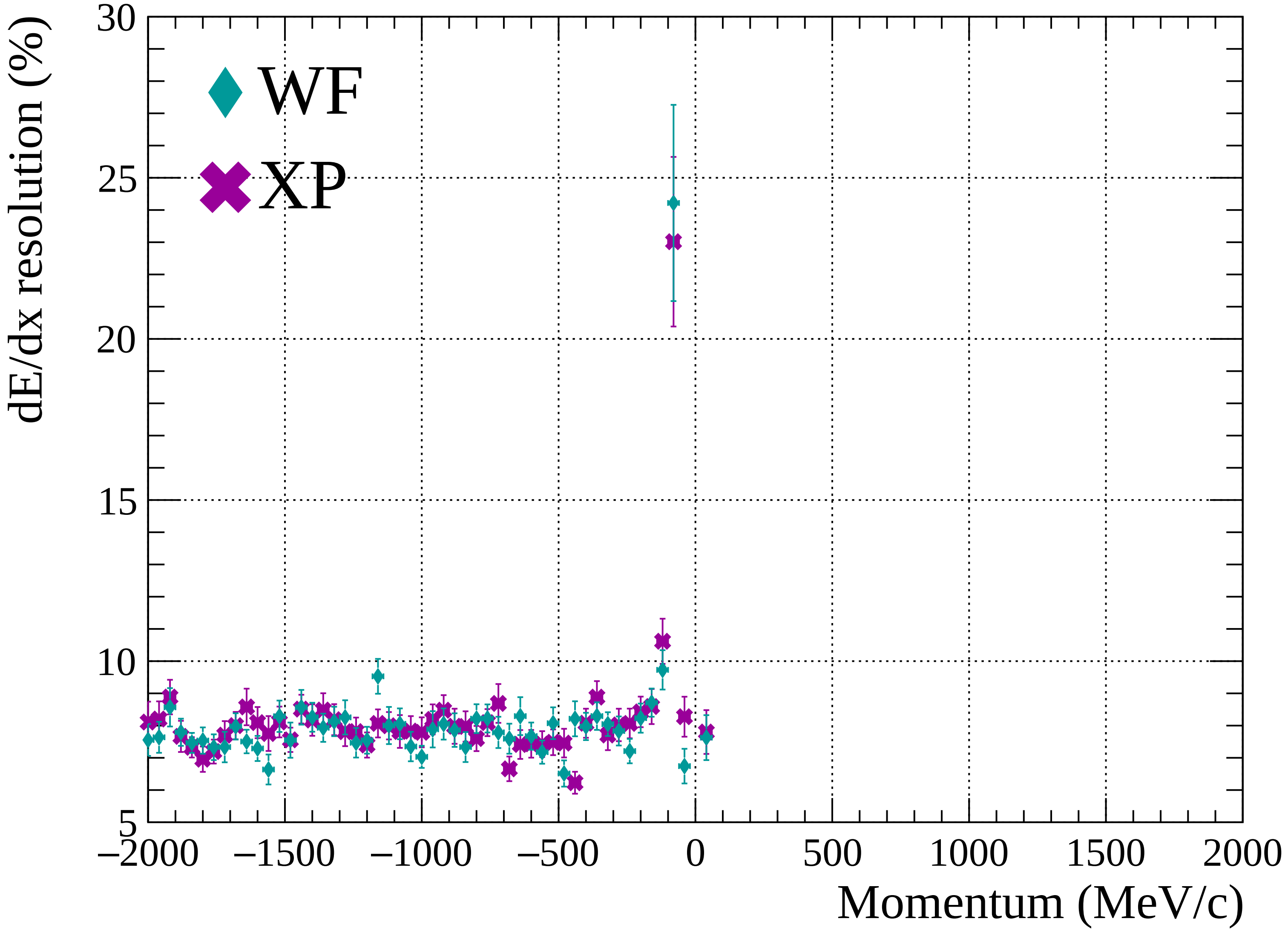

dE/dx resolution (%)
WF
XP
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000



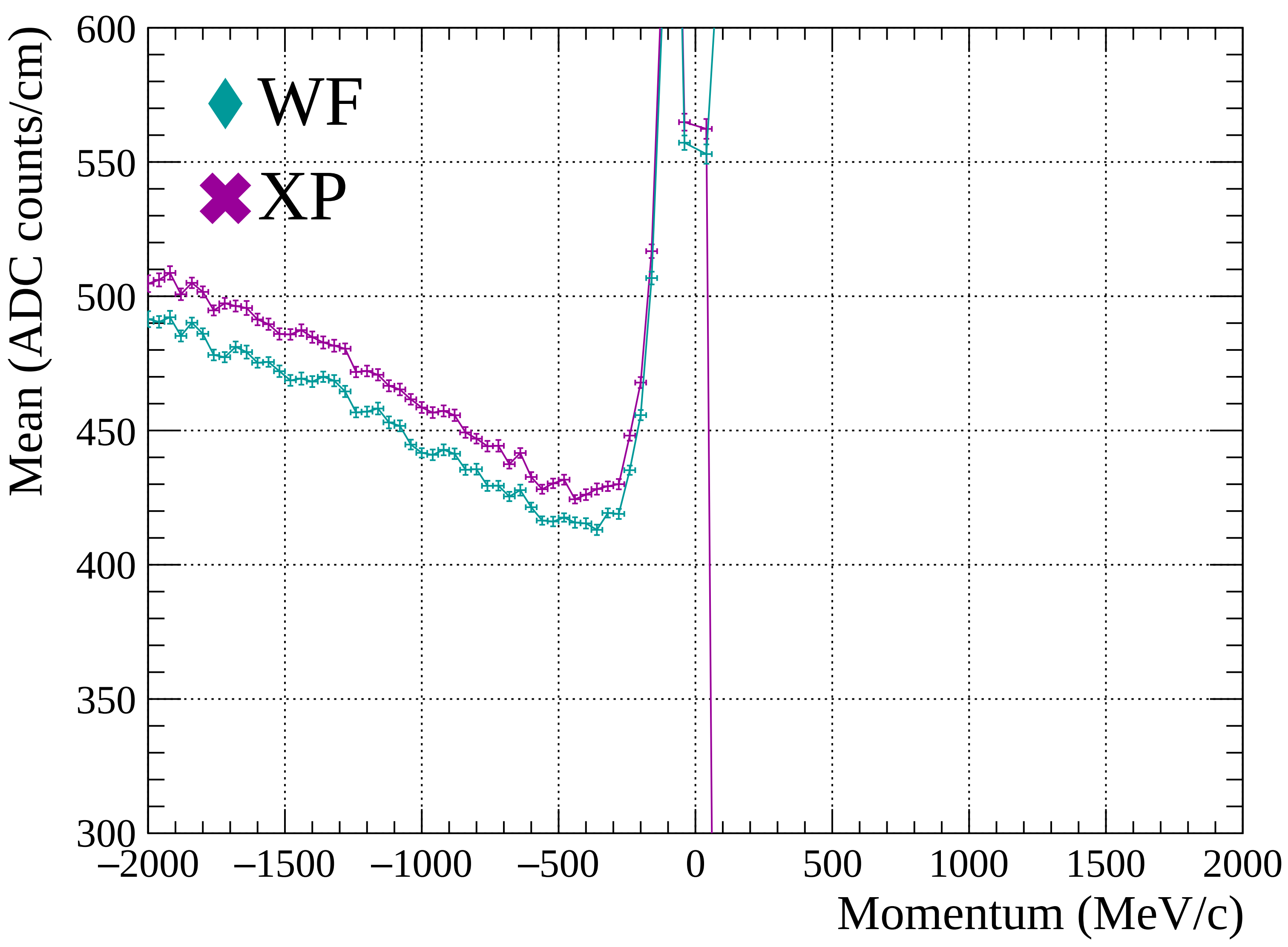
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)
WF
XP



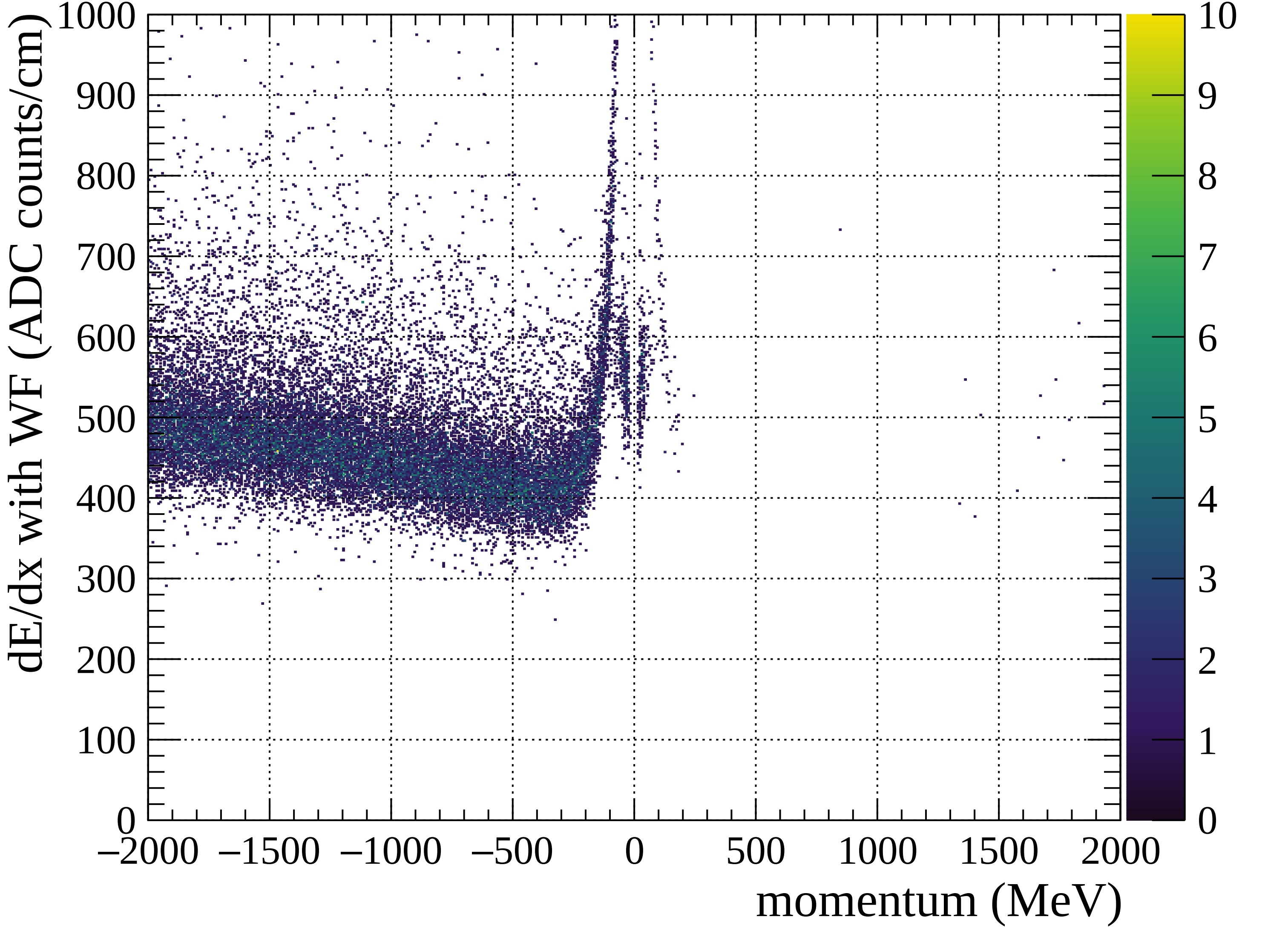
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)
22
20
18
16
14
12
10
8
6
4
2



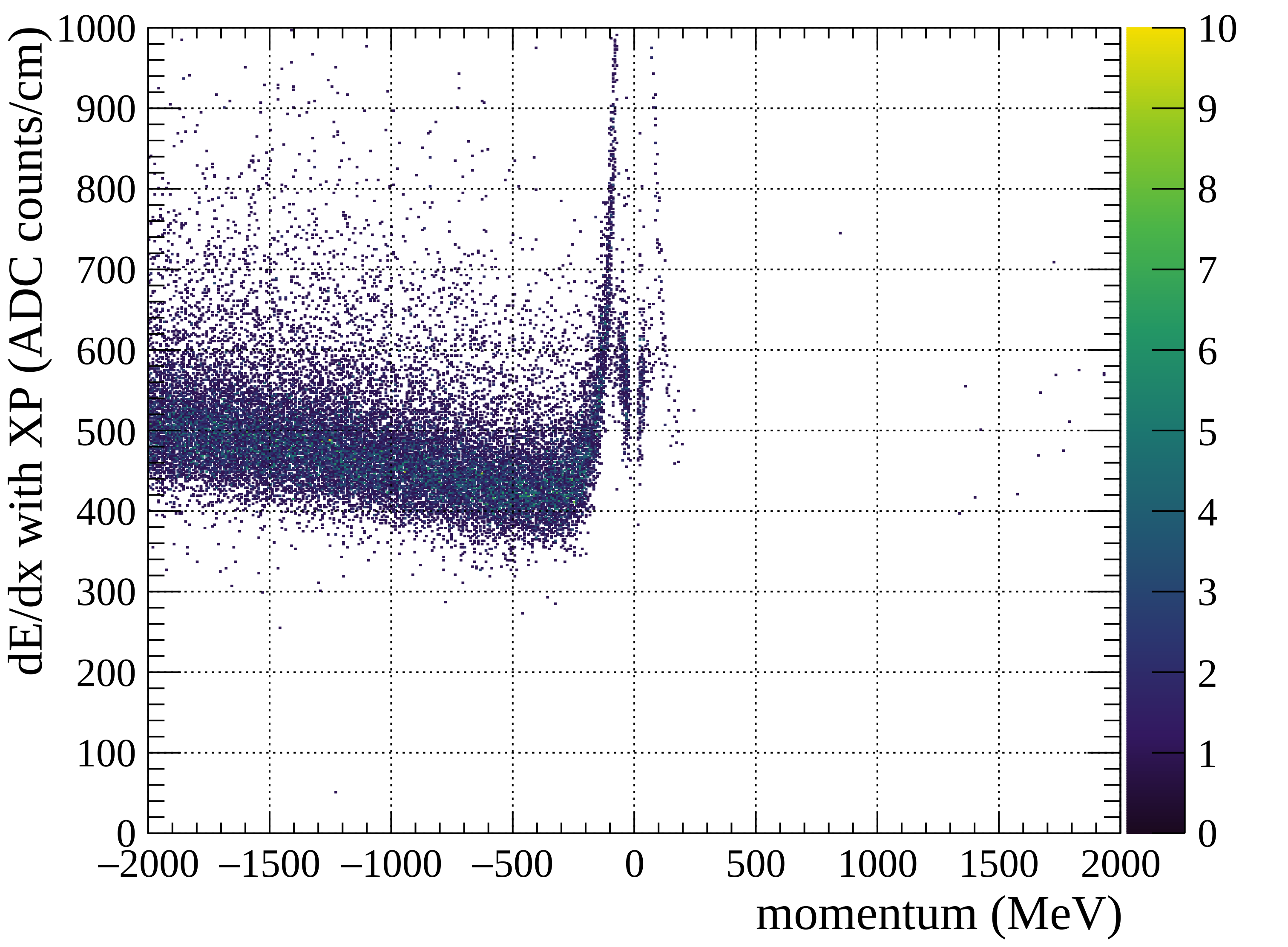
dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



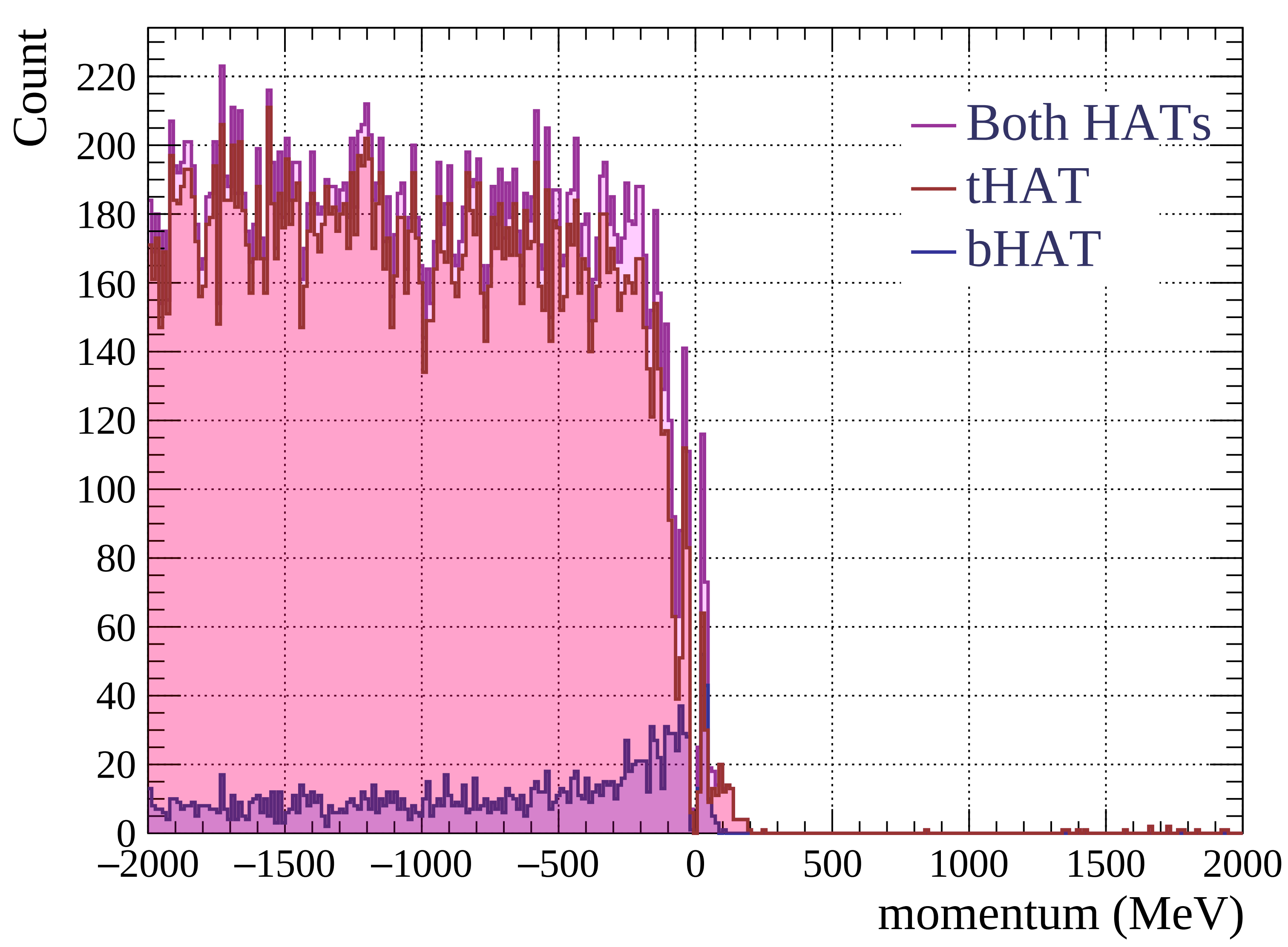
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

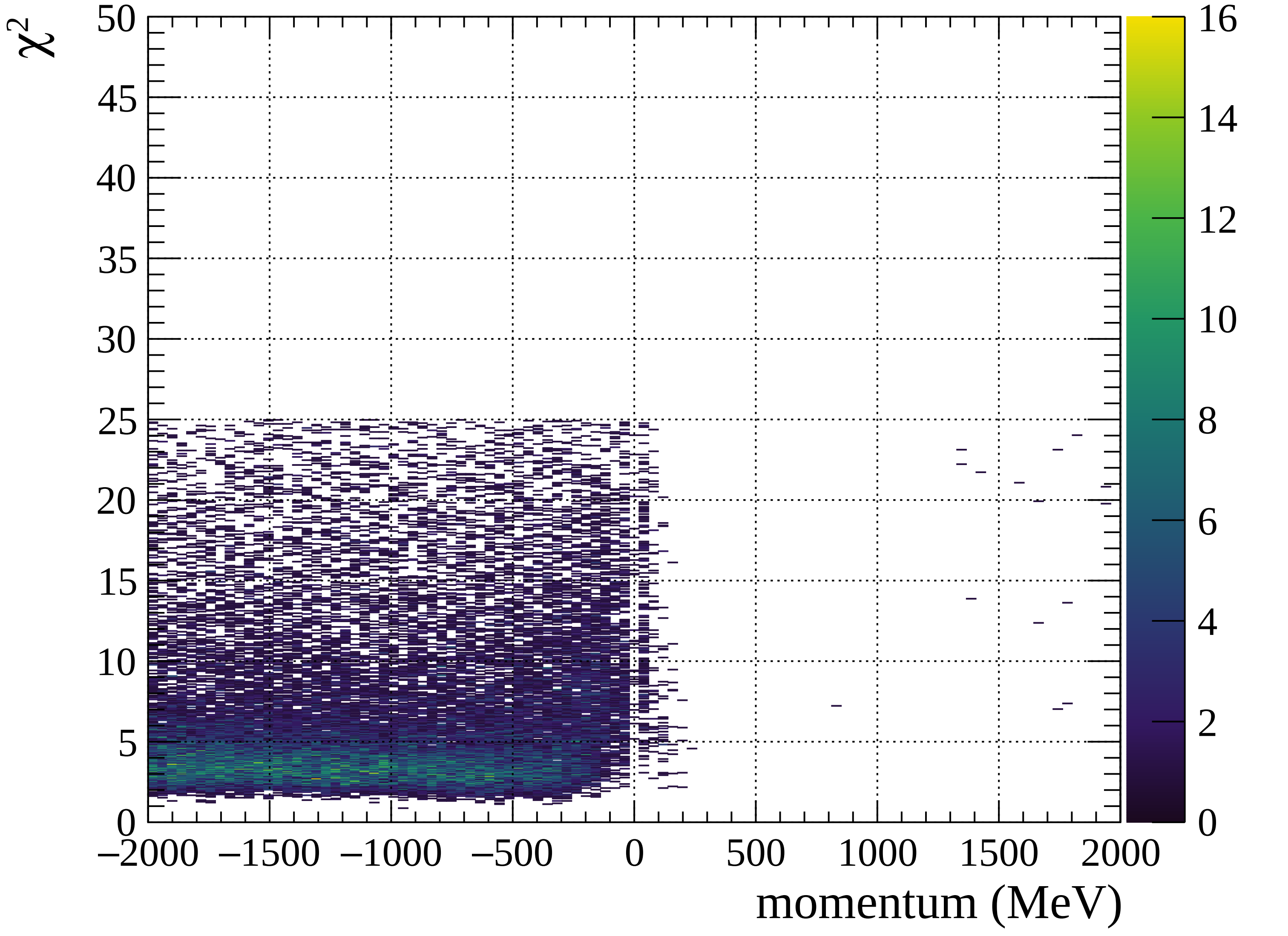


dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
1
2
3
4
5
6
7
8
9
10

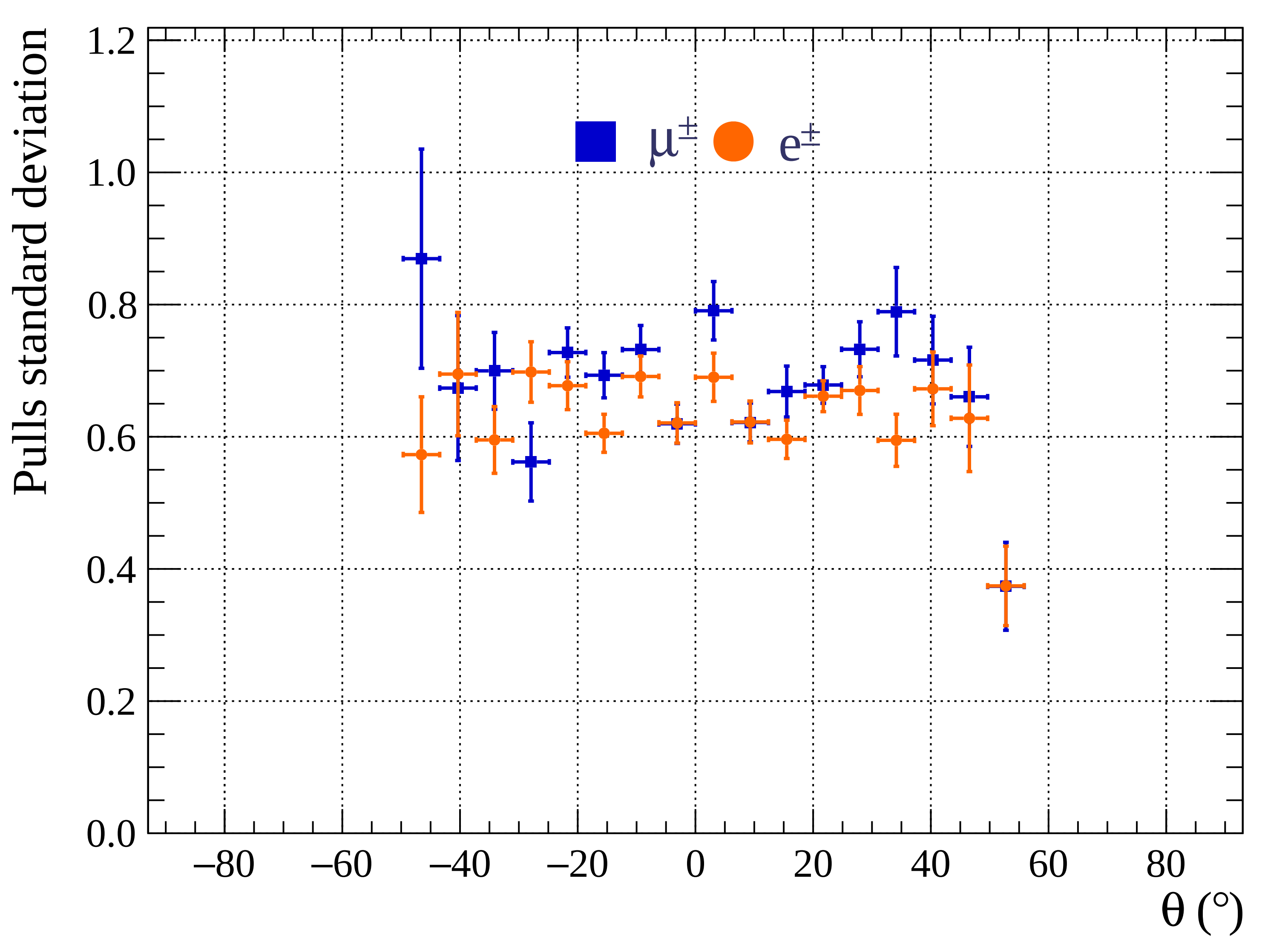


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
10
9
8
7
6
5
4
3
2
1
0

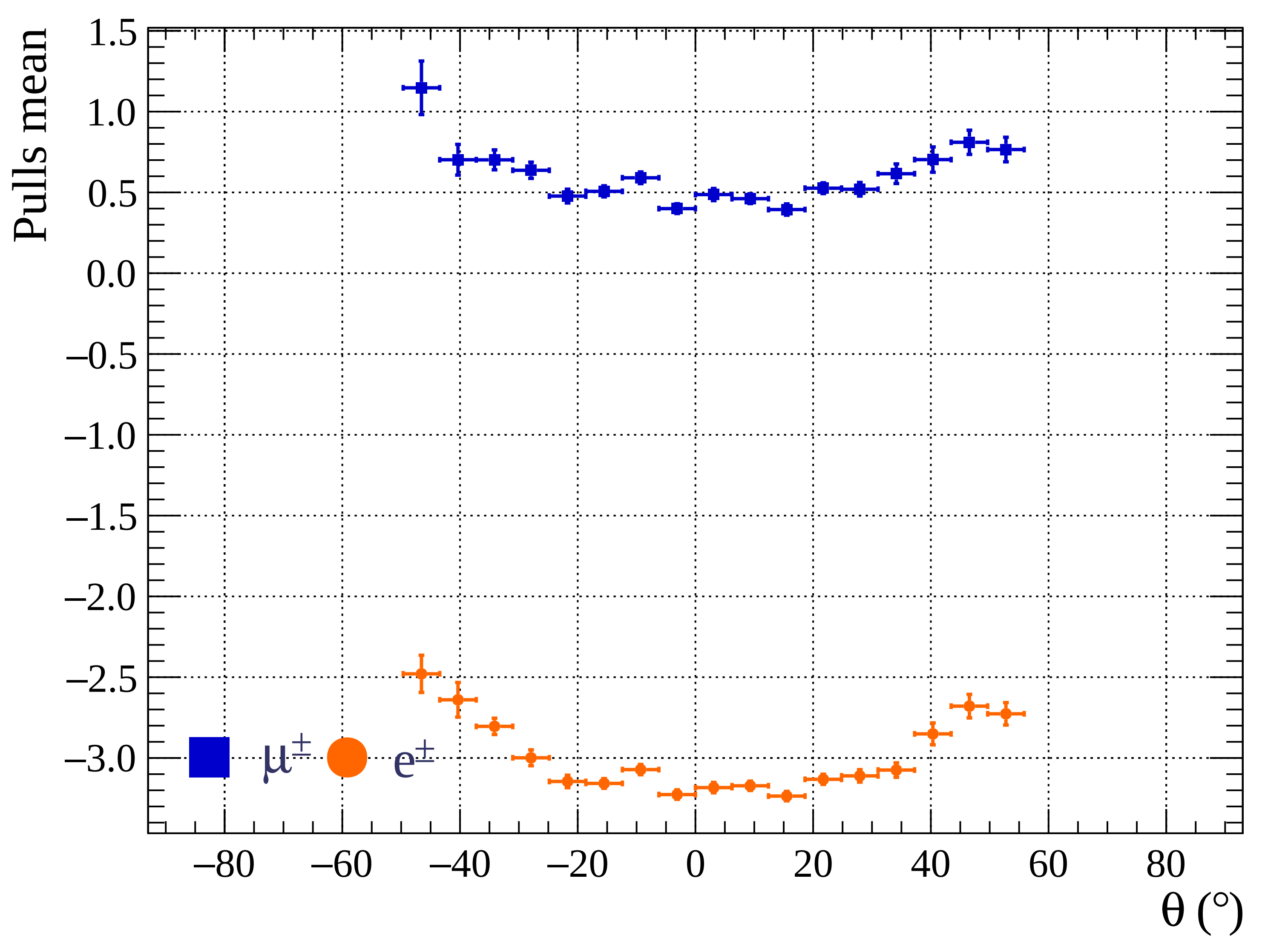

Count
220
200
180
160
140
120
100
80
60
40
20
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

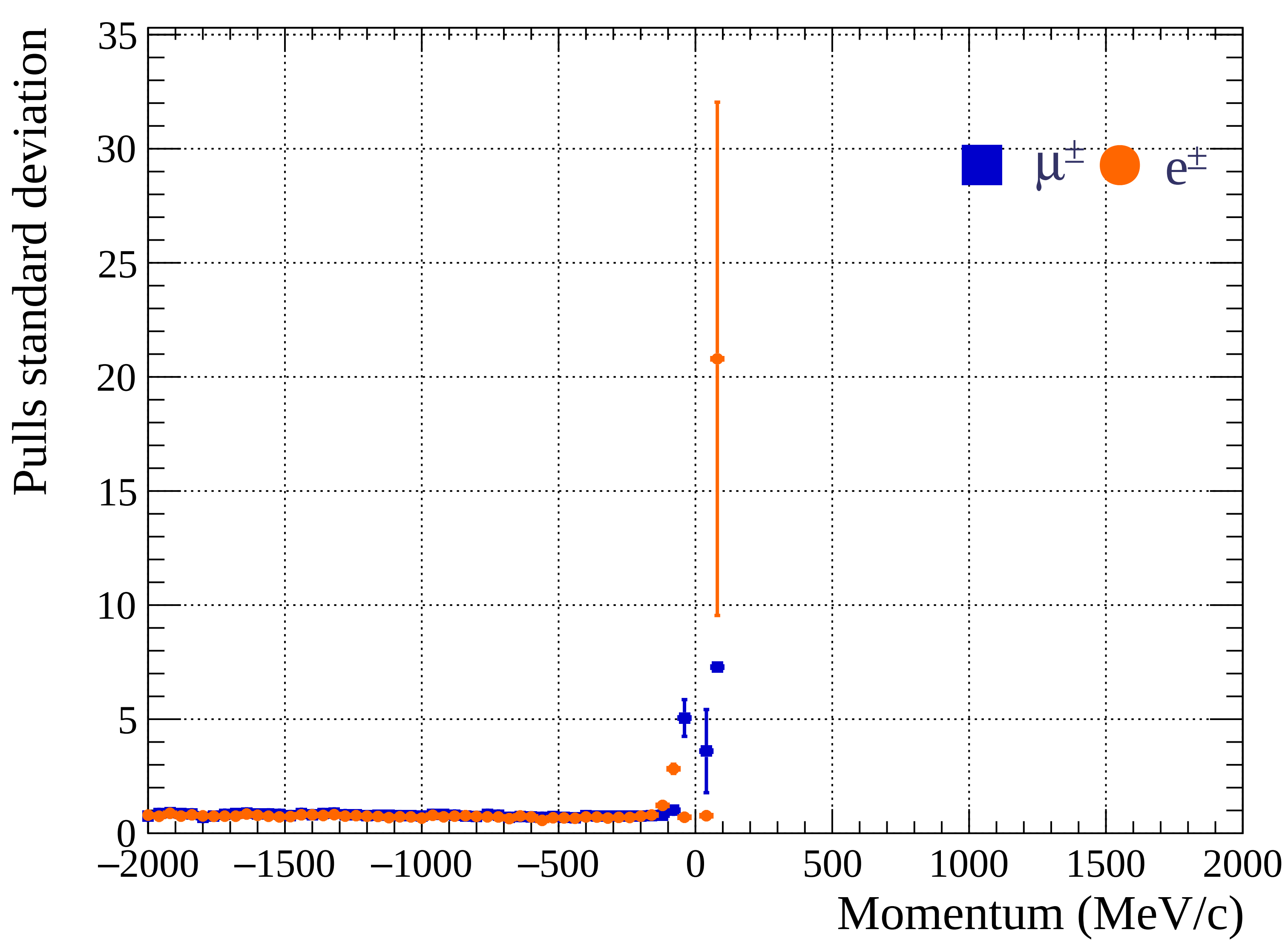


χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
16
14
12
10
8
6
4
2
0

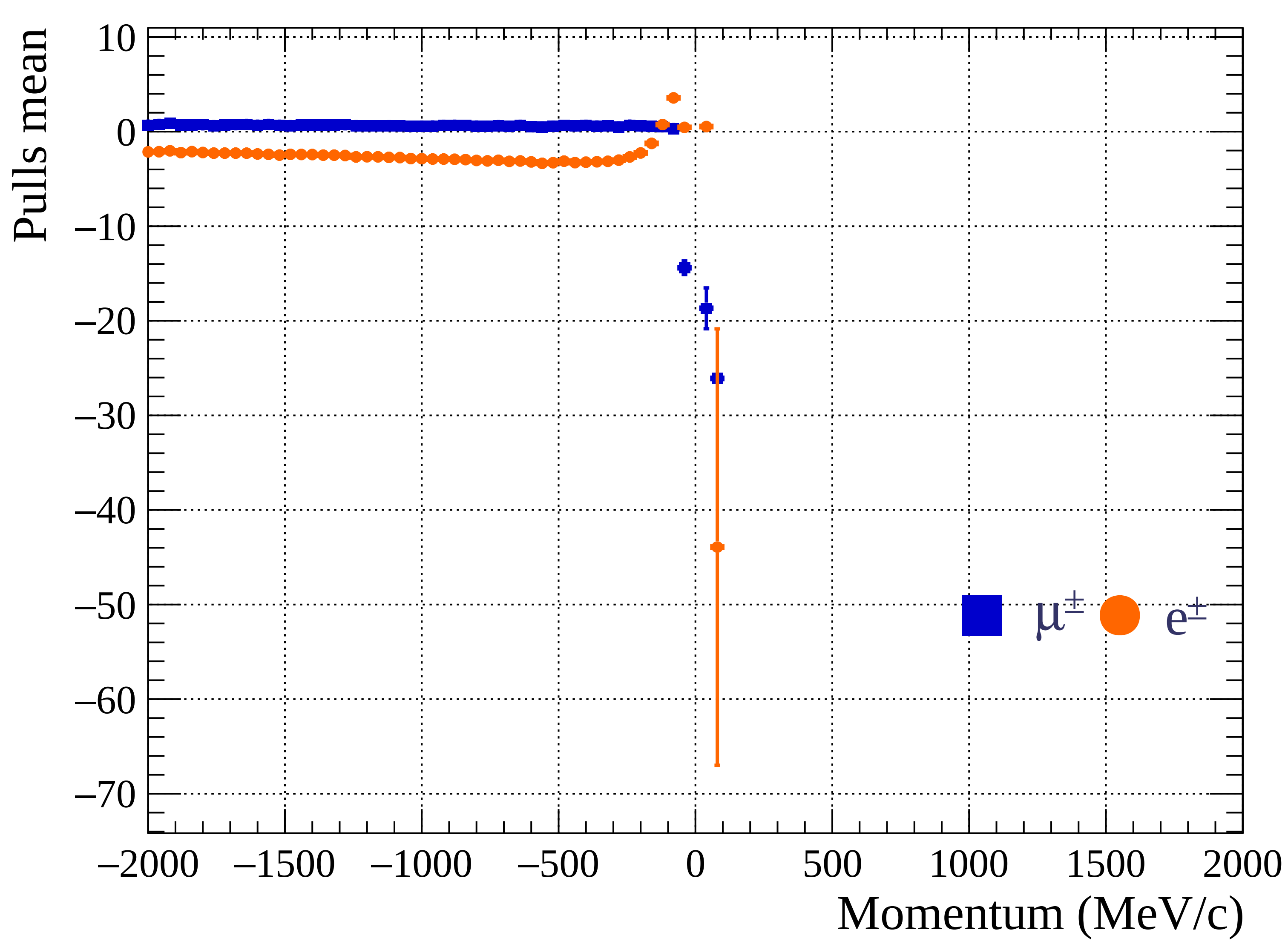


Pulls standard deviation
μ±
e±
θ (°)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80

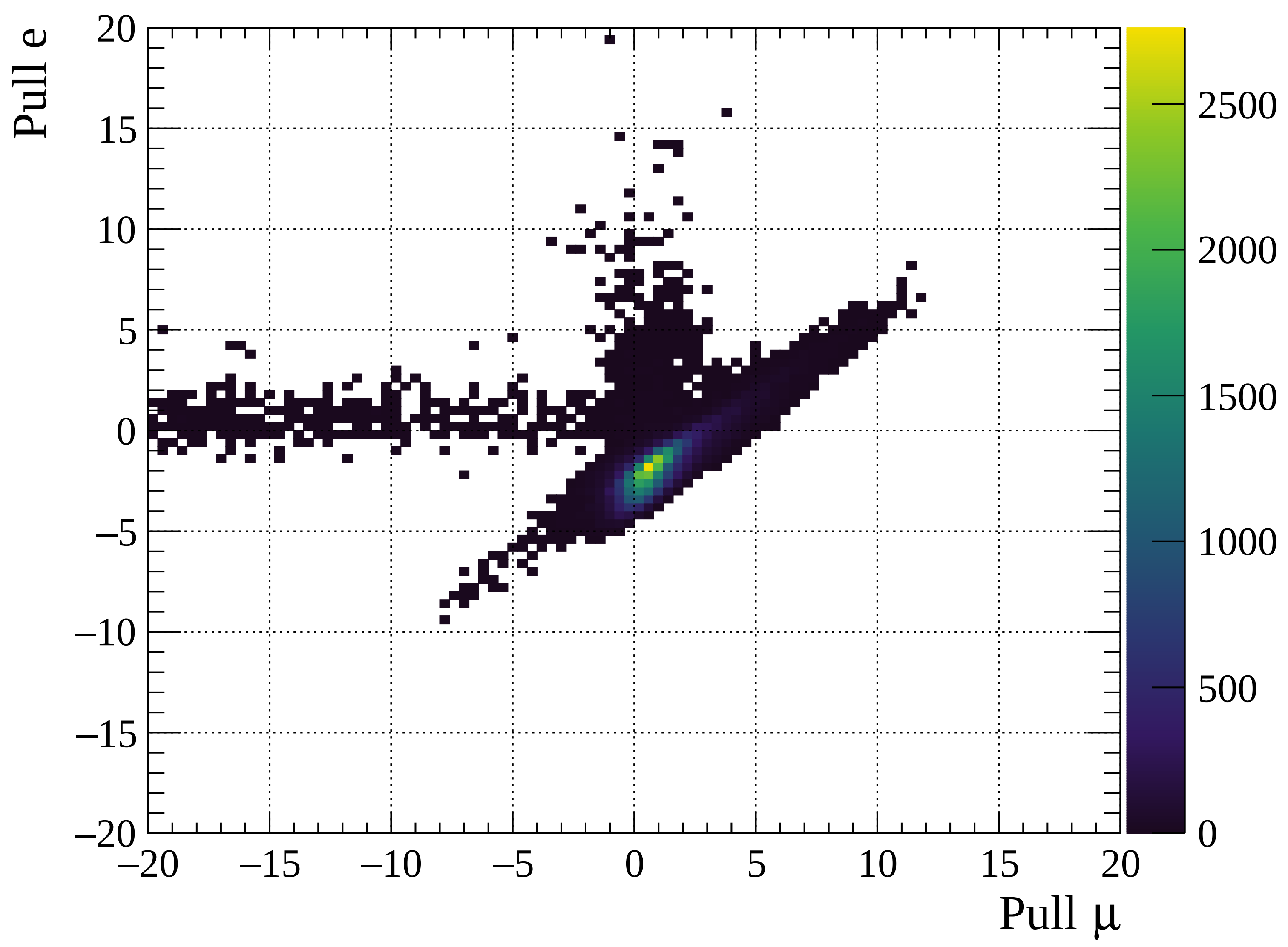

Pulls mean
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



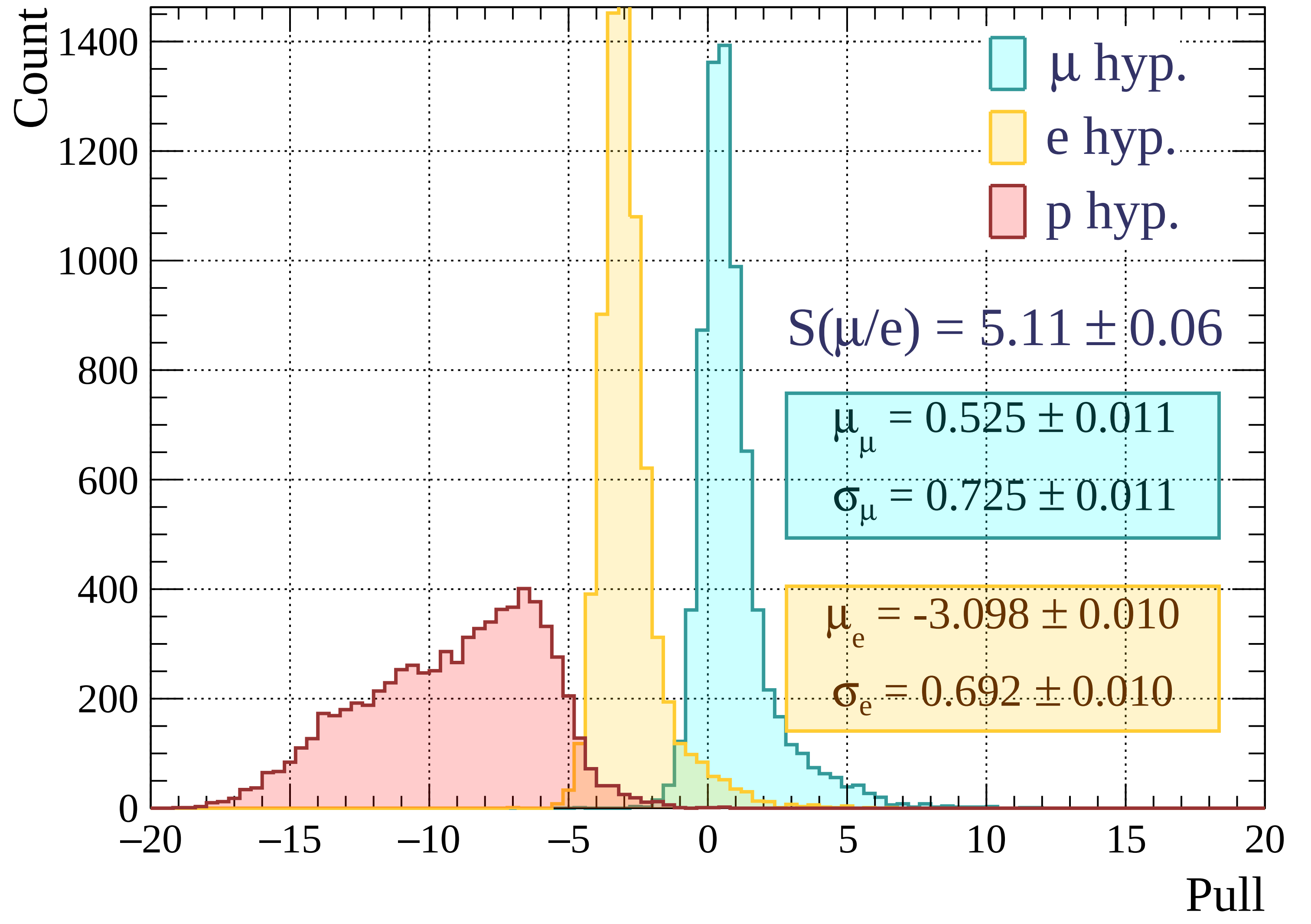
Pulls standard deviation
Momentum (MeV/c)
0
5
10
15
20
25
30
35
−2000
−1500
−1000
−500
0
500
1000
1500
2000
$\mu^{\pm}$
$e^{\pm}$



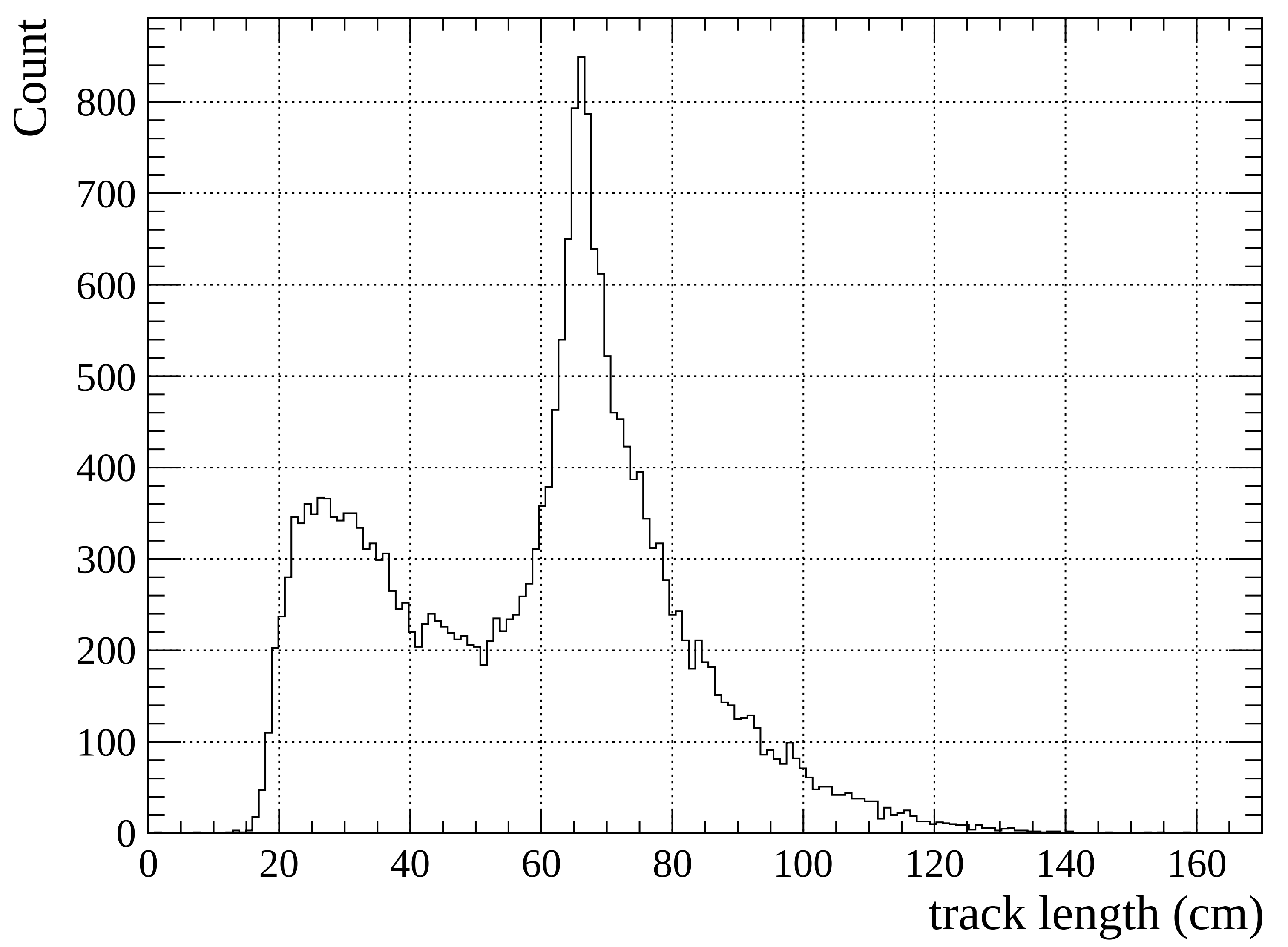
Pulls mean
10
0
−10
−20
−30
−40
−50
−60
−70
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$



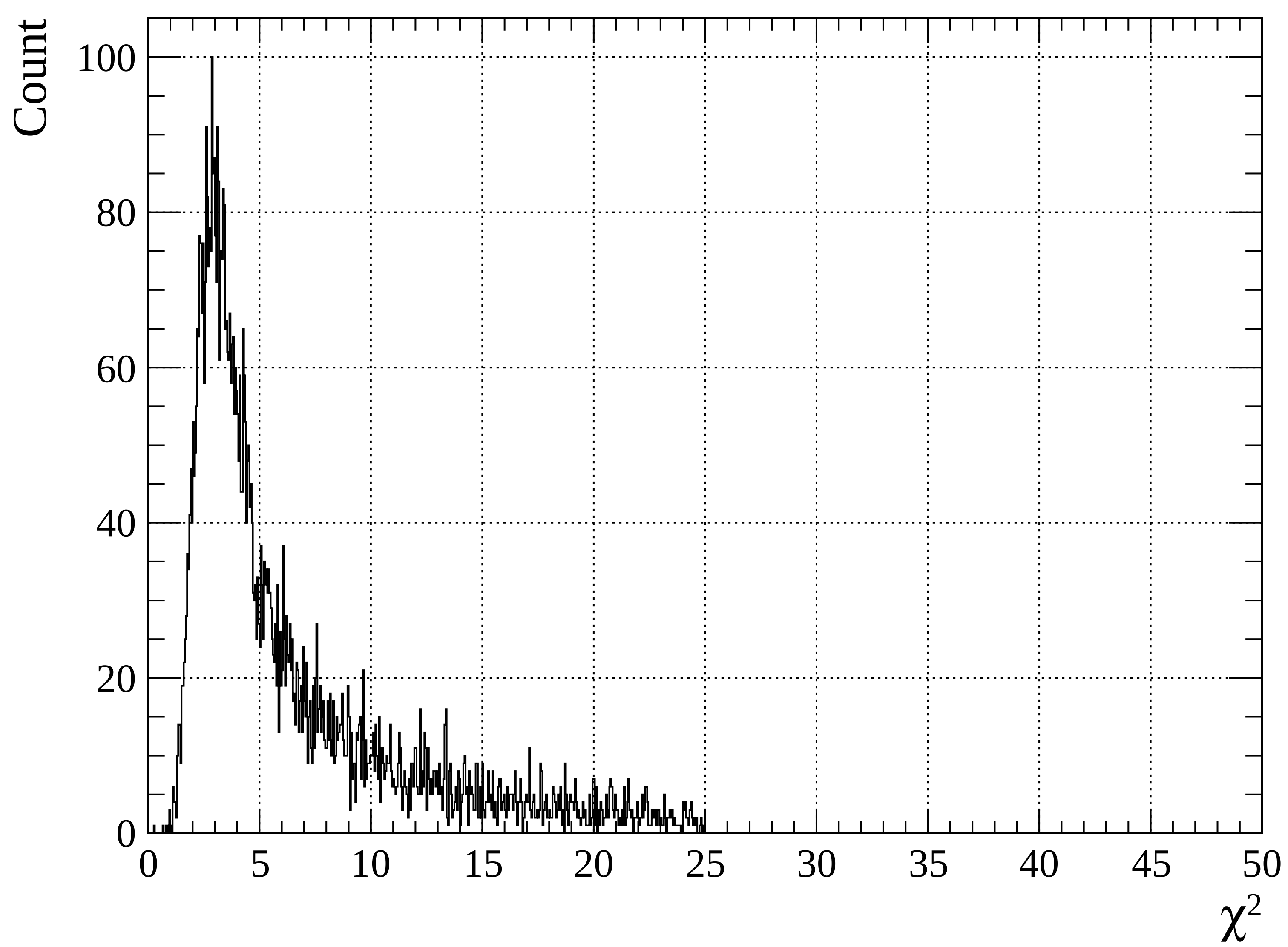
Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
2500
2000
1500
1000
500
0

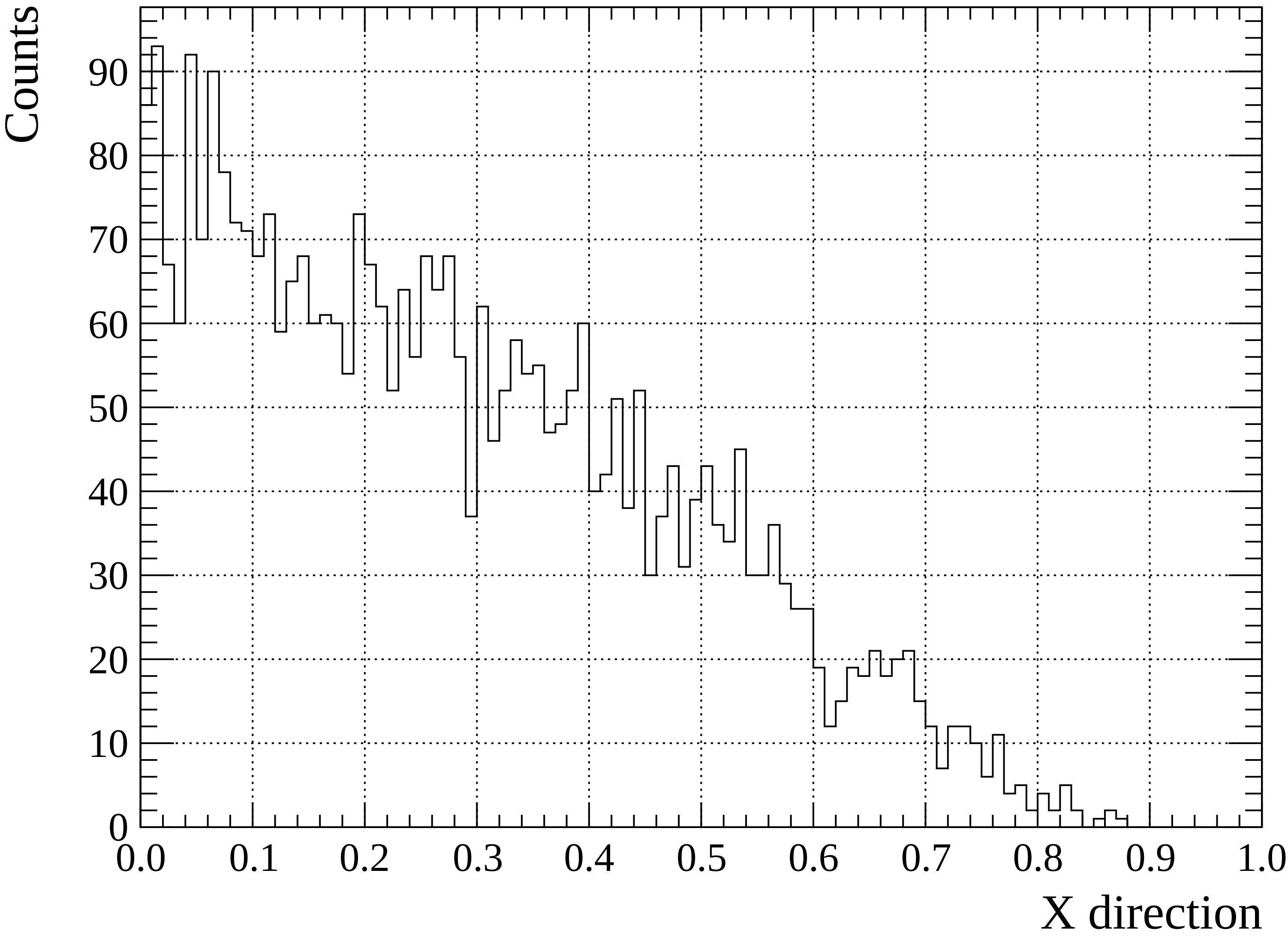


Count
1400
1200
1000
800
600
400
200
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.11 ± 0.06
μ_μ = 0.525 ± 0.011
σ_μ = 0.725 ± 0.011
μ_e = -3.098 ± 0.010
σ_e = 0.692 ± 0.010

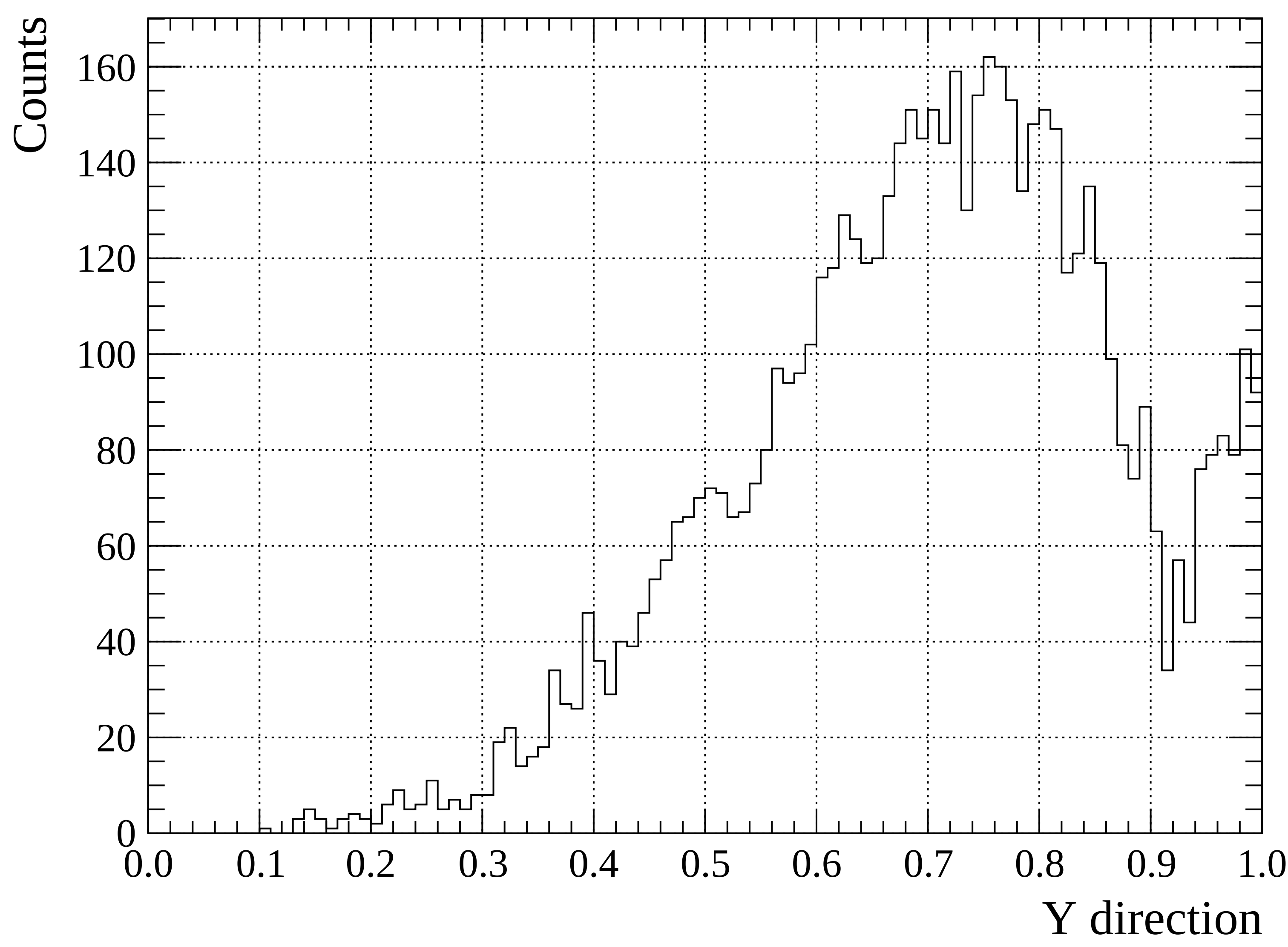


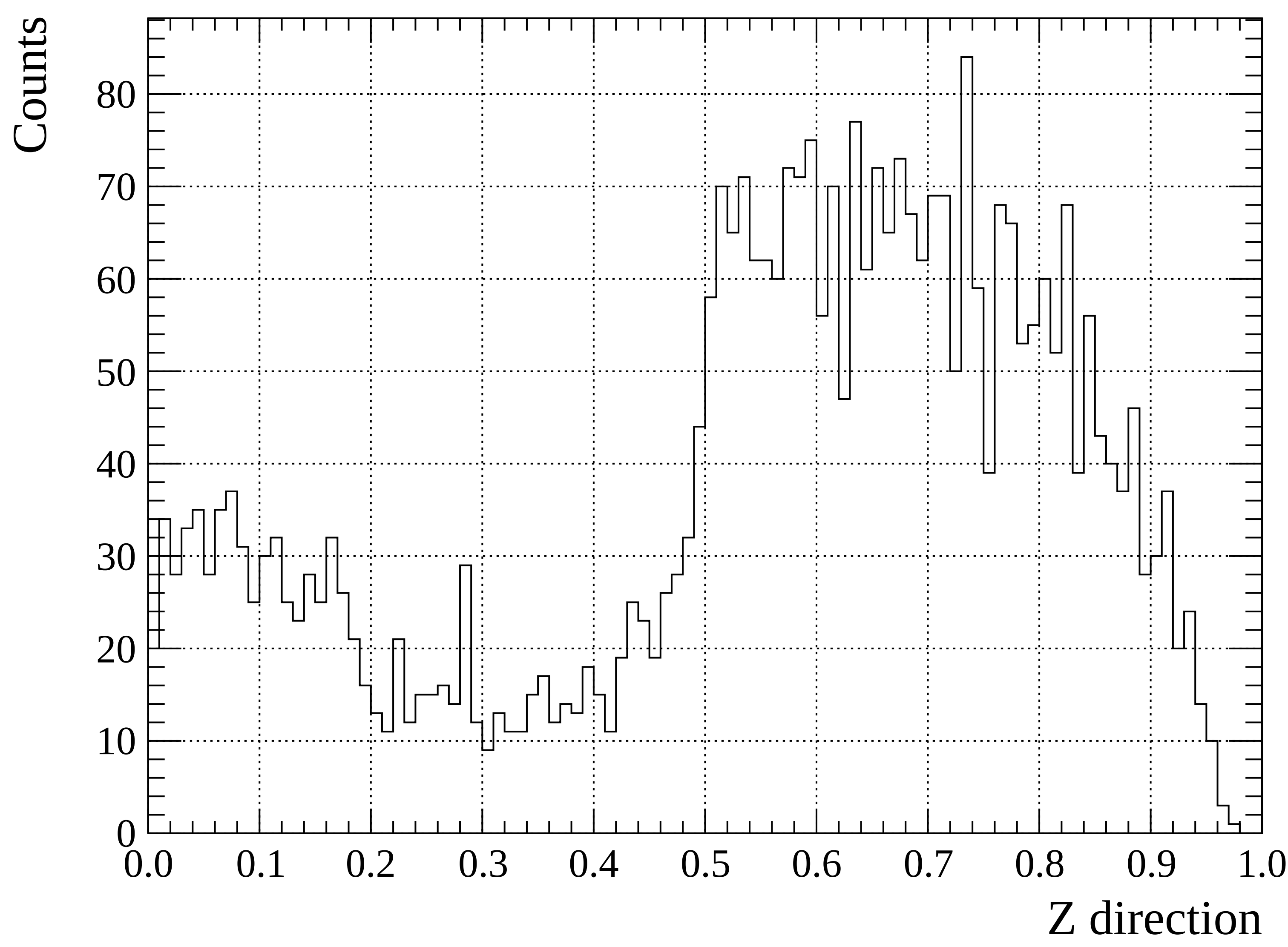
Count
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)

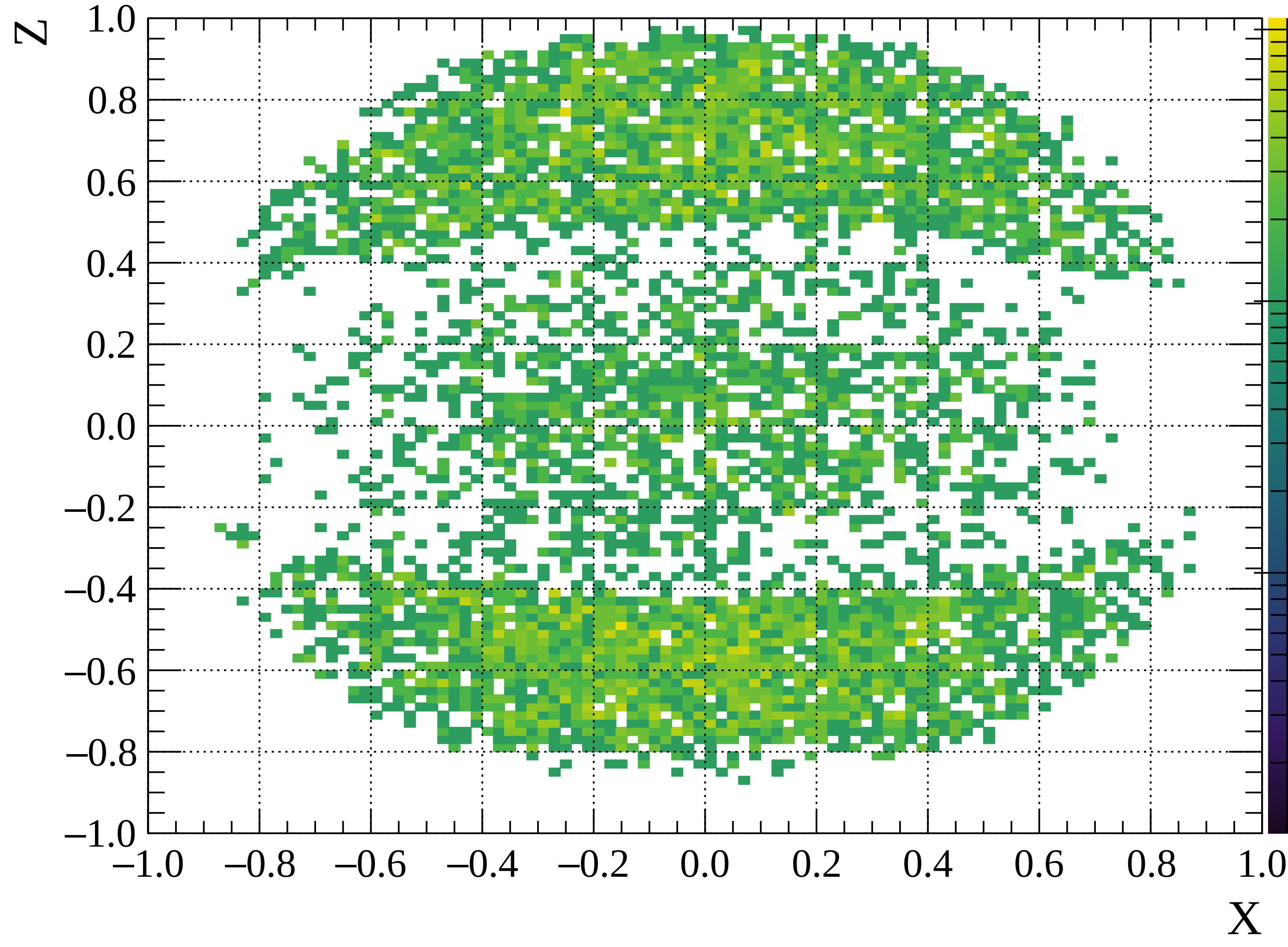

Count
100
80
60
40
20
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

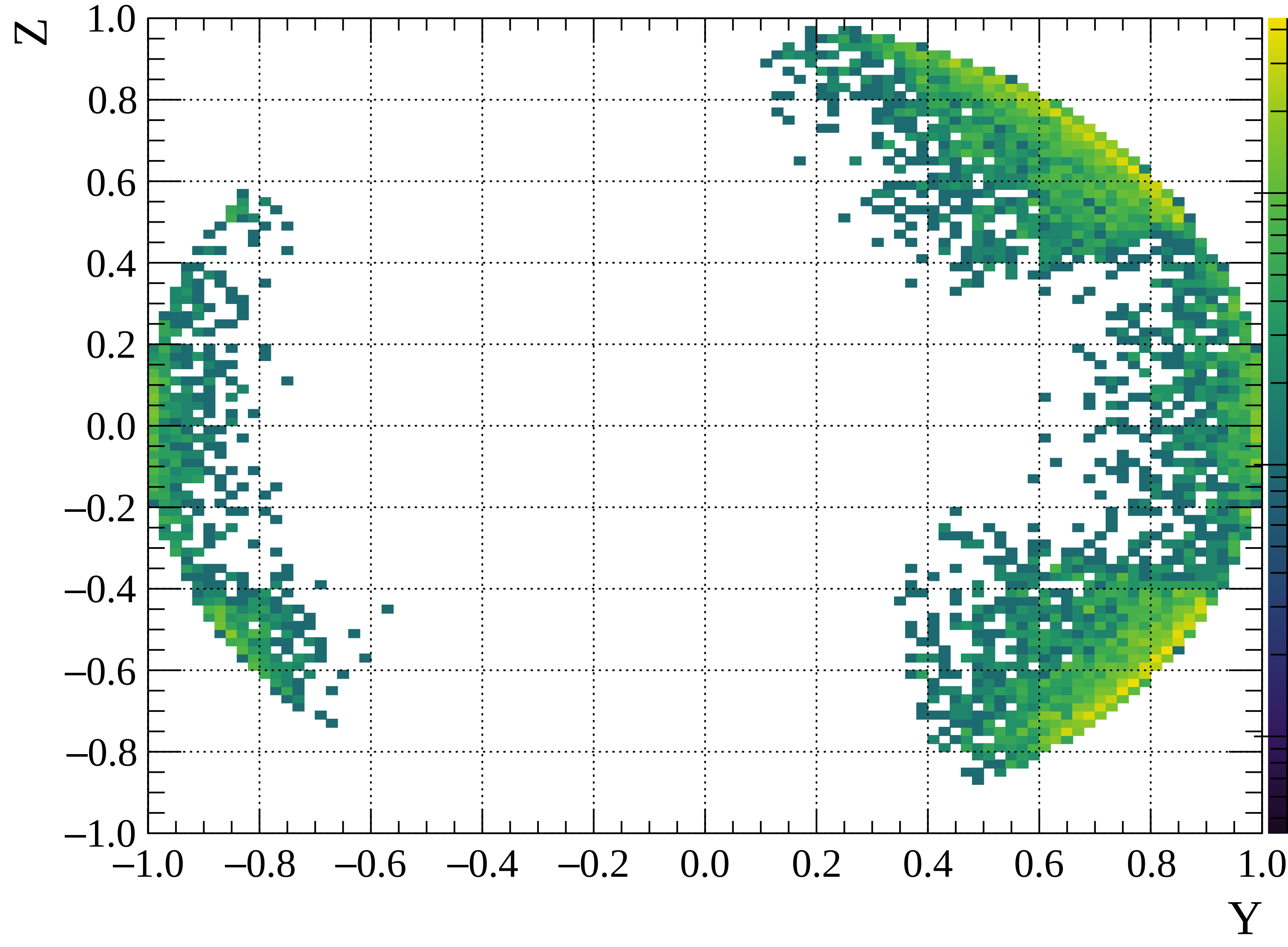


Counts
90
80
70
60
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction

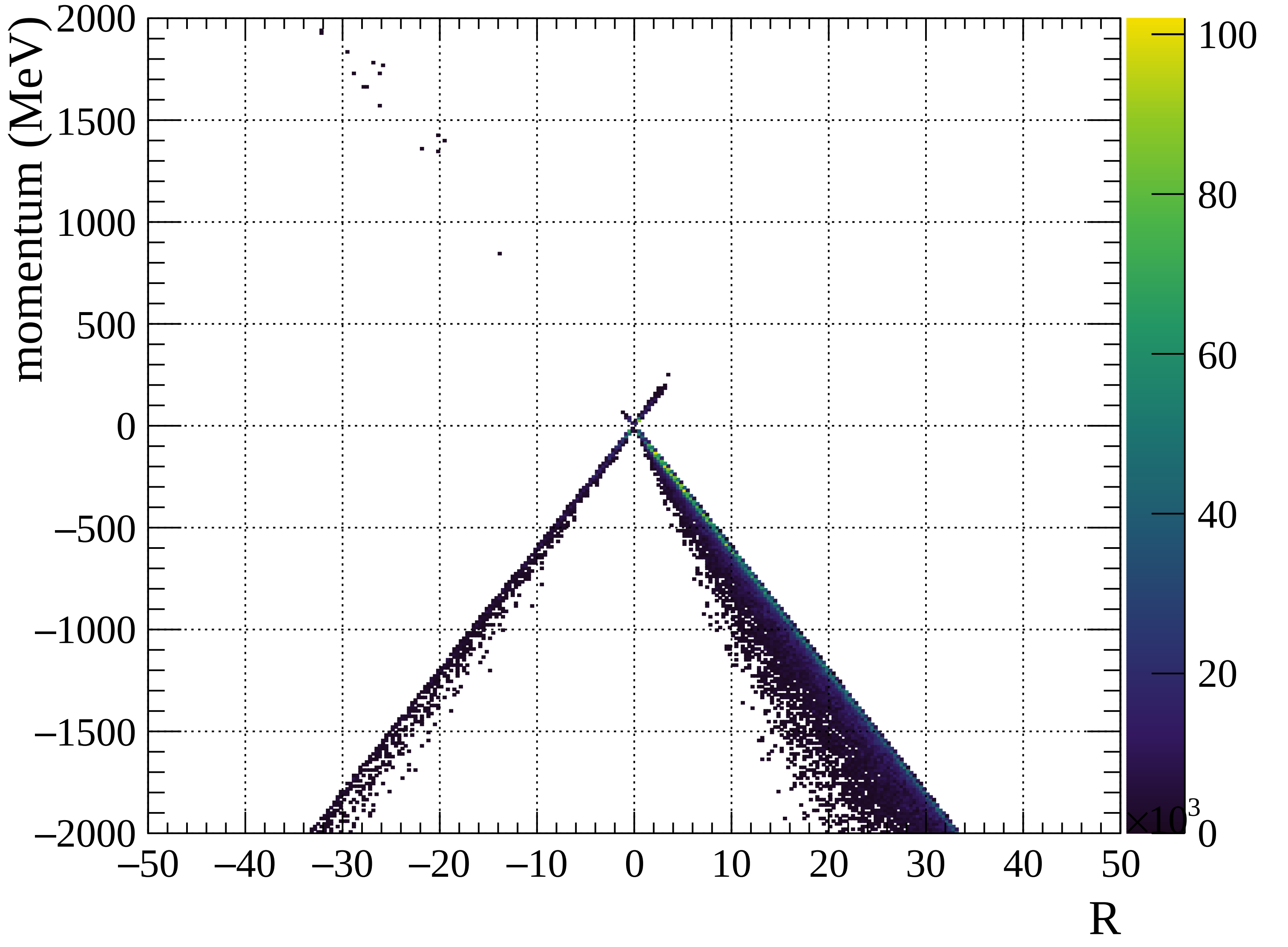

Counts
160
140
120
100
80
60
40
20
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction


Counts
0
10
20
30
40
50
60
70
80
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction

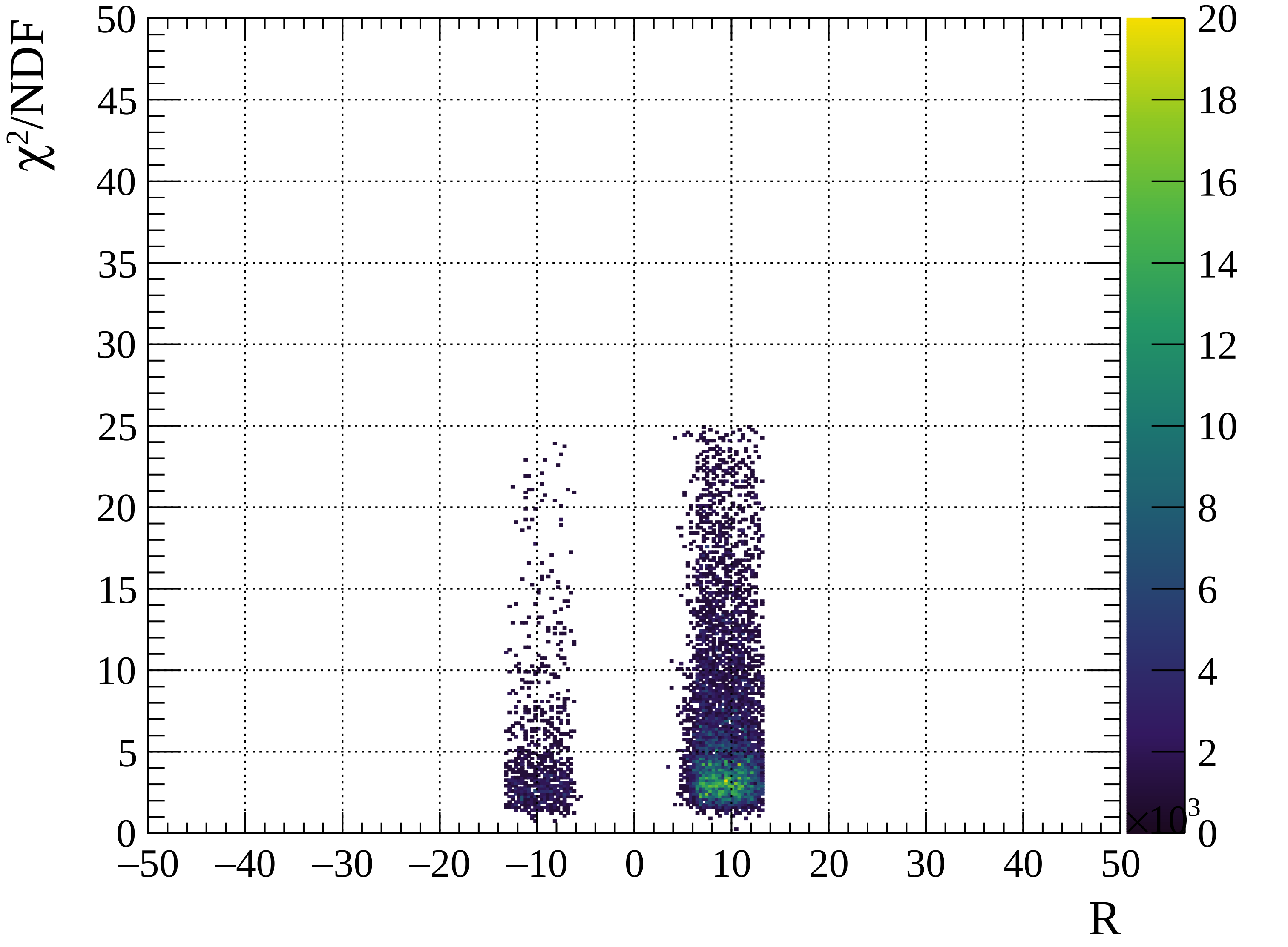

Z
X
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0



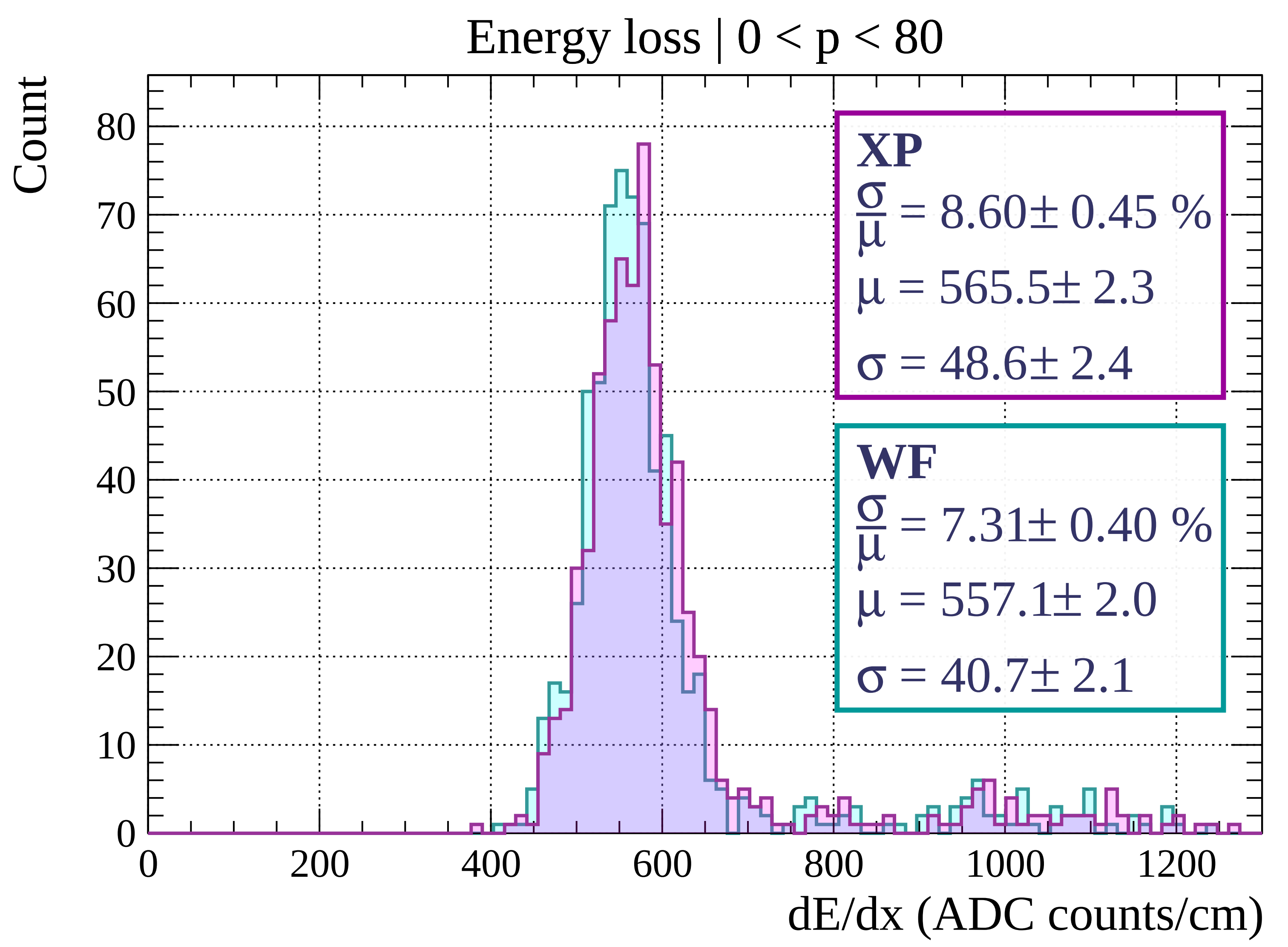
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y



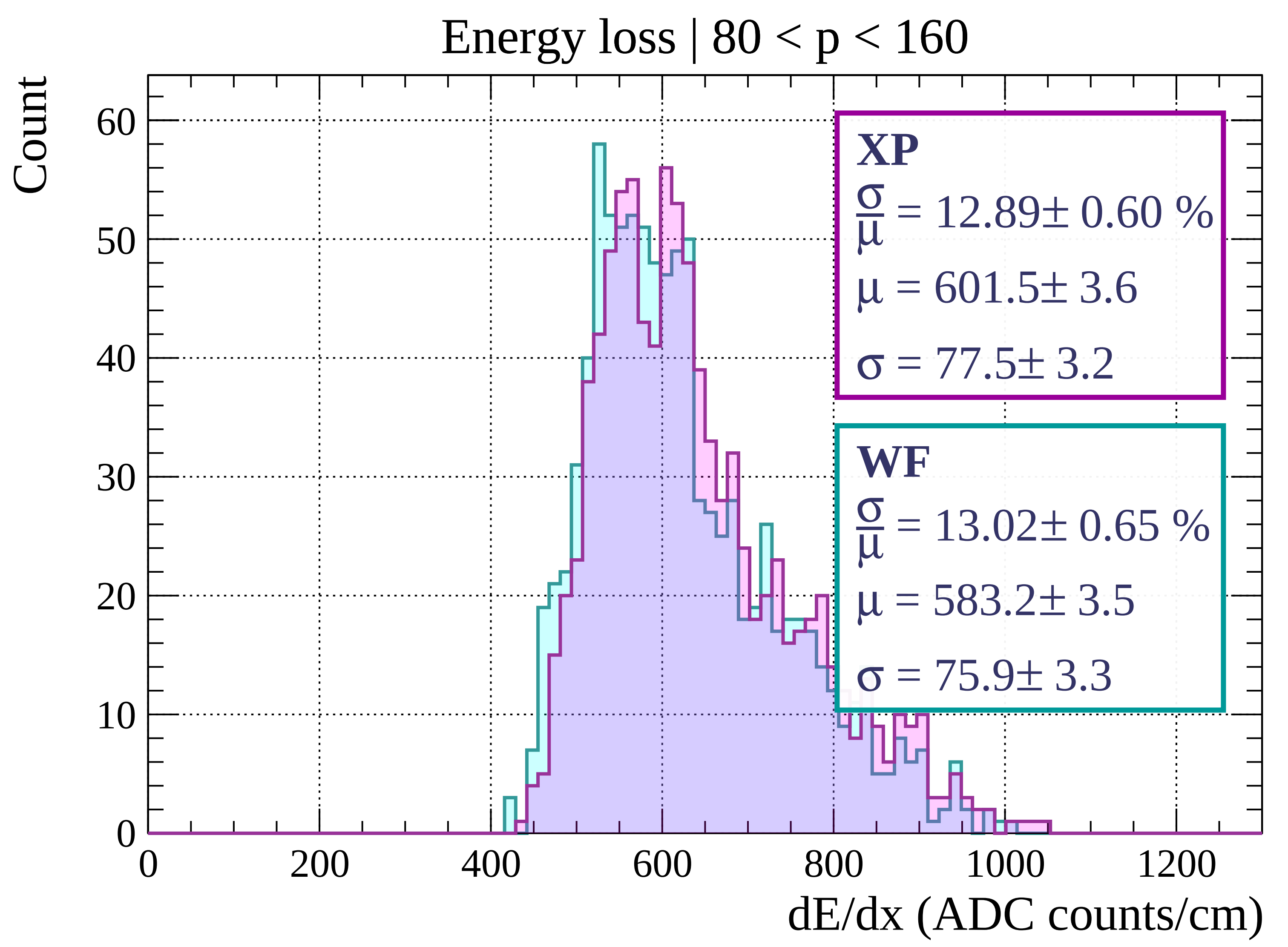
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
100
80
60
40
20
0
×10^3



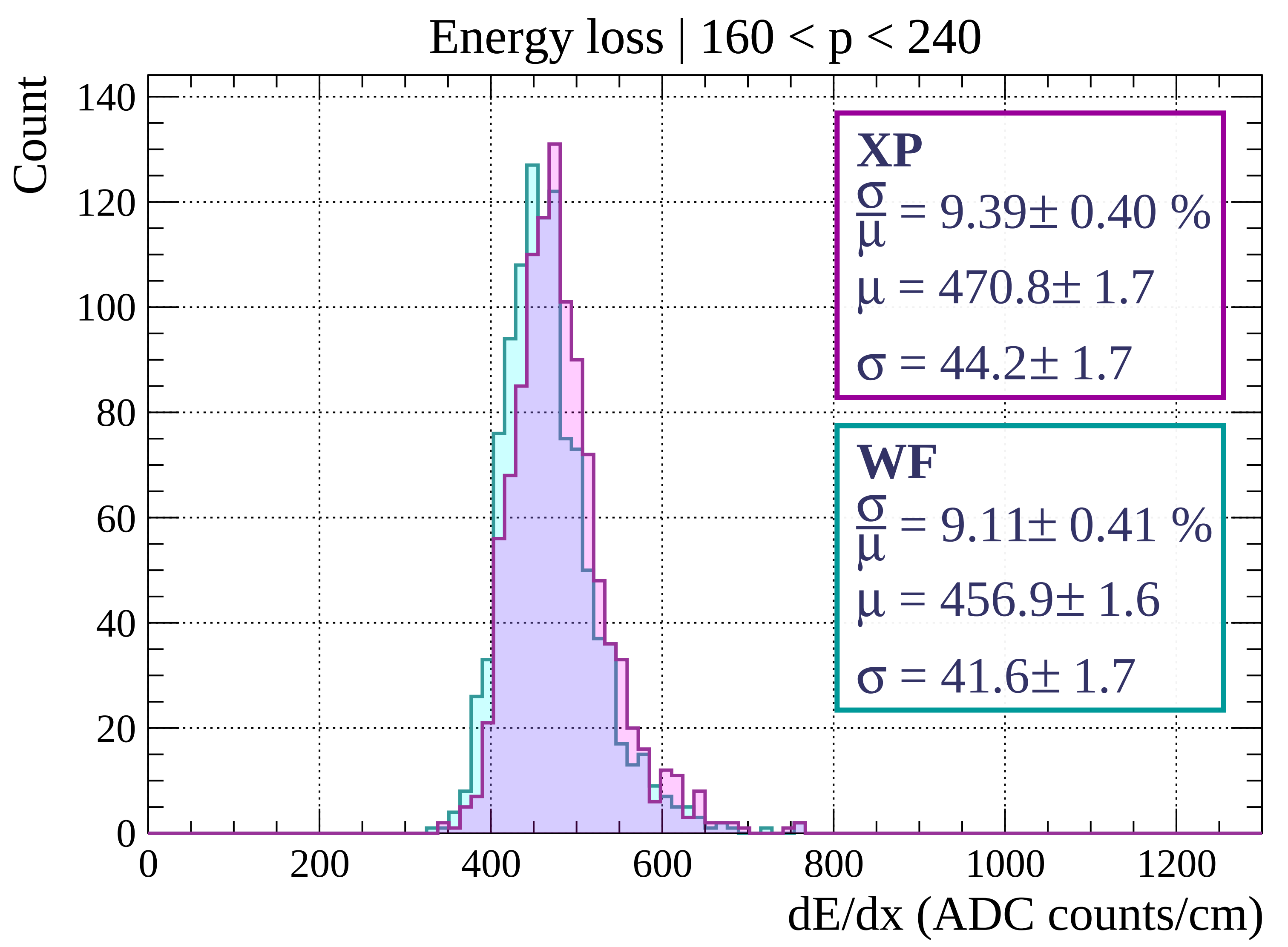
χ²/NDF
R
×10³



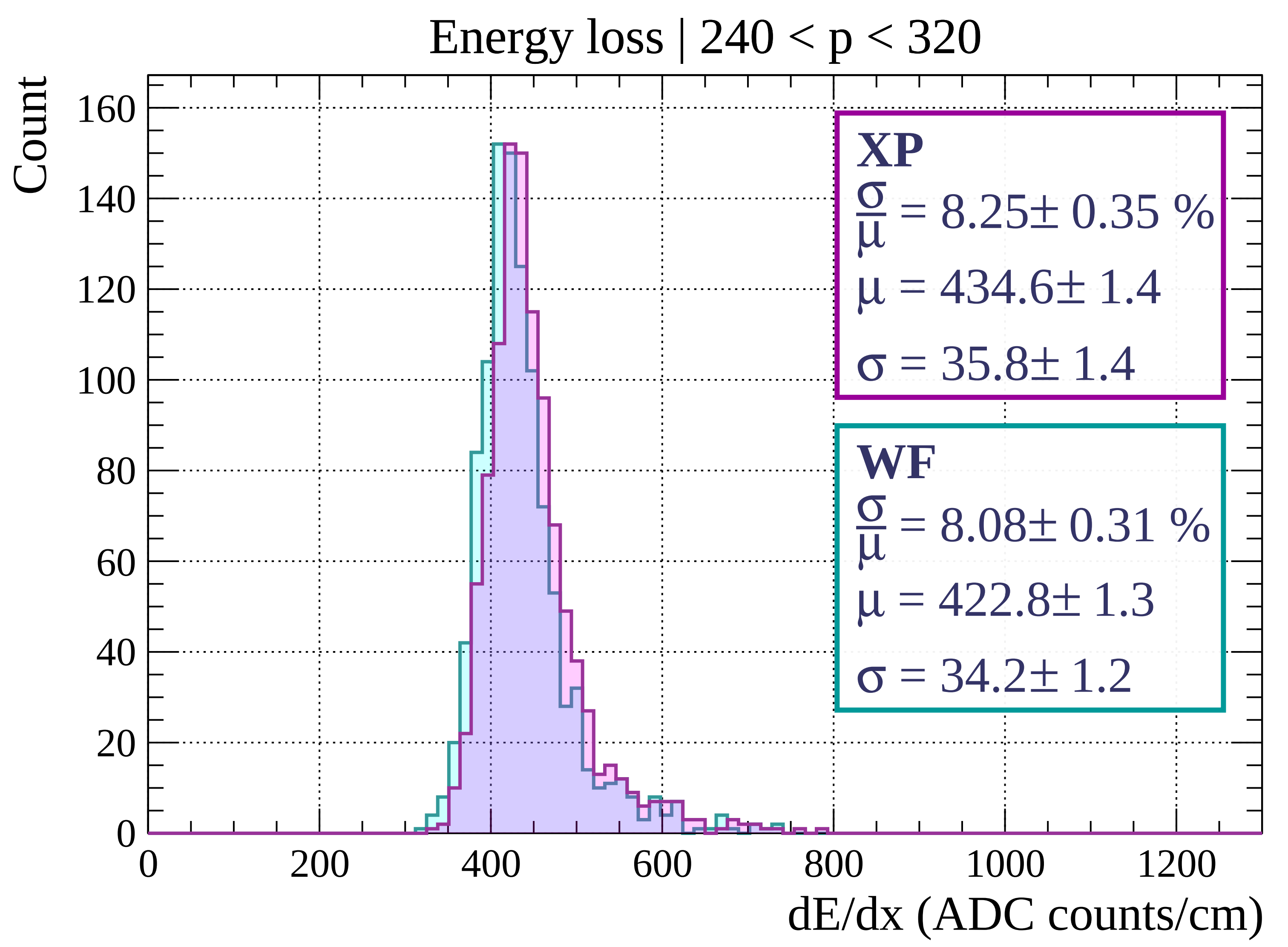
Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 8.60± 0.45 %
μ = 565.5± 2.3
σ = 48.6± 2.4
WF
σ/μ = 7.31± 0.40 %
μ = 557.1± 2.0
σ = 40.7± 2.1



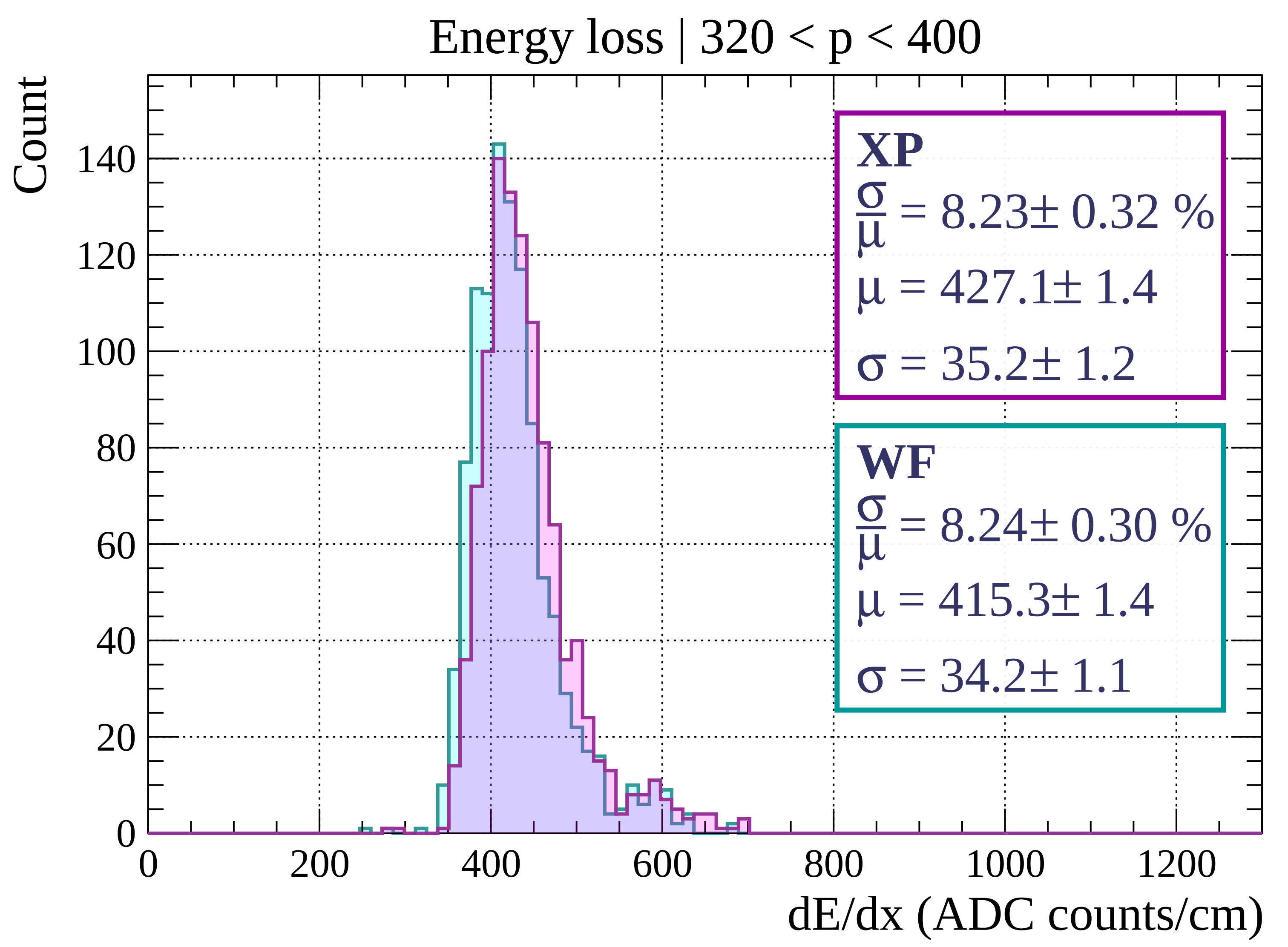
Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.89± 0.60 %
μ = 601.5± 3.6
σ = 77.5± 3.2
WF
$\frac{\sigma}{\mu}$ = 13.02± 0.65 %
μ = 583.2± 3.5
σ = 75.9± 3.3



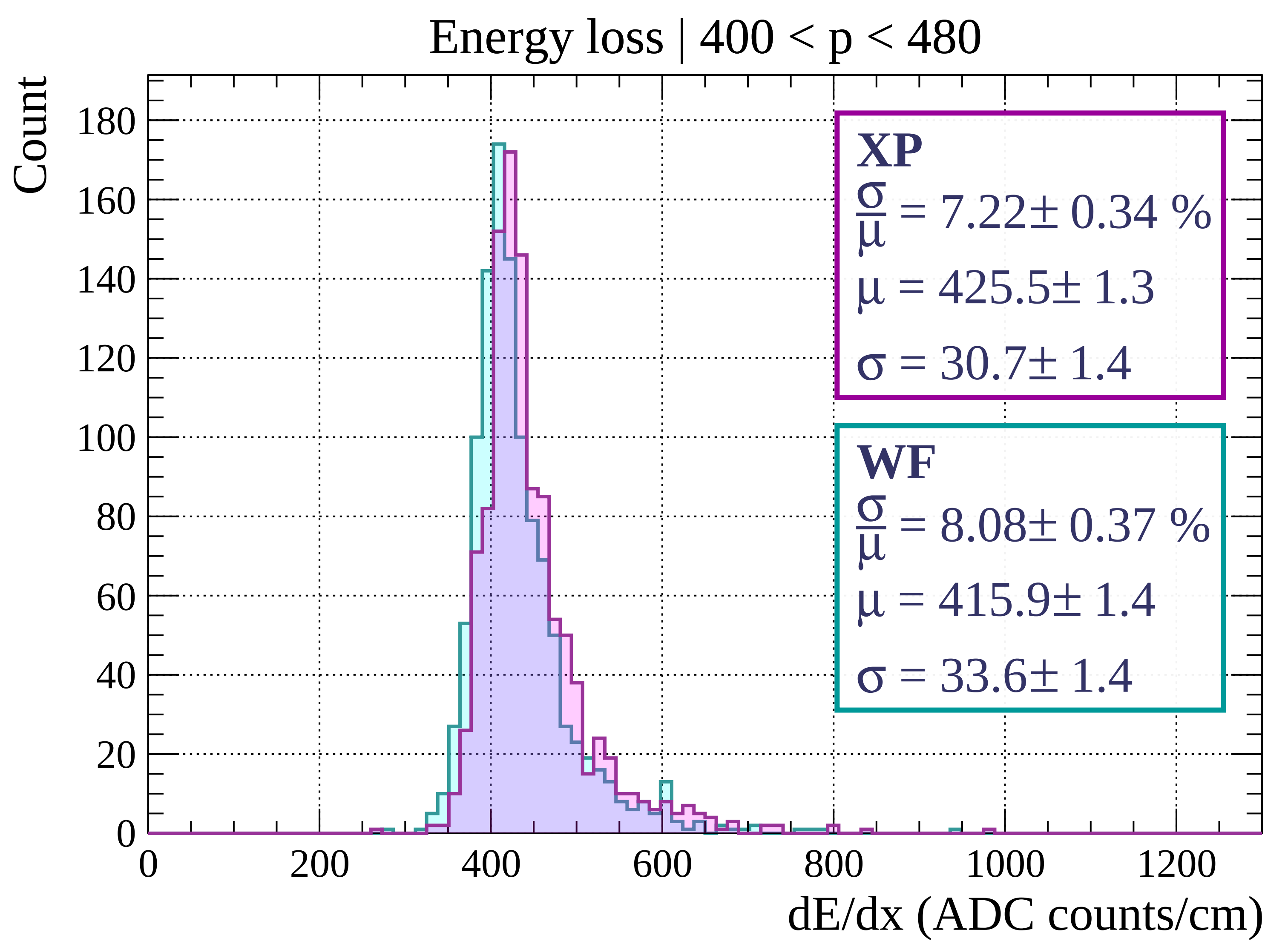
Energy loss | 160 < p < 240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.39± 0.40 %
μ = 470.8± 1.7
σ = 44.2± 1.7
WF
$\frac{\sigma}{\mu}$ = 9.11± 0.41 %
μ = 456.9± 1.6
σ = 41.6± 1.7



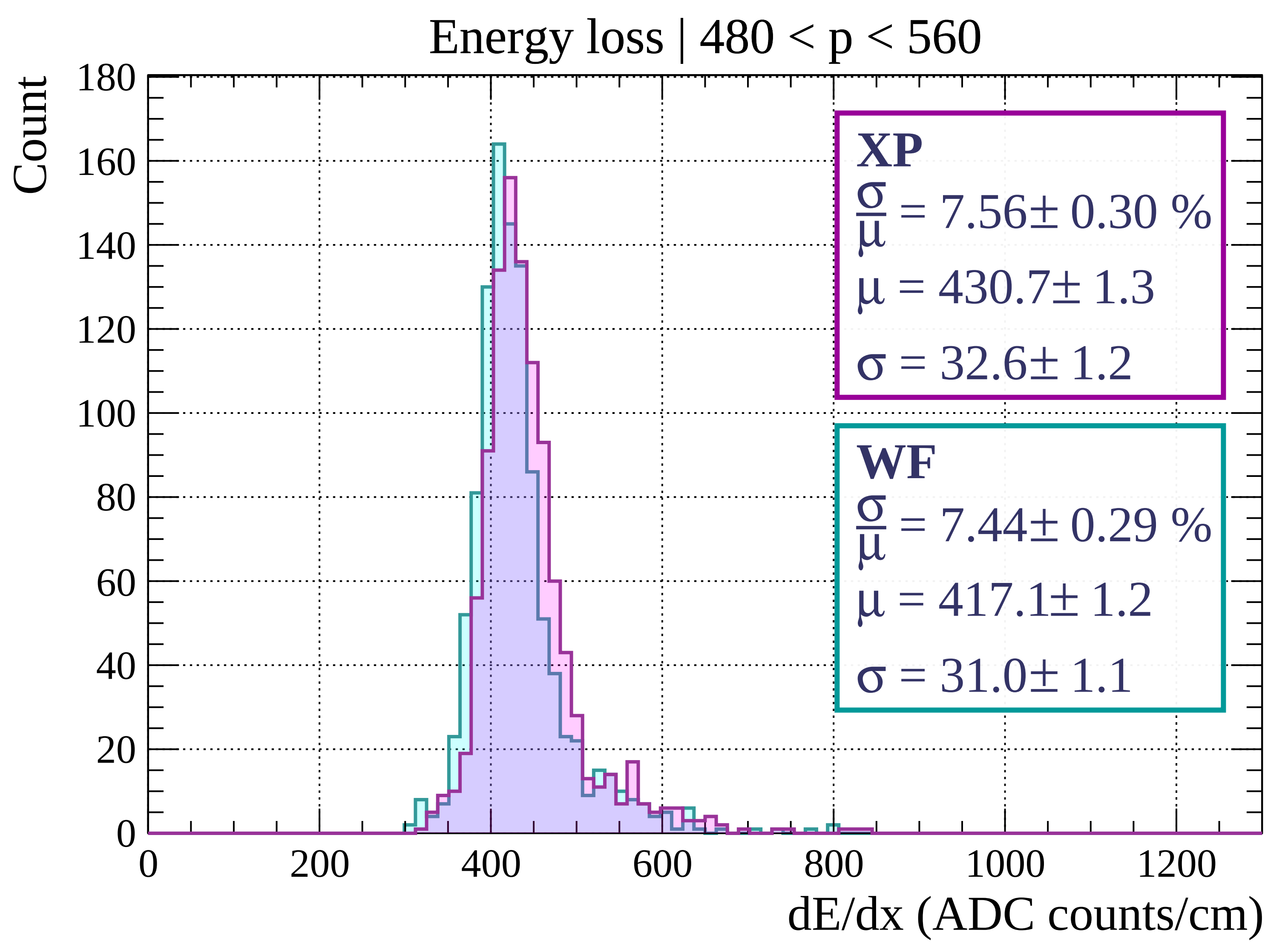
Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 8.25± 0.35 %
μ = 434.6± 1.4
σ = 35.8± 1.4
WF
σ/μ = 8.08± 0.31 %
μ = 422.8± 1.3
σ = 34.2± 1.2



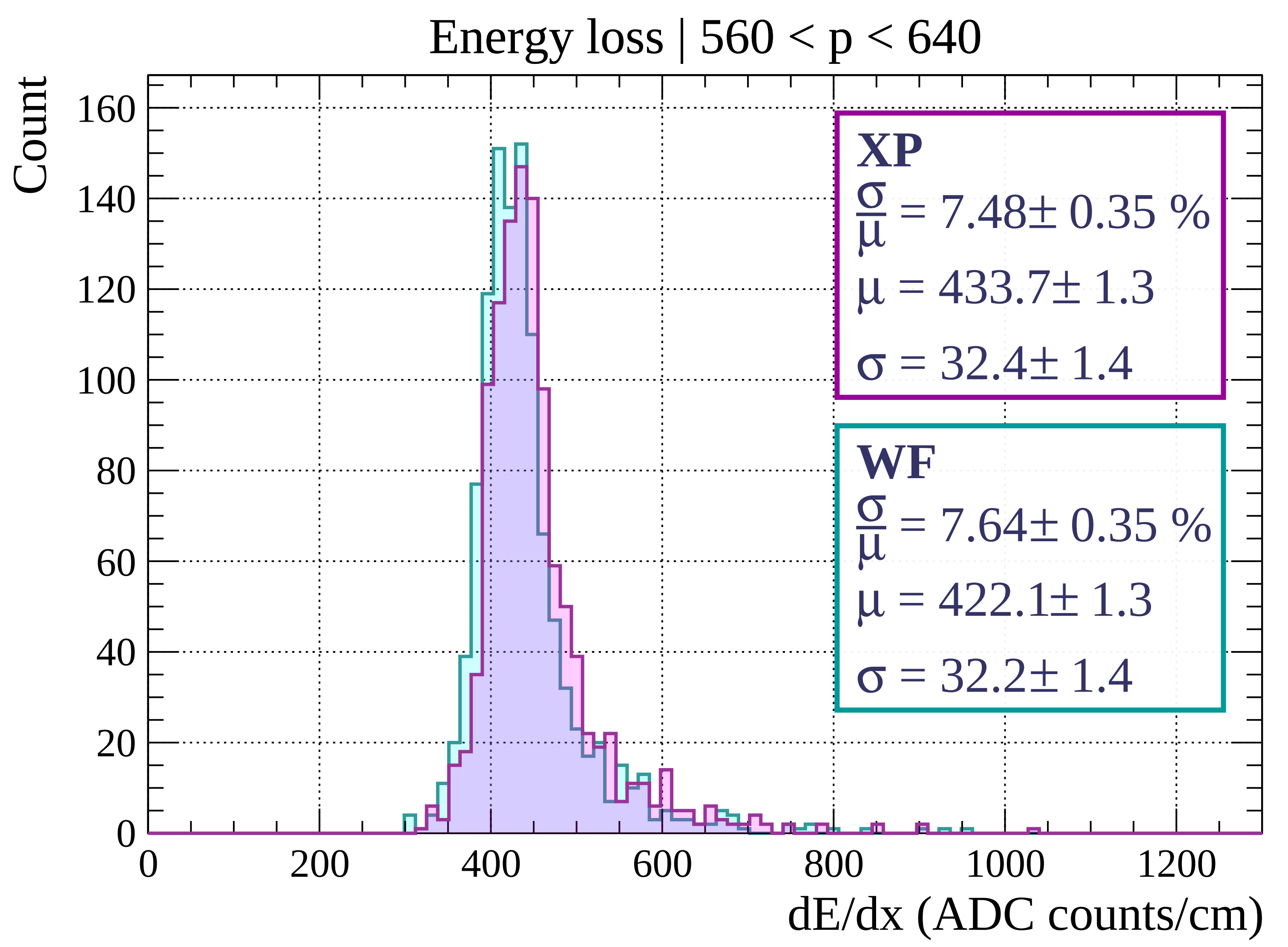
Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 8.23± 0.32 %
μ = 427.1± 1.4
σ = 35.2± 1.2
WF
σ/μ = 8.24± 0.30 %
μ = 415.3± 1.4
σ = 34.2± 1.1



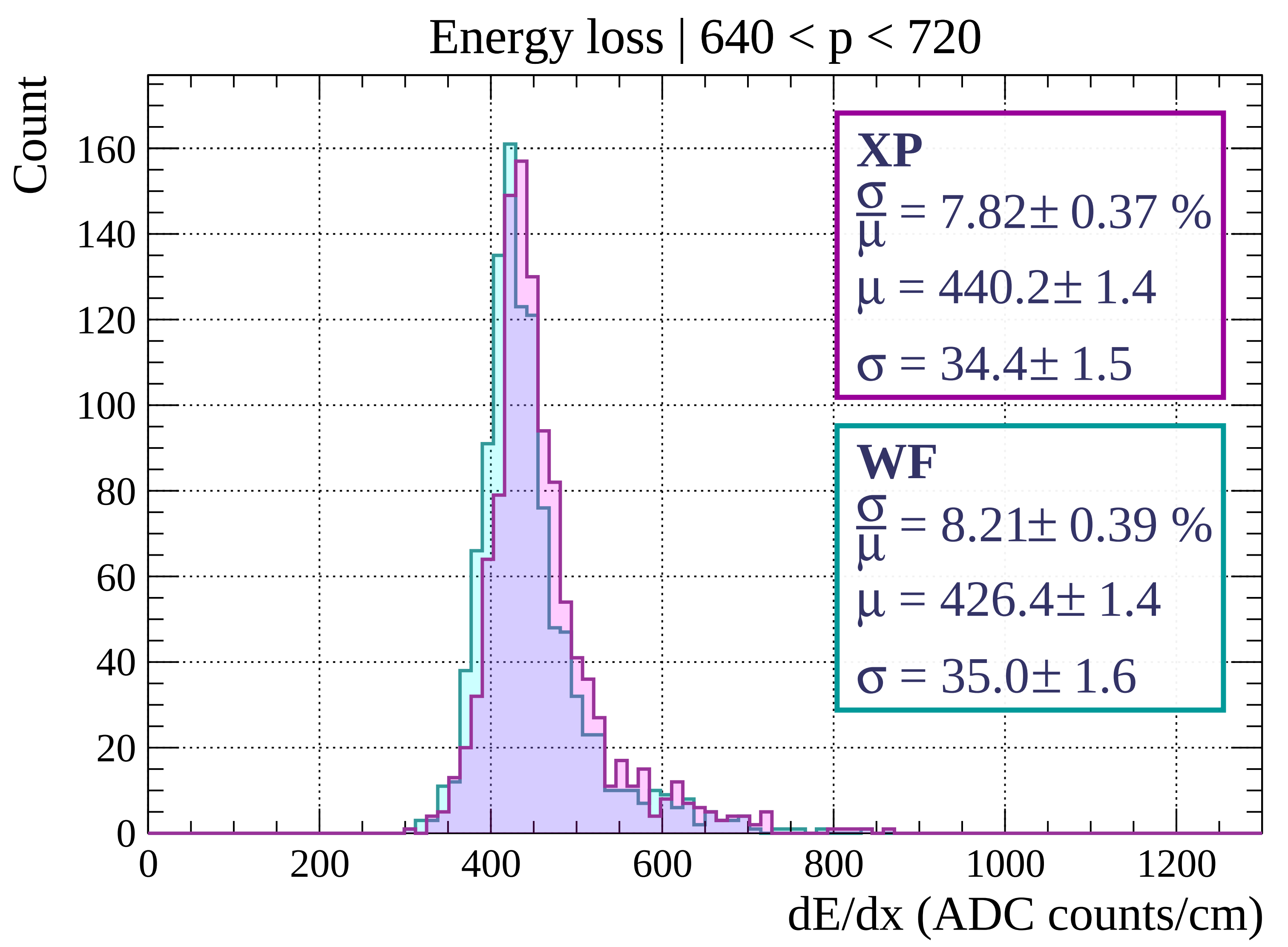
Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 7.22 \pm 0.34$ %
$\mu = 425.5 \pm 1.3$
$\sigma = 30.7 \pm 1.4$
WF
$\frac{\sigma}{\mu} = 8.08 \pm 0.37$ %
$\mu = 415.9 \pm 1.4$
$\sigma = 33.6 \pm 1.4$



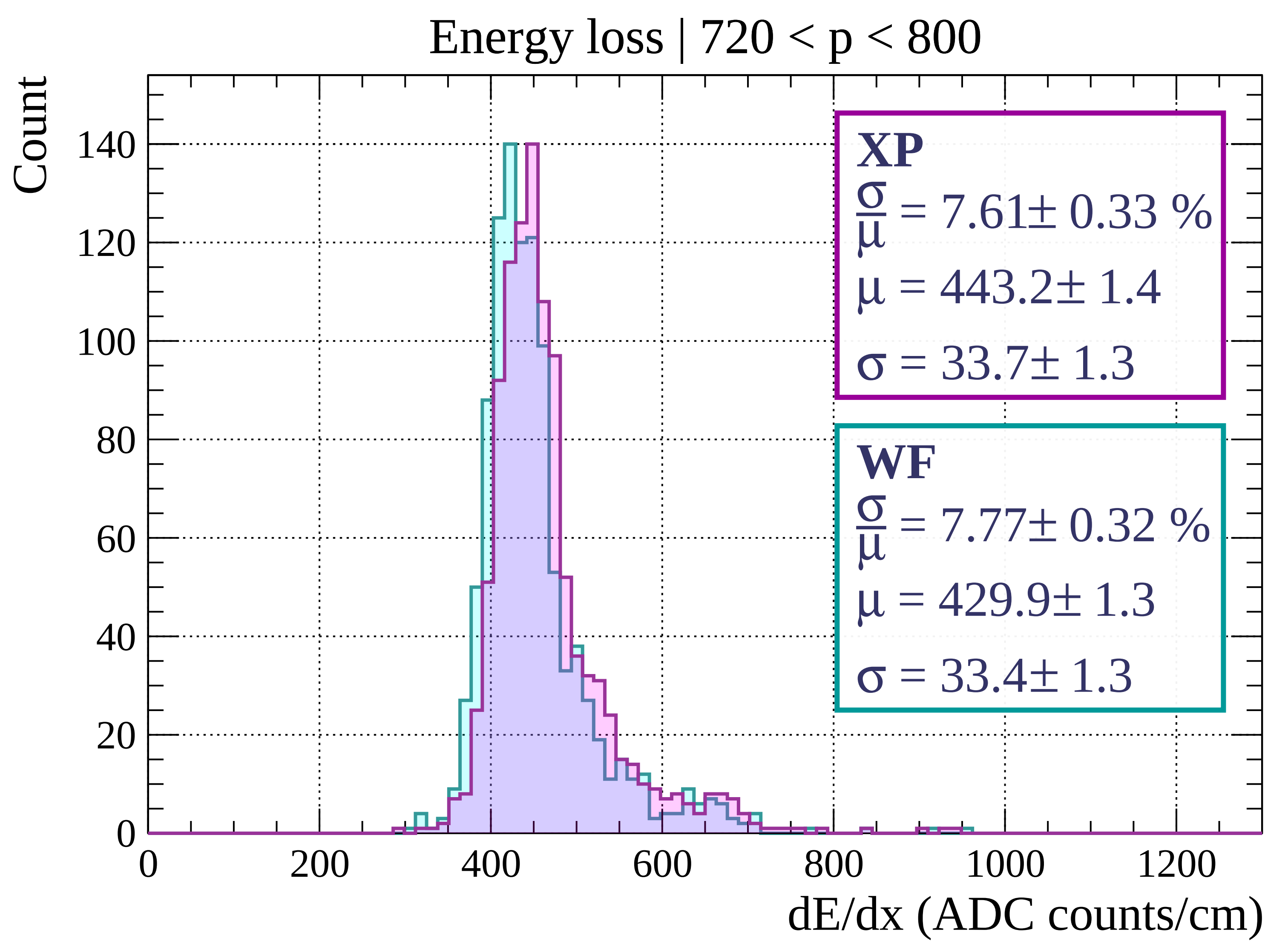
Energy loss | 480 < p < 560
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.56± 0.30 %
μ = 430.7± 1.3
σ = 32.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 7.44± 0.29 %
μ = 417.1± 1.2
σ = 31.0± 1.1



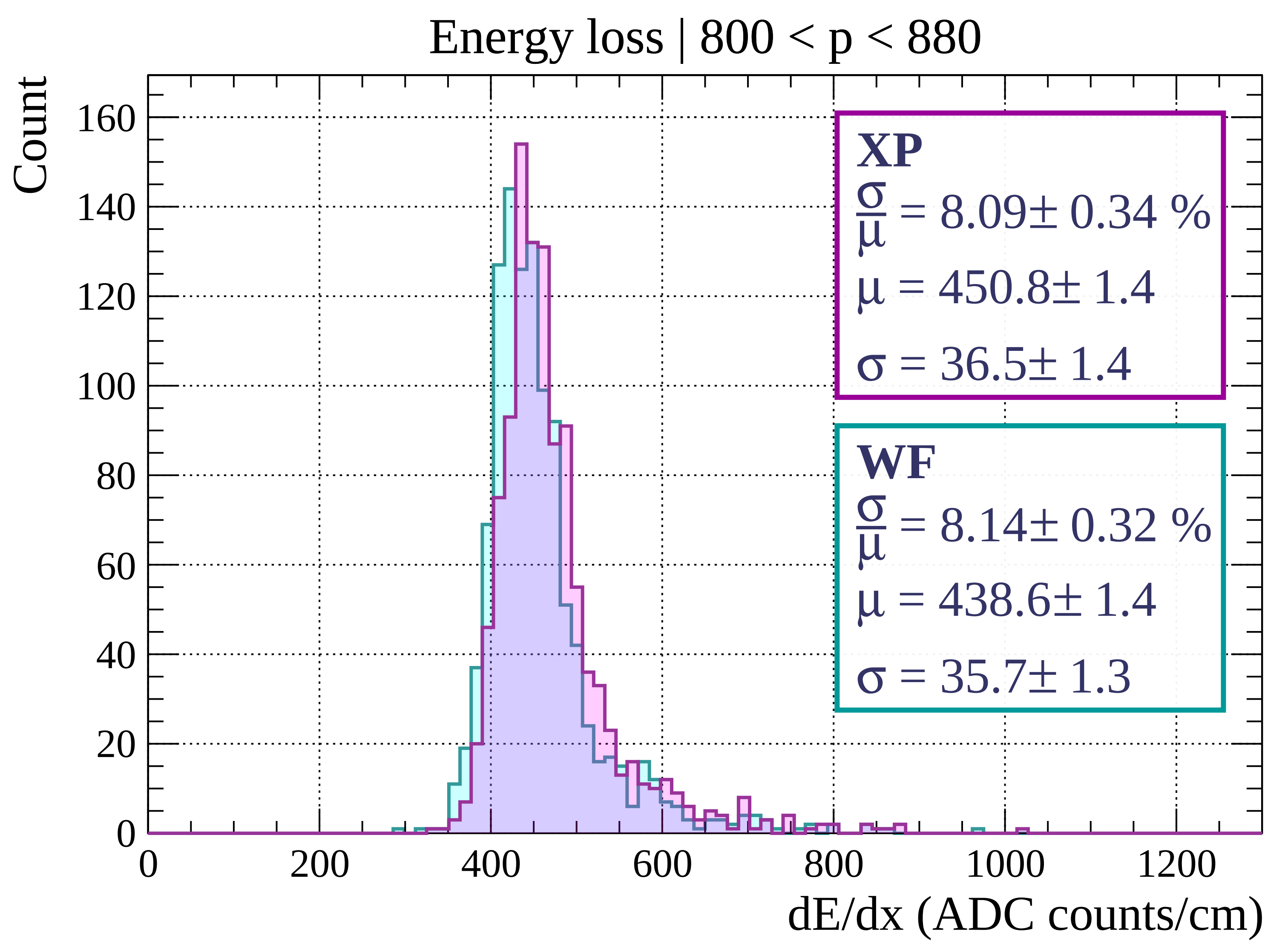
Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 7.48± 0.35 %
μ = 433.7± 1.3
σ = 32.4± 1.4
WF
σ/μ = 7.64± 0.35 %
μ = 422.1± 1.3
σ = 32.2± 1.4



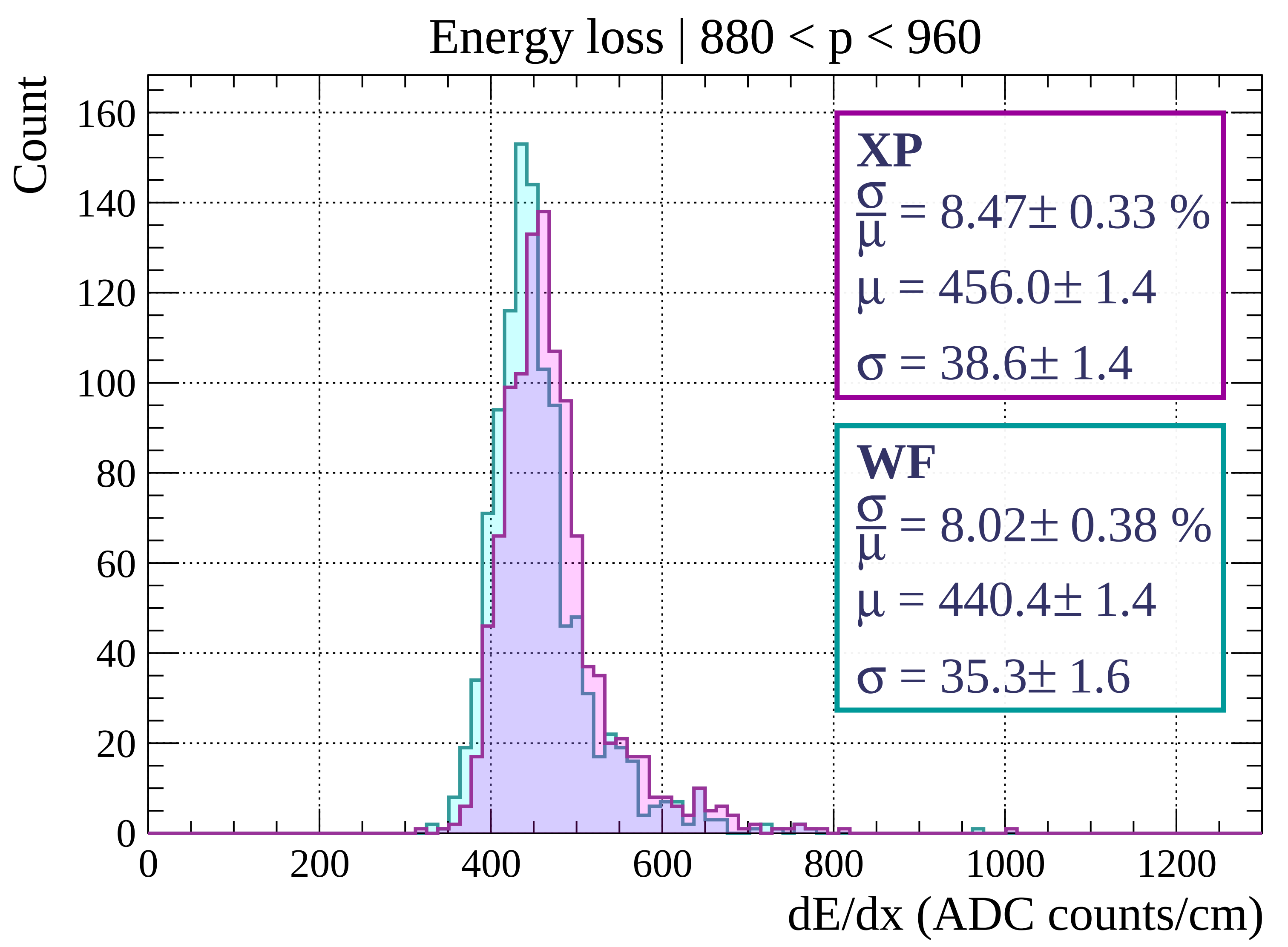
Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 7.82± 0.37 %
μ = 440.2± 1.4
σ = 34.4± 1.5
WF
σ/μ = 8.21± 0.39 %
μ = 426.4± 1.4
σ = 35.0± 1.6



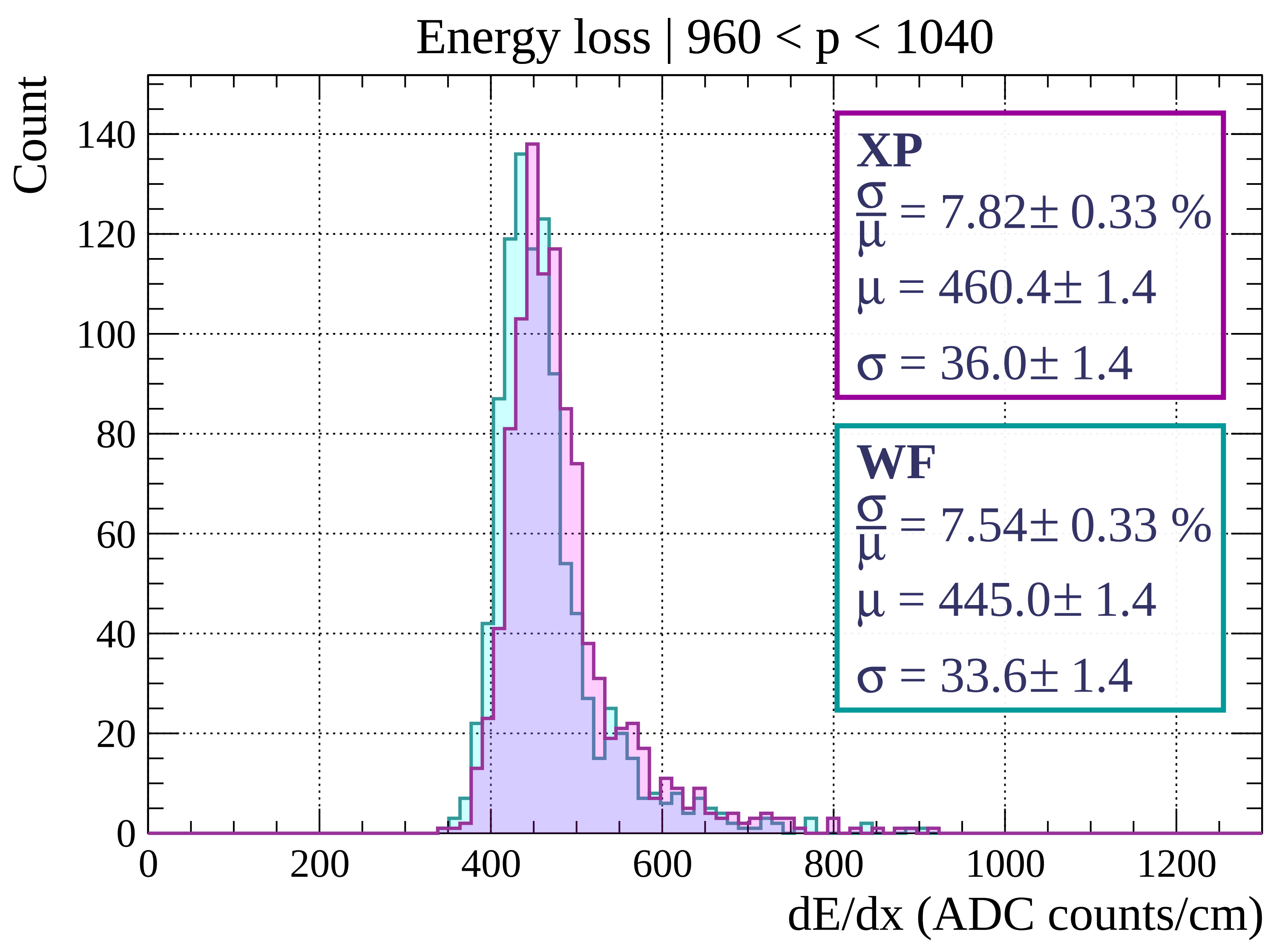
Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.61± 0.33 %
μ = 443.2± 1.4
σ = 33.7± 1.3
WF
$\frac{\sigma}{\mu}$ = 7.77± 0.32 %
μ = 429.9± 1.3
σ = 33.4± 1.3



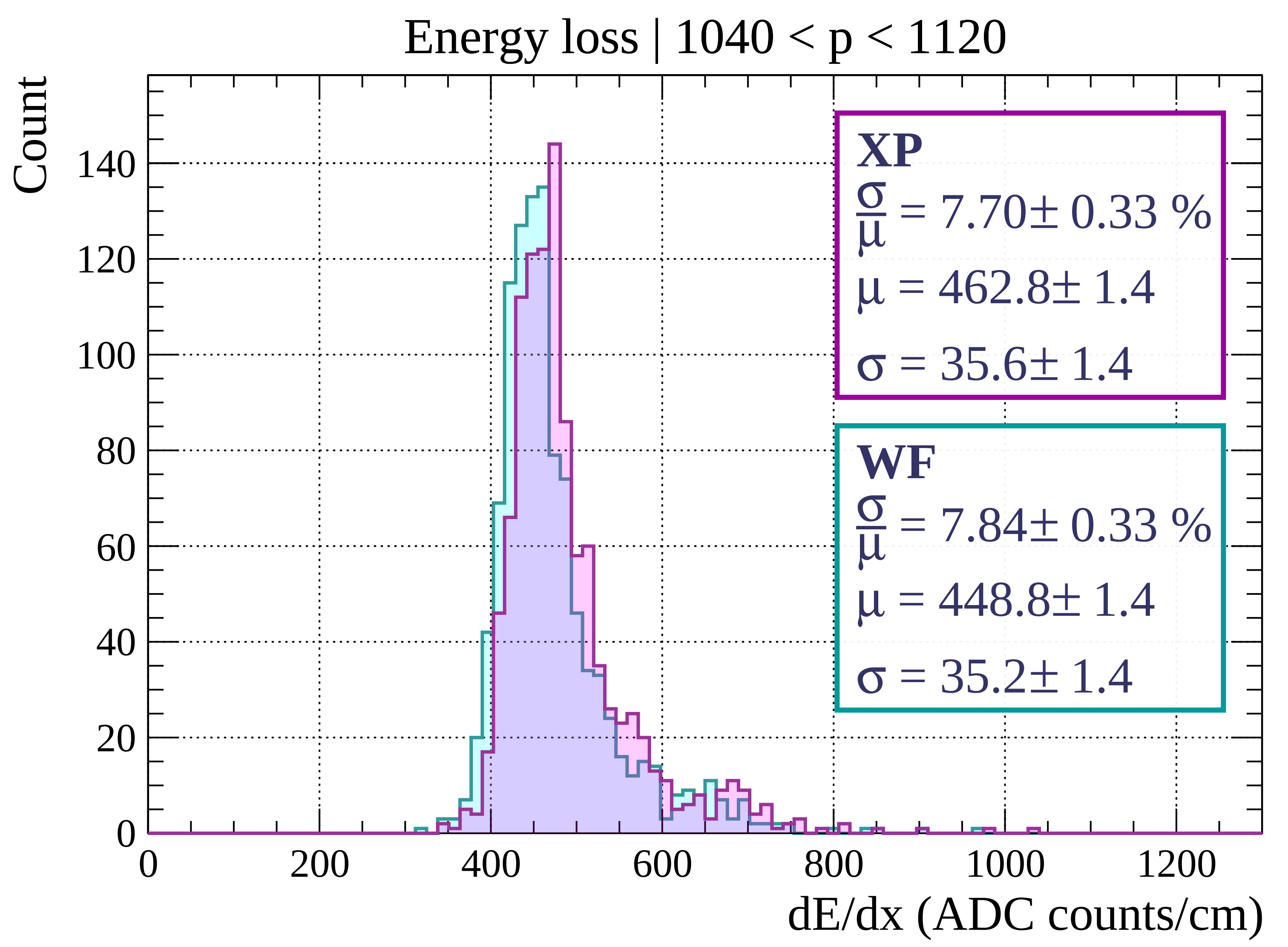
Energy loss | 800 < p < 880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.09± 0.34 %
μ = 450.8± 1.4
σ = 36.5± 1.4
WF
σ/μ = 8.14± 0.32 %
μ = 438.6± 1.4
σ = 35.7± 1.3
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



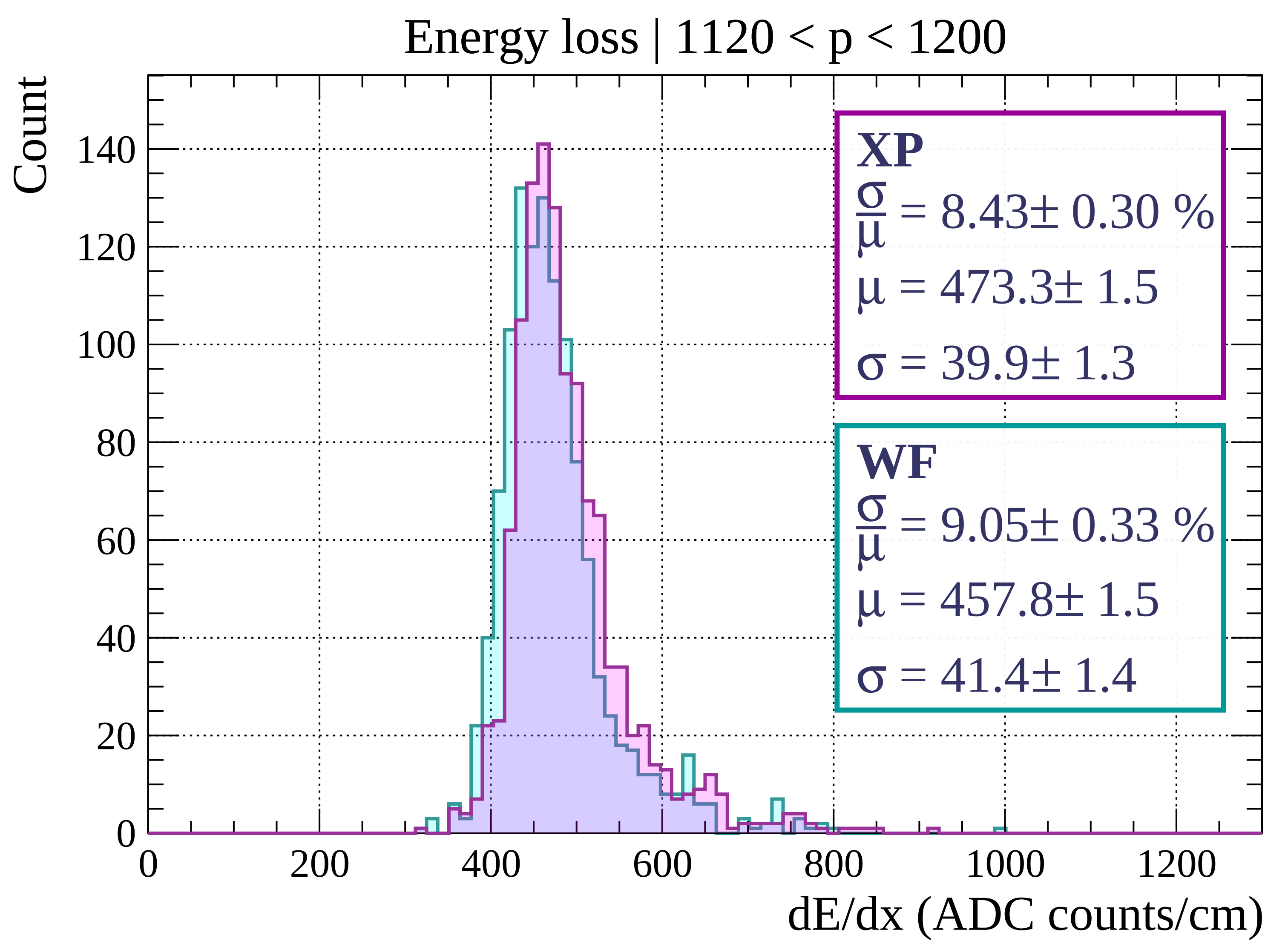
Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP
σ/μ = 8.47± 0.33 %
μ = 456.0± 1.4
σ = 38.6± 1.4
WF
σ/μ = 8.02± 0.38 %
μ = 440.4± 1.4
σ = 35.3± 1.6



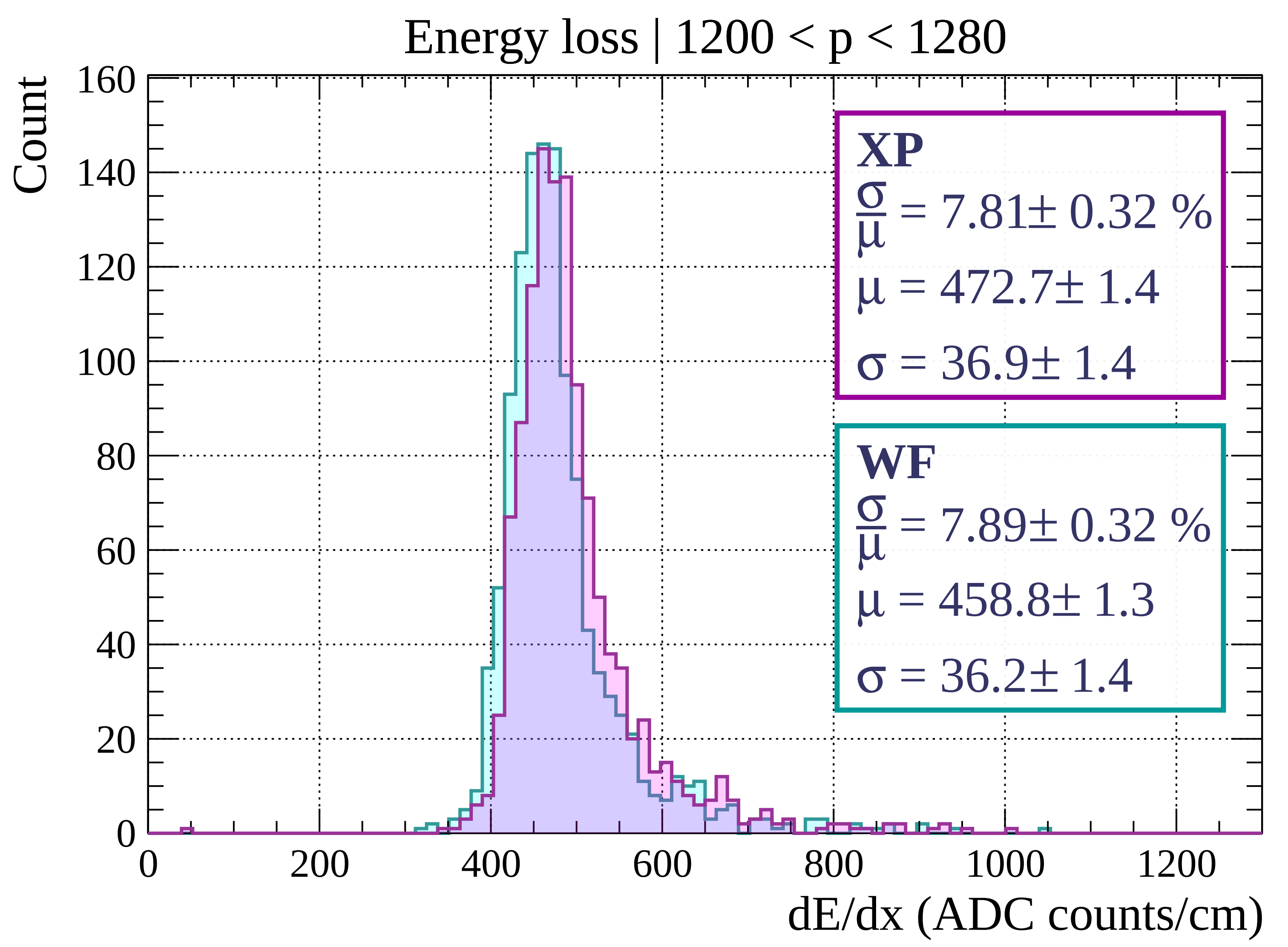
Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 7.82± 0.33 %
μ = 460.4± 1.4
σ = 36.0± 1.4
WF
σ/μ = 7.54± 0.33 %
μ = 445.0± 1.4
σ = 33.6± 1.4



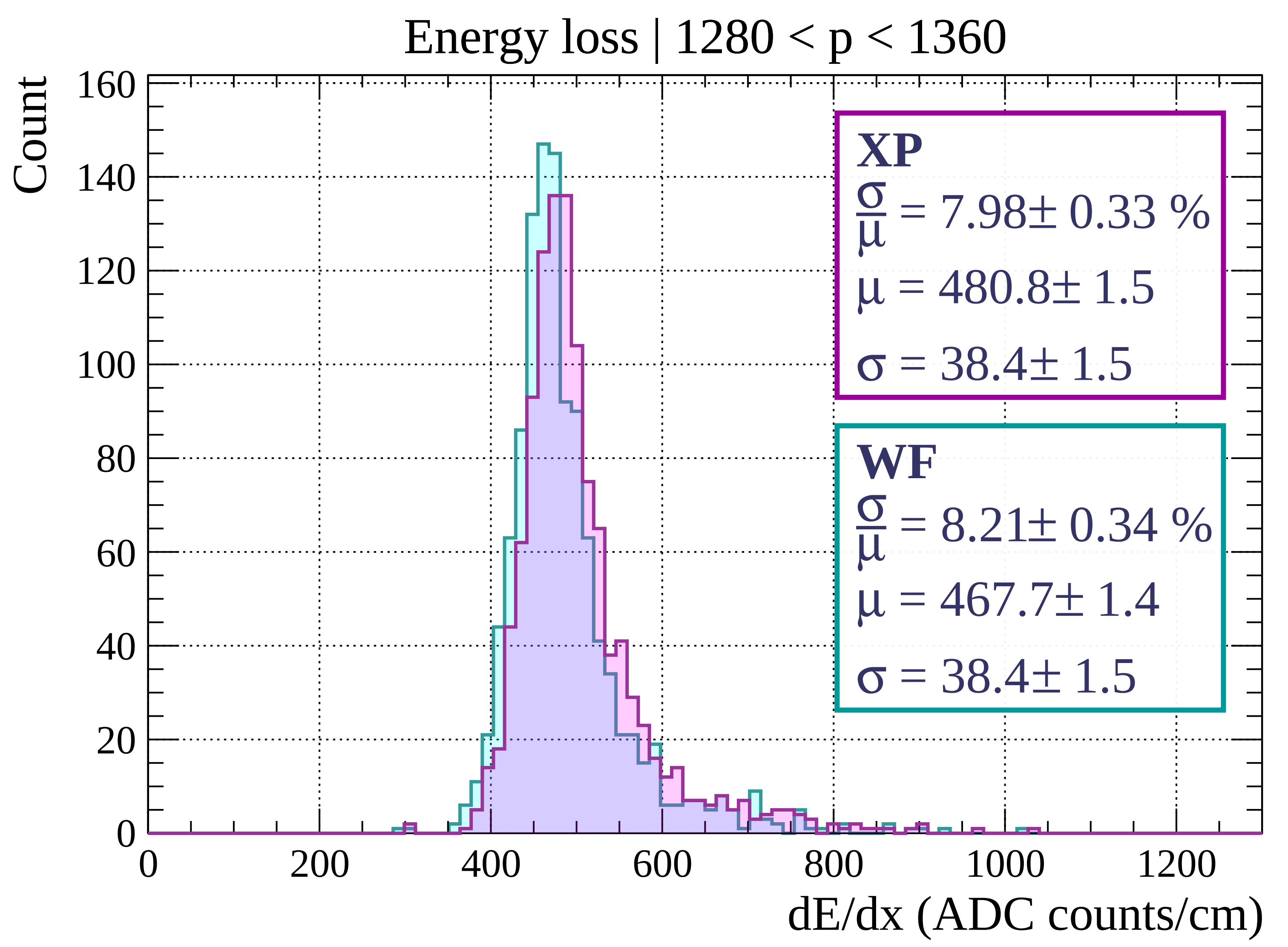
Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.70 \pm 0.33$ %
$\mu = 462.8 \pm 1.4$
$\sigma = 35.6 \pm 1.4$
WF
$\frac{\sigma}{\mu} = 7.84 \pm 0.33$ %
$\mu = 448.8 \pm 1.4$
$\sigma = 35.2 \pm 1.4$
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



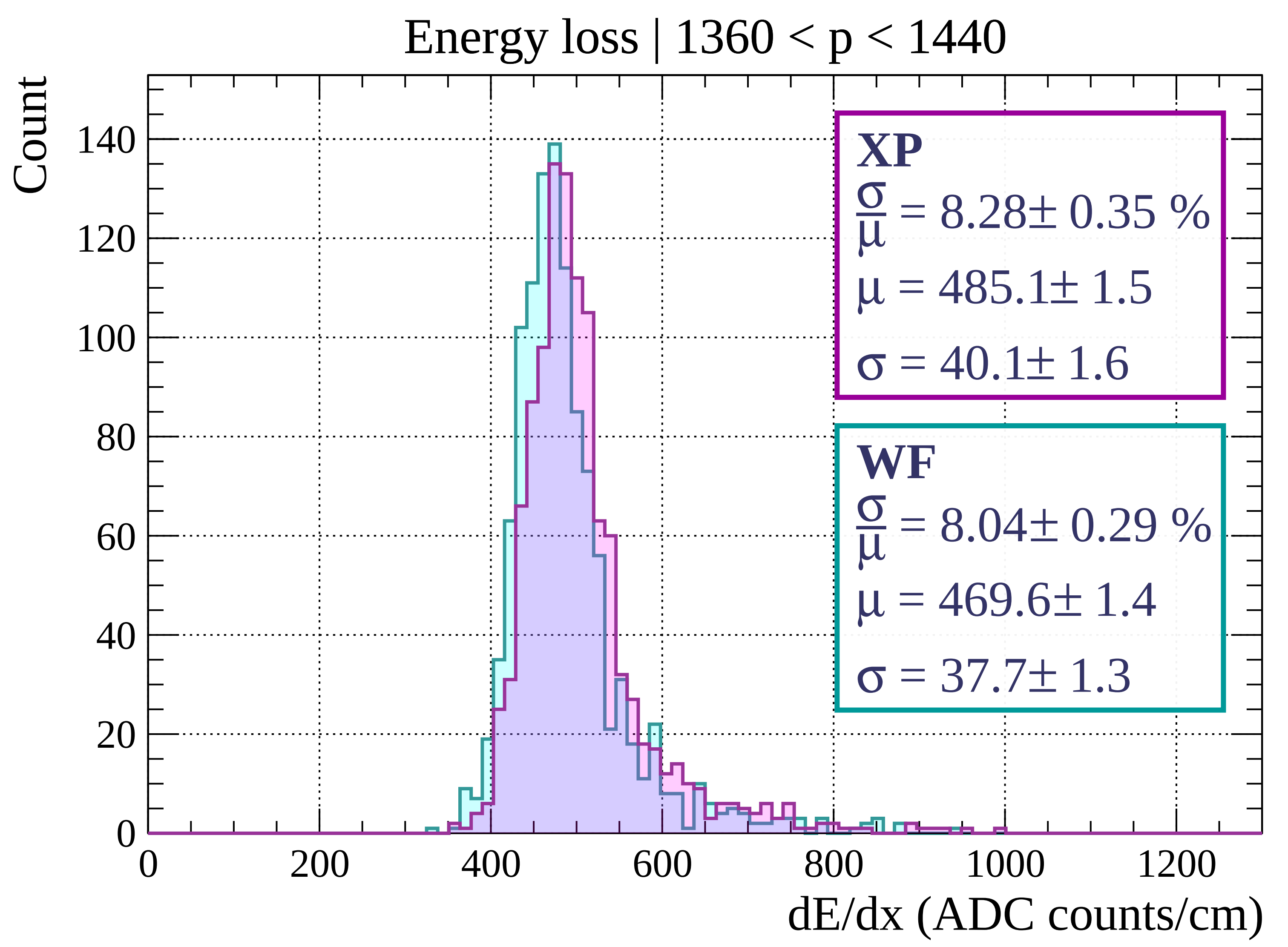
Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 8.43± 0.30 %
μ = 473.3± 1.5
σ = 39.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 9.05± 0.33 %
μ = 457.8± 1.5
σ = 41.4± 1.4



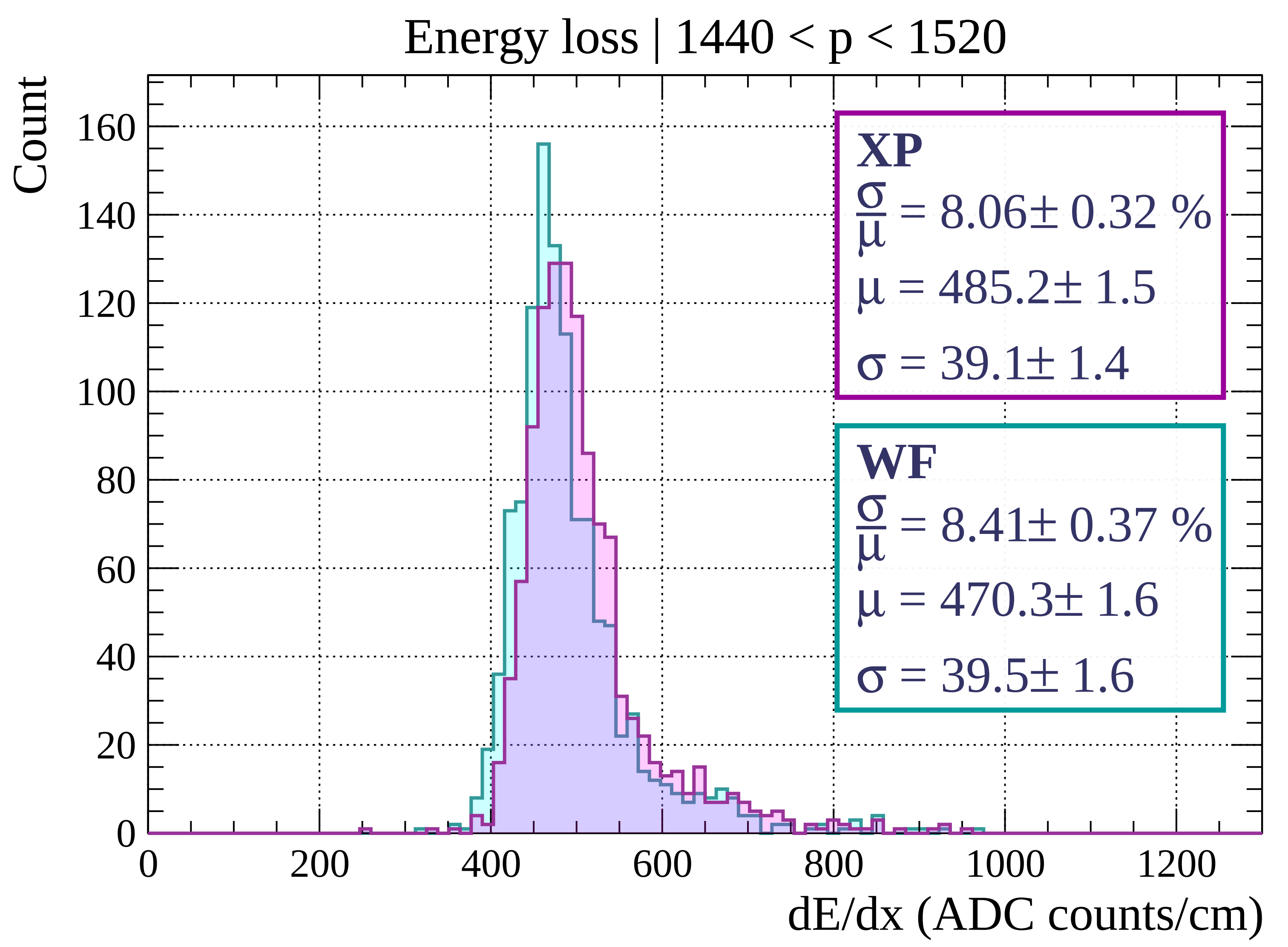
Energy loss | 1200 < p < 1280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 7.81± 0.32 %
μ = 472.7± 1.4
σ = 36.9± 1.4
WF
σ/μ = 7.89± 0.32 %
μ = 458.8± 1.3
σ = 36.2± 1.4



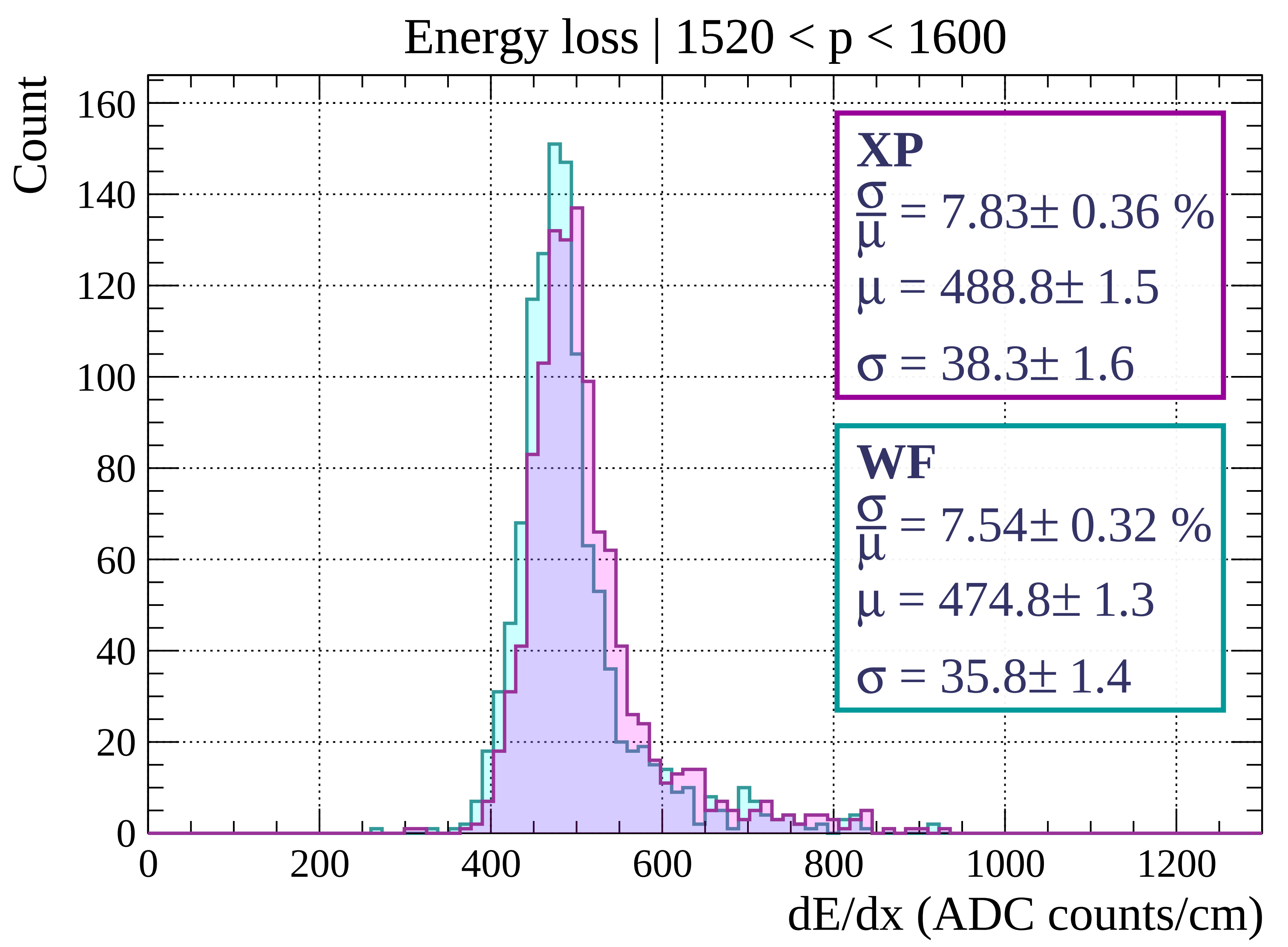
Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 7.98± 0.33 %
μ = 480.8± 1.5
σ = 38.4± 1.5
WF
σ/μ = 8.21± 0.34 %
μ = 467.7± 1.4
σ = 38.4± 1.5



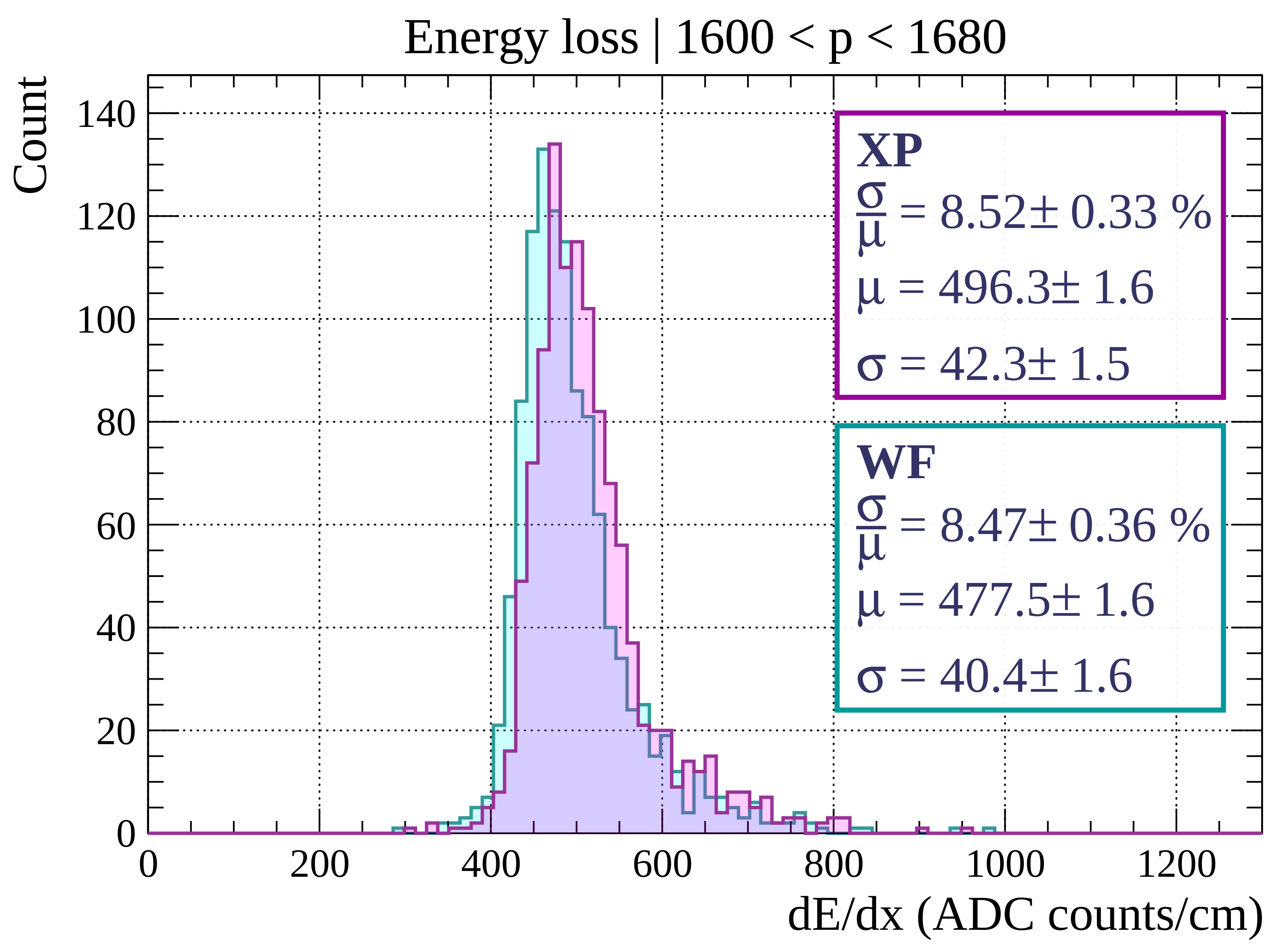
Energy loss | 1360 < p < 1440
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.28± 0.35 %
μ = 485.1± 1.5
σ = 40.1± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.04± 0.29 %
μ = 469.6± 1.4
σ = 37.7± 1.3



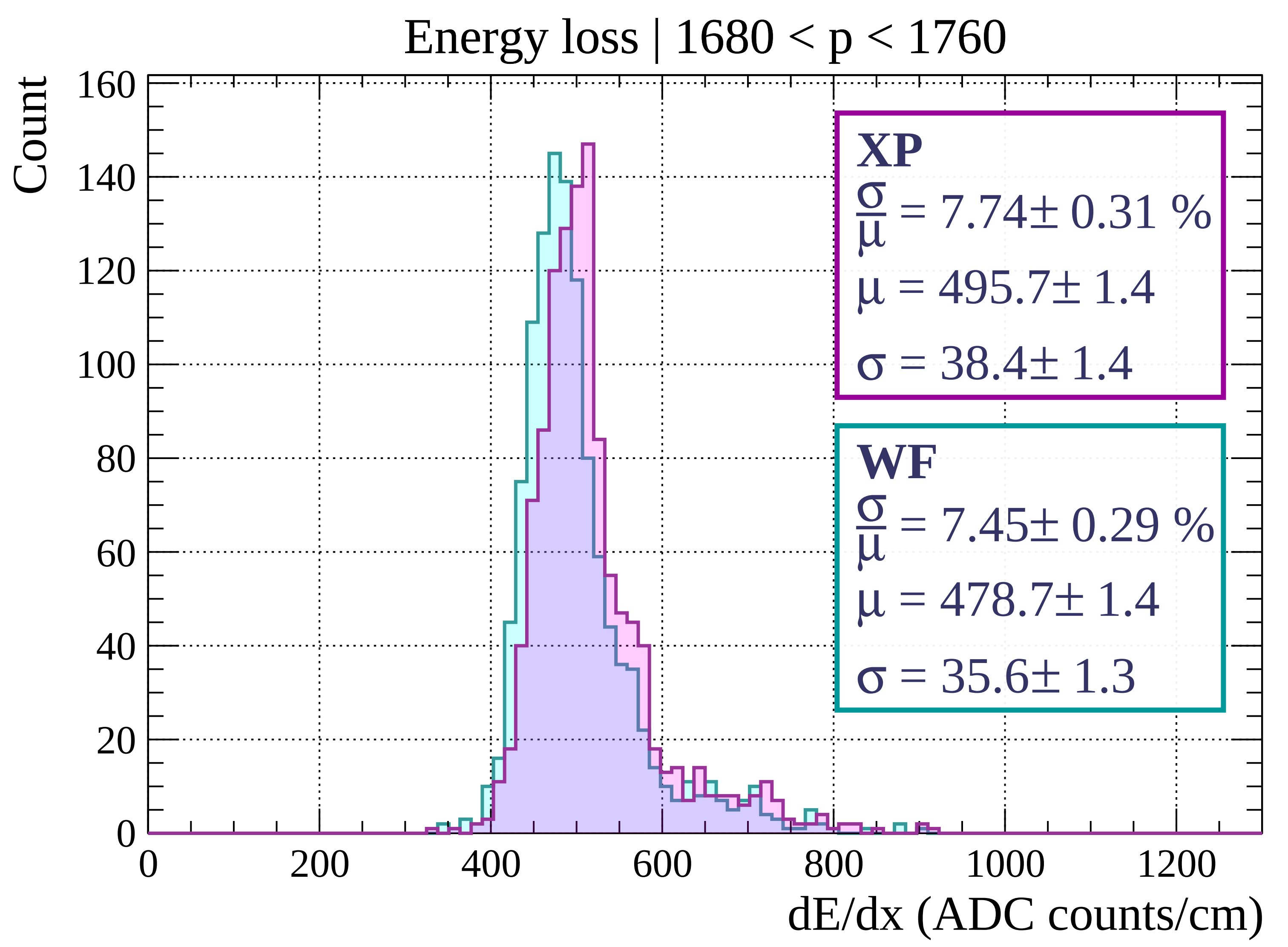
Energy loss | 1440 < p < 1520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 8.06± 0.32 %
μ = 485.2± 1.5
σ = 39.1± 1.4
WF
σ/μ = 8.41± 0.37 %
μ = 470.3± 1.6
σ = 39.5± 1.6

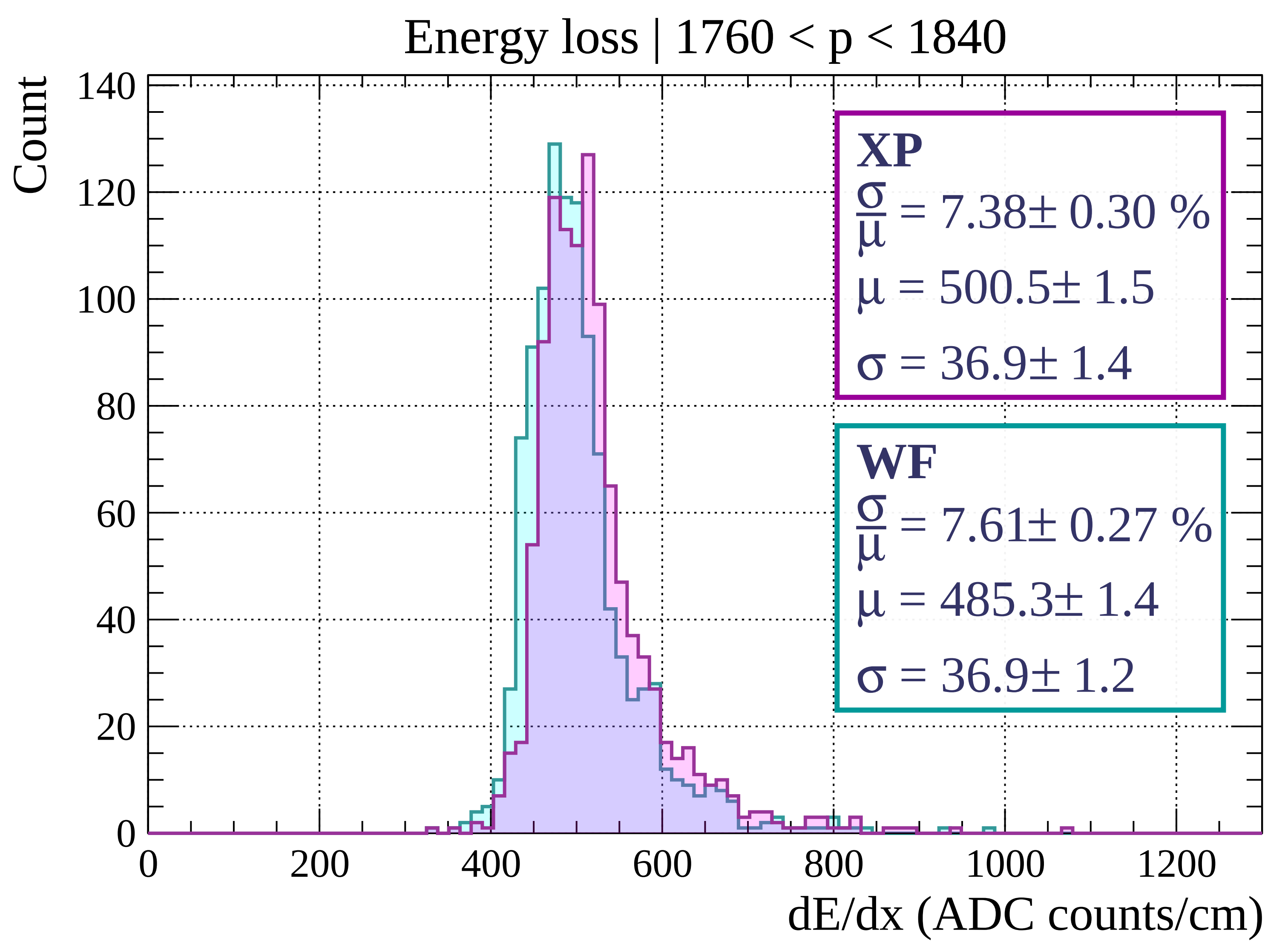


Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.83± 0.36 %
μ = 488.8± 1.5
σ = 38.3± 1.6
WF
$\frac{\sigma}{\mu}$ = 7.54± 0.32 %
μ = 474.8± 1.3
σ = 35.8± 1.4
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160

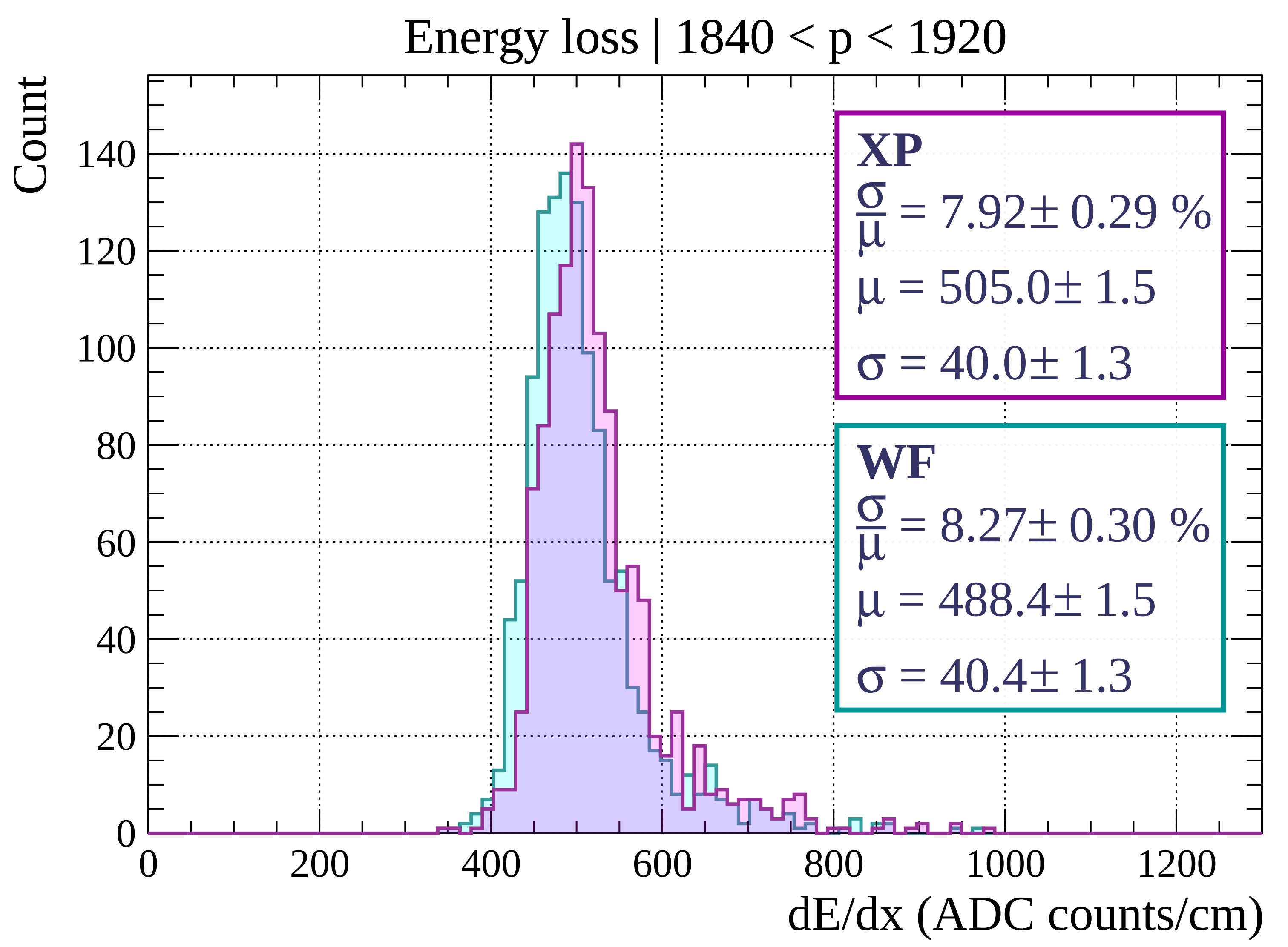


Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.52± 0.33 %
μ = 496.3± 1.6
σ = 42.3± 1.5
WF
σ/μ = 8.47± 0.36 %
μ = 477.5± 1.6
σ = 40.4± 1.6

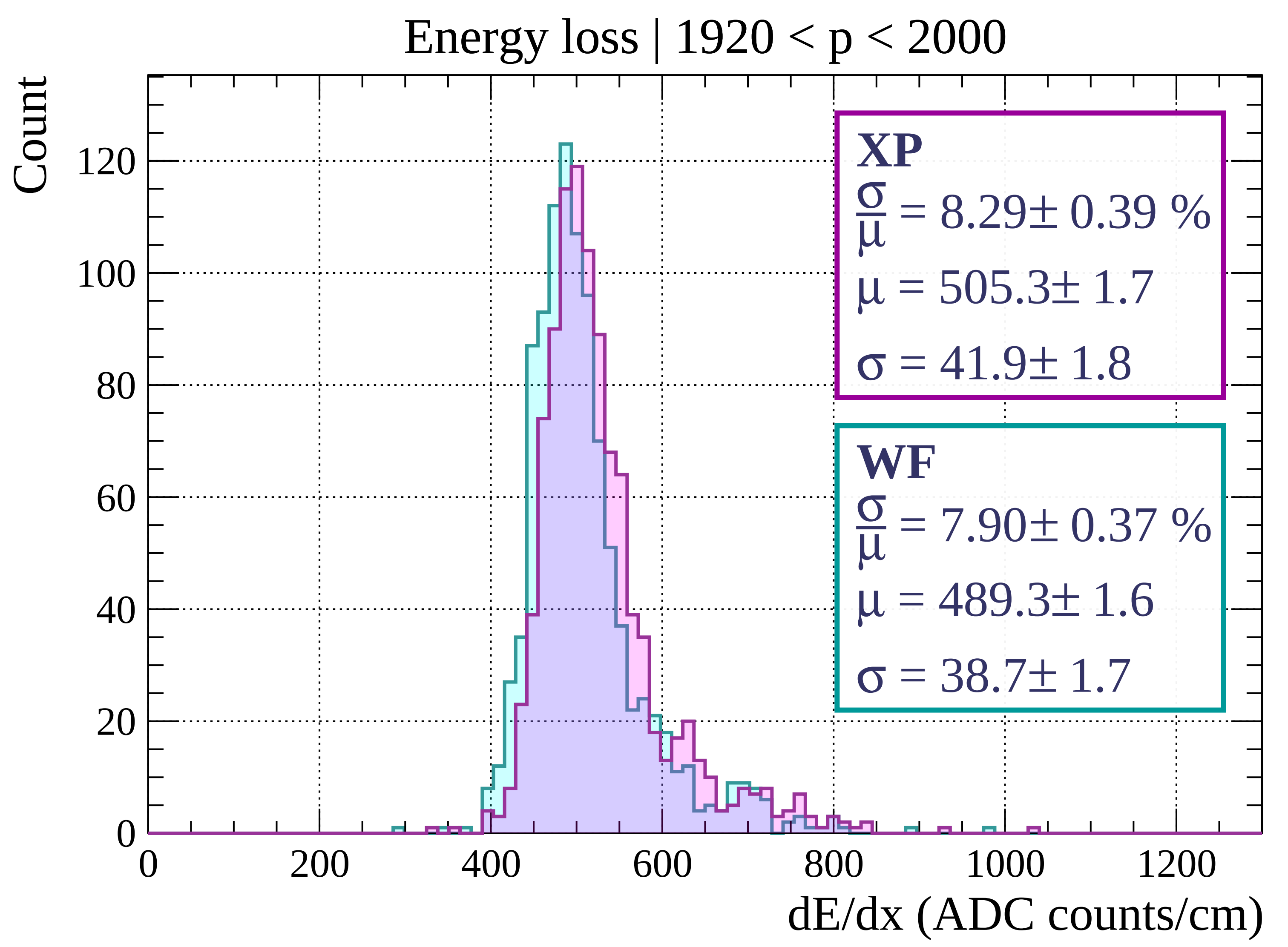

Energy loss | 1680 < p < 1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 7.74± 0.31 %
μ = 495.7± 1.4
σ = 38.4± 1.4
WF
$\frac{\sigma}{\mu}$ = 7.45± 0.29 %
μ = 478.7± 1.4
σ = 35.6± 1.3



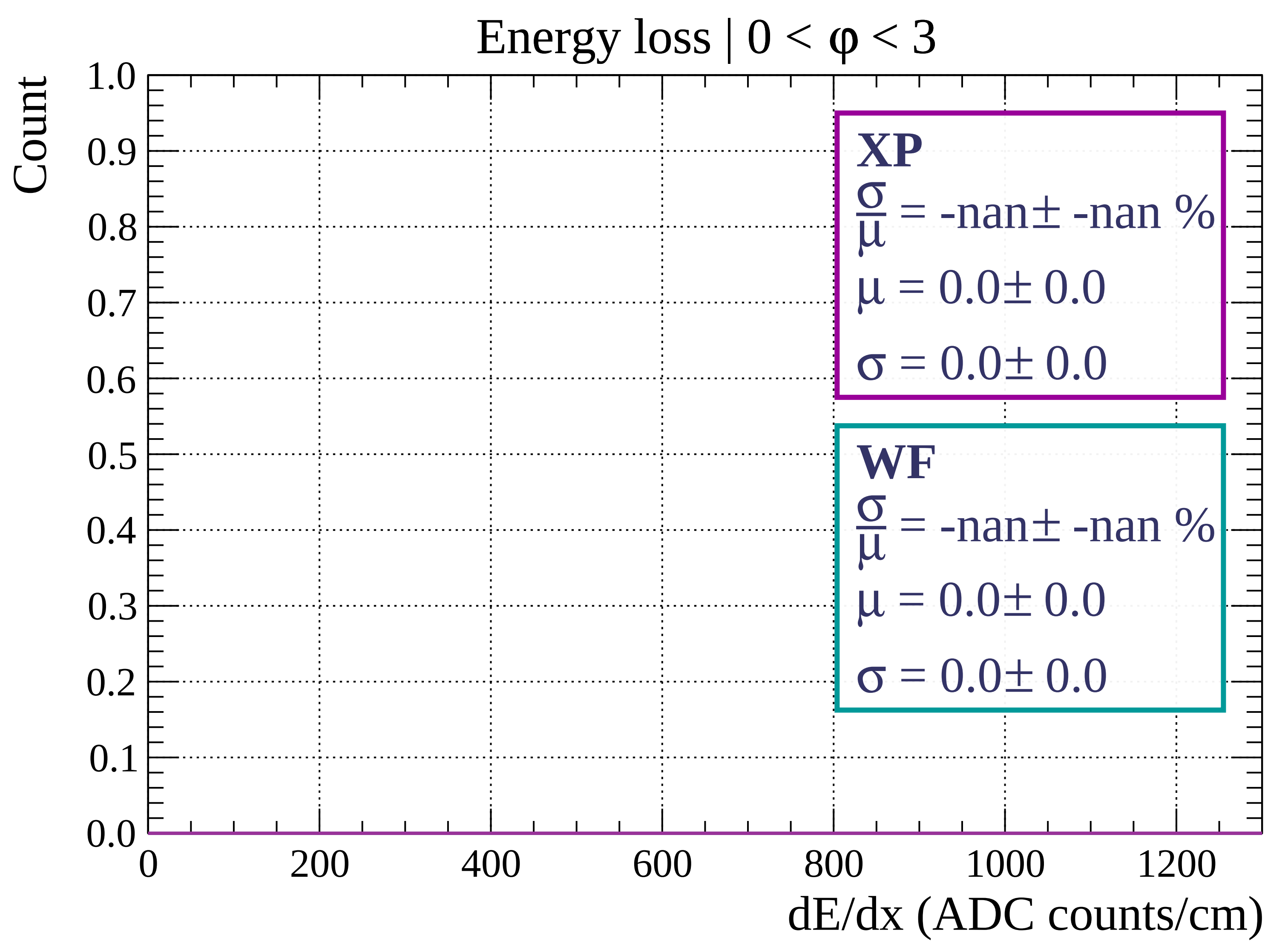
Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu} = 7.38 \pm 0.30$ %
$\mu = 500.5 \pm 1.5$
$\sigma = 36.9 \pm 1.4$
WF
$\frac{\sigma}{\mu} = 7.61 \pm 0.27$ %
$\mu = 485.3 \pm 1.4$
$\sigma = 36.9 \pm 1.2$



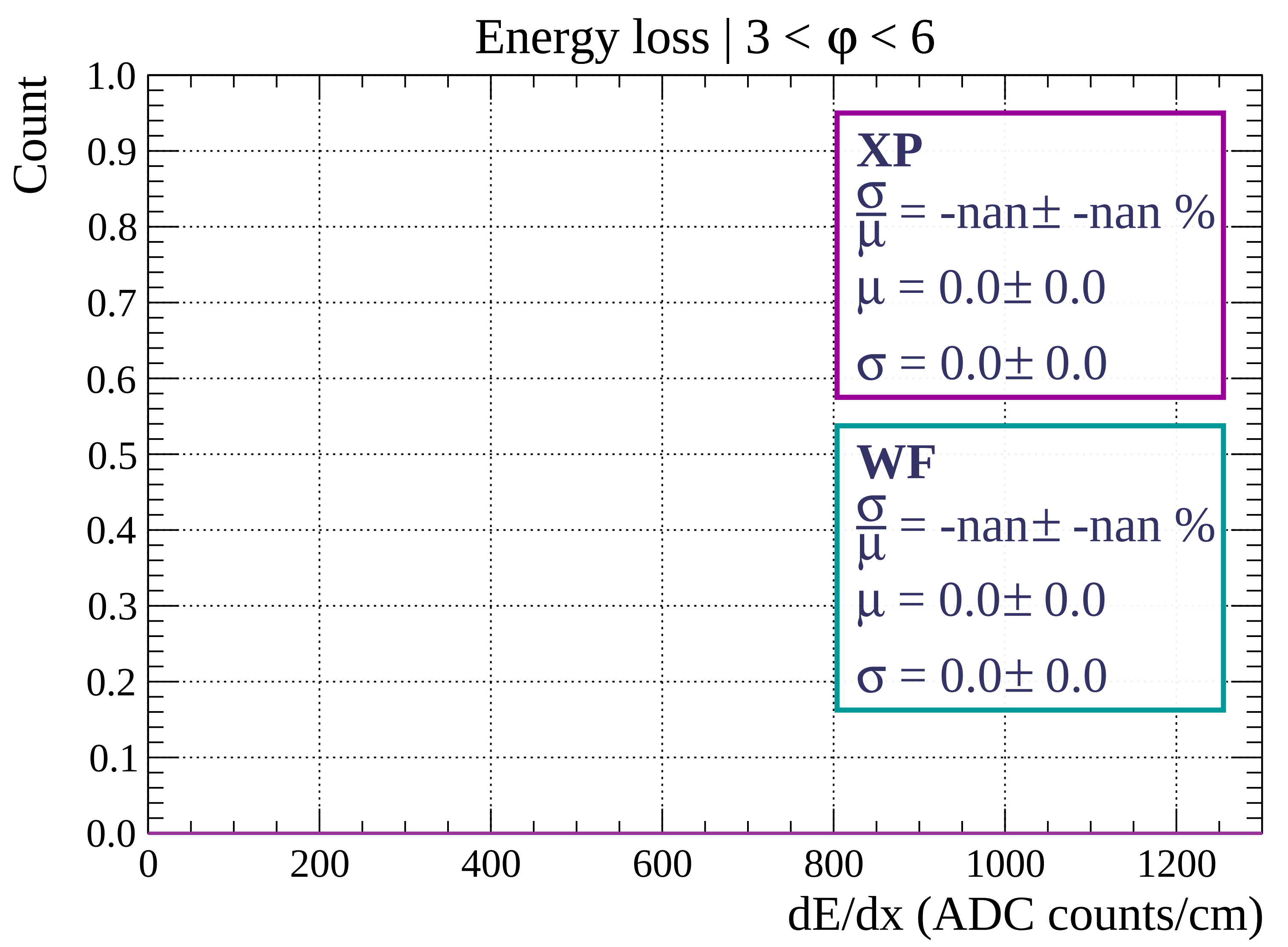
Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 7.92± 0.29 %
μ = 505.0± 1.5
σ = 40.0± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.27± 0.30 %
μ = 488.4± 1.5
σ = 40.4± 1.3

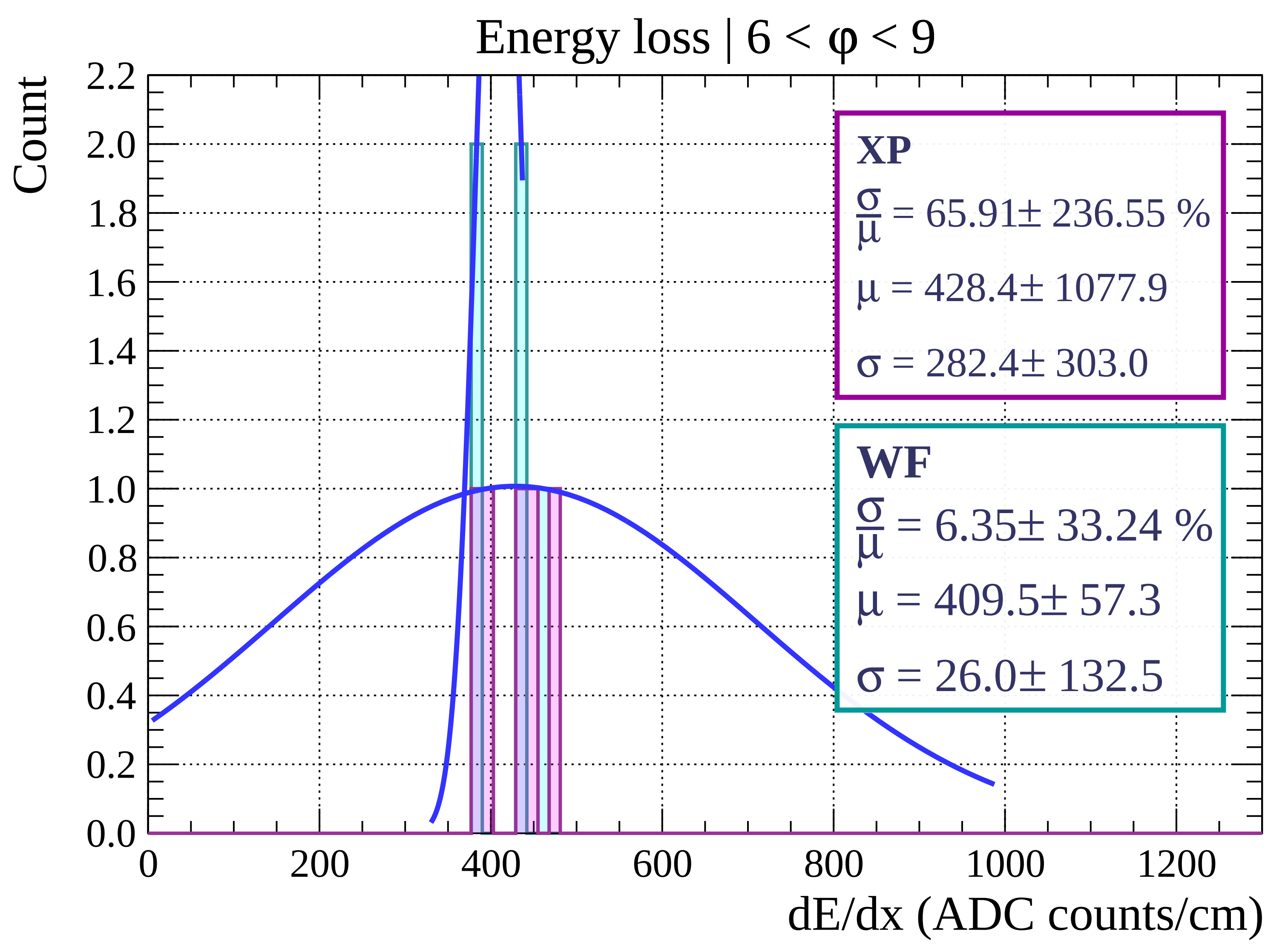


Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.29± 0.39 %
μ = 505.3± 1.7
σ = 41.9± 1.8
WF
σ/μ = 7.90± 0.37 %
μ = 489.3± 1.6
σ = 38.7± 1.7

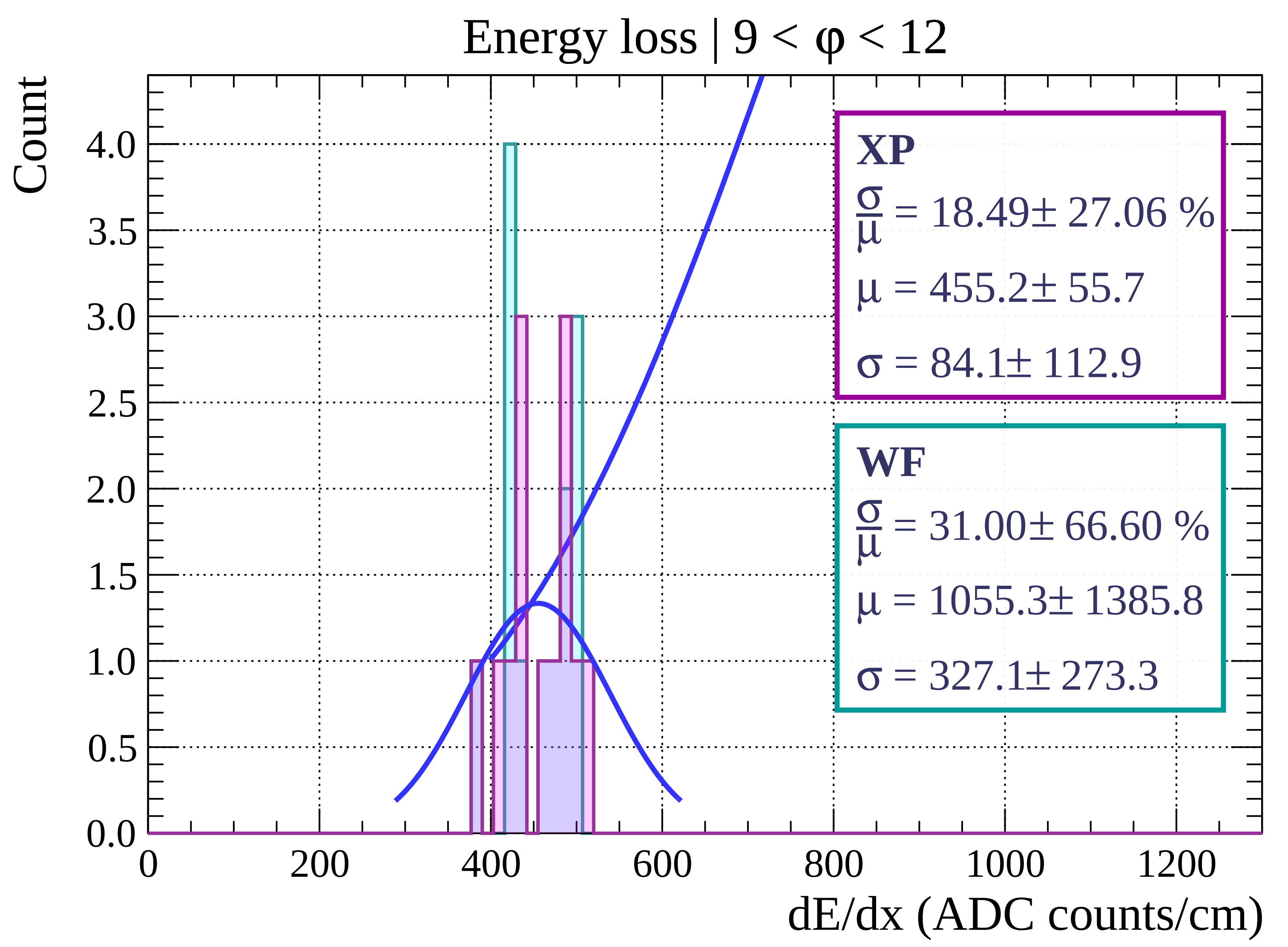


Energy loss | 0 < φ < 3
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

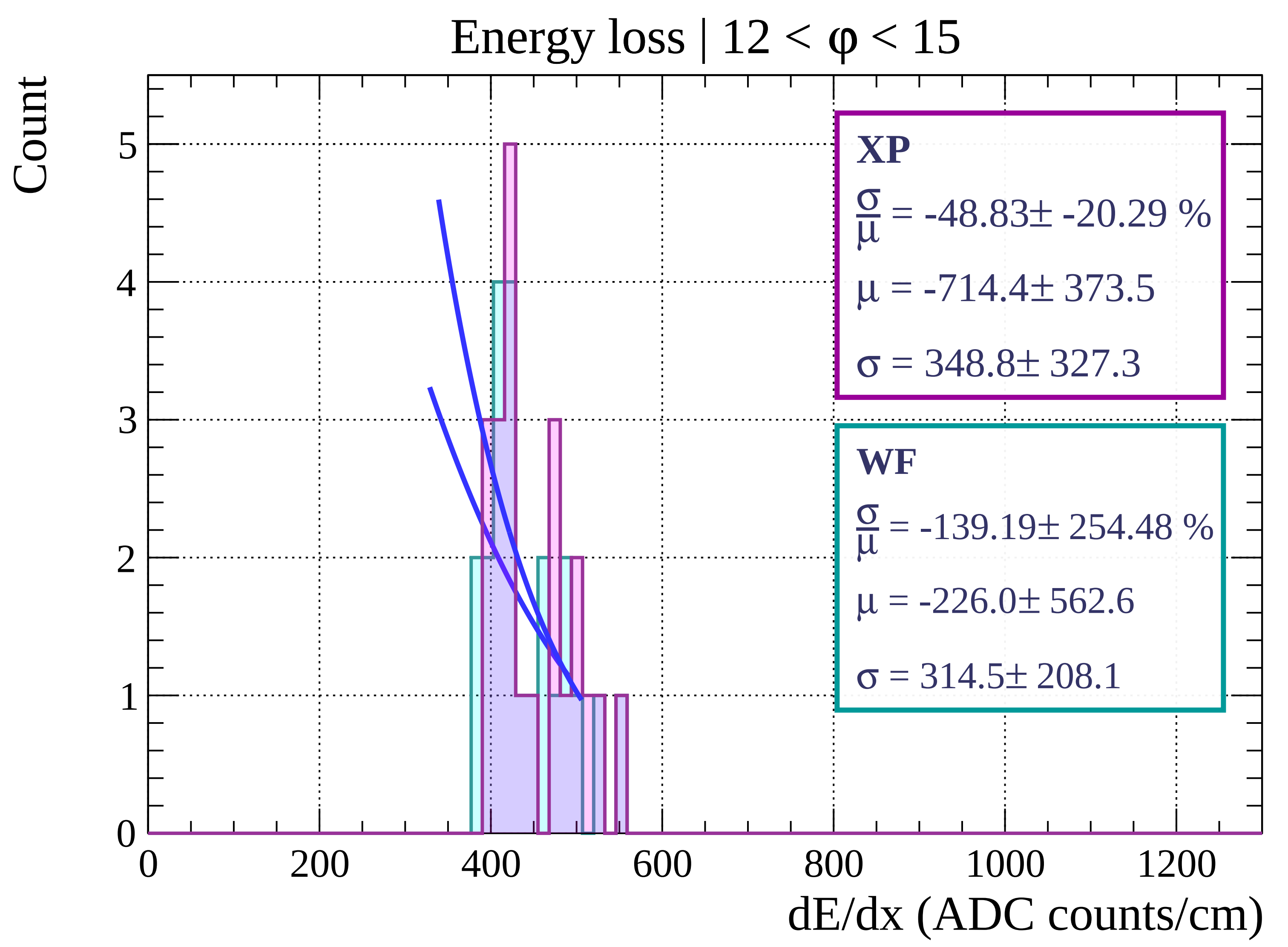

Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



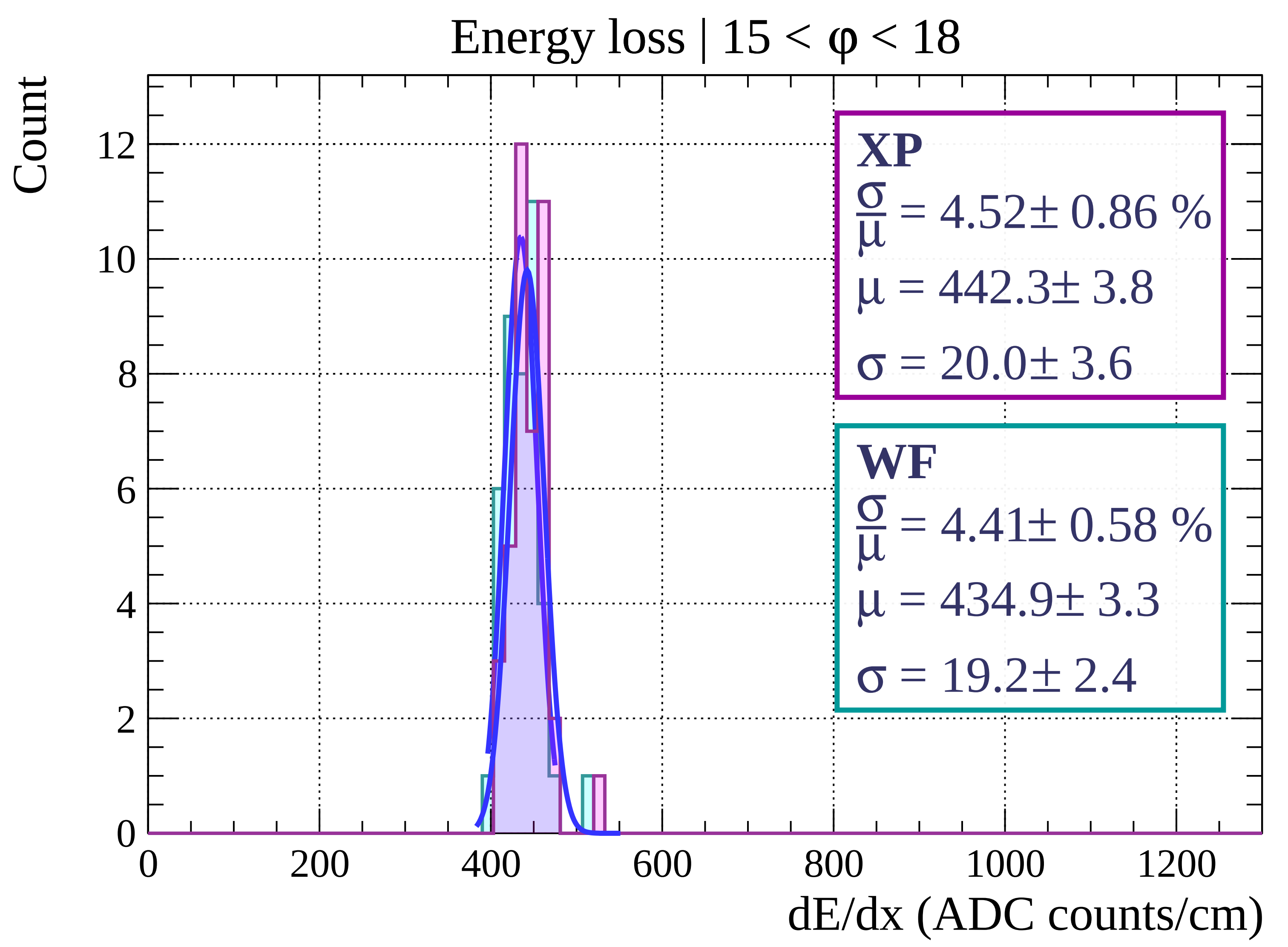
Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 65.91± 236.55 %
μ = 428.4± 1077.9
σ = 282.4± 303.0
WF
$\frac{\sigma}{\mu}$ = 6.35± 33.24 %
μ = 409.5± 57.3
σ = 26.0± 132.5



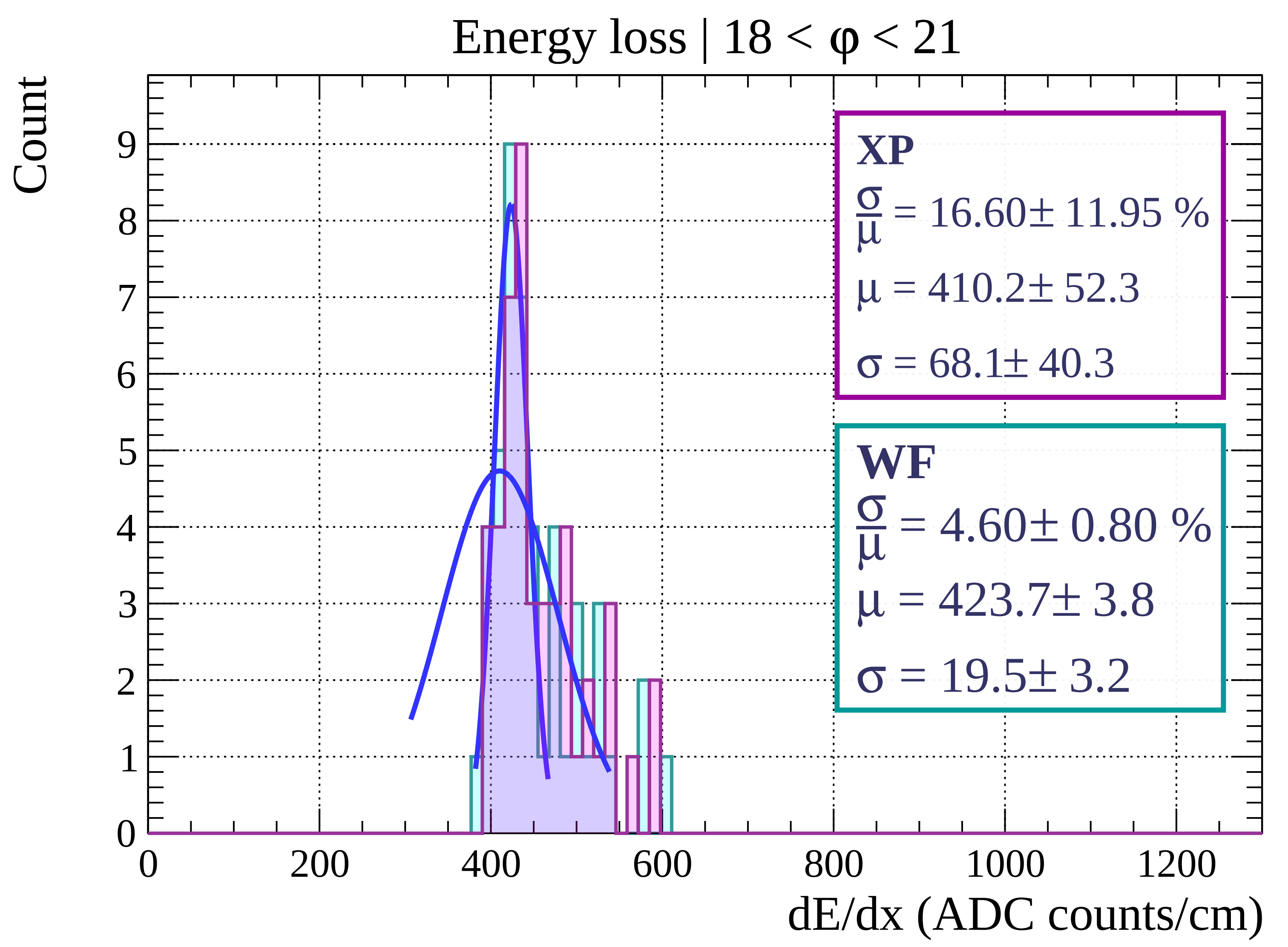
Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 18.49± 27.06 %
μ = 455.2± 55.7
σ = 84.1± 112.9
WF
σ/μ = 31.00± 66.60 %
μ = 1055.3± 1385.8
σ = 327.1± 273.3
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200



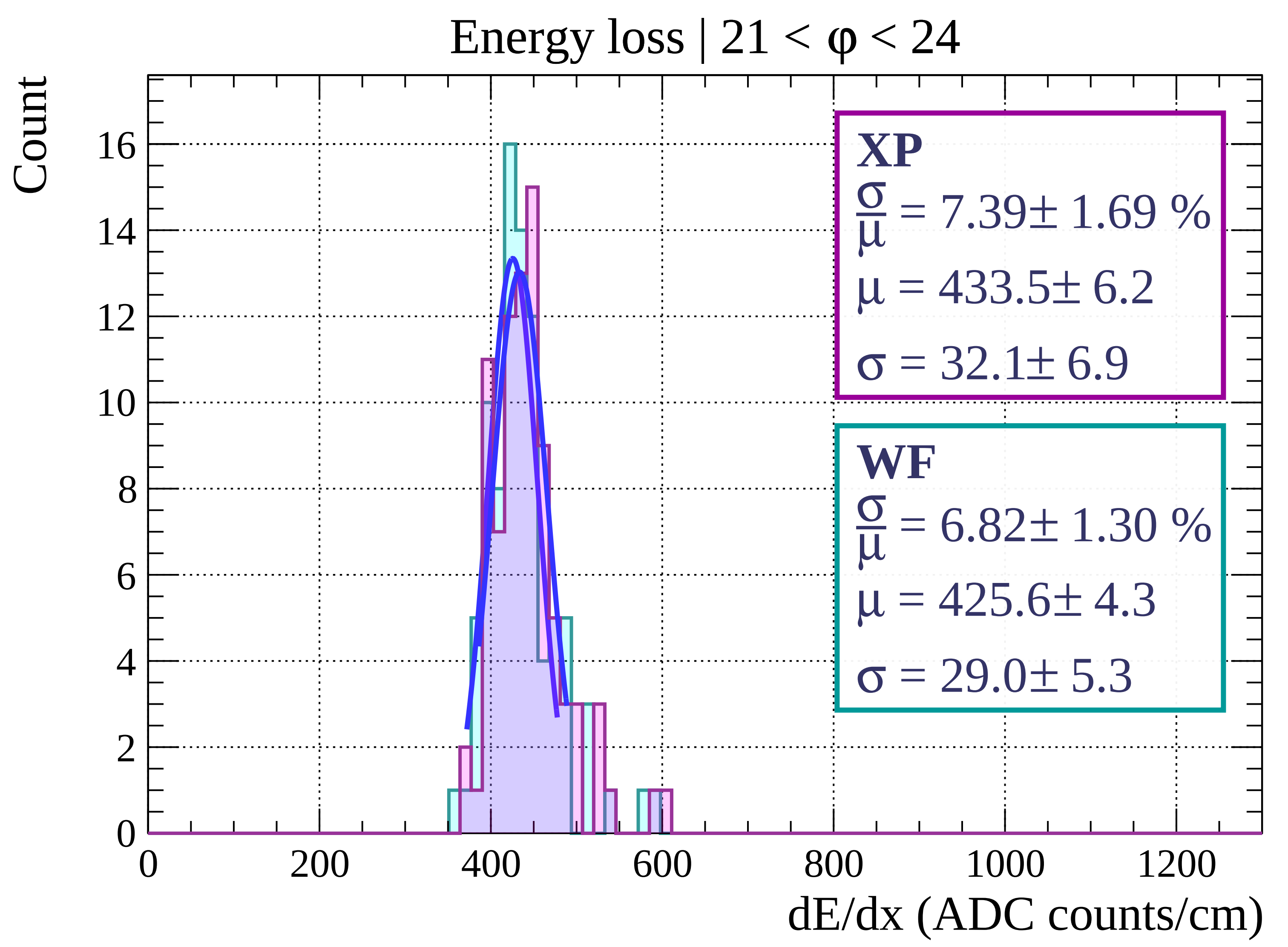
Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = -48.83± -20.29 %
μ = -714.4± 373.5
σ = 348.8± 327.3
WF
σ/μ = -139.19± 254.48 %
μ = -226.0± 562.6
σ = 314.5± 208.1



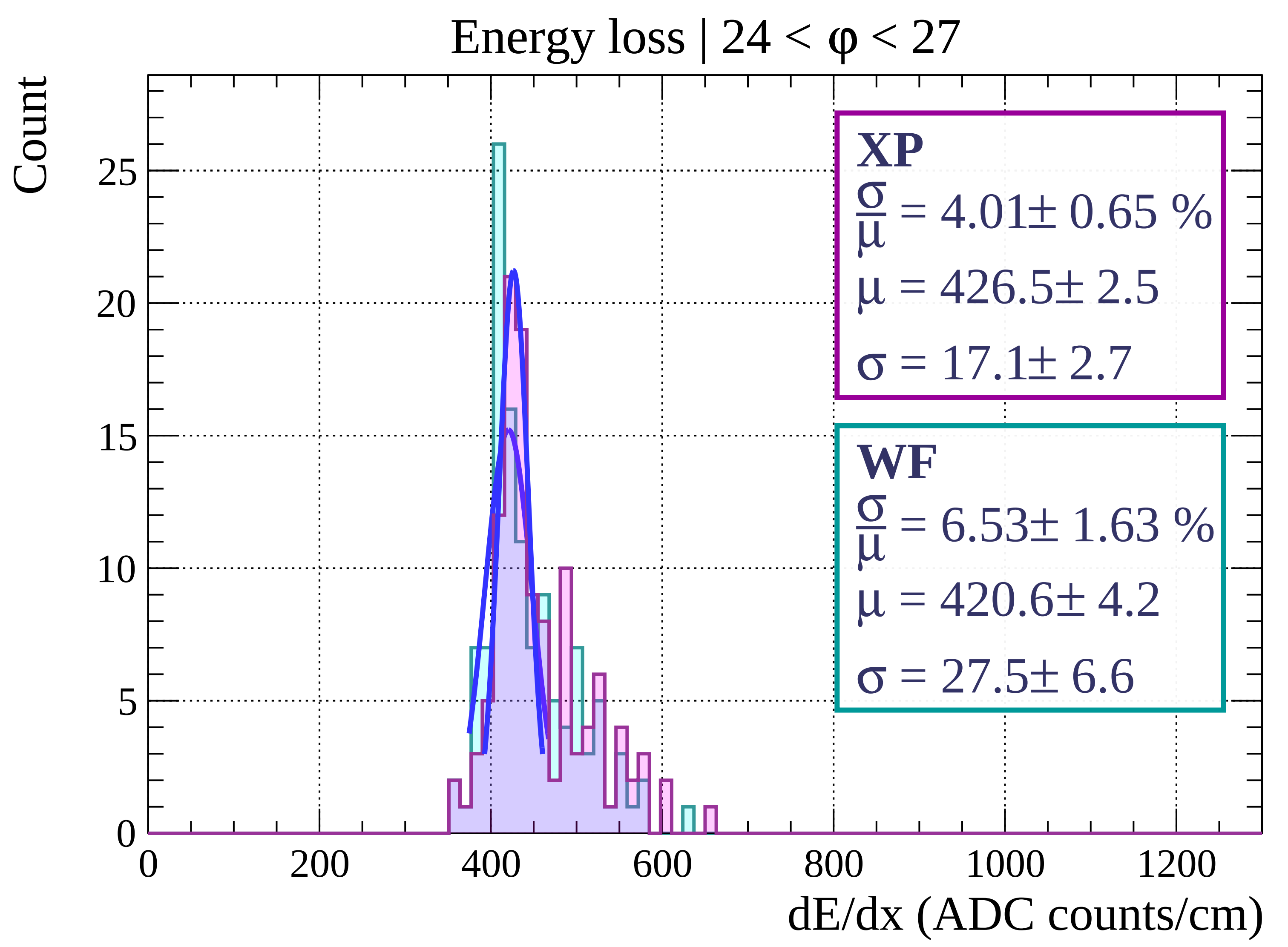
Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
σ/μ = 4.52± 0.86 %
μ = 442.3± 3.8
σ = 20.0± 3.6
WF
σ/μ = 4.41± 0.58 %
μ = 434.9± 3.3
σ = 19.2± 2.4



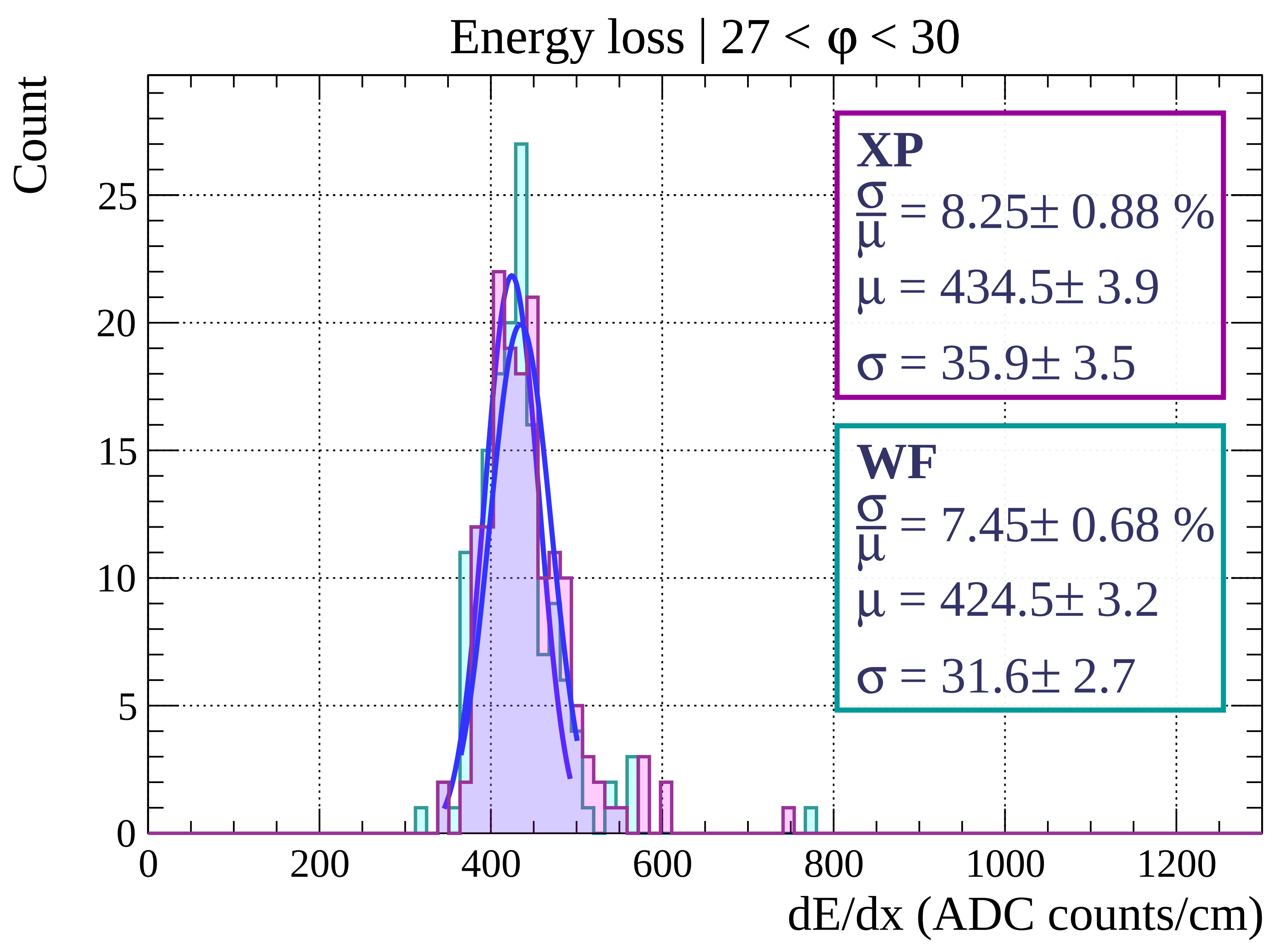
Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 16.60± 11.95 %
μ = 410.2± 52.3
σ = 68.1± 40.3
WF
σ/μ = 4.60± 0.80 %
μ = 423.7± 3.8
σ = 19.5± 3.2



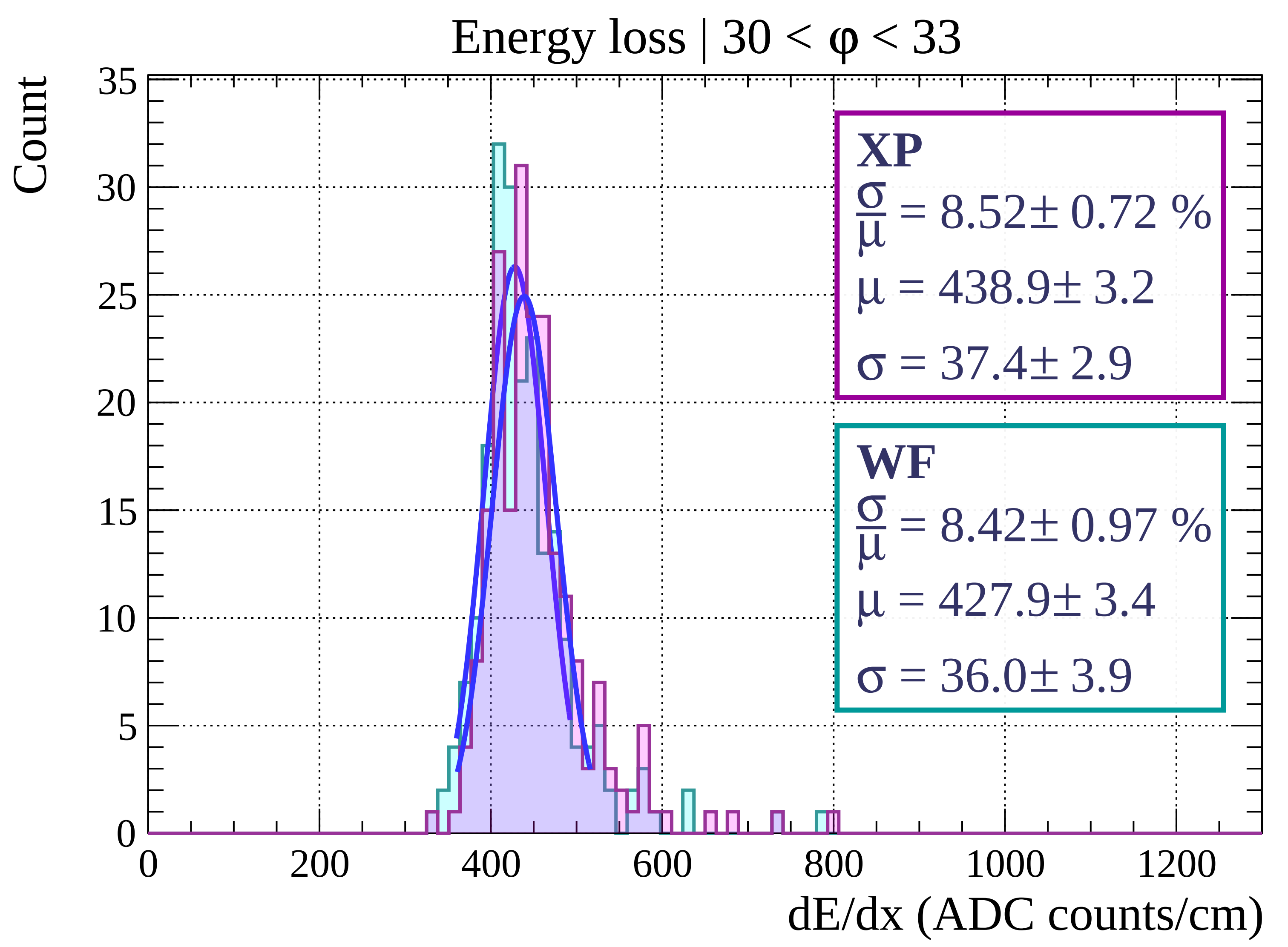
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
XP
$\frac{\sigma}{\mu}$ = 7.39± 1.69 %
μ = 433.5± 6.2
σ = 32.1± 6.9
WF
$\frac{\sigma}{\mu}$ = 6.82± 1.30 %
μ = 425.6± 4.3
σ = 29.0± 5.3



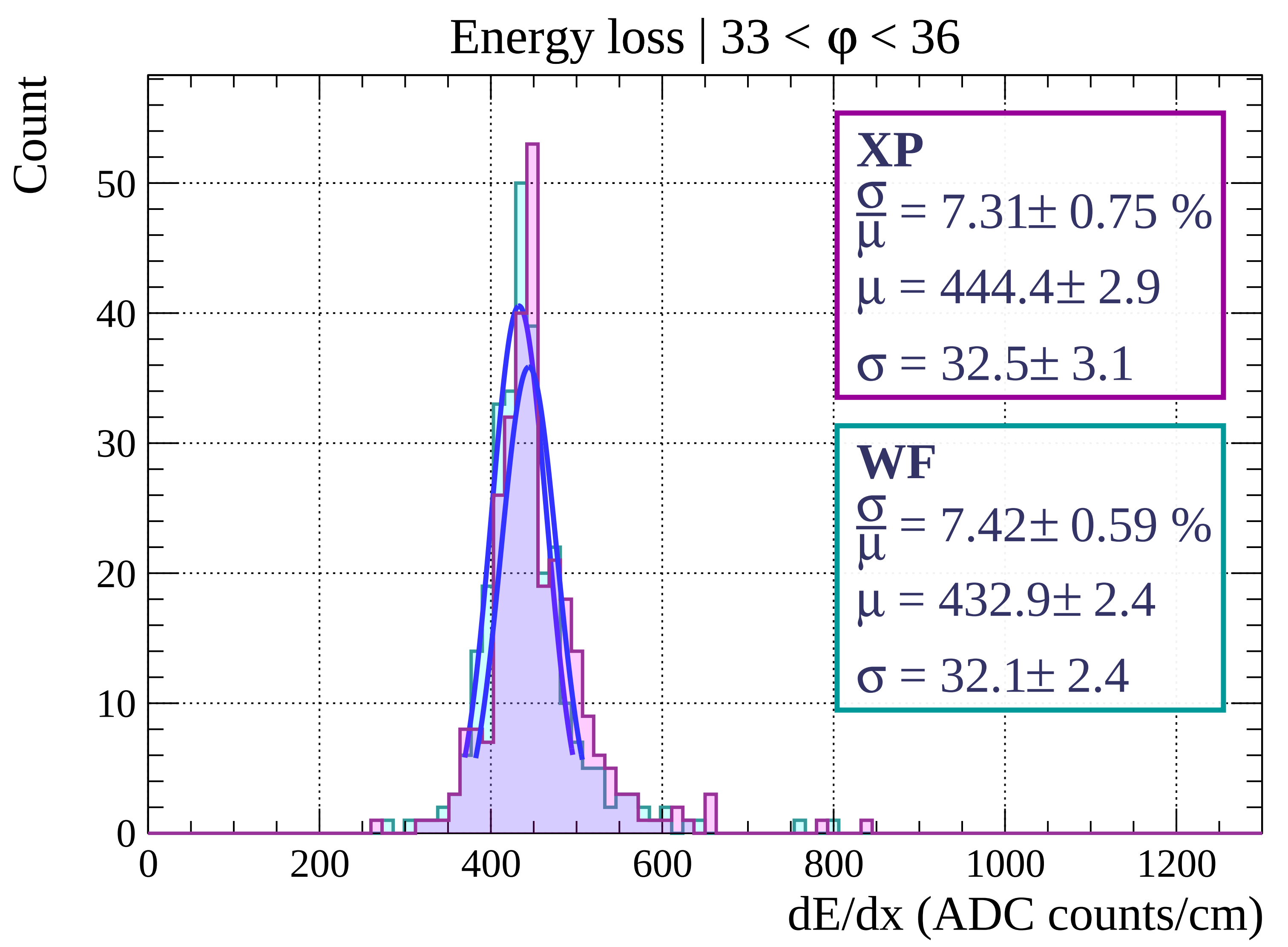
Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 4.01± 0.65 %
μ = 426.5± 2.5
σ = 17.1± 2.7
WF
σ/μ = 6.53± 1.63 %
μ = 420.6± 4.2
σ = 27.5± 6.6



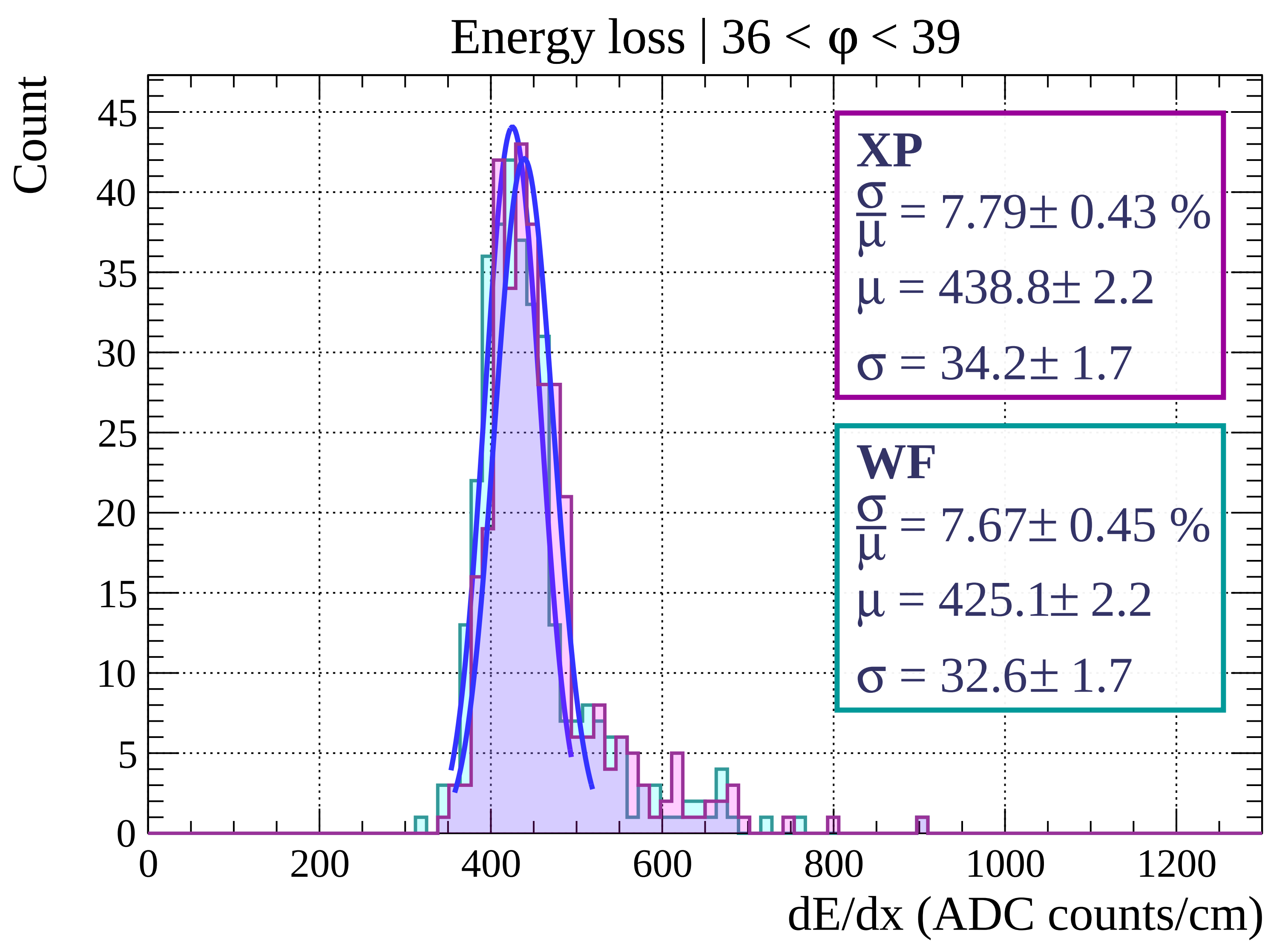
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.25± 0.88 %
μ = 434.5± 3.9
σ = 35.9± 3.5
WF
σ/μ = 7.45± 0.68 %
μ = 424.5± 3.2
σ = 31.6± 2.7
0
5
10
15
20
25
0
200
400
600
800
1000
1200



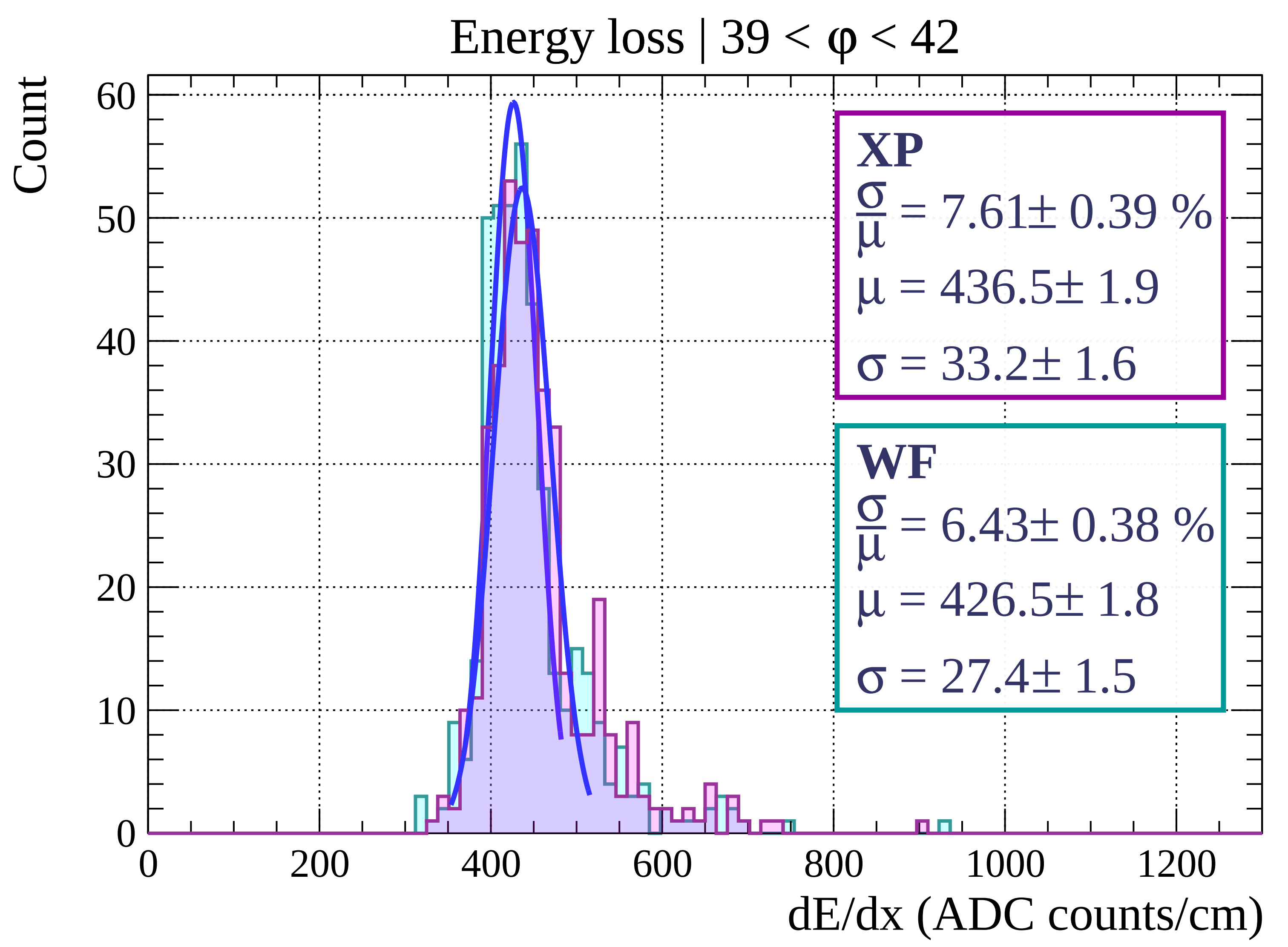
Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 8.52± 0.72 %
μ = 438.9± 3.2
σ = 37.4± 2.9
WF
σ/μ = 8.42± 0.97 %
μ = 427.9± 3.4
σ = 36.0± 3.9



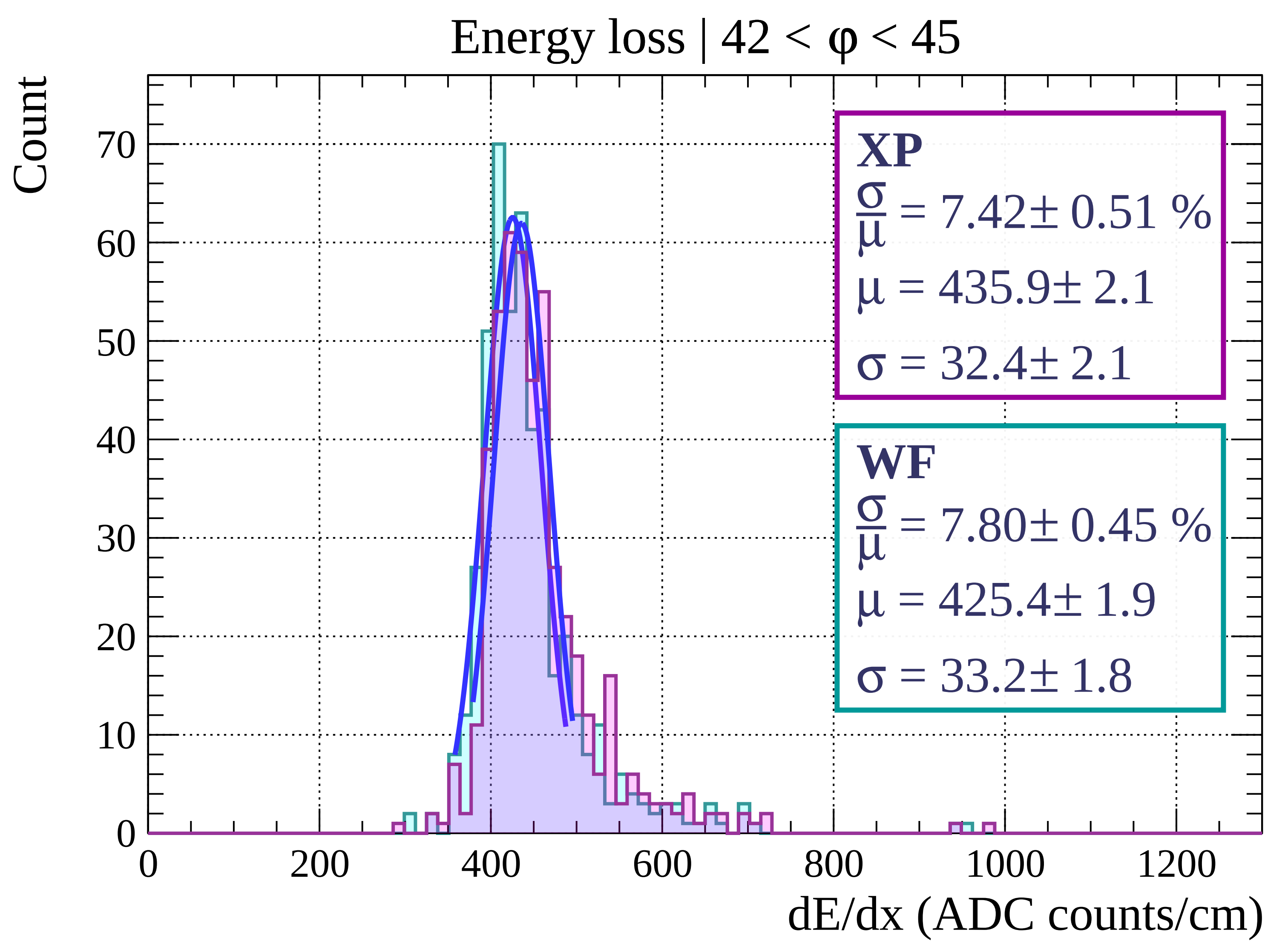
Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 7.31± 0.75 %
μ = 444.4± 2.9
σ = 32.5± 3.1
WF
σ/μ = 7.42± 0.59 %
μ = 432.9± 2.4
σ = 32.1± 2.4



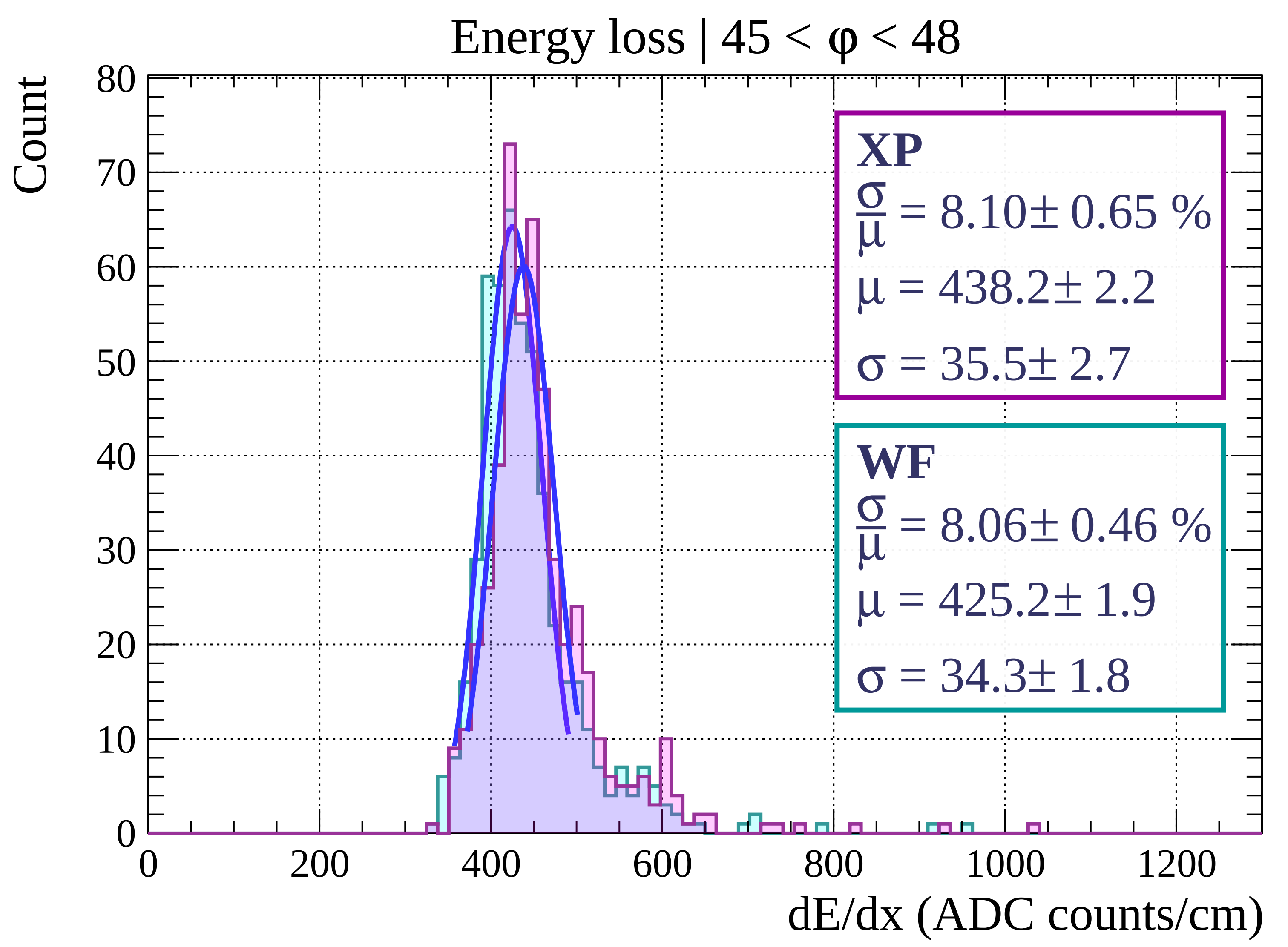
Energy loss | 36 < φ < 39
Count
XP
σ/μ = 7.79± 0.43 %
μ = 438.8± 2.2
σ = 34.2± 1.7
WF
σ/μ = 7.67± 0.45 %
μ = 425.1± 2.2
σ = 32.6± 1.7
dE/dx (ADC counts/cm)

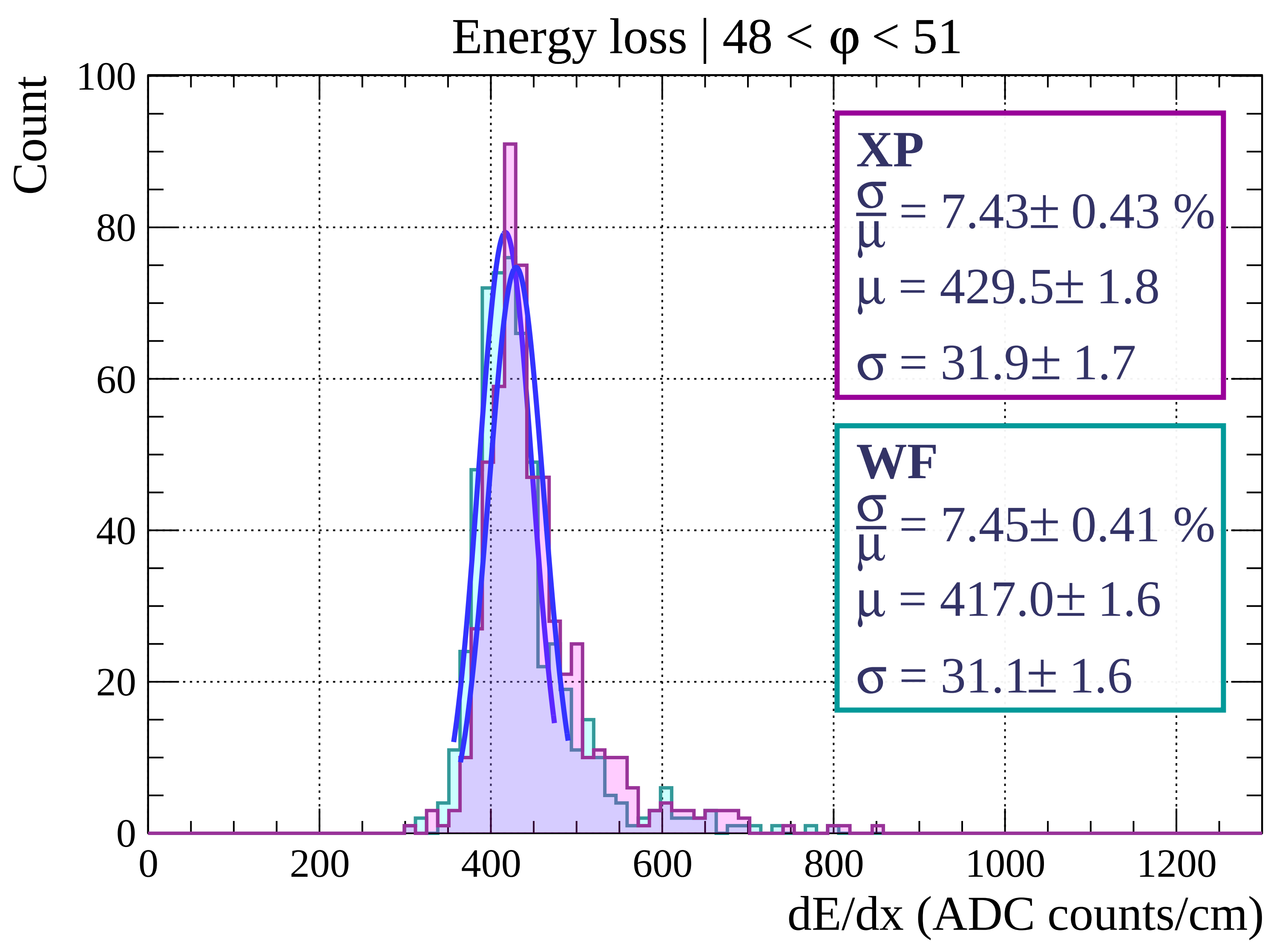


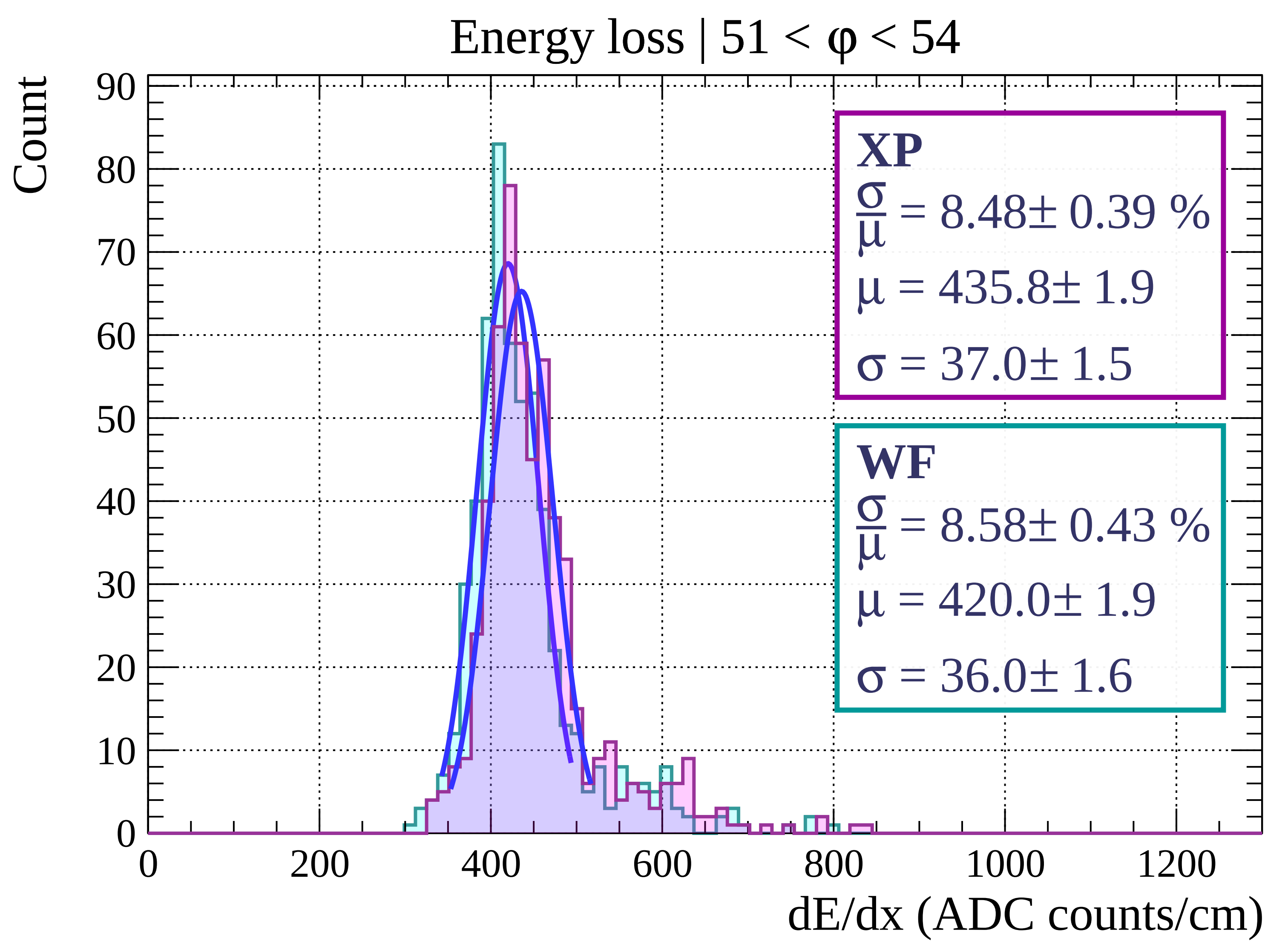
Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 7.61± 0.39 %
μ = 436.5± 1.9
σ = 33.2± 1.6
WF
σ/μ = 6.43± 0.38 %
μ = 426.5± 1.8
σ = 27.4± 1.5



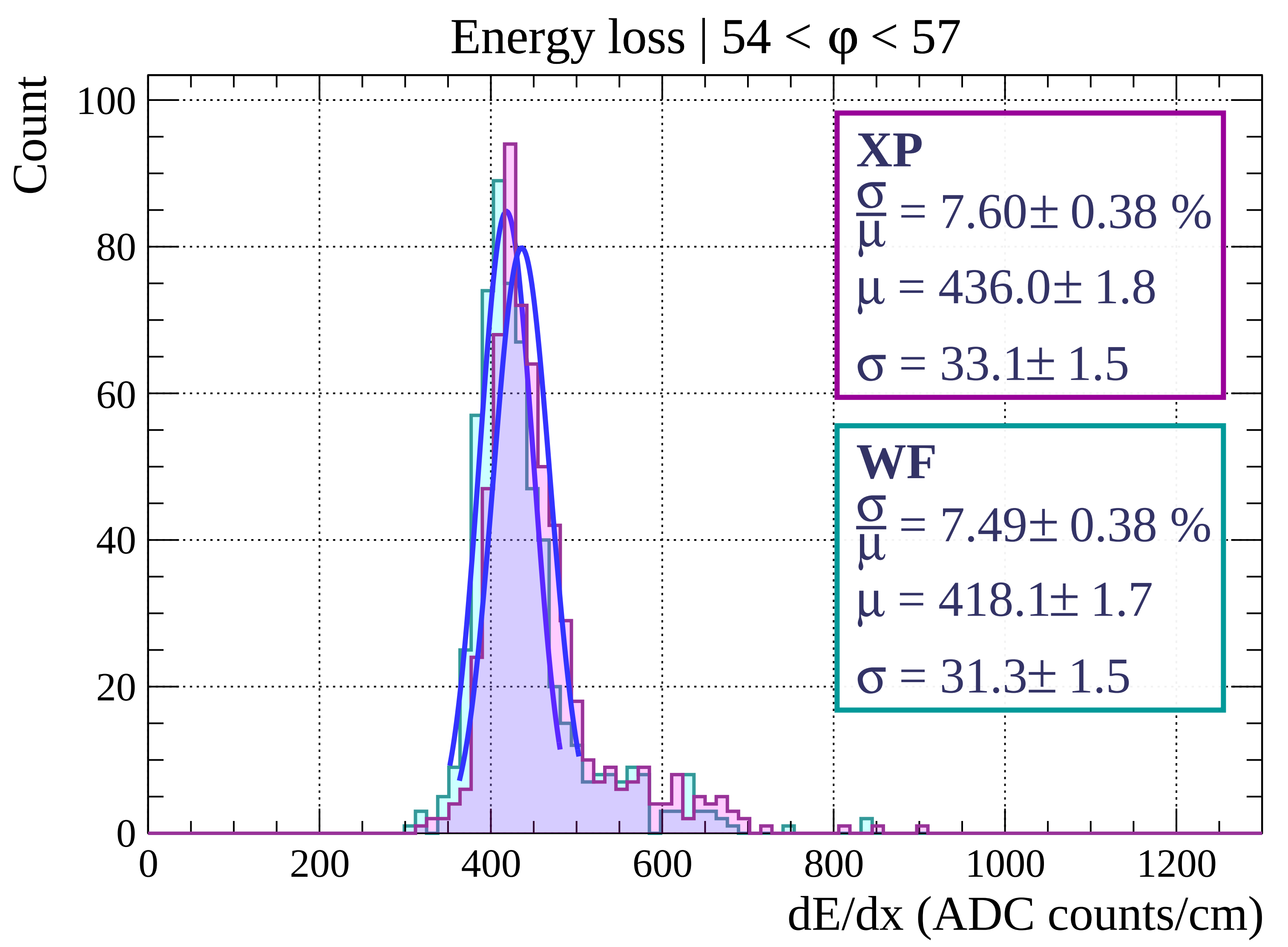
Energy loss | 42 < φ < 45
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 7.42± 0.51 %
μ = 435.9± 2.1
σ = 32.4± 2.1
WF
σ/μ = 7.80± 0.45 %
μ = 425.4± 1.9
σ = 33.2± 1.8


Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 8.10± 0.65 %
μ = 438.2± 2.2
σ = 35.5± 2.7
WF
σ/μ = 8.06± 0.46 %
μ = 425.2± 1.9
σ = 34.3± 1.8

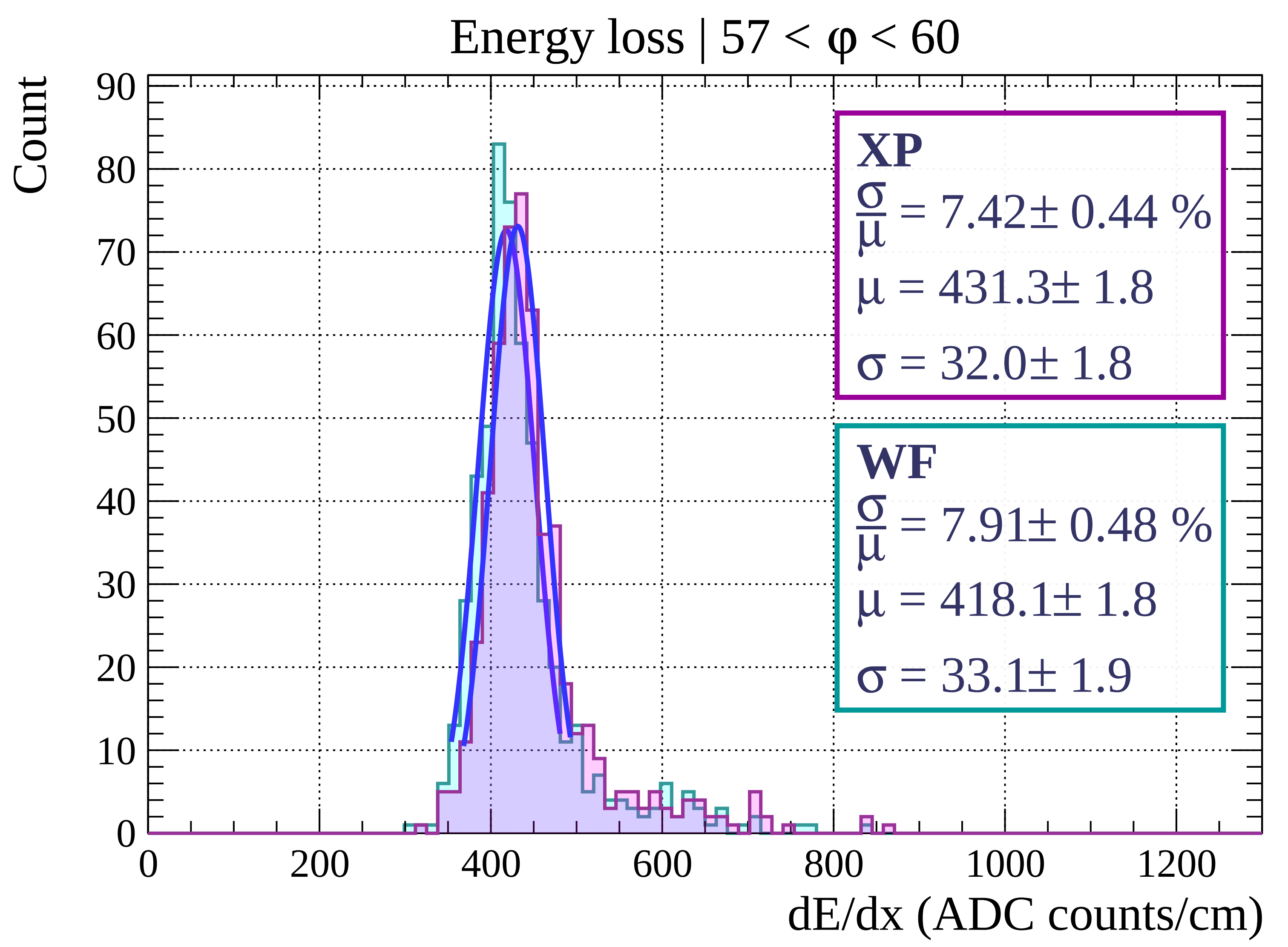

Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.43± 0.43 %
μ = 429.5± 1.8
σ = 31.9± 1.7
WF
σ/μ = 7.45± 0.41 %
μ = 417.0± 1.6
σ = 31.1± 1.6



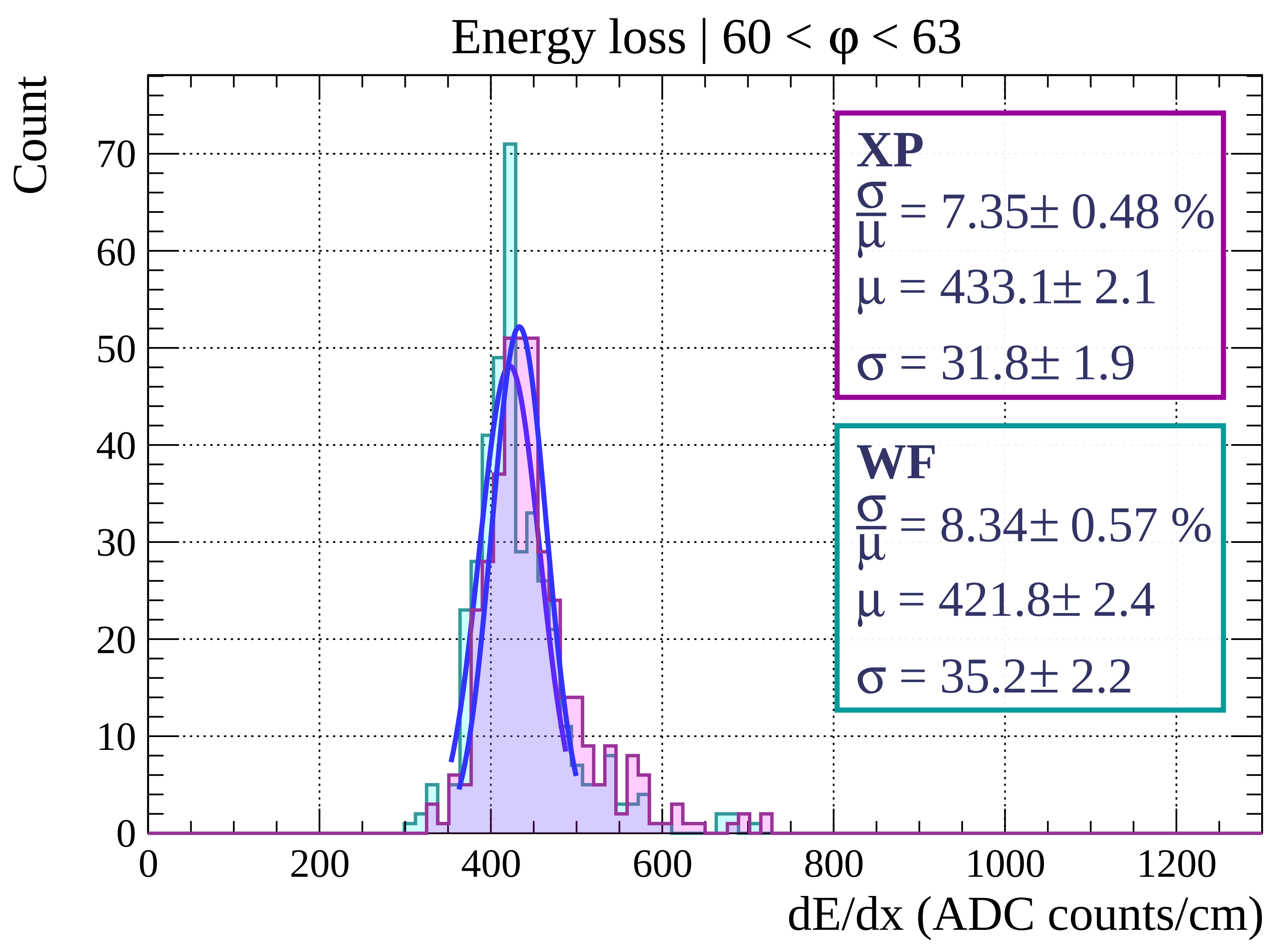
Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu}$ = 8.48± 0.39 %
μ = 435.8± 1.9
σ = 37.0± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.58± 0.43 %
μ = 420.0± 1.9
σ = 36.0± 1.6



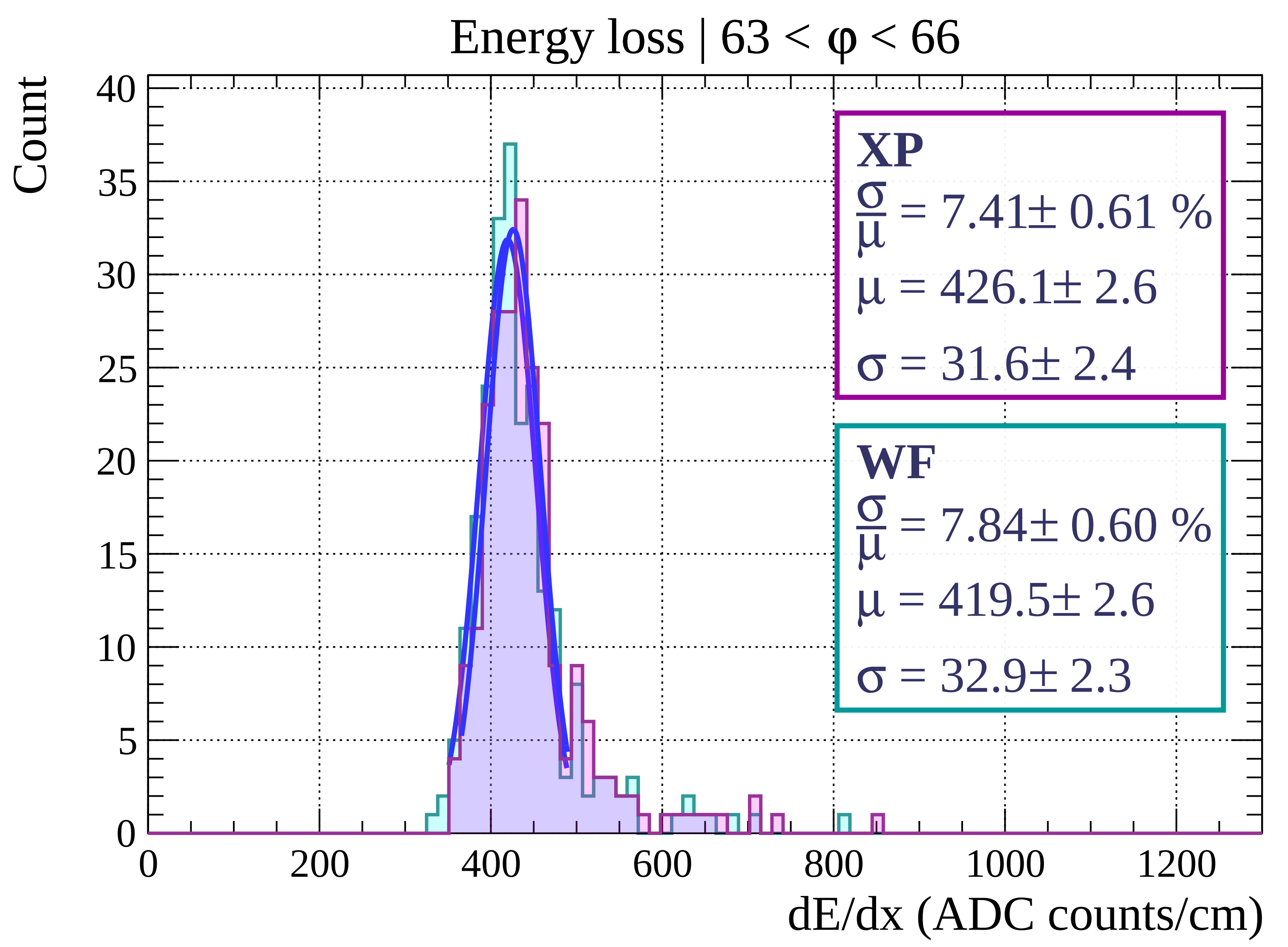
Energy loss | 54 < φ < 57
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.60± 0.38 %
μ = 436.0± 1.8
σ = 33.1± 1.5
WF
σ/μ = 7.49± 0.38 %
μ = 418.1± 1.7
σ = 31.3± 1.5



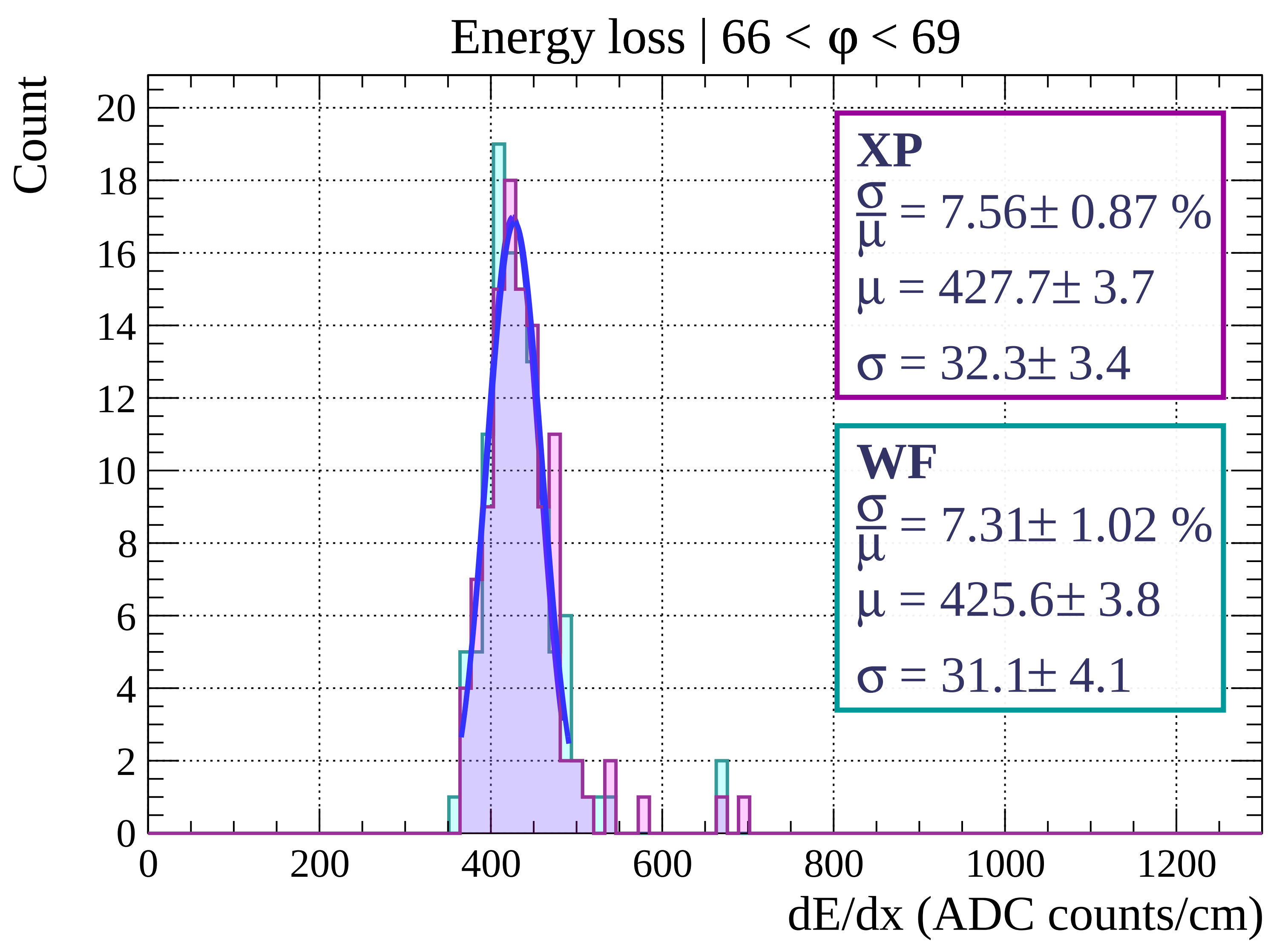
Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 7.42± 0.44 %
μ = 431.3± 1.8
σ = 32.0± 1.8
WF
σ/μ = 7.91± 0.48 %
μ = 418.1± 1.8
σ = 33.1± 1.9

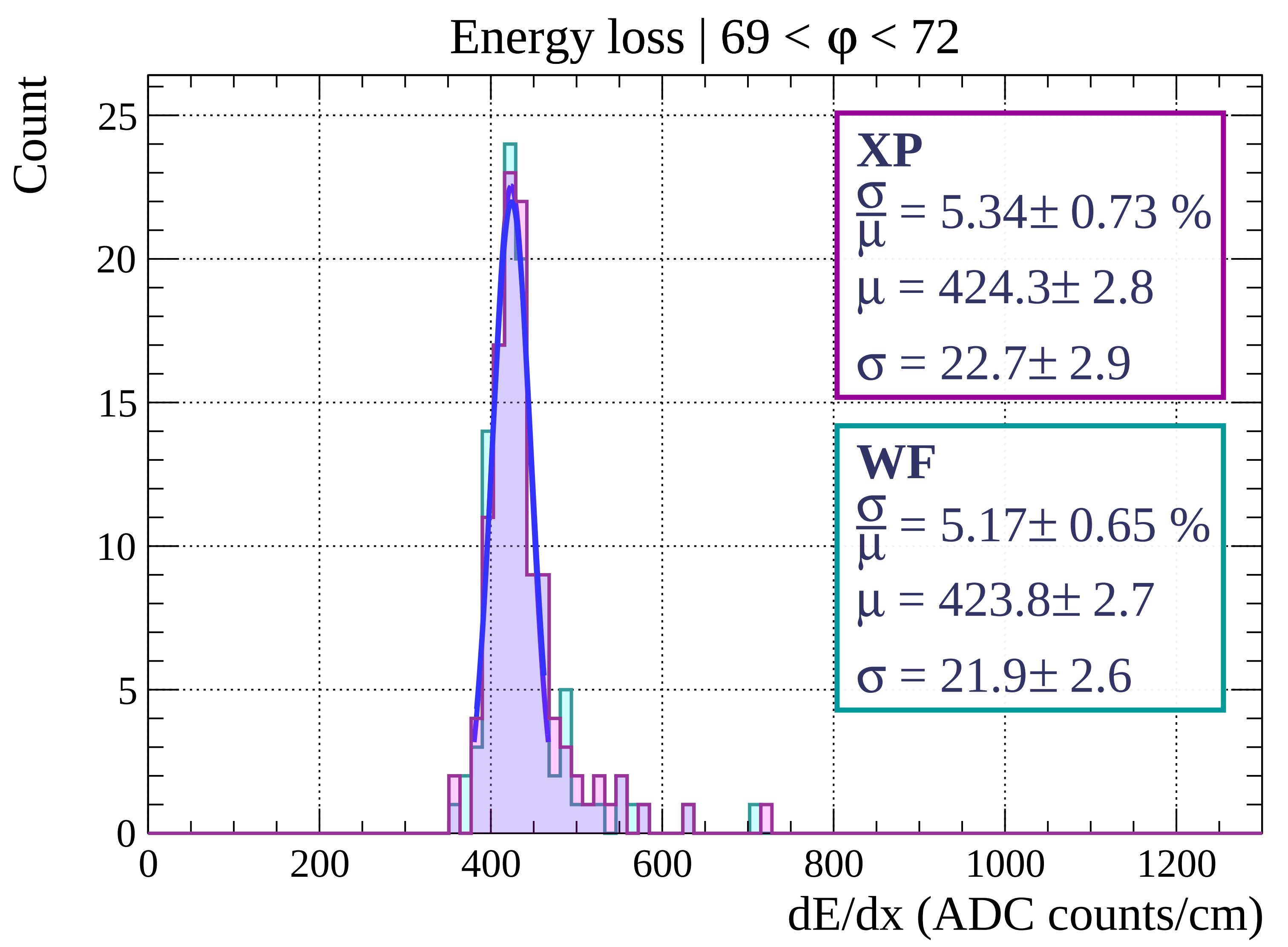


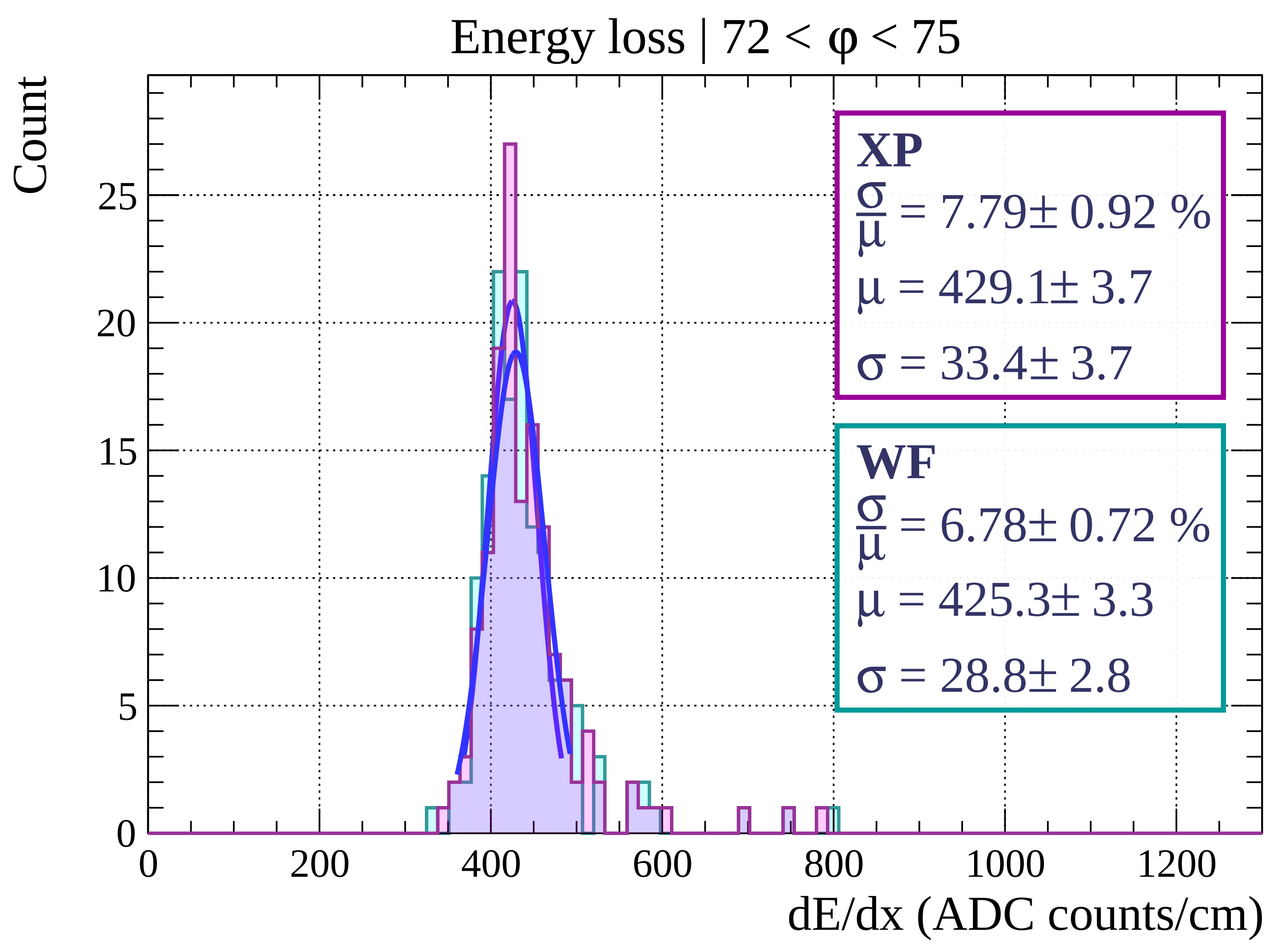
Energy loss | 60 < φ < 63
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
σ/μ = 7.35± 0.48 %
μ = 433.1± 2.1
σ = 31.8± 1.9
WF
σ/μ = 8.34± 0.57 %
μ = 421.8± 2.4
σ = 35.2± 2.2



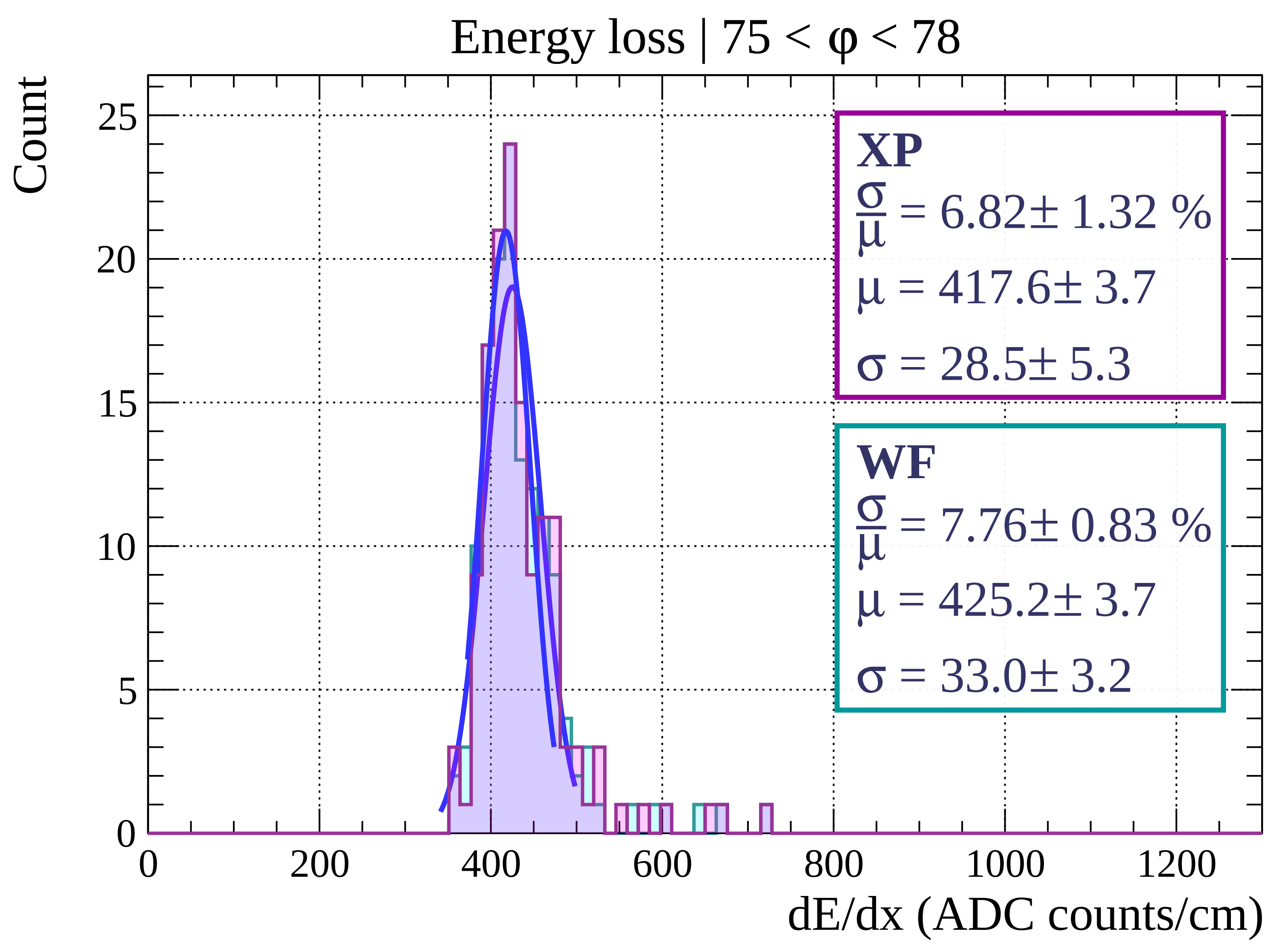
Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
σ/μ = 7.41± 0.61 %
μ = 426.1± 2.6
σ = 31.6± 2.4
WF
σ/μ = 7.84± 0.60 %
μ = 419.5± 2.6
σ = 32.9± 2.3


Energy loss | 66 < φ < 69
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.56± 0.87 %
μ = 427.7± 3.7
σ = 32.3± 3.4
WF
σ/μ = 7.31± 1.02 %
μ = 425.6± 3.8
σ = 31.1± 4.1
0
2
4
6
8
10
12
14
16
18
20
0
200
400
600
800
1000
1200

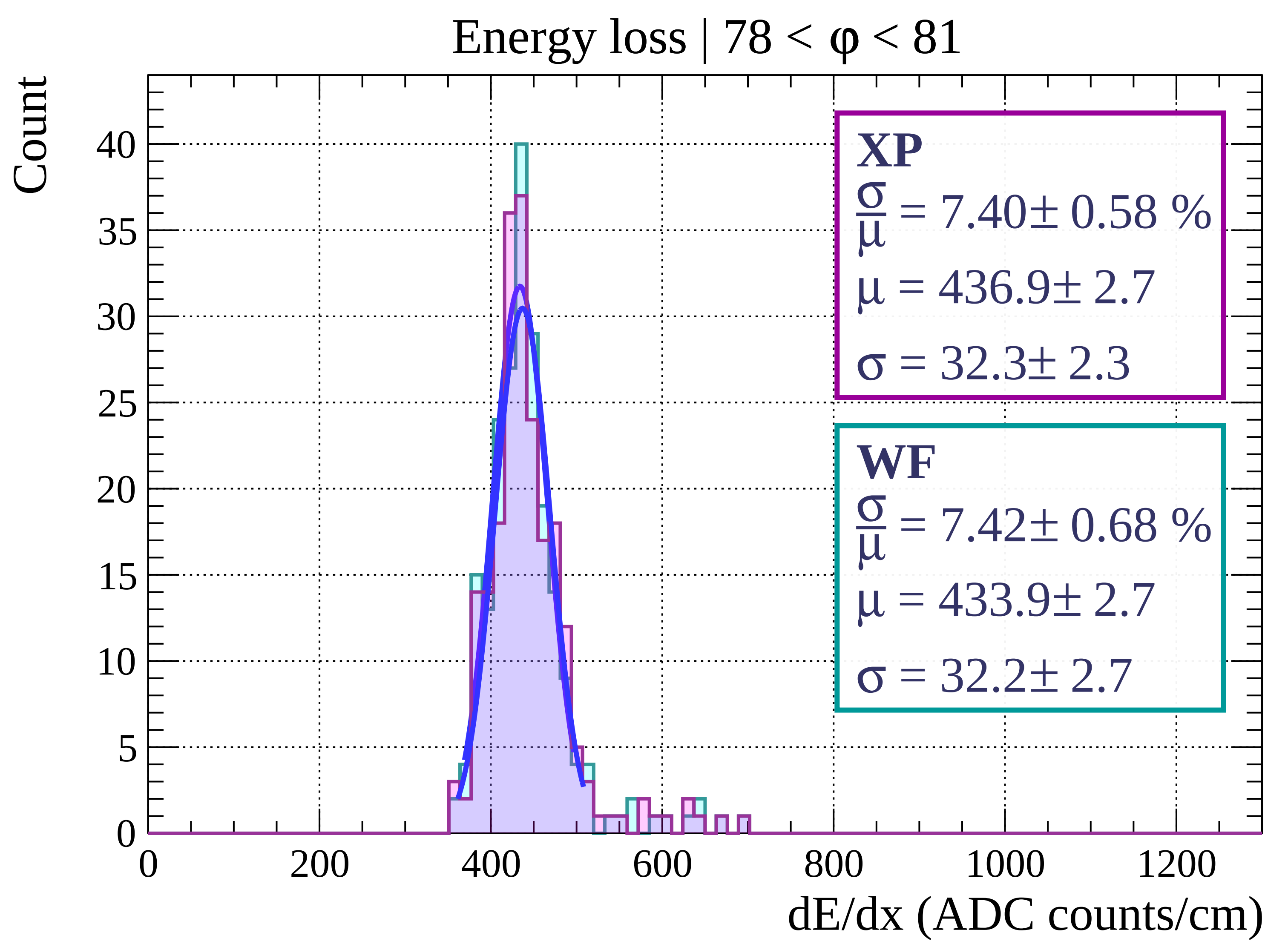

Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 5.34± 0.73 %
μ = 424.3± 2.8
σ = 22.7± 2.9
WF
σ/μ = 5.17± 0.65 %
μ = 423.8± 2.7
σ = 21.9± 2.6



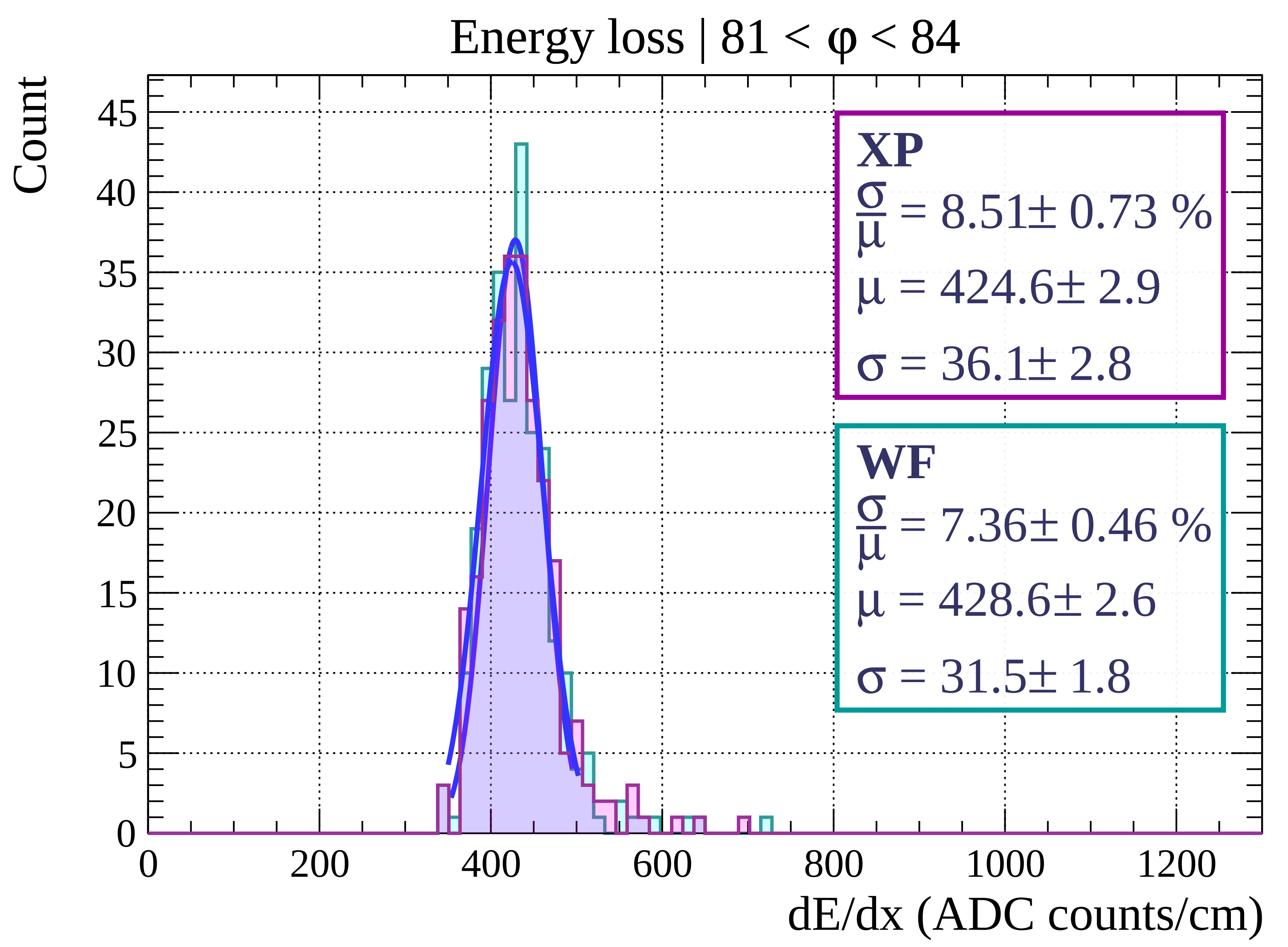
Energy loss | 72 < φ < 75
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 7.79± 0.92 %
μ = 429.1± 3.7
σ = 33.4± 3.7
WF
$\frac{\sigma}{\mu}$ = 6.78± 0.72 %
μ = 425.3± 3.3
σ = 28.8± 2.8

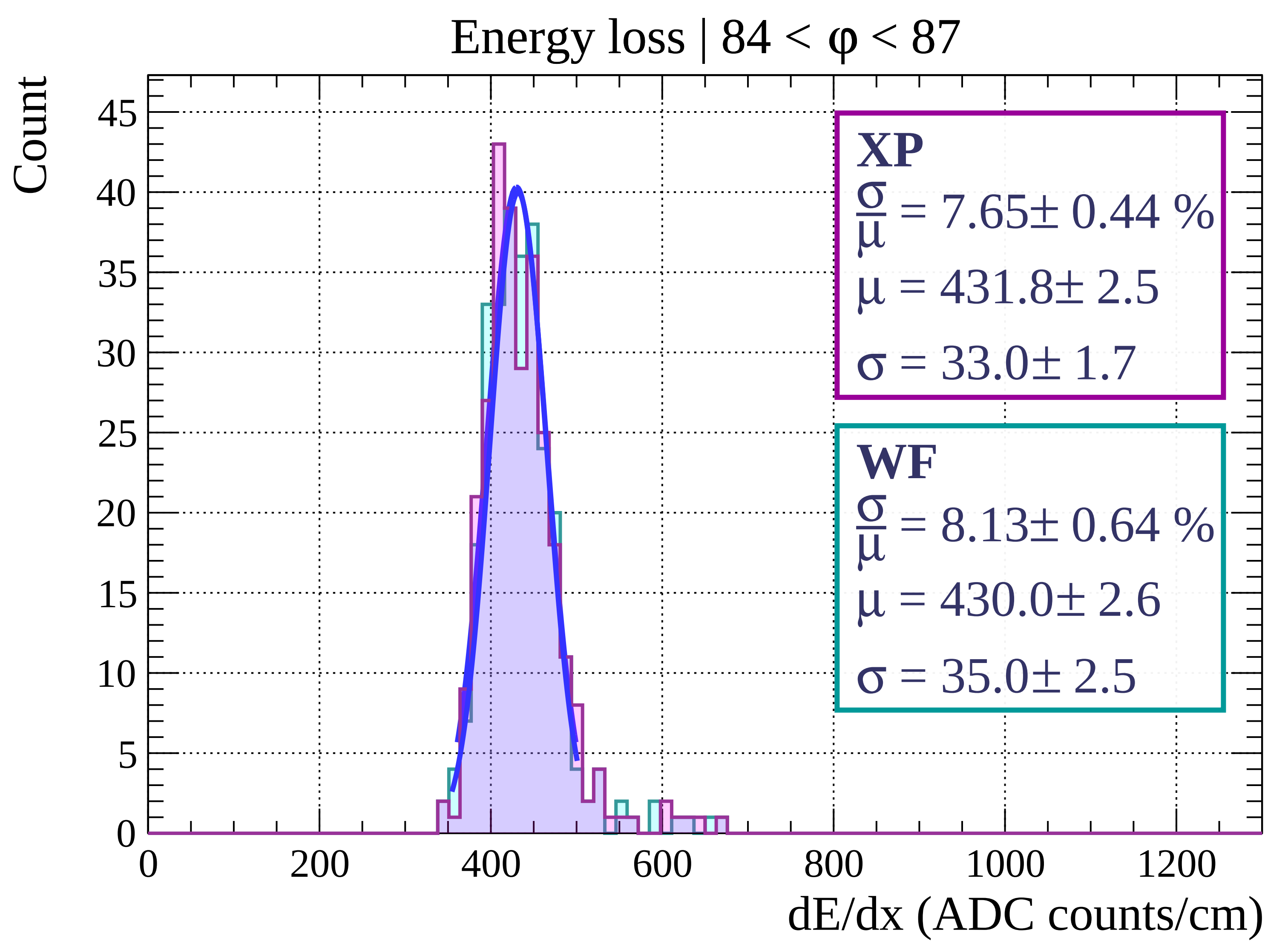


Energy loss | 75 < φ < 78
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.82± 1.32 %
μ = 417.6± 3.7
σ = 28.5± 5.3
WF
$\frac{\sigma}{\mu}$ = 7.76± 0.83 %
μ = 425.2± 3.7
σ = 33.0± 3.2

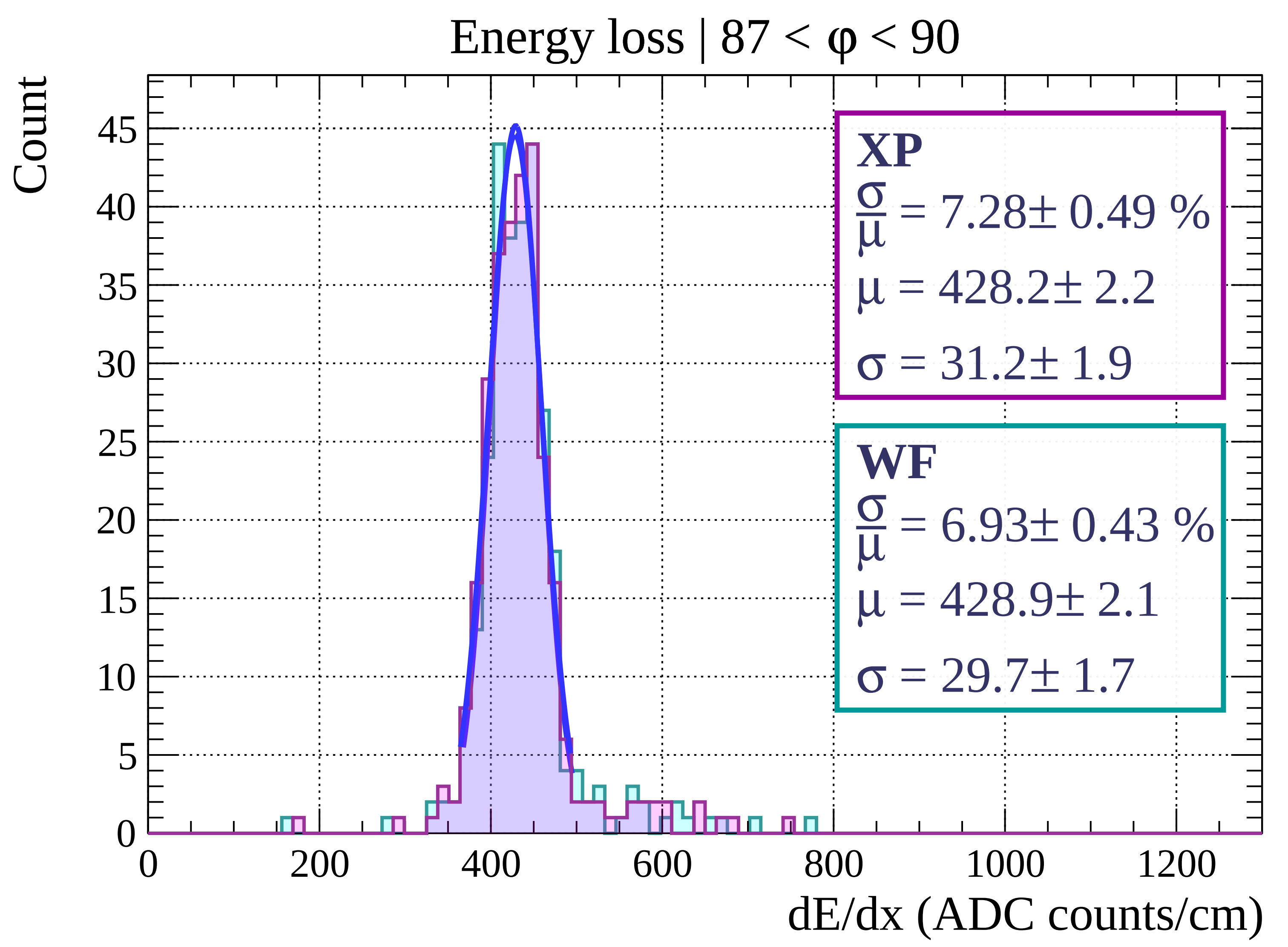


Energy loss | 78 < φ < 81
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 7.40± 0.58 %
μ = 436.9± 2.7
σ = 32.3± 2.3
WF
σ/μ = 7.42± 0.68 %
μ = 433.9± 2.7
σ = 32.2± 2.7

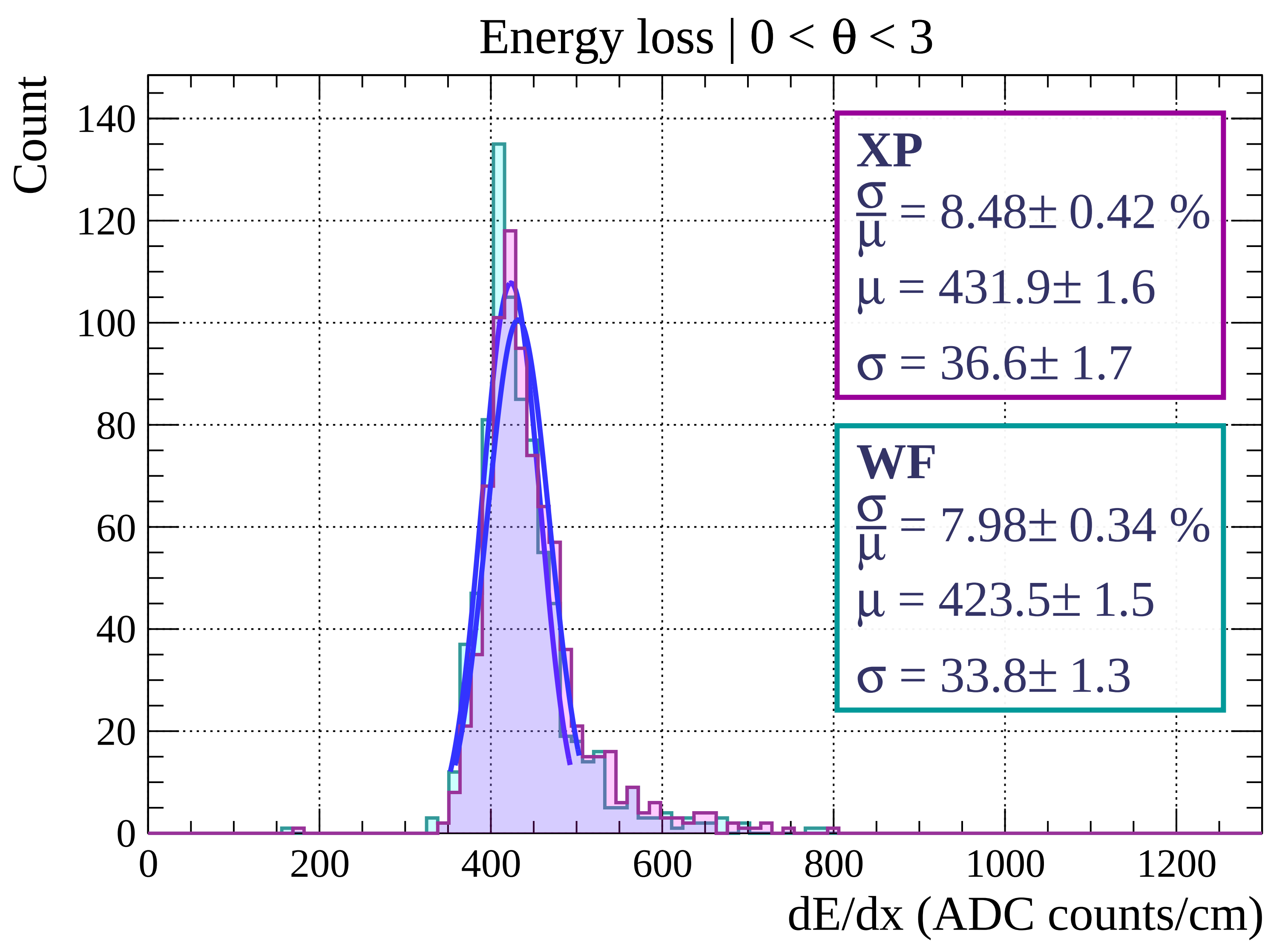

Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP
σ/μ = 8.51± 0.73 %
μ = 424.6± 2.9
σ = 36.1± 2.8
WF
σ/μ = 7.36± 0.46 %
μ = 428.6± 2.6
σ = 31.5± 1.8



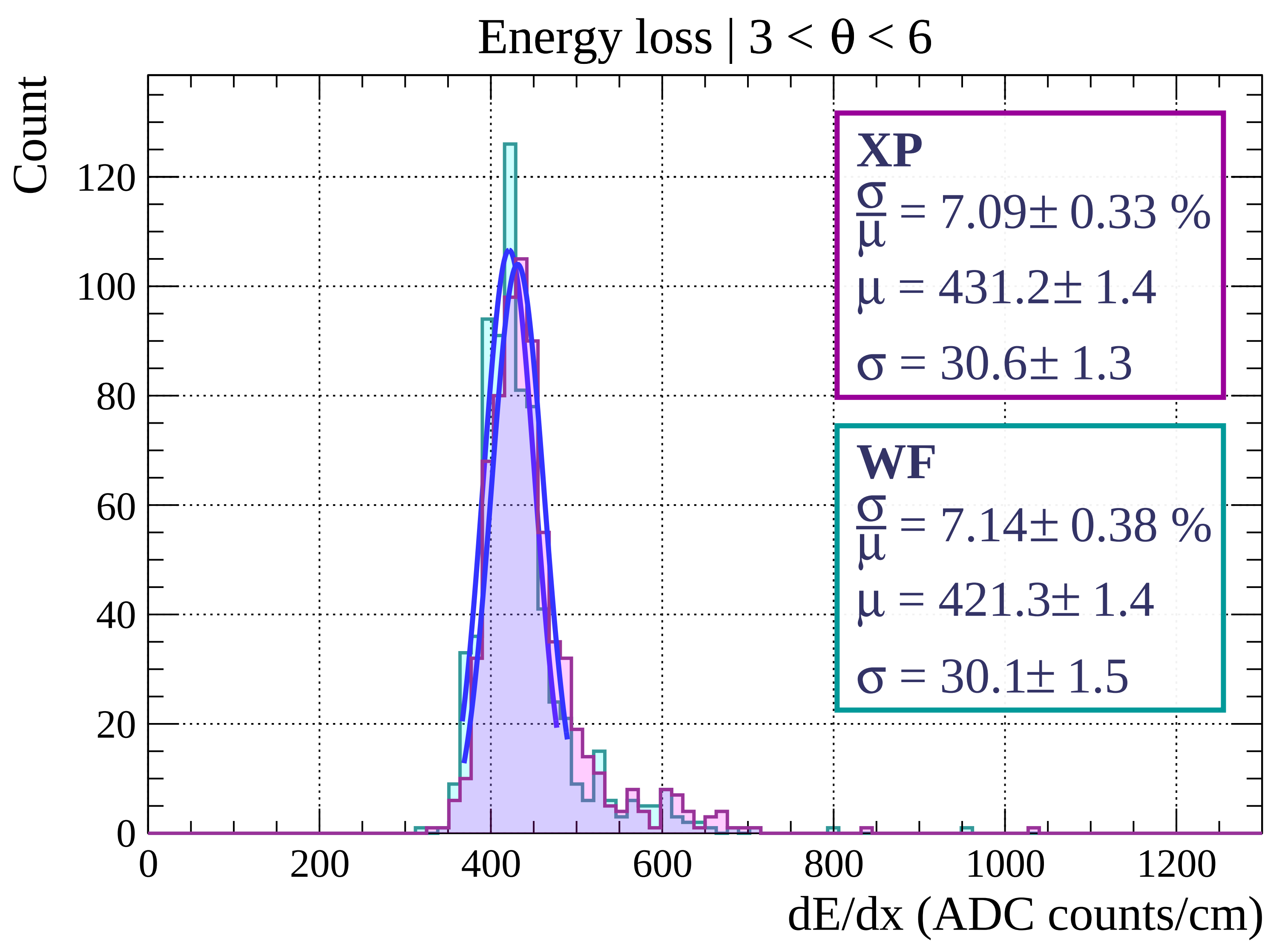
Energy loss | 84 < φ < 87
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.65± 0.44 %
μ = 431.8± 2.5
σ = 33.0± 1.7
WF
σ/μ = 8.13± 0.64 %
μ = 430.0± 2.6
σ = 35.0± 2.5



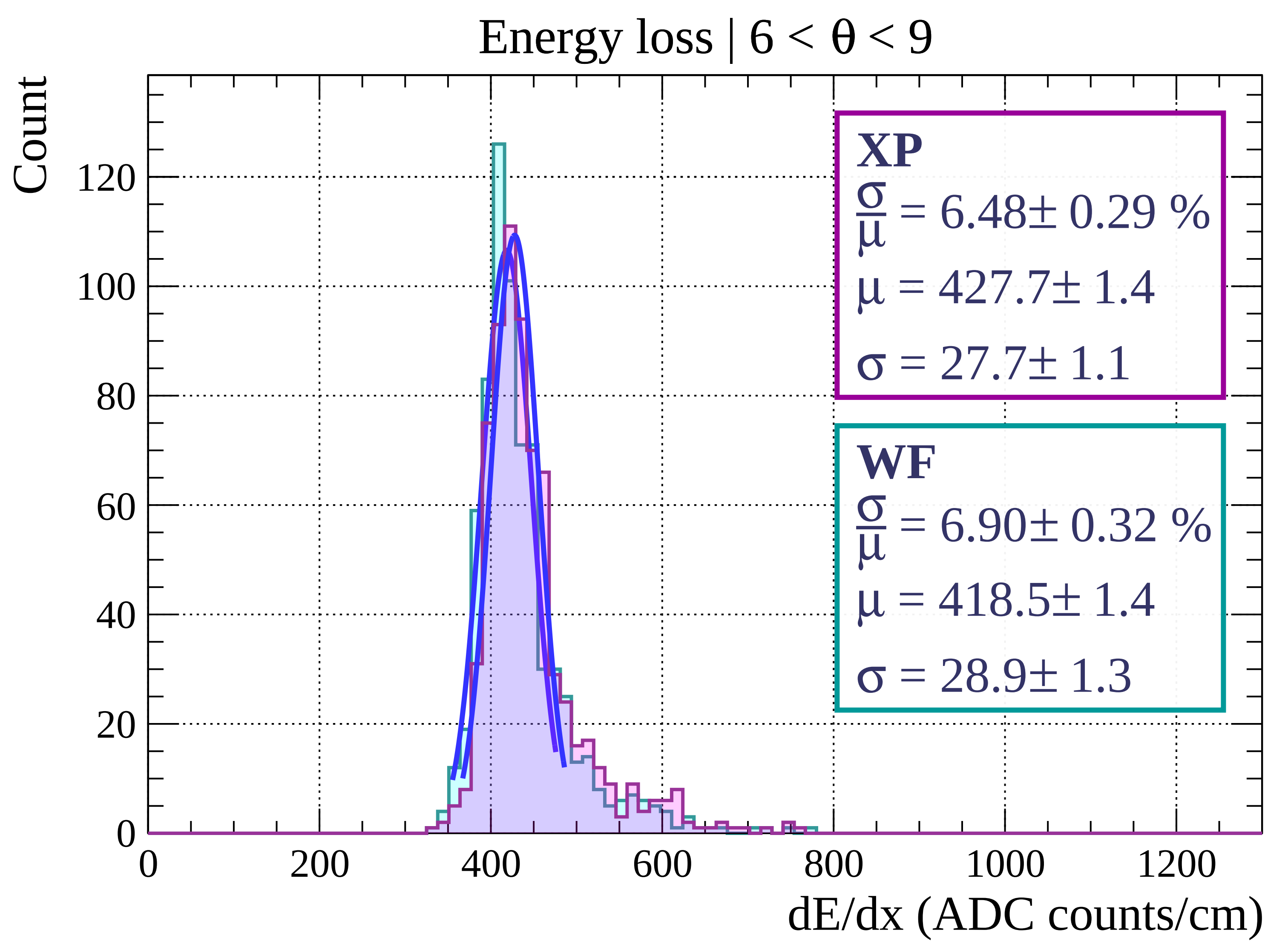
Energy loss | 87 < φ < 90
Count
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.28± 0.49 %
μ = 428.2± 2.2
σ = 31.2± 1.9
WF
σ/μ = 6.93± 0.43 %
μ = 428.9± 2.1
σ = 29.7± 1.7



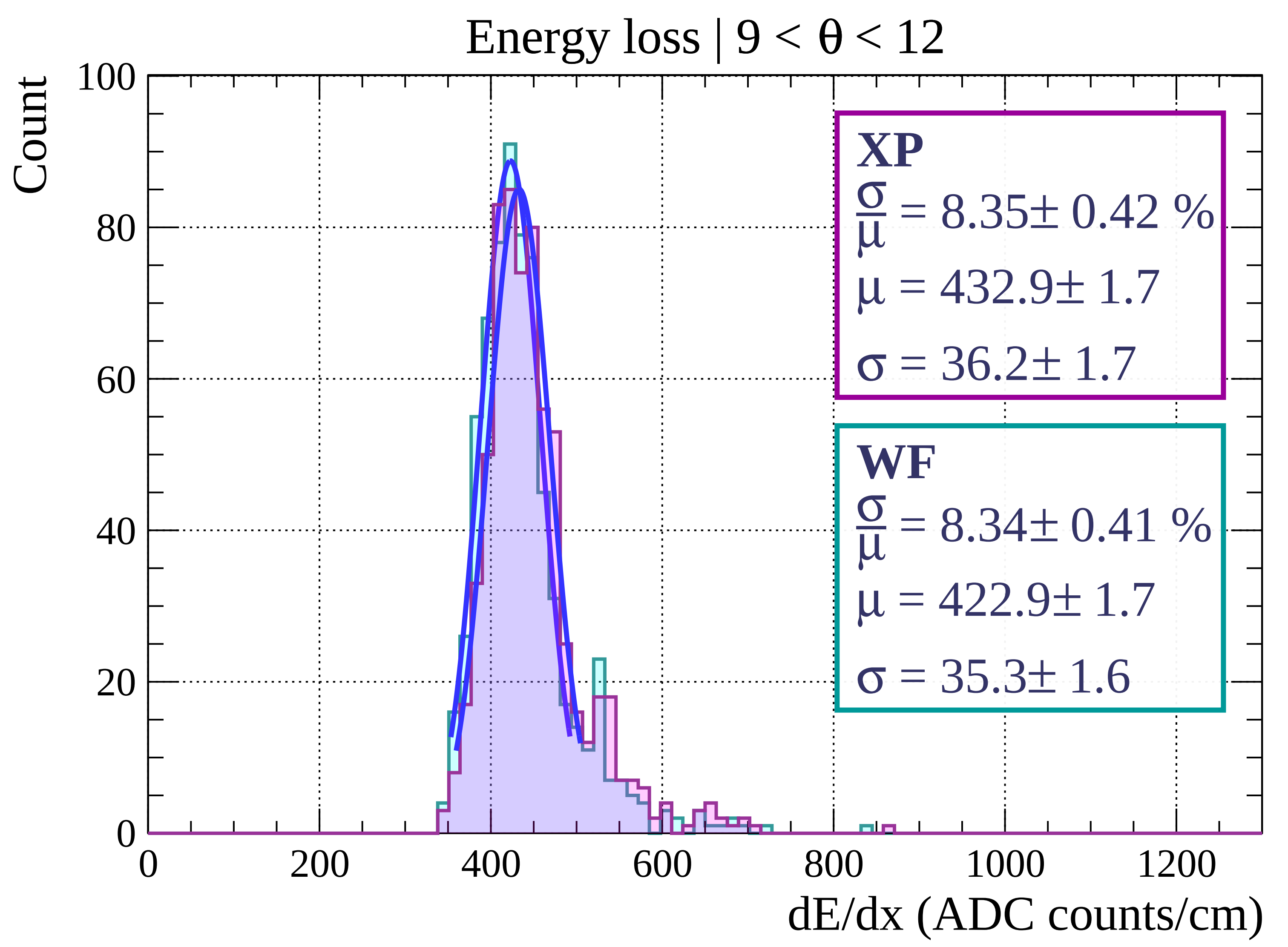
Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 8.48± 0.42 %
μ = 431.9± 1.6
σ = 36.6± 1.7
WF
σ/μ = 7.98± 0.34 %
μ = 423.5± 1.5
σ = 33.8± 1.3



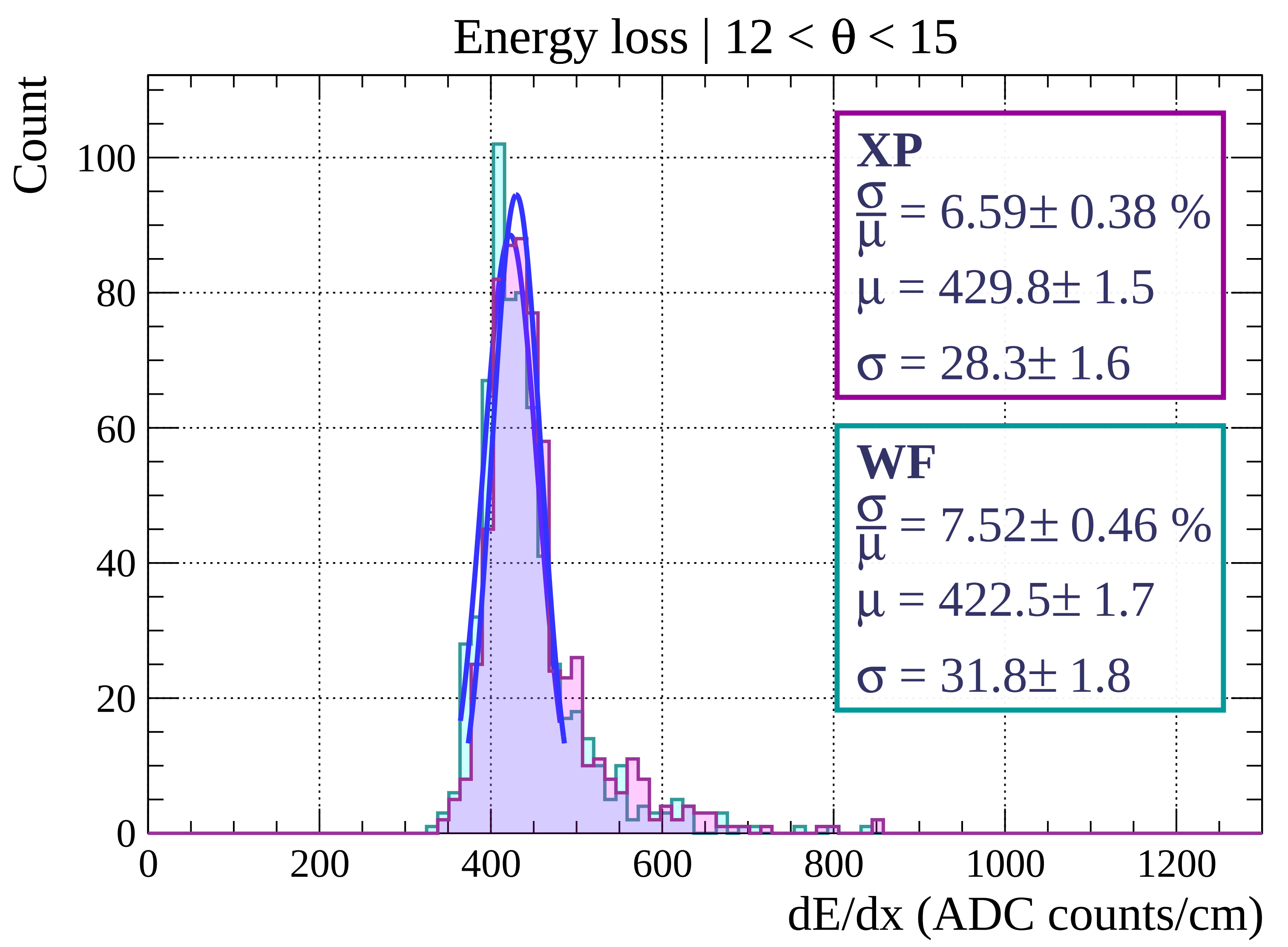
Energy loss | 3 < θ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 7.09± 0.33 %
μ = 431.2± 1.4
σ = 30.6± 1.3
WF
σ/μ = 7.14± 0.38 %
μ = 421.3± 1.4
σ = 30.1± 1.5



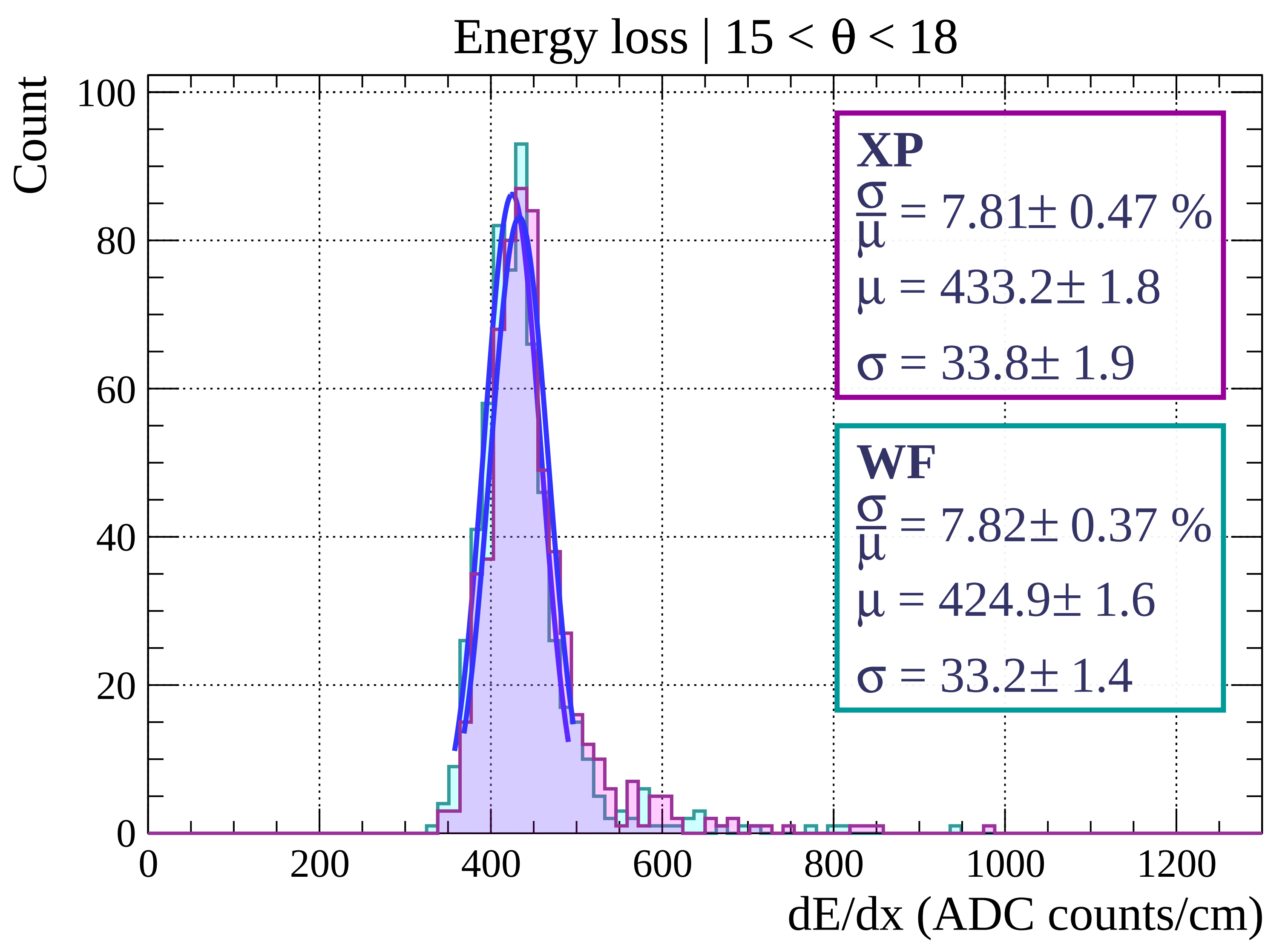
Energy loss | 6 < θ < 9
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.48± 0.29 %
μ = 427.7± 1.4
σ = 27.7± 1.1
WF
σ/μ = 6.90± 0.32 %
μ = 418.5± 1.4
σ = 28.9± 1.3



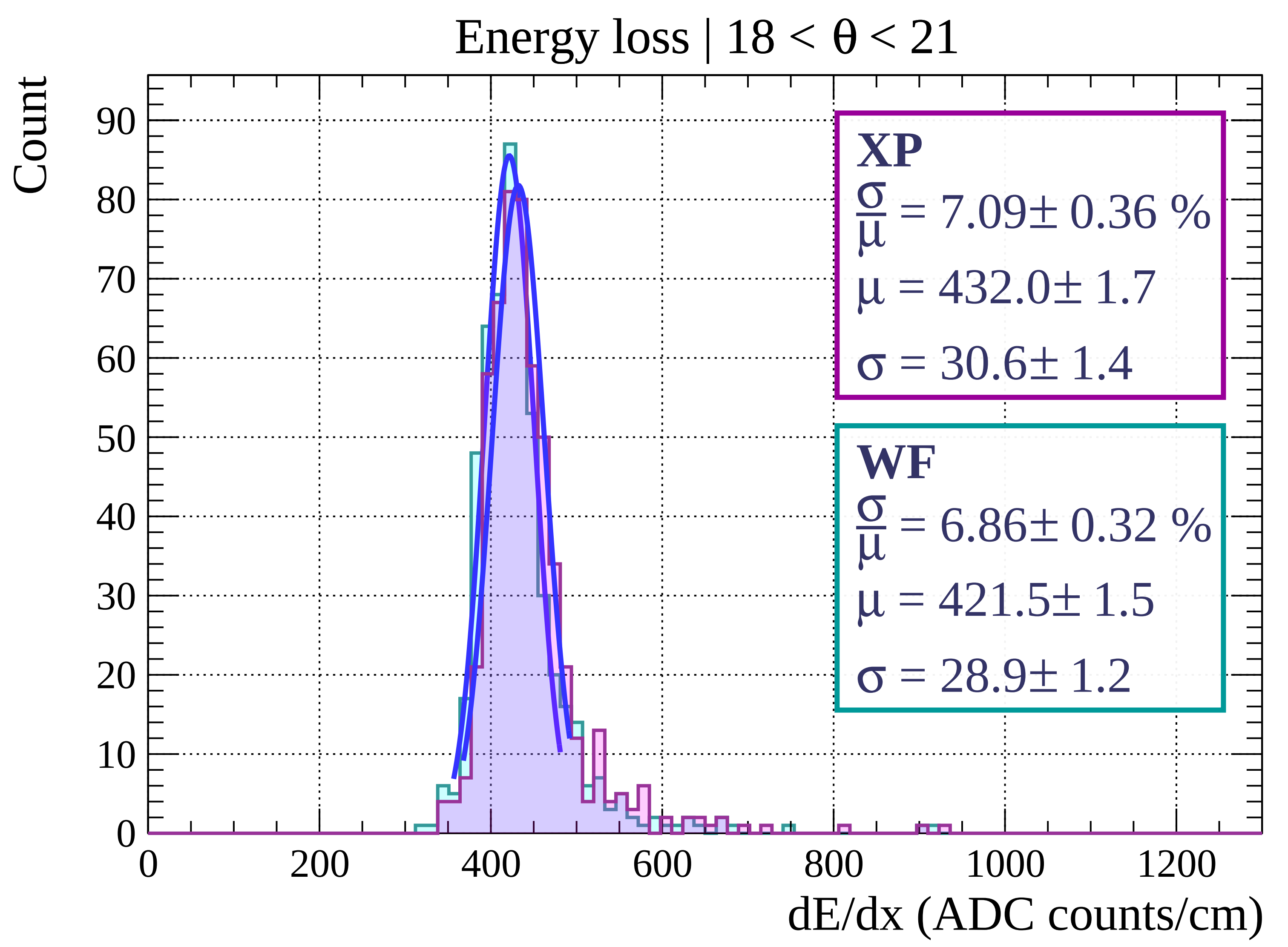
Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 8.35± 0.42 %
μ = 432.9± 1.7
σ = 36.2± 1.7
WF
σ/μ = 8.34± 0.41 %
μ = 422.9± 1.7
σ = 35.3± 1.6



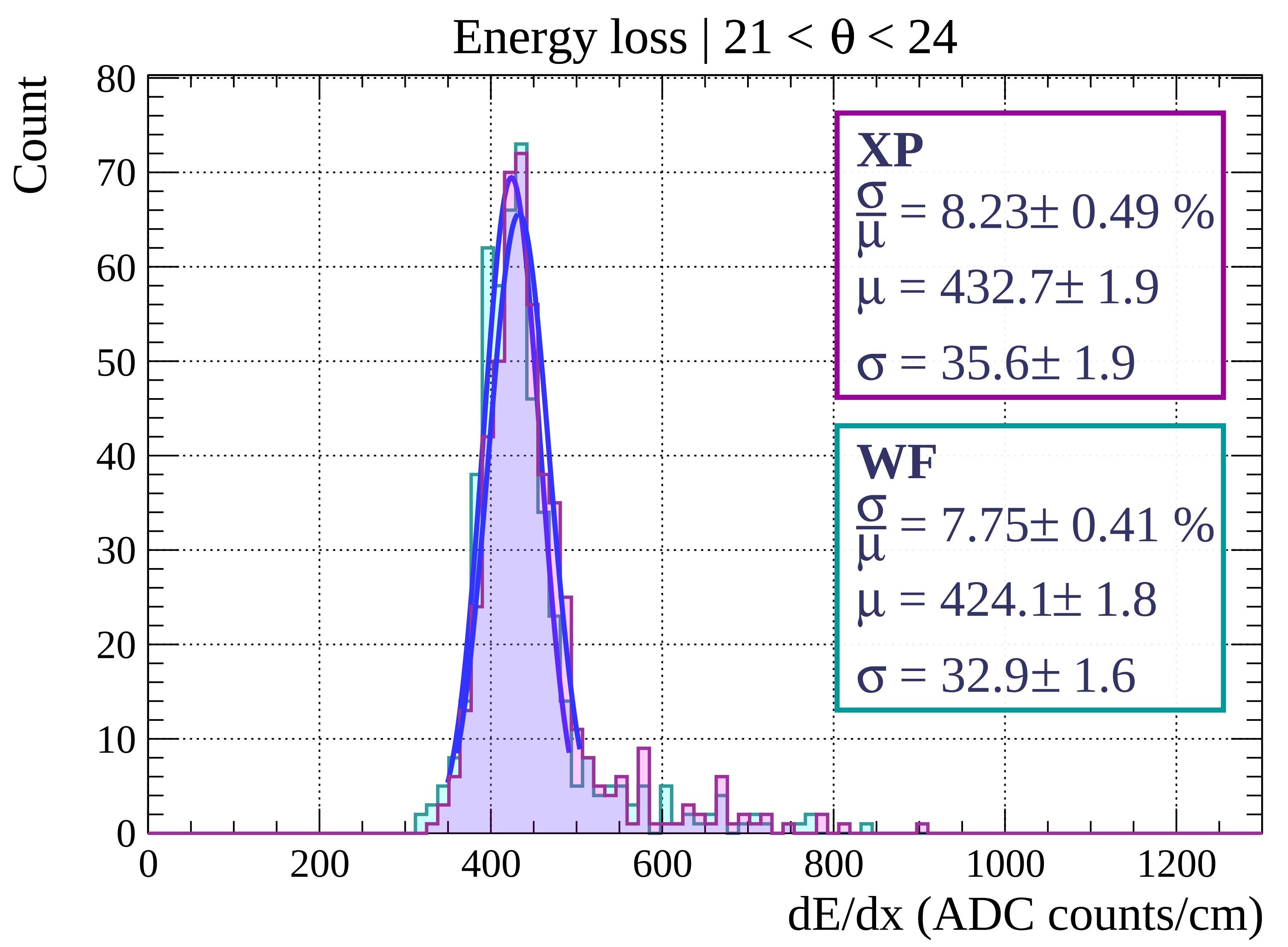
Energy loss | 12 < θ < 15
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 6.59± 0.38 %
μ = 429.8± 1.5
σ = 28.3± 1.6
WF
σ/μ = 7.52± 0.46 %
μ = 422.5± 1.7
σ = 31.8± 1.8



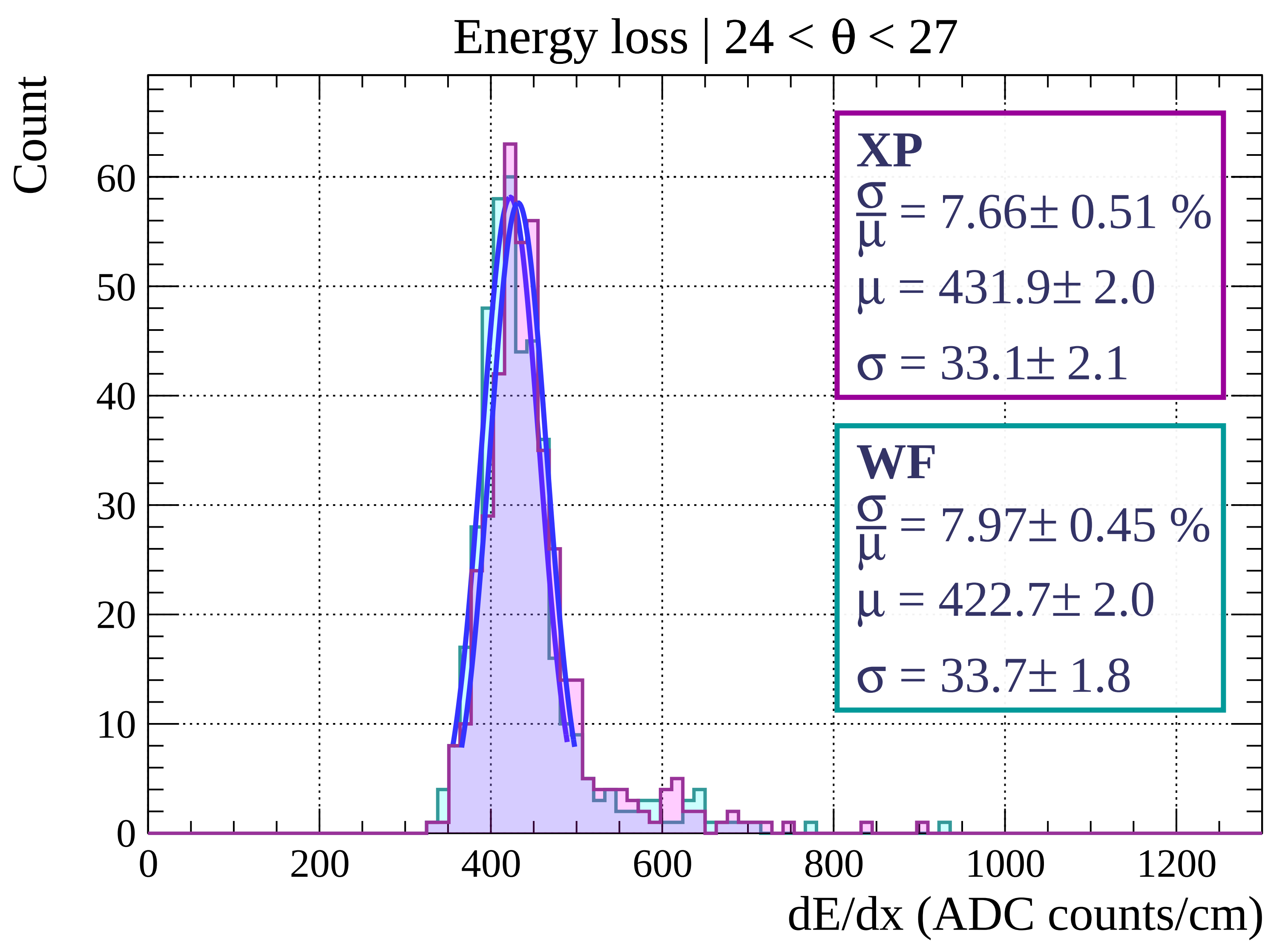
Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 7.81± 0.47 %
μ = 433.2± 1.8
σ = 33.8± 1.9
WF
σ/μ = 7.82± 0.37 %
μ = 424.9± 1.6
σ = 33.2± 1.4



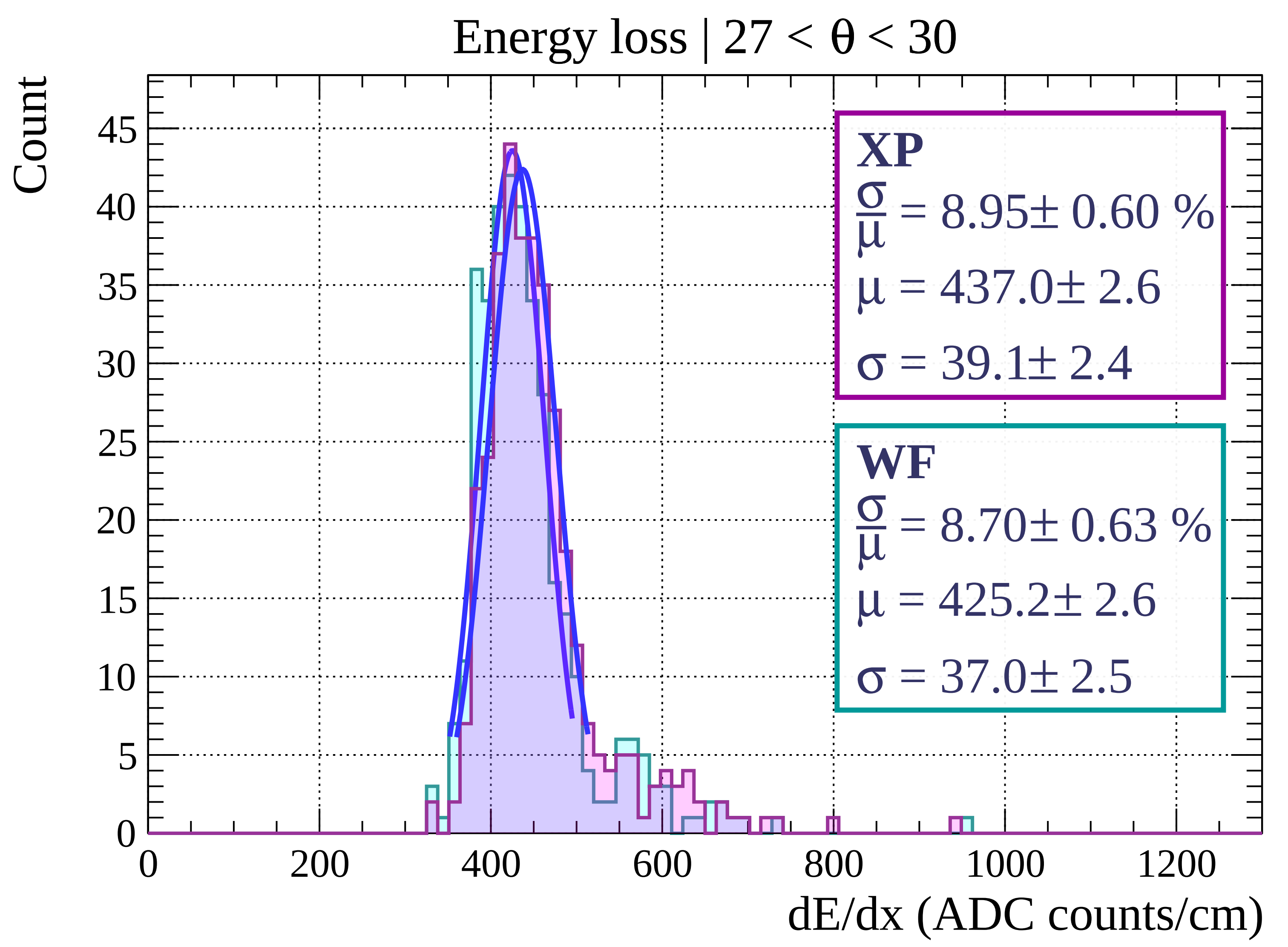
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
$\frac{\sigma}{\mu}$ = 7.09± 0.36 %
μ = 432.0± 1.7
σ = 30.6± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.86± 0.32 %
μ = 421.5± 1.5
σ = 28.9± 1.2



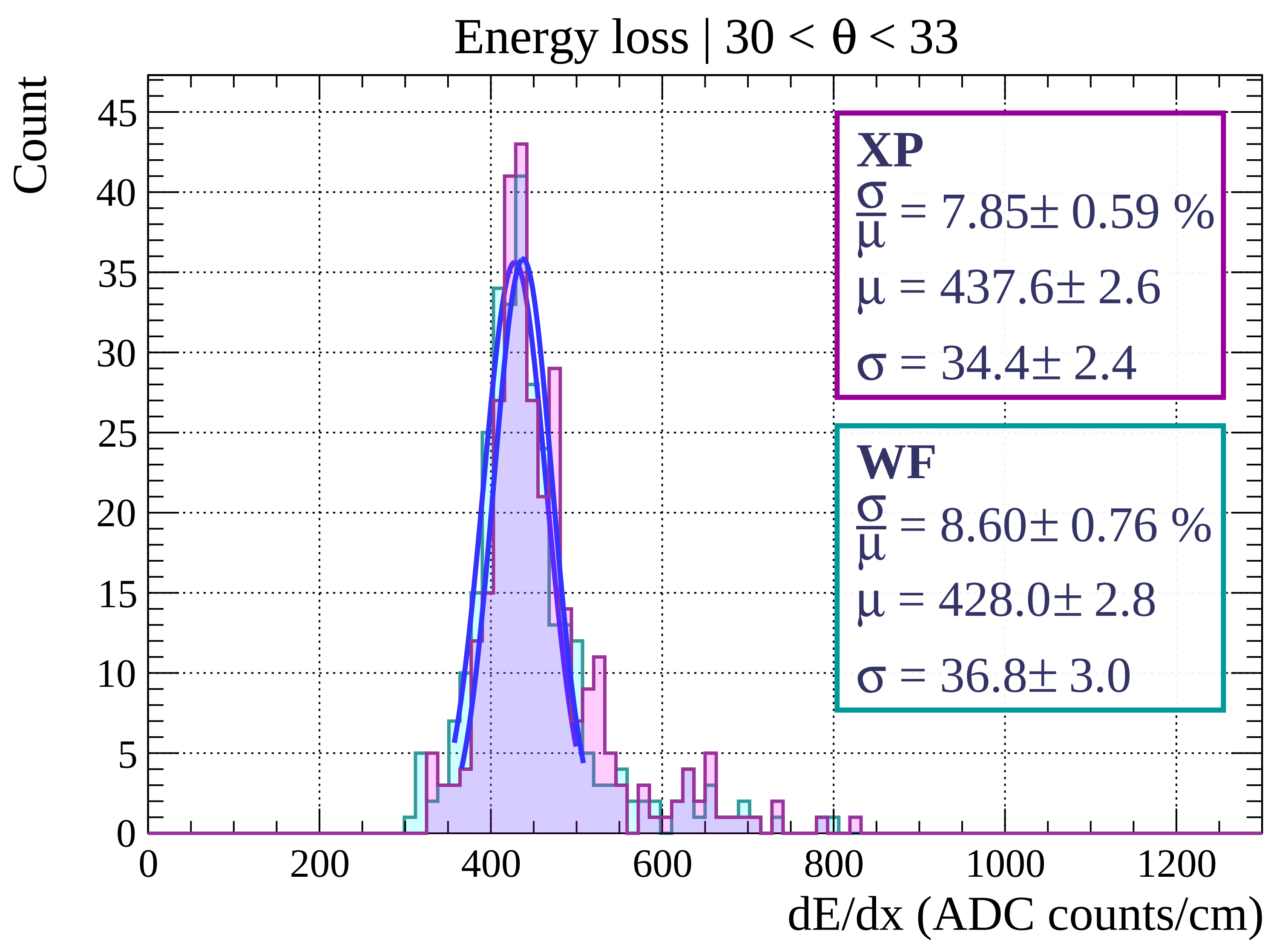
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
σ/μ = 8.23± 0.49 %
μ = 432.7± 1.9
σ = 35.6± 1.9
WF
σ/μ = 7.75± 0.41 %
μ = 424.1± 1.8
σ = 32.9± 1.6



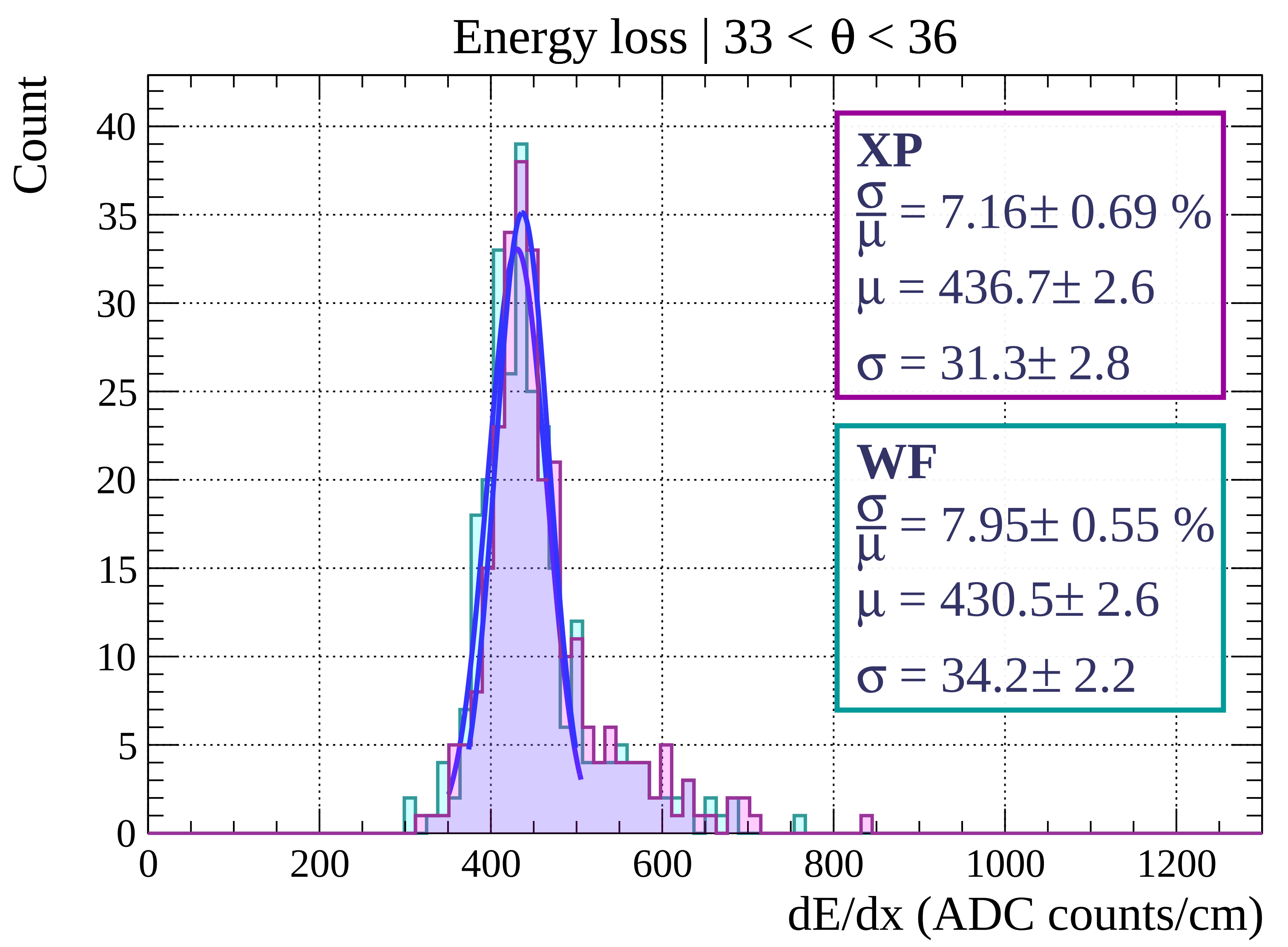
Energy loss | 24 < θ < 27
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 7.66± 0.51 %
μ = 431.9± 2.0
σ = 33.1± 2.1
WF
σ/μ = 7.97± 0.45 %
μ = 422.7± 2.0
σ = 33.7± 1.8



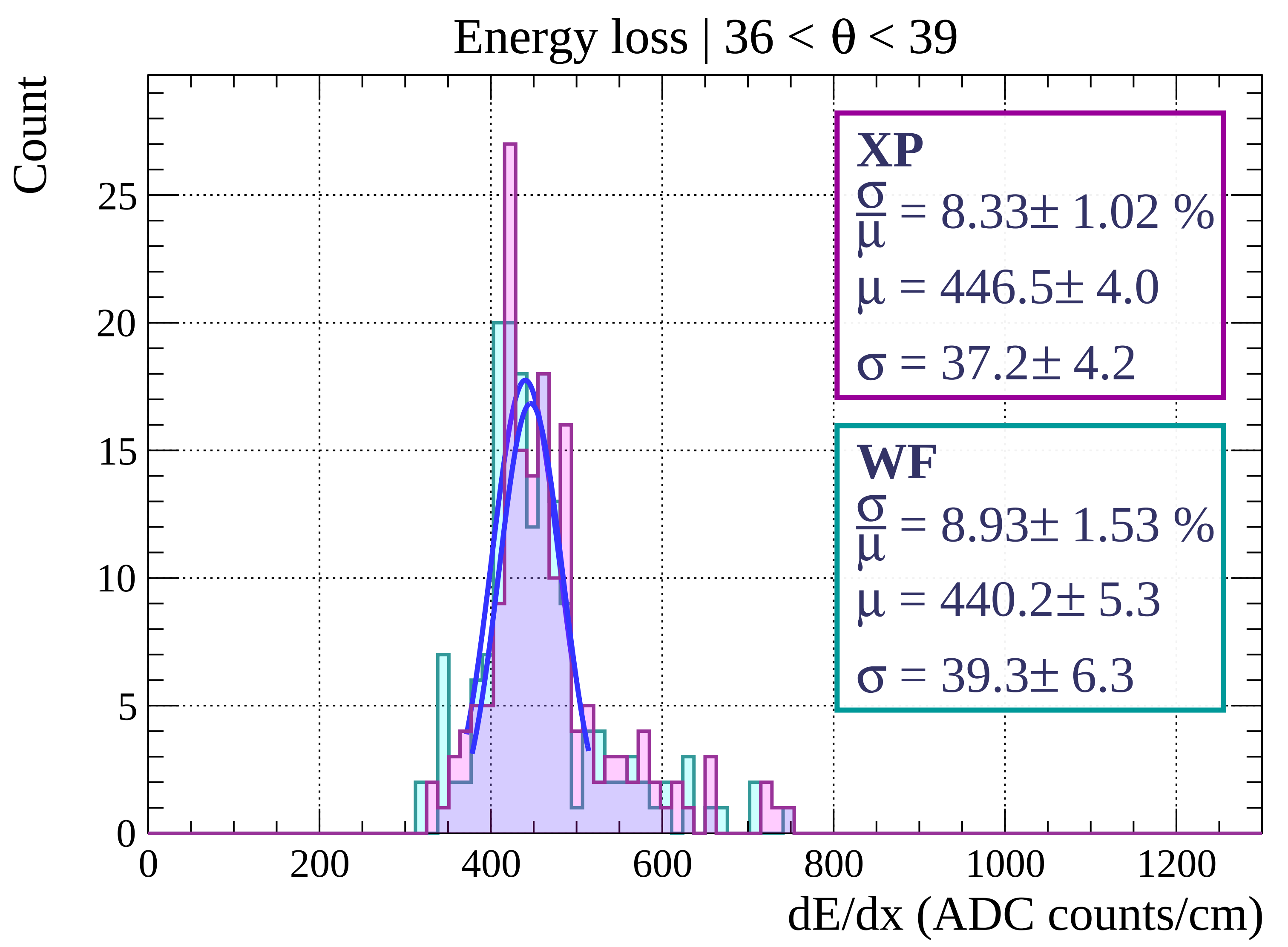
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 8.95± 0.60 %
μ = 437.0± 2.6
σ = 39.1± 2.4
WF
σ/μ = 8.70± 0.63 %
μ = 425.2± 2.6
σ = 37.0± 2.5



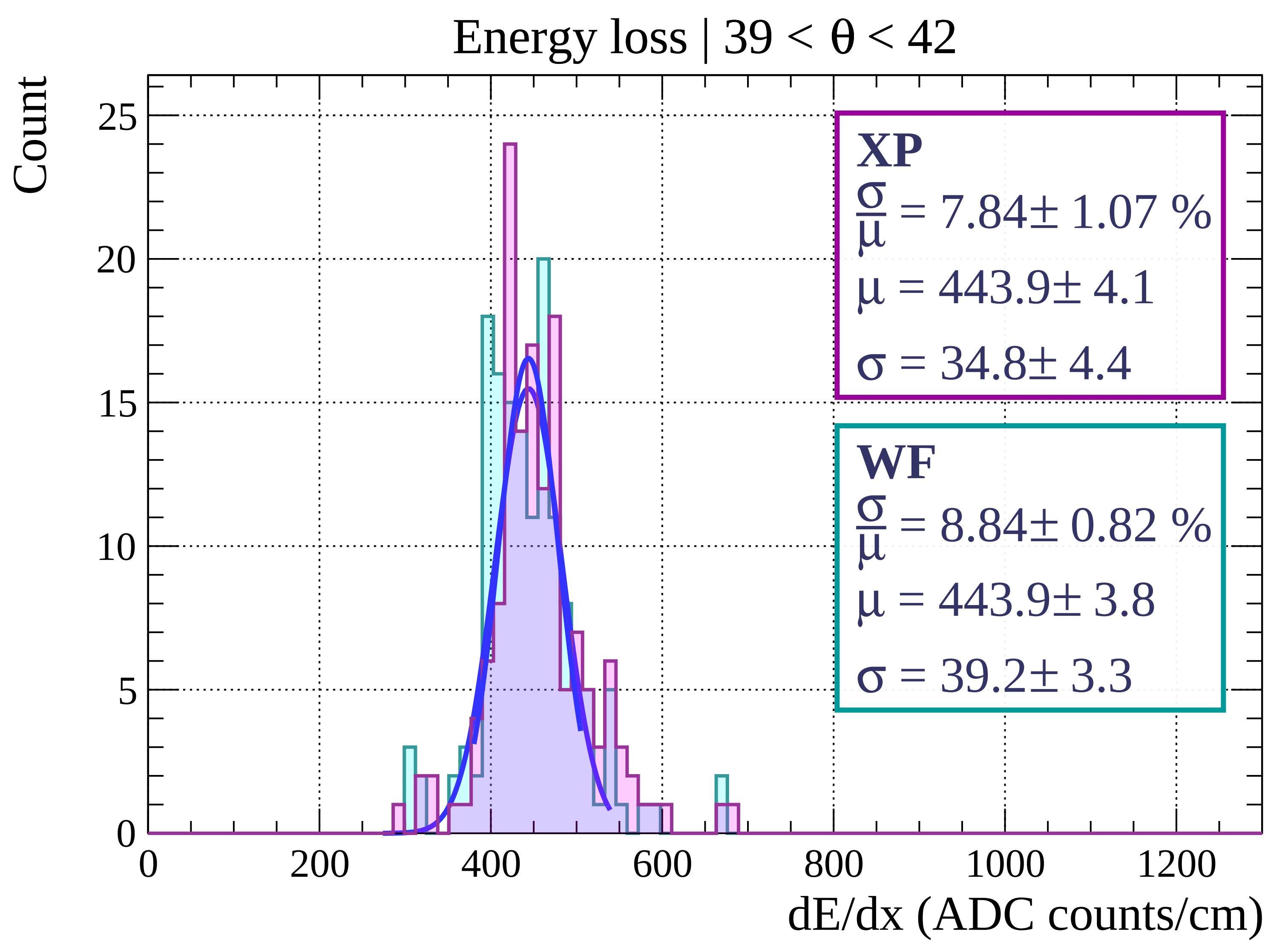
Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP
σ/μ = 7.85± 0.59 %
μ = 437.6± 2.6
σ = 34.4± 2.4
WF
σ/μ = 8.60± 0.76 %
μ = 428.0± 2.8
σ = 36.8± 3.0



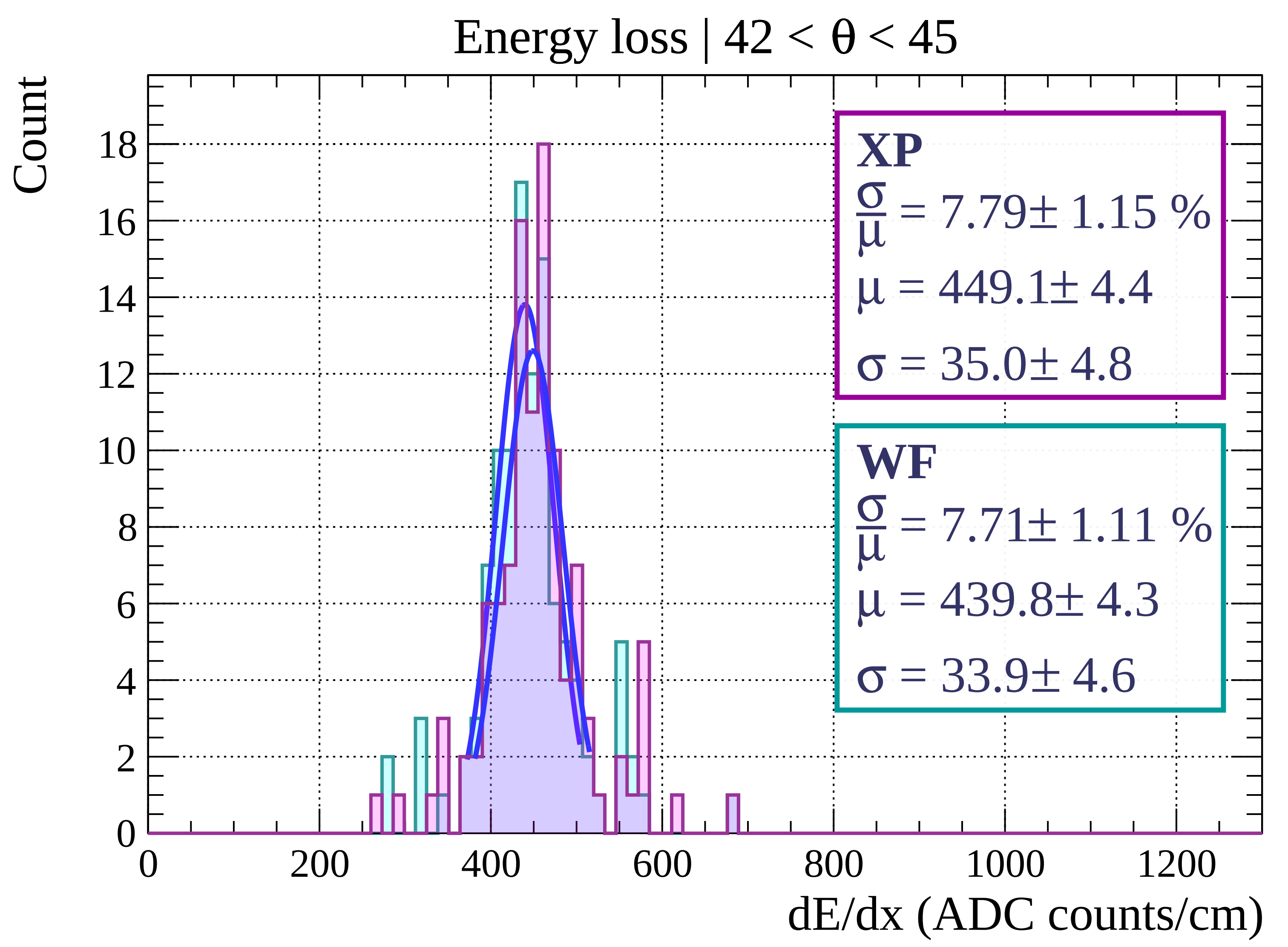
Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
σ/μ = 7.16± 0.69 %
μ = 436.7± 2.6
σ = 31.3± 2.8
WF
σ/μ = 7.95± 0.55 %
μ = 430.5± 2.6
σ = 34.2± 2.2

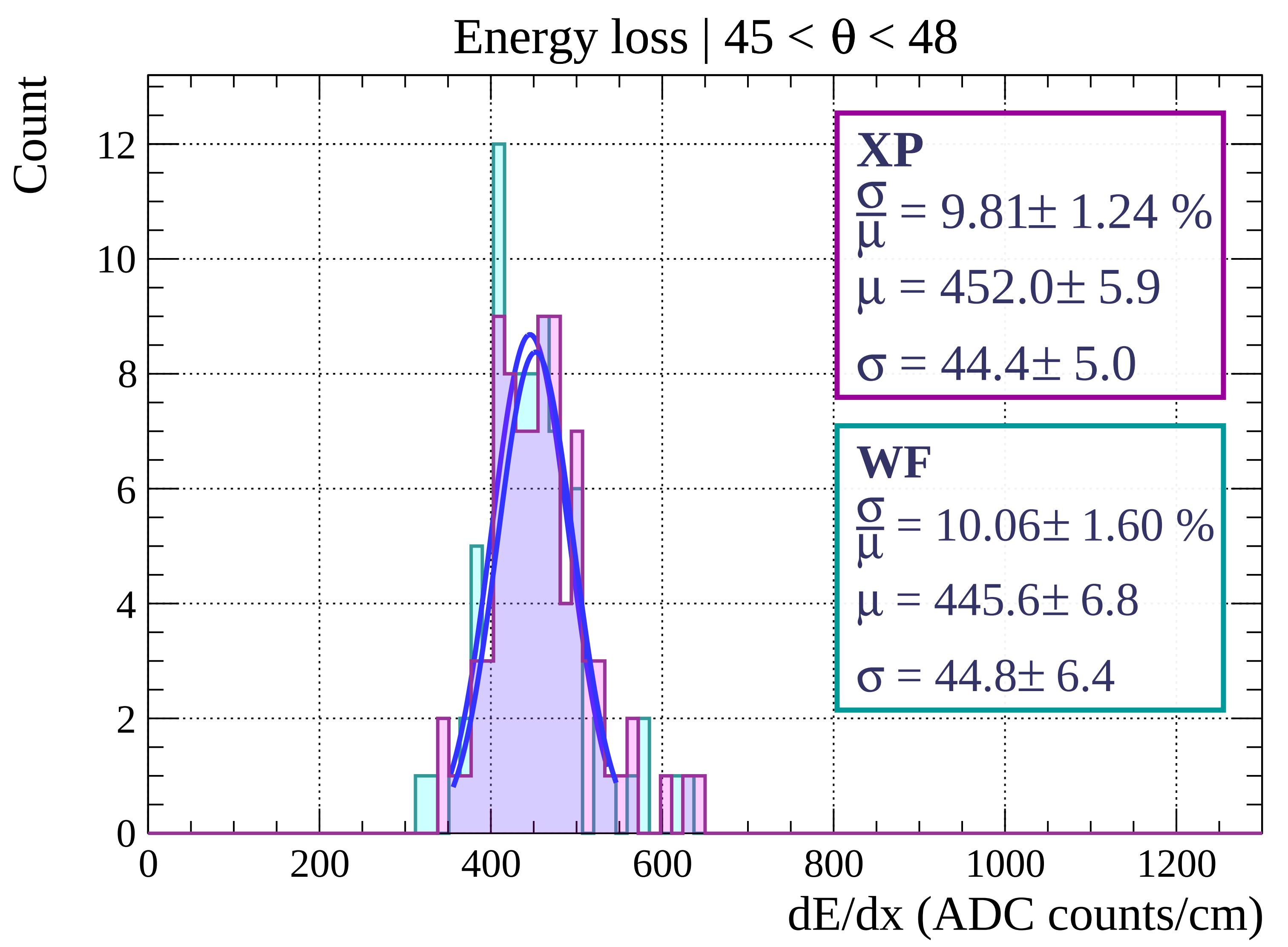


Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 8.33± 1.02 %
μ = 446.5± 4.0
σ = 37.2± 4.2
WF
σ/μ = 8.93± 1.53 %
μ = 440.2± 5.3
σ = 39.3± 6.3

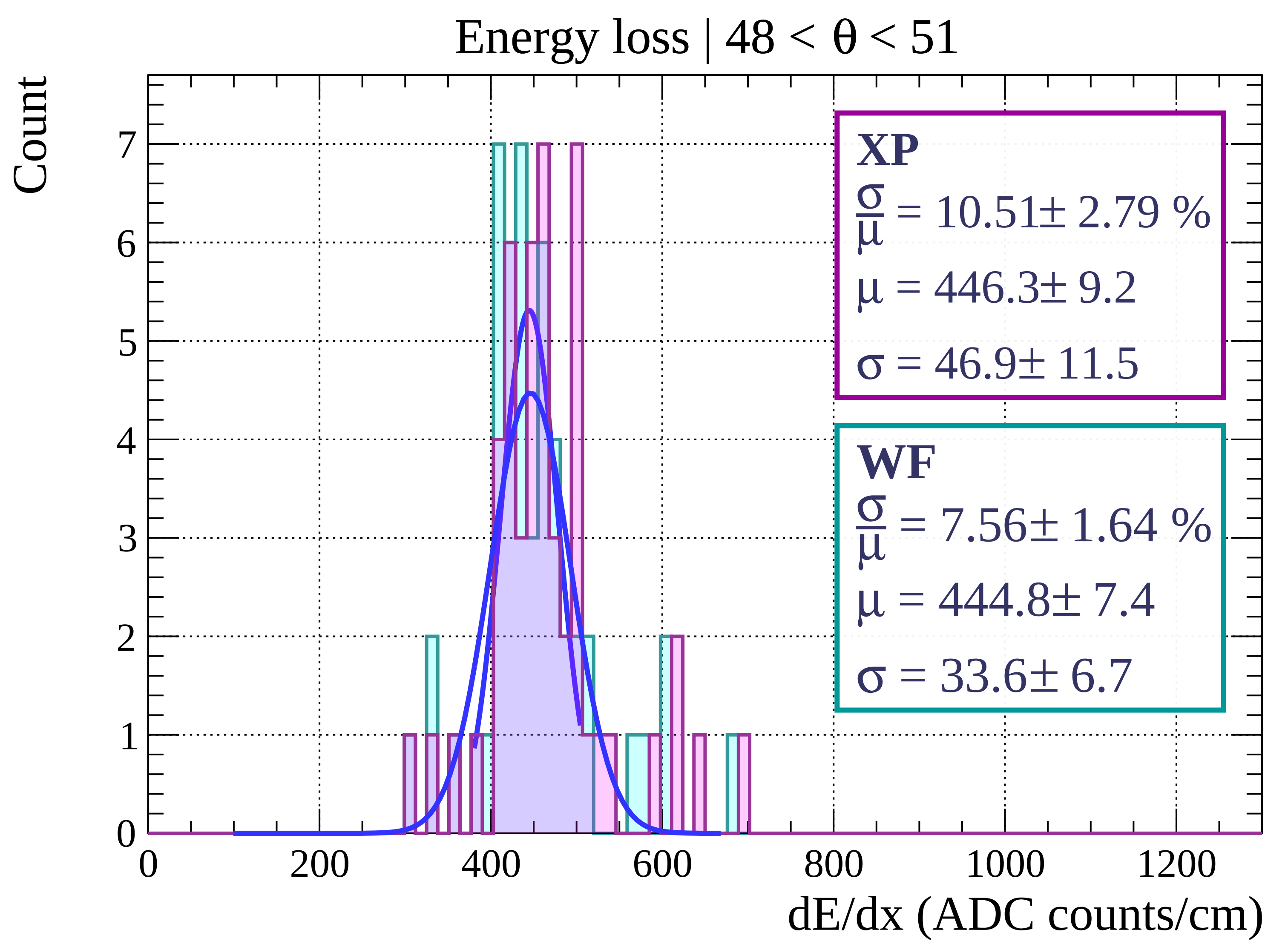


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 7.84± 1.07 %
μ = 443.9± 4.1
σ = 34.8± 4.4
WF
$\frac{\sigma}{\mu}$ = 8.84± 0.82 %
μ = 443.9± 3.8
σ = 39.2± 3.3

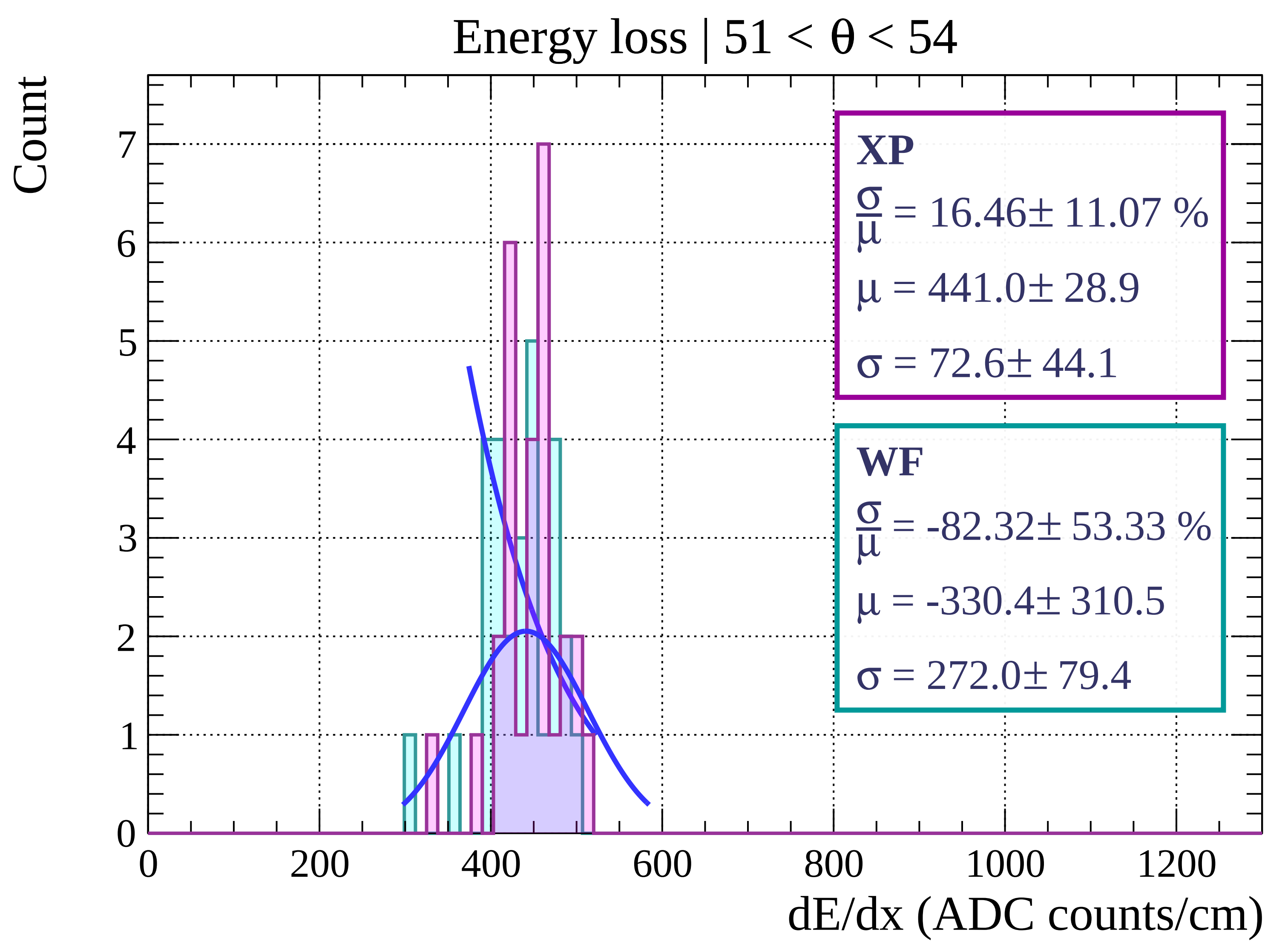

Energy loss | 42 < θ < 45
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 7.79± 1.15 %
μ = 449.1± 4.4
σ = 35.0± 4.8
WF
σ/μ = 7.71± 1.11 %
μ = 439.8± 4.3
σ = 33.9± 4.6



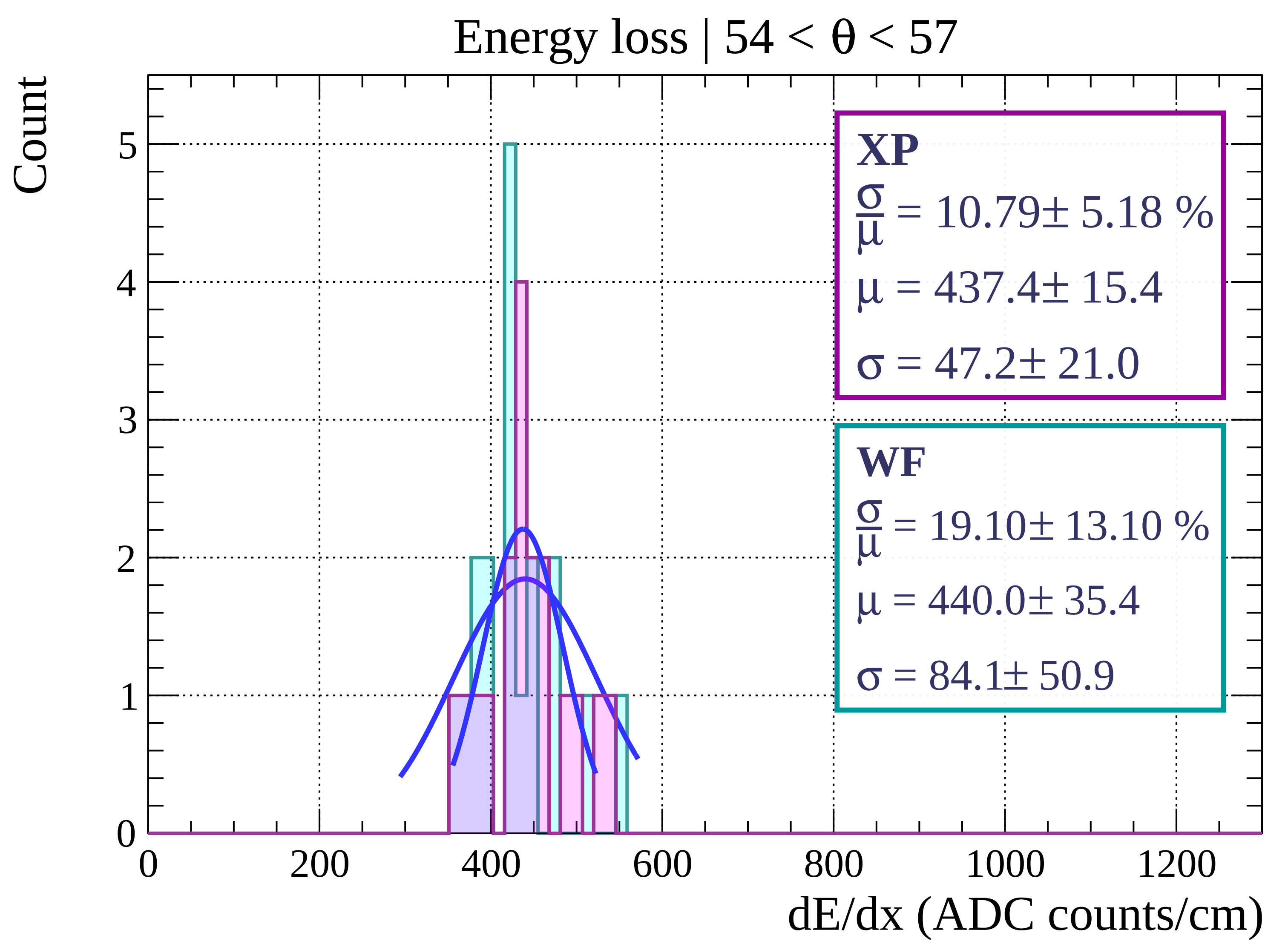
Energy loss | 45 < θ < 48
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.81± 1.24 %
μ = 452.0± 5.9
σ = 44.4± 5.0
WF
σ/μ = 10.06± 1.60 %
μ = 445.6± 6.8
σ = 44.8± 6.4



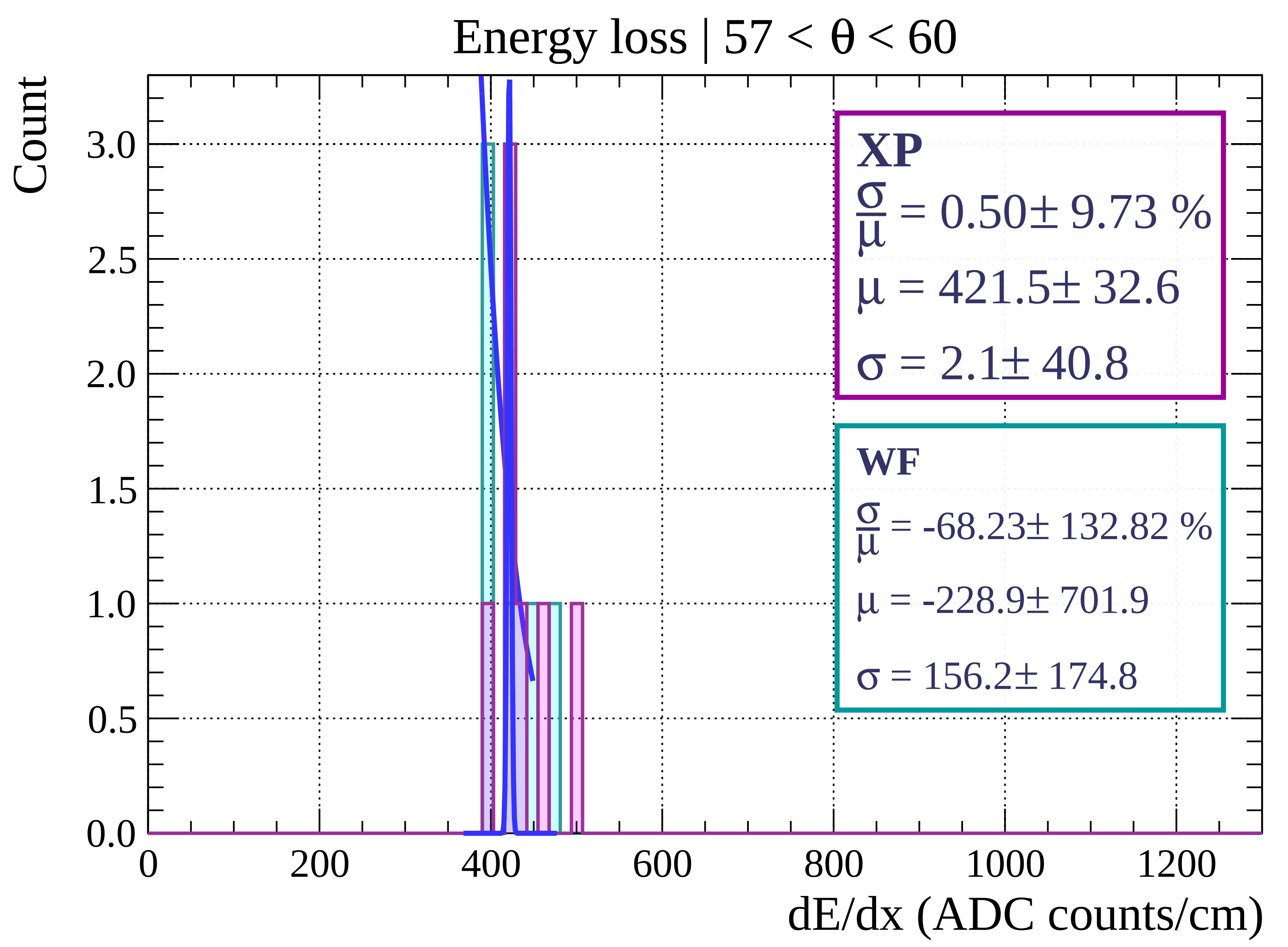
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 10.51± 2.79 %
μ = 446.3± 9.2
σ = 46.9± 11.5
WF
σ/μ = 7.56± 1.64 %
μ = 444.8± 7.4
σ = 33.6± 6.7



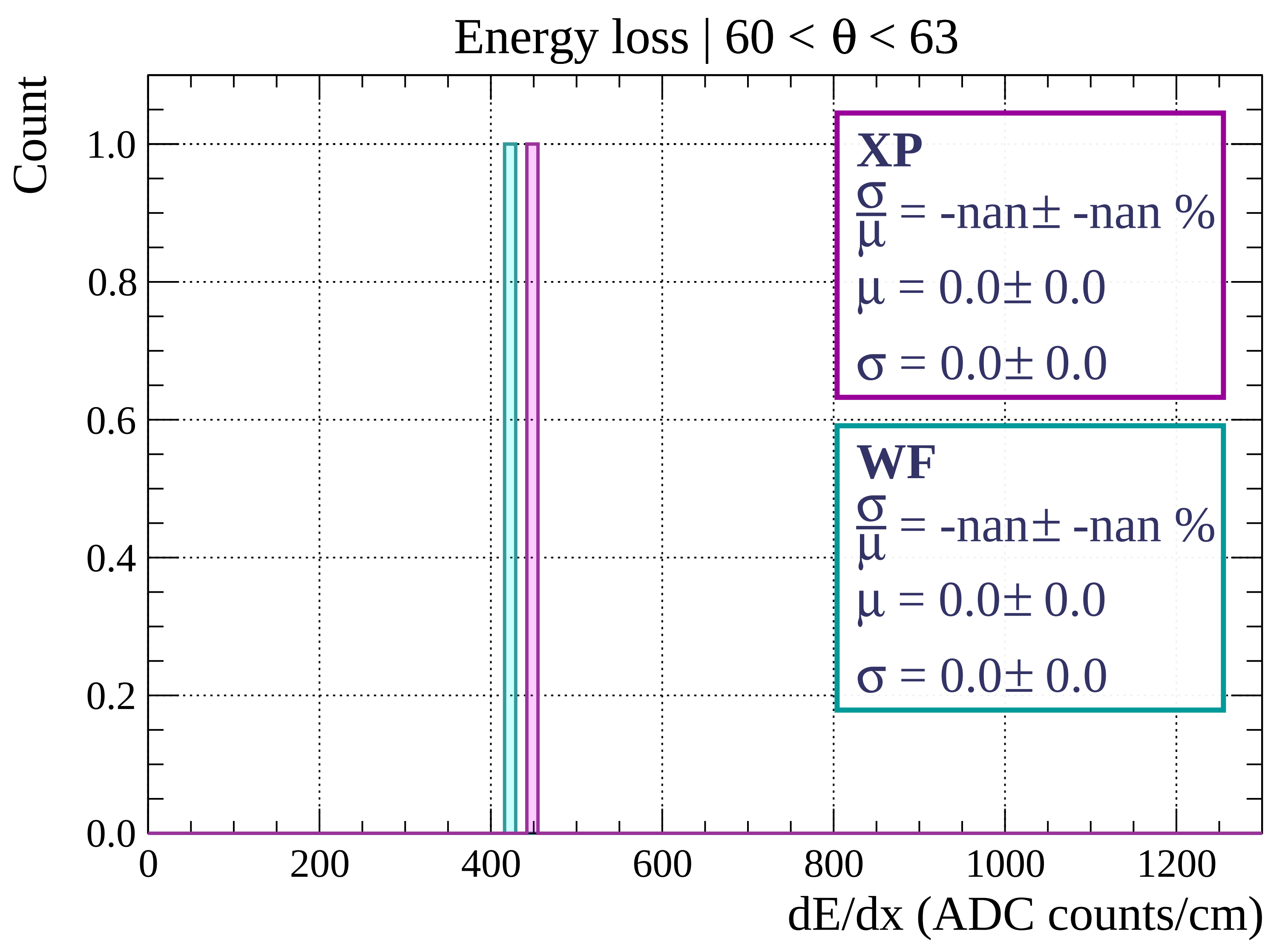
Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
XP
σ/μ = 16.46± 11.07 %
μ = 441.0± 28.9
σ = 72.6± 44.1
WF
σ/μ = -82.32± 53.33 %
μ = -330.4± 310.5
σ = 272.0± 79.4



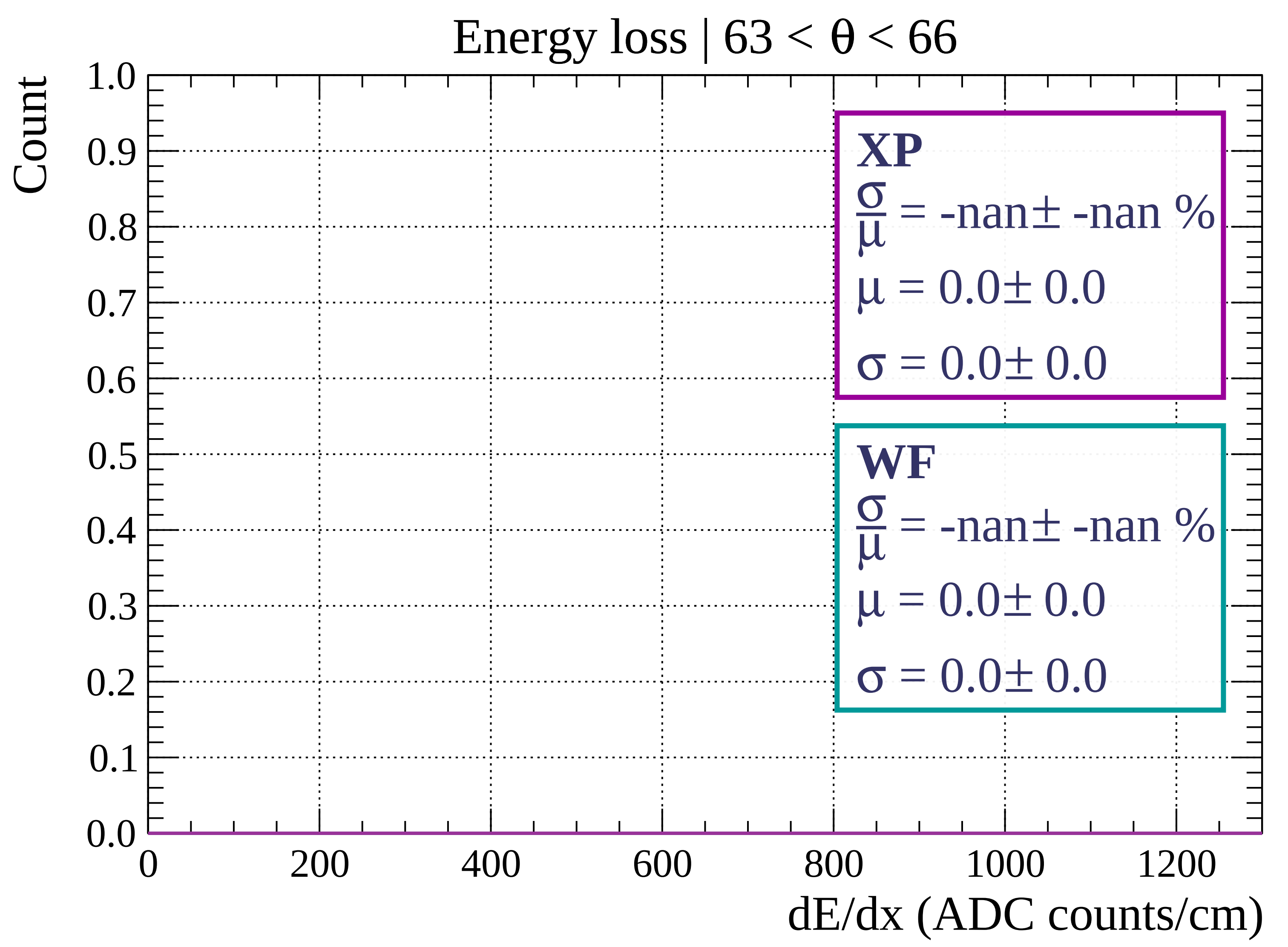
Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 10.79± 5.18 %
μ = 437.4± 15.4
σ = 47.2± 21.0
WF
σ/μ = 19.10± 13.10 %
μ = 440.0± 35.4
σ = 84.1± 50.9



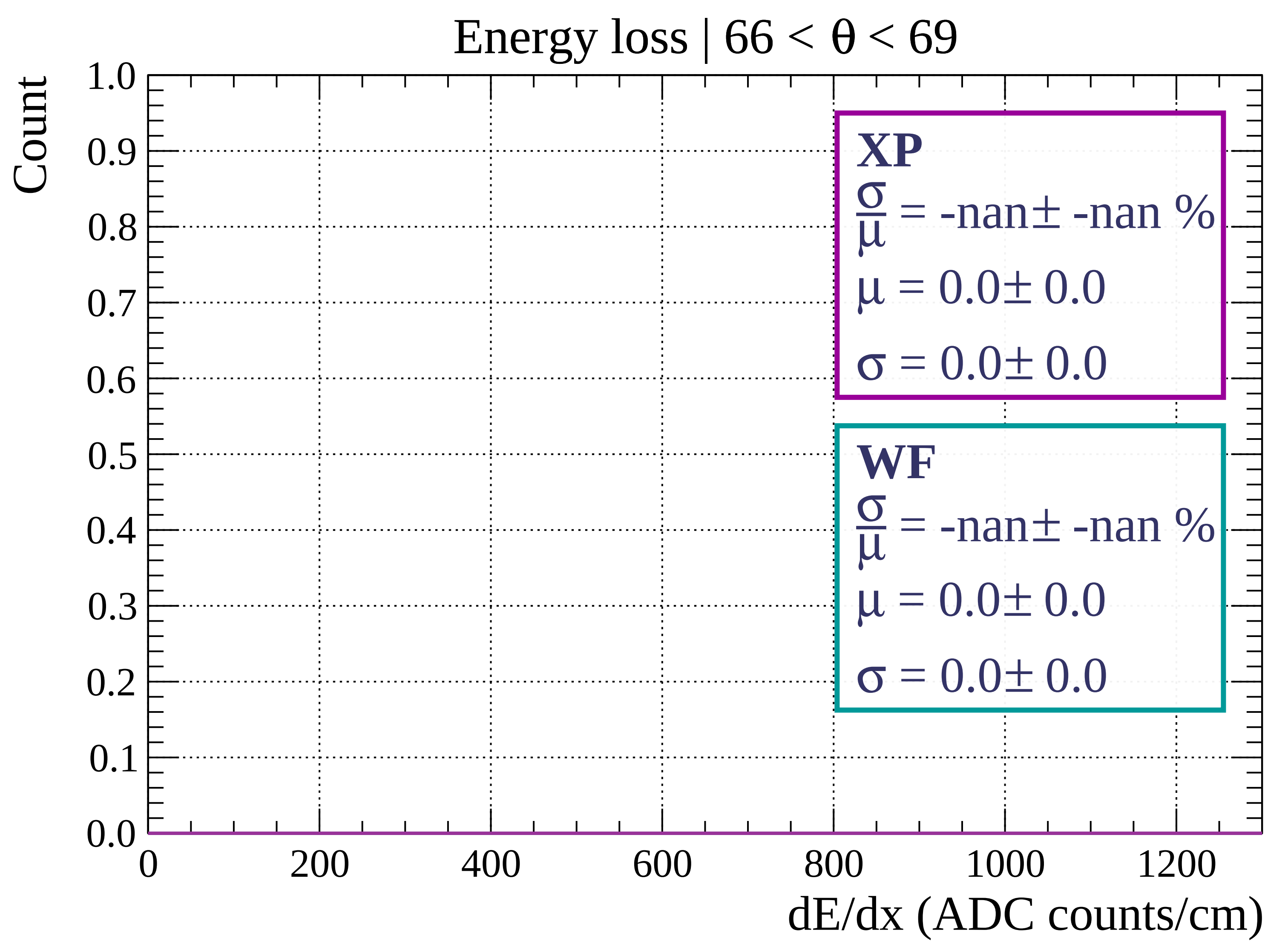
Energy loss | 57 < θ < 60
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.50± 9.73 %
μ = 421.5± 32.6
σ = 2.1± 40.8
WF
σ/μ = -68.23± 132.82 %
μ = -228.9± 701.9
σ = 156.2± 174.8



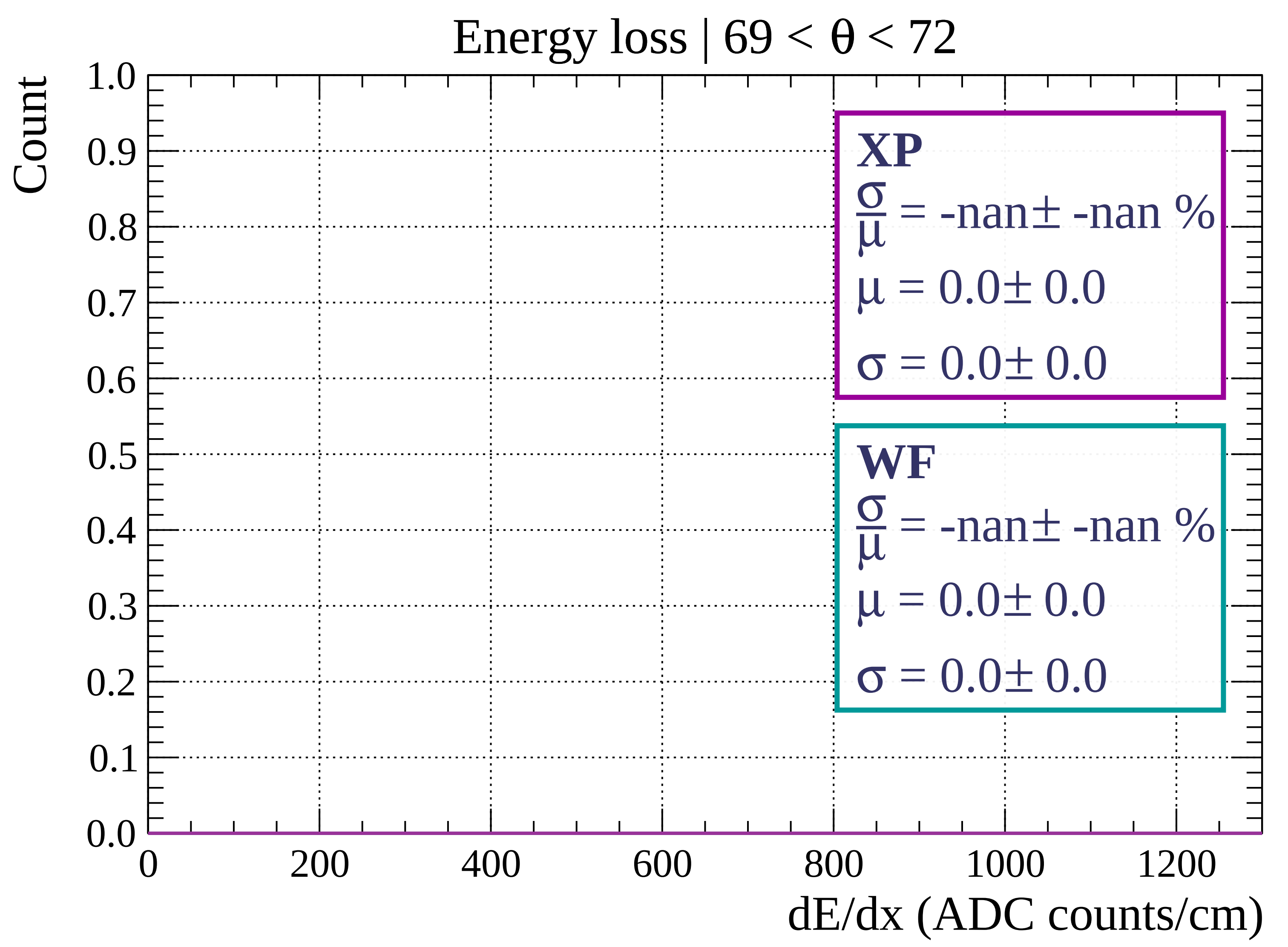
Energy loss | 60 < θ < 63
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



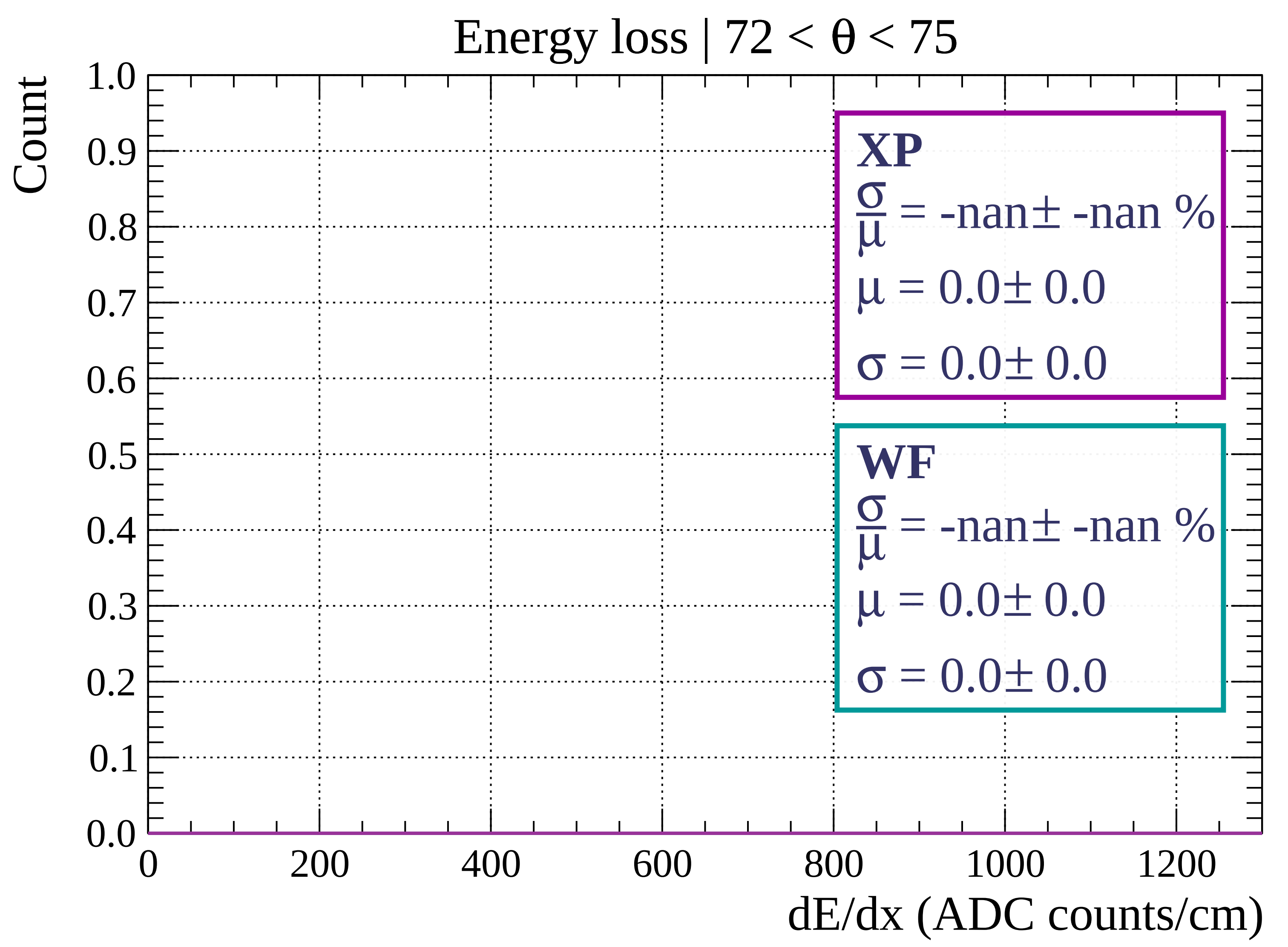
Energy loss | 63 < θ < 66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



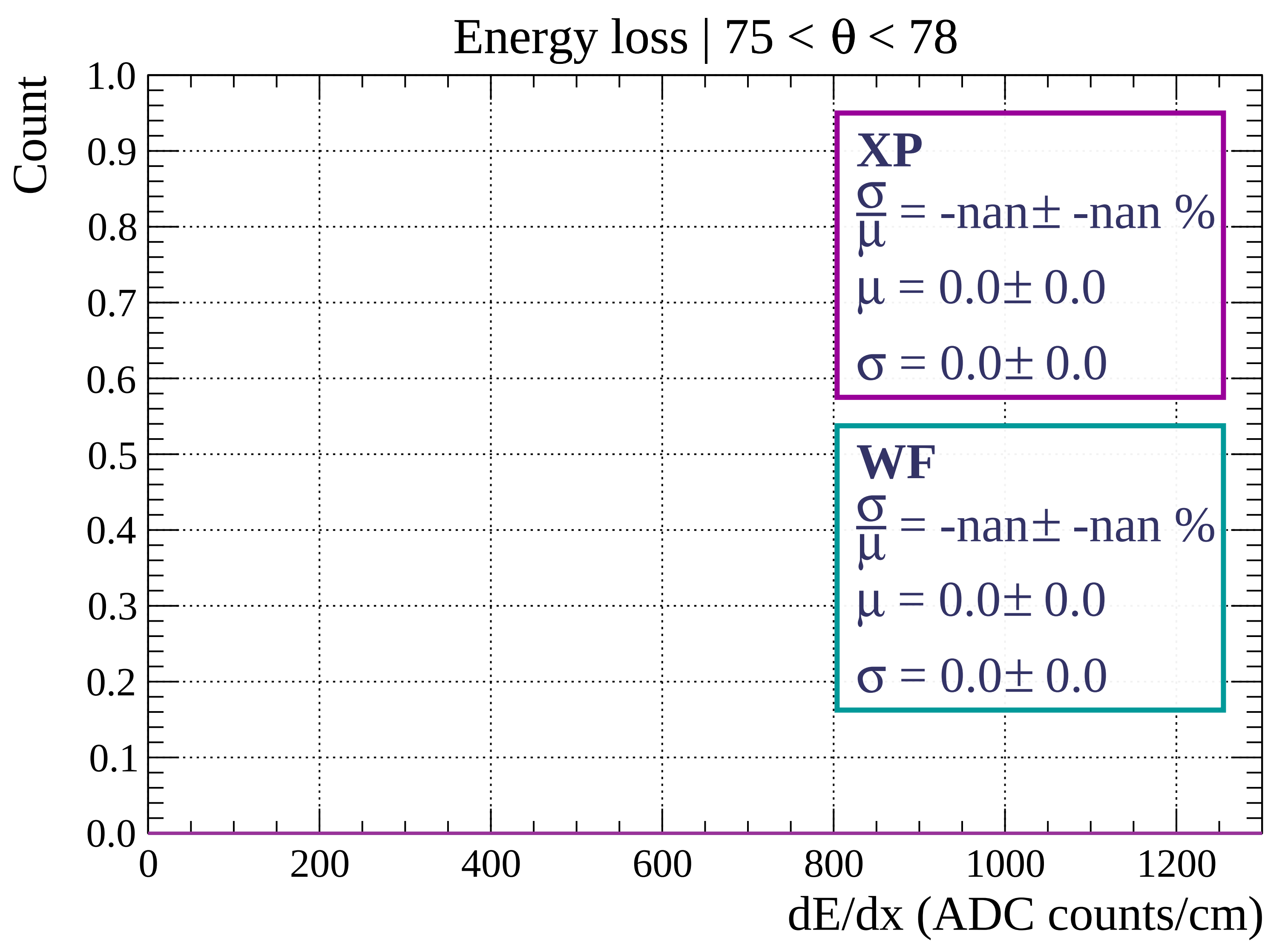
Energy loss | 66 < θ < 69
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



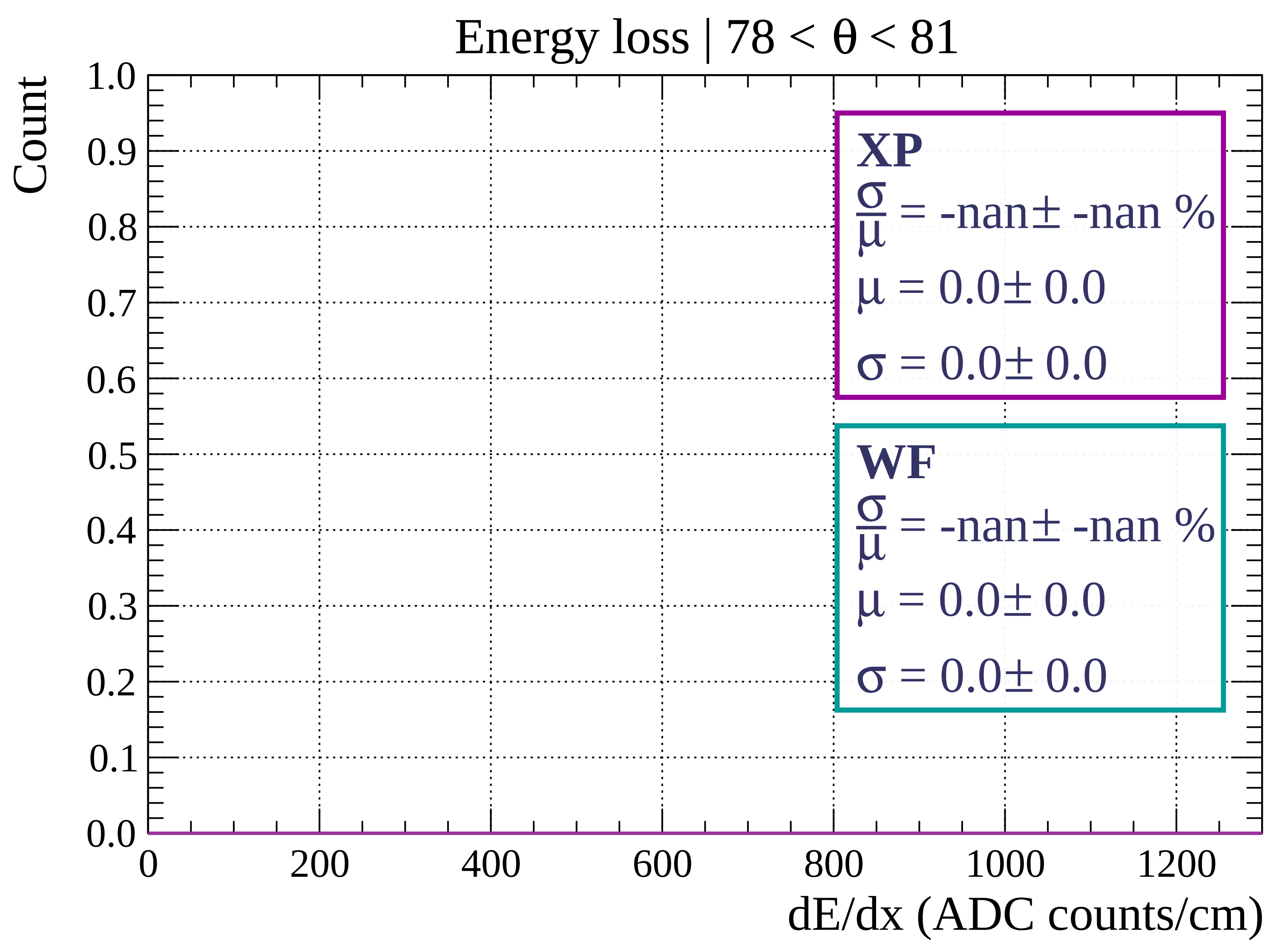
Energy loss | 69 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



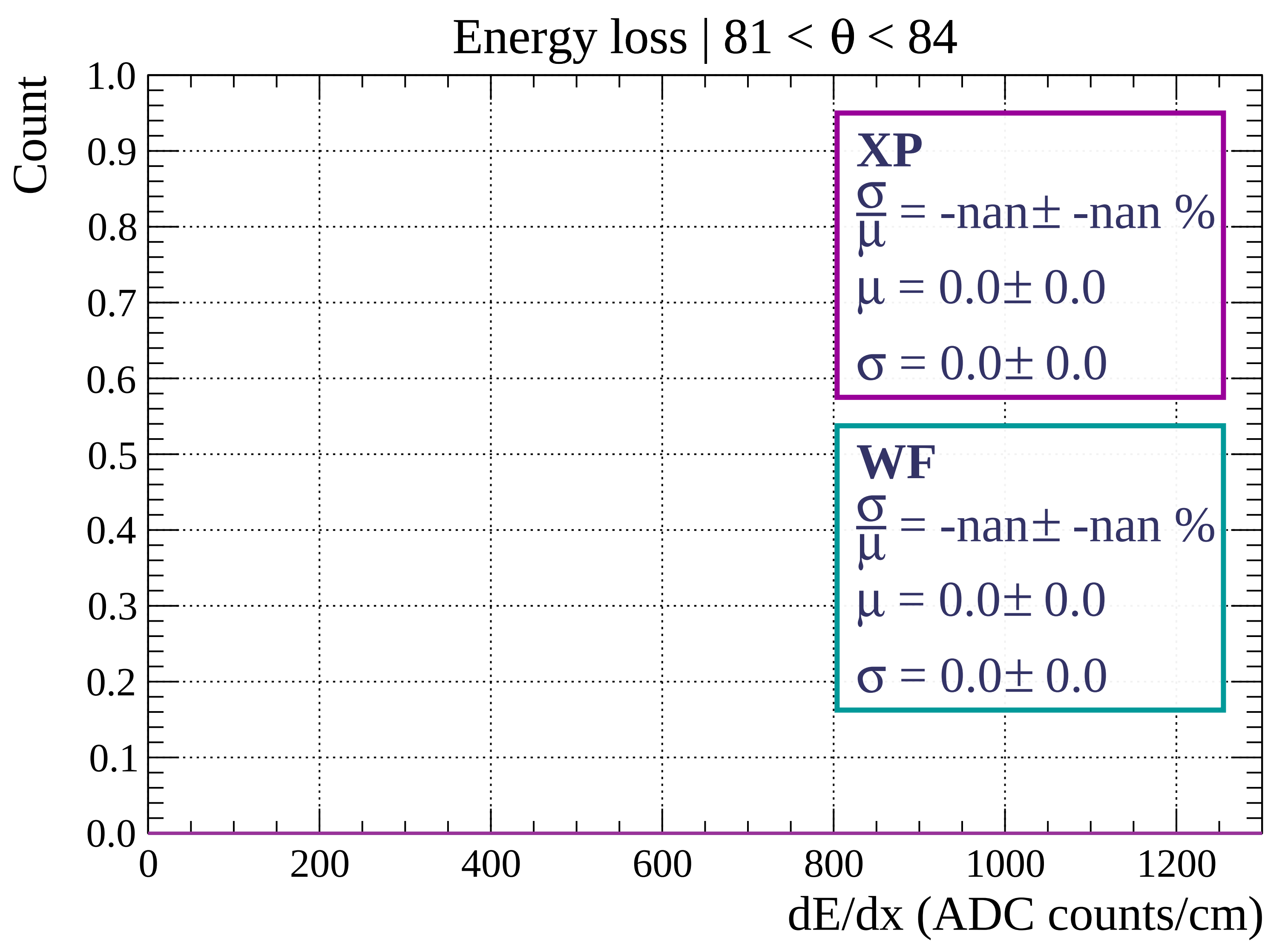
Energy loss | 72 < θ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



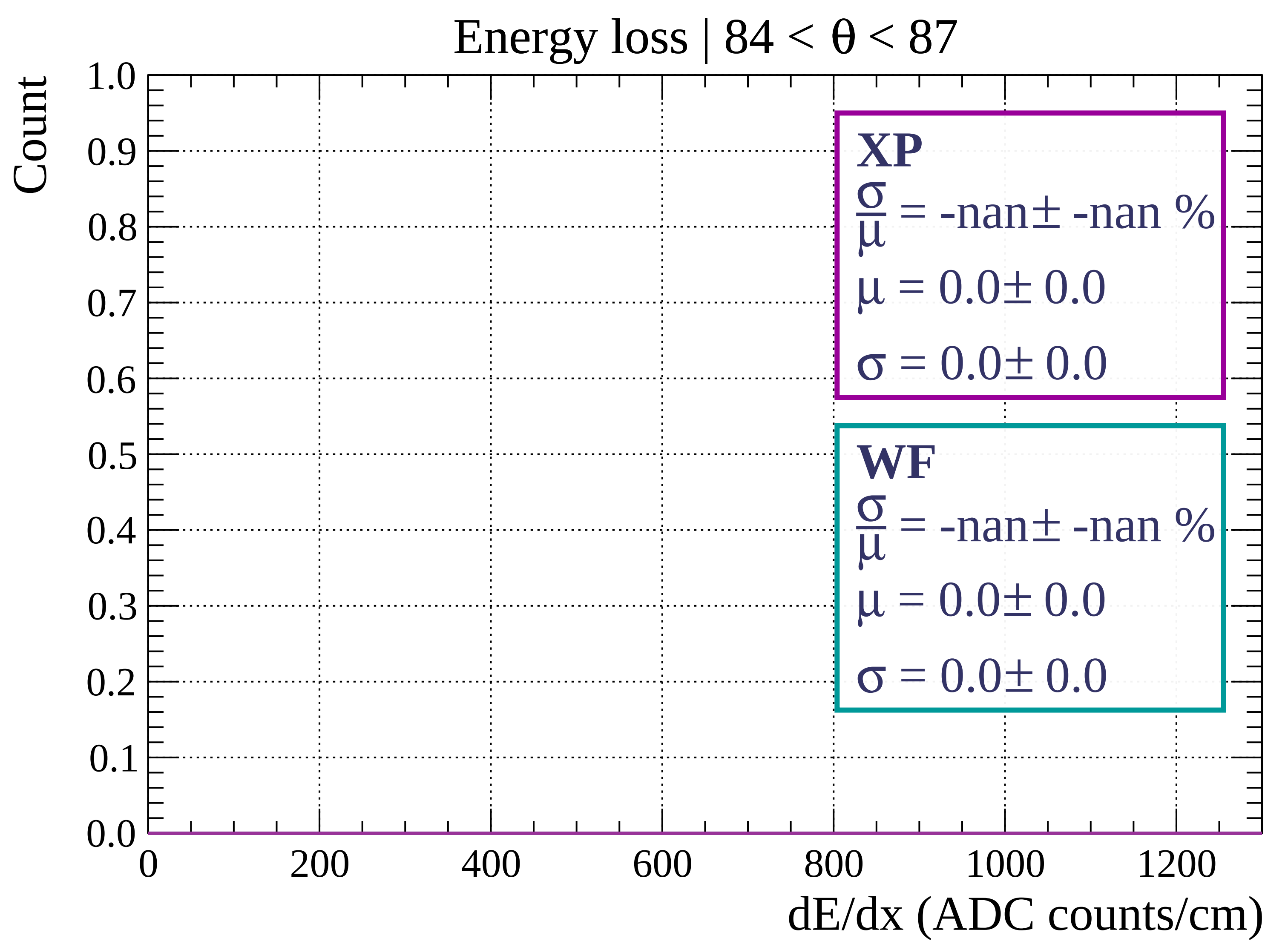
Energy loss | 75 < θ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



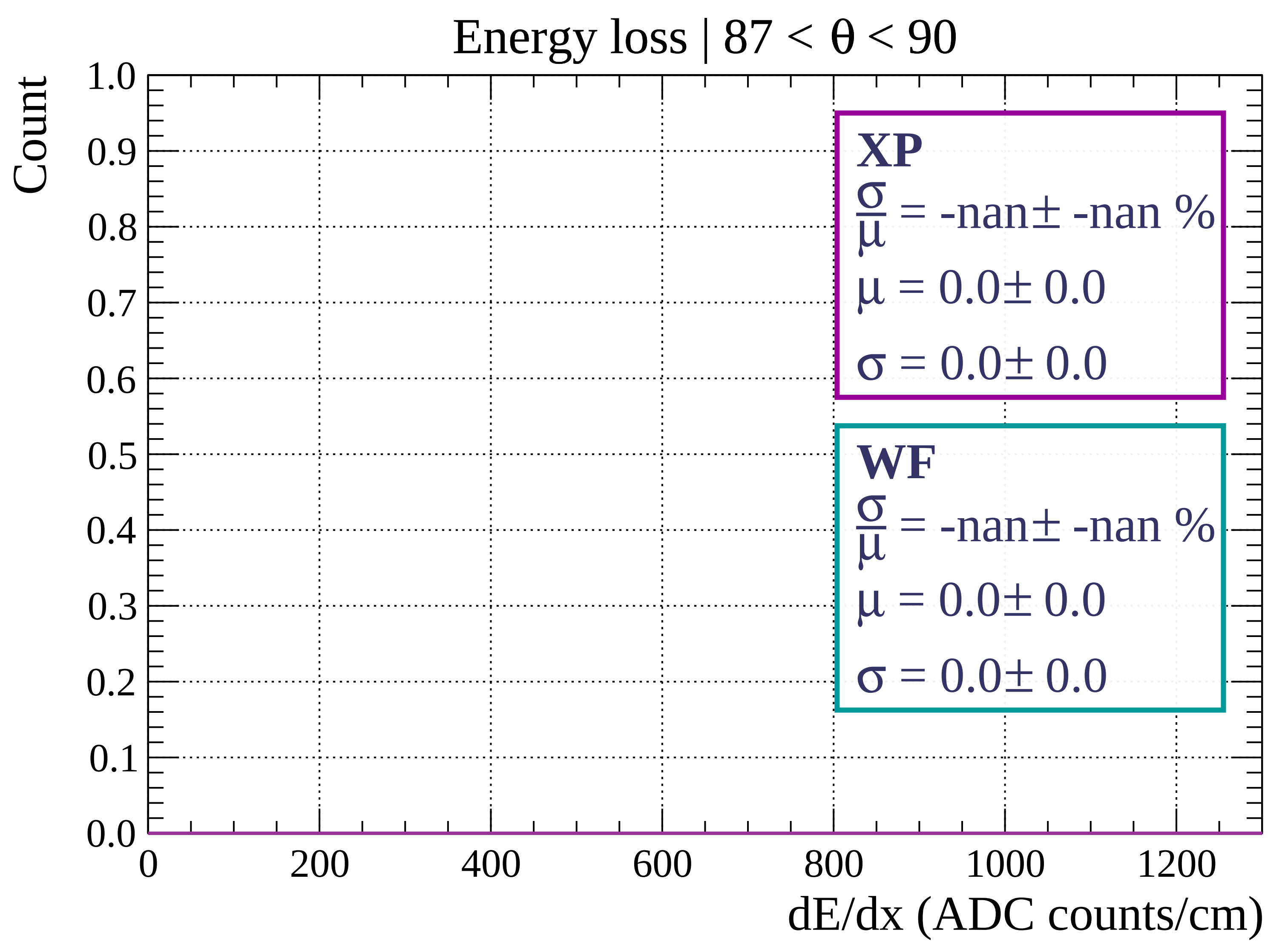
Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

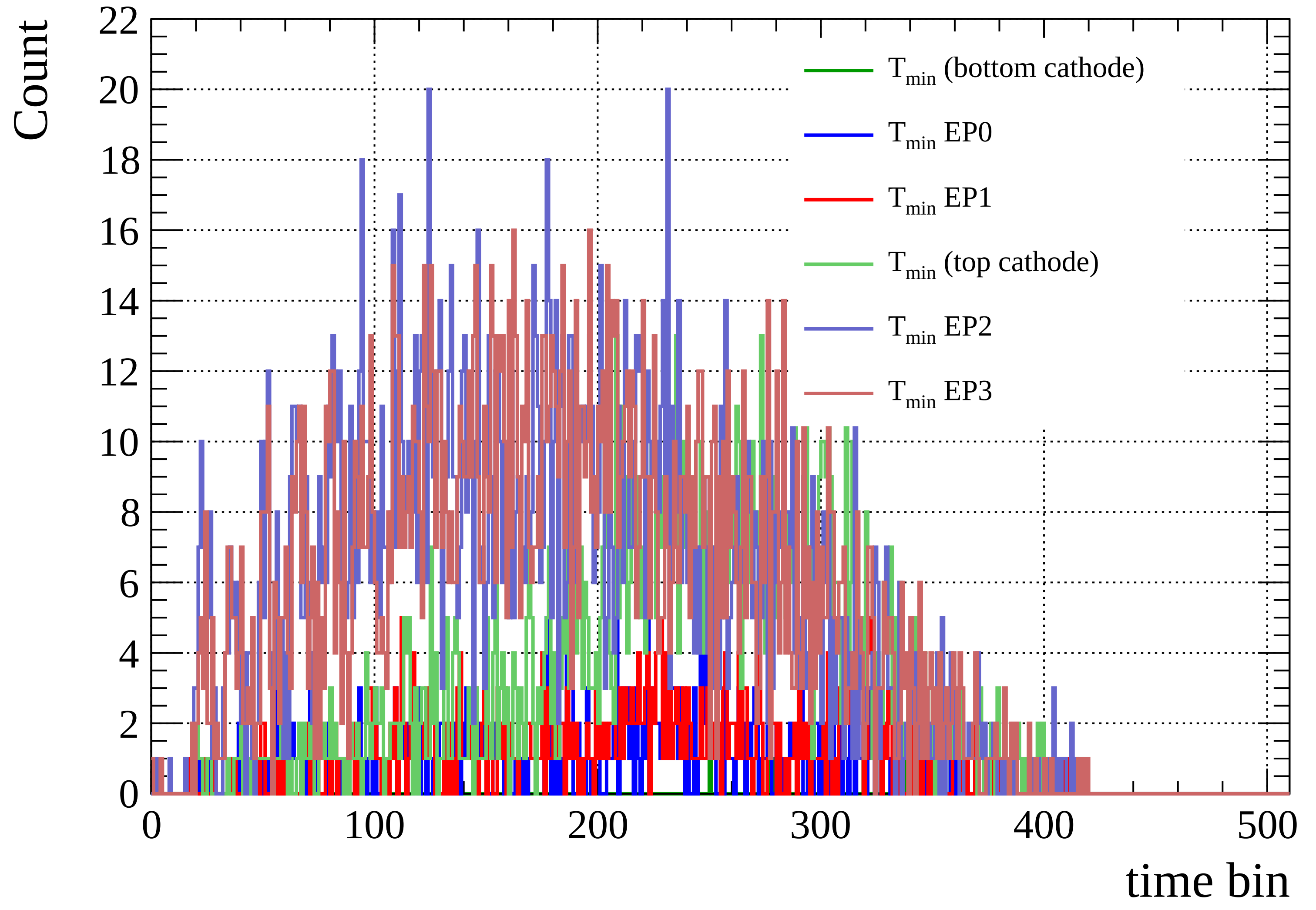


Energy loss | 81 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

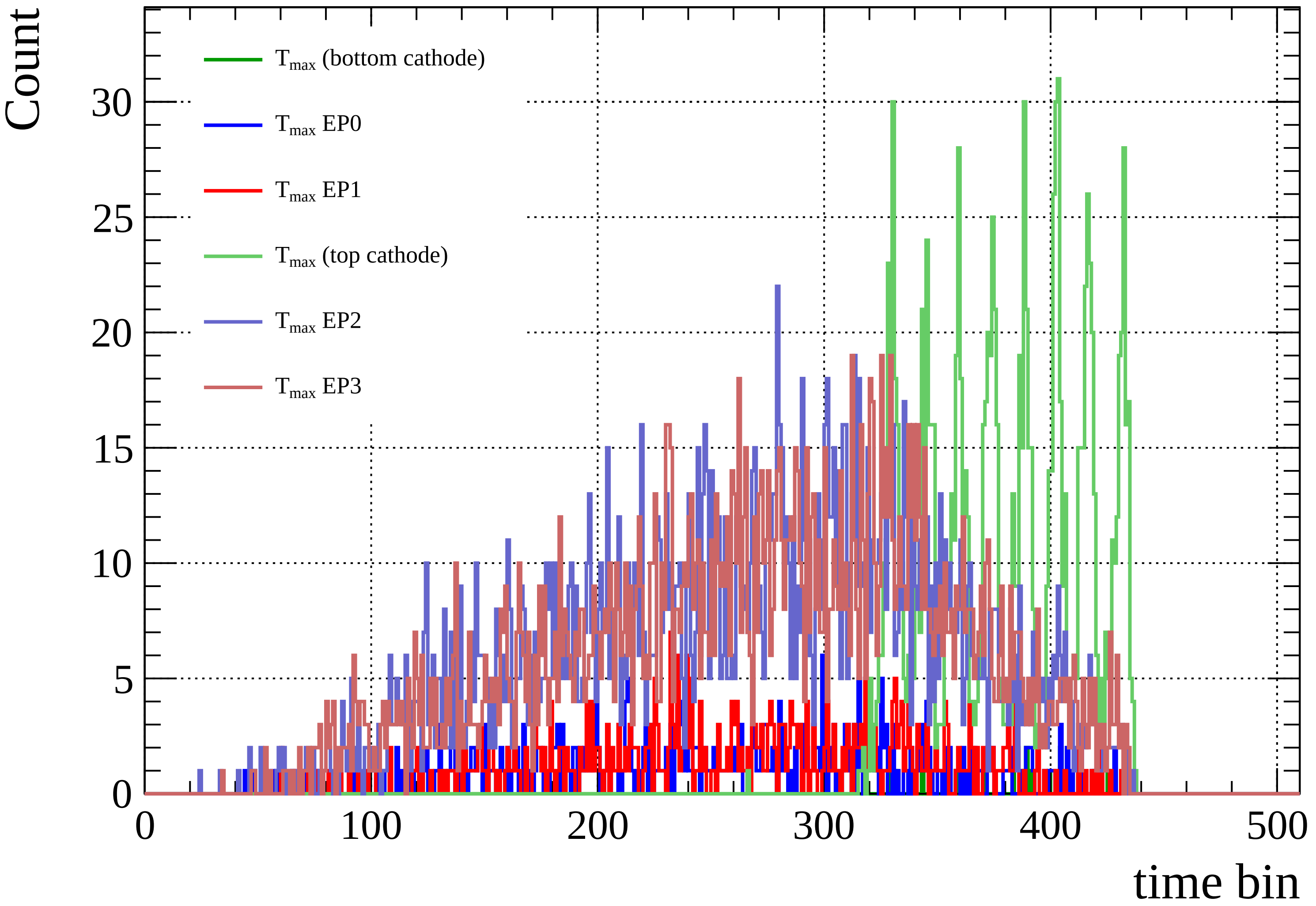


Energy loss | 84 < θ < 87
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 87 < θ < 90
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
2
4
6
8
10
12
14
16
18
20
22
0
100
200
300
400
500


Count
time bin
0
100
200
300
400
500
0
5
10
15
20
25
30
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3